

Fermi Notes on Physics

Volume Four

Introductory Undergraduate
Lectures on Mathematical
Methods for Physics

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Foundations of Mathematical Language

1.1 Series, Limits, and Convergence

Often, we will have occasion in physics to replace a quantity by a sum of simpler pieces, such as the first few terms of a Taylor series, and then ask whether the replacement improves as more pieces are added. In this section we first recall the sums and expansions you met at school, then make precise what it means for a sum with infinitely many terms to have a value, and finally use the answer to find the range of x over which the familiar Taylor series can be used.

Sums and Expansions from School

An ordered list of numbers $u_1, u_2, u_3, \dots$ is called a sequence, and the sum of the terms of a sequence is called a series. An arithmetic sequence has a constant difference d between successive terms, and the sum of its first N terms, starting from a , is

$$S_N = \sum_{n=0}^{N-1} (a + nd) = \frac{N}{2} [2a + (N-1)d] . \quad (1.1)$$

A geometric sequence, on the other hand, has a constant ratio r

between successive terms, and the sum of its first N terms is

$$S_N = \sum_{n=0}^{N-1} ar^n = a \frac{1 - r^N}{1 - r}, \quad r \neq 1, \quad (1.2)$$

a result you should be familiar with from school, which follows from subtracting rS_N from S_N , since every term except a and ar^N cancels. When $|r| < 1$ the power r^N becomes smaller as N grows, and the sum of infinitely many terms is $a/(1 - r)$.

For a positive integer p , the binomial expansion is

$$(1 + x)^p = \sum_{k=0}^p \binom{p}{k} x^k, \quad \binom{p}{k} = \frac{p!}{k!(p-k)!}, \quad (1.3)$$

and for any other real power p , negative or fractional, it becomes a series without end,

$$(1 + x)^p = 1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \dots, \quad (1.4)$$

which holds for $|x| < 1$. Both expansions are special cases of the Taylor series of a function f about the point $x = a$,

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n, \quad (1.5)$$

in which $f^{(n)}$ is the n th derivative of f and $f^{(0)}$ is f itself. A Taylor series about $a = 0$ is also called a Maclaurin series. At school these series are used as approximations, keeping the first few terms when $x - a$ is small, as in the approximation $\sin \theta \approx \theta$ for a pendulum swinging through small angles. What remains open is when a series with infinitely many terms has a sum, and whether the sum of a Taylor series is equal to the function it came from.

Limits of Sequences

A sequence u_n is said to tend to the limit L if, for every positive number ϵ , however small, there is a number N such that

$$|u_n - L| < \epsilon \quad \text{for every } n > N. \quad (1.6)$$

In words, from some point onwards every term lies within a band of half-width ϵ around L , however narrow the band is made. We write $u_n \rightarrow L$, or $\lim_{n \rightarrow \infty} u_n = L$. As an example, consider the

sequence $u_n = n/(n+1)$. Its distance from 1 is $1/(n+1)$, which is less than ϵ whenever $n > 1/\epsilon$, and therefore $u_n \rightarrow 1$. A sequence that has no limit, such as $u_n = (-1)^n$, is said to diverge.

We will need one further property of the real numbers: a sequence whose terms never decrease, and which never exceeds some fixed number, has a limit. The reason is that the real numbers have no gaps, so the terms crowd towards the smallest number that none of them exceeds.

Convergence of a Series

Let

$$S_N = \sum_{n=1}^N u_n \quad (1.7)$$

be the sum of the first N terms of a sequence, which is called the N th partial sum. The series $\sum u_n$ is said to converge to the sum S if the partial sums tend to S , and is said to diverge otherwise. The geometric series converges to $a/(1-r)$ when $|r| < 1$, and diverges when $|r| \geq 1$.

If a series converges, its terms must tend to zero, because $u_N = S_N - S_{N-1}$, and S_N and S_{N-1} tend to the same limit. The converse is false, and the standard example is the harmonic series,

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots \quad (1.8)$$

In Figure 1 the terms of the harmonic series fall towards zero, and yet the partial sums keep rising, more and more slowly. To show that they rise without limit, we group the terms as follows:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \left(\frac{1}{9} + \cdots + \frac{1}{16}\right) + \cdots \quad (1.9)$$

The bracket that ends at $1/2^k$ contains 2^{k-1} terms, each at least $1/2^k$, so the terms in the bracket add up to at least $\frac{1}{2}$, as Figure 2 shows. Counting the first two terms and the $k-1$ brackets up to $1/2^k$, we find

$$S_{2^k} \geq 1 + \frac{k}{2}, \quad (1.10)$$

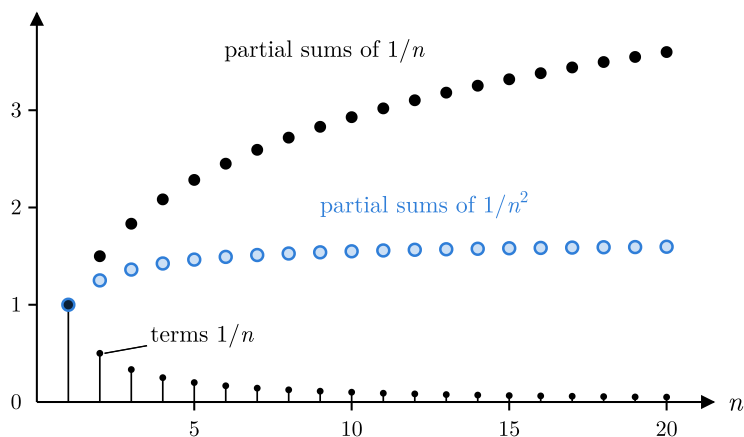


Figure 1. The terms $1/n$ of the harmonic series, drawn as stems, with the partial sums of the harmonic series (black dots) and of the series $\sum 1/n^2$ (blue circles). The terms of both series tend to zero. The partial sums of $\sum 1/n^2$ level off towards a limit, while those of the harmonic series keep rising.

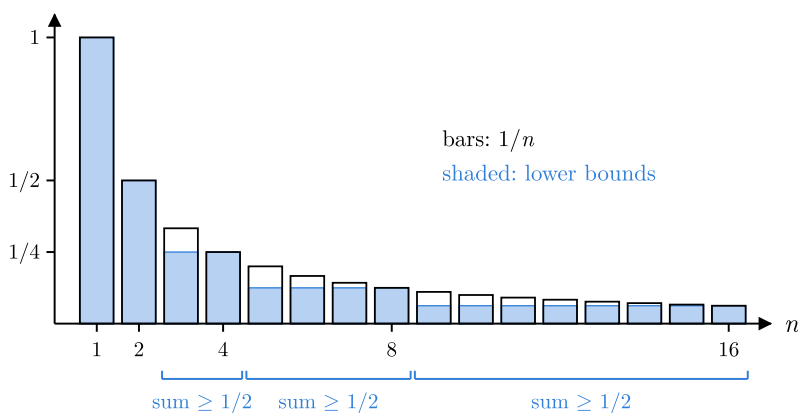


Figure 2. The grouping argument for the harmonic series. Each bar has height $1/n$. The first two bars are shaded in full, and for $n \geq 3$ the shaded bar inside each bar has the height of the last term of its group. The groups end at $n = 4, 8, 16, \dots$, and in each group the shaded bars add up to exactly $\frac{1}{2}$.

which exceeds any given number once k is large enough, and the harmonic series therefore diverges.

Tests for Convergence

Let $\sum u_n$ and $\sum v_n$ be two series whose terms are positive or zero, with $u_n \leq v_n$ for every n . The comparison test states that if $\sum v_n$ converges, then so does $\sum u_n$, and that if $\sum u_n$ diverges, then so does $\sum v_n$. The first statement holds because the partial sums of $\sum u_n$ never decrease and never exceed the sum of $\sum v_n$, so by the property of the real numbers given above they have a limit. The second statement follows from the first, since if $\sum v_n$ converged, then $\sum u_n$ would converge as well.

As an example, consider the series $\sum 1/n^2$, whose partial sums level off in Figure 1. Since $n+1 \leq 2n$, we have

$$\frac{1}{n^2} \leq \frac{2}{n(n+1)} = 2 \left(\frac{1}{n} - \frac{1}{n+1} \right), \quad (1.11)$$

where the equality can be checked by putting the bracket over a common denominator. In the sum of the right-hand side from $n = 1$ to $n = N$, the term $-1/(n+1)$ of each bracket cancels the term $1/(n+1)$ of the next bracket, and the sum is $2[1 - 1/(N+1)]$, which tends to 2. The series $\sum 1/n^2$ therefore converges, by comparison with the series $\sum 2/[n(n+1)]$, and its sum is at most 2. In the other direction, $1/\sqrt{n} \geq 1/n$, and the series $\sum 1/\sqrt{n}$ diverges by comparison with the harmonic series.

A second test for convergence compares a series with a geometric series. The modulus $|u_n|$ of a term is its size regardless of its sign. Let

$$q = \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right|. \quad (1.12)$$

The ratio test states that the series $\sum u_n$ converges if $q < 1$ and diverges if $q > 1$. To see why, suppose that $q < 1$ and choose a number r with $q < r < 1$. From some term u_M onwards the ratio of successive moduli is less than r , so that $|u_n| < |u_M| r^{n-M}$ for $n > M$. Leaving out the first M terms does not affect whether a series converges, so the series of moduli converges by comparison with a geometric series, and we show in the next paragraph that

the series itself then converges. If $q > 1$, the moduli of the terms eventually grow, and the terms cannot tend to zero. When $q = 1$ the test gives no answer. The harmonic series and the series $\sum 1/n^2$ both have $q = 1$, and the harmonic series diverges while the series $\sum 1/n^2$ converges.

When the terms of a series have both signs, the series is said to converge absolutely if the series of moduli $\sum |u_n|$ converges. An absolutely convergent series converges, because $0 \leq u_n + |u_n| \leq 2|u_n|$, so that $\sum(u_n + |u_n|)$ converges by the comparison test, and subtracting $\sum |u_n|$ from it leaves $\sum u_n$. The ratio test, which is stated in terms of moduli, tests for absolute convergence. A series that converges, but not absolutely, is said to converge conditionally. The terms of an absolutely convergent series may be regrouped or taken in any order without changing the sum it evaluates to.

For series whose terms alternate in sign there is a separate test. Let b_n be positive numbers that decrease and tend to zero. The alternating-series test states that the series

$$\sum_{n=1}^{\infty} (-1)^{n+1} b_n = b_1 - b_2 + b_3 - b_4 + \cdots \quad (1.13)$$

converges. The partial sum $S_{2N} = (b_1 - b_2) + (b_3 - b_4) + \cdots + (b_{2N-1} - b_{2N})$ never decreases as N grows, since each bracket is positive or zero, and grouping the same terms as $b_1 - (b_2 - b_3) - \cdots - b_{2N}$ shows that it never exceeds b_1 . The even partial sums therefore have a limit S , and the odd partial sums $S_{2N+1} = S_{2N} + b_{2N+1}$ have the same limit, because $b_{2N+1} \rightarrow 0$. The alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \cdots$ converges by this test, while its series of moduli is the harmonic series, so it converges conditionally. Its sum is $\ln 2$, as you will show in Problem 4.

Power Series

A series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots, \quad (1.14)$$

in which the coefficients c_n are constants, is called a power series in x . The ratio test gives $q = |x| \lim |c_{n+1}/c_n|$, so the series converges

absolutely when $|x| < R$ and diverges when $|x| > R$, where

$$R = \lim_{n \rightarrow \infty} \left| \frac{c_n}{c_{n+1}} \right|. \quad (1.15)$$

The number R is called the radius of convergence. At the points $x = \pm R$ the ratio test gives no answer, and the series has to be examined by another test. A power series in $x - a$ converges in the same way for $|x - a| < R$.

For the Maclaurin series of the exponential, $c_n = 1/n!$, the ratio $c_n/c_{n+1} = n + 1$ grows without limit, and the series converges for every x . For the series of $\ln(1 + x)$, whose coefficients are $c_n = (-1)^{n+1}/n$ for $n \geq 1$, the ratio $|c_n/c_{n+1}| = (n + 1)/n$ tends to 1, so $R = 1$. At $x = 1$ this series is the alternating harmonic series, which converges, and at $x = -1$ it is minus the harmonic series, which diverges. The binomial series with a power p that is not a positive integer has $|c_n/c_{n+1}| = (n + 1)/|p - n|$, which also tends to 1. The standard series, with their ranges of validity, are

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, & \text{all } x, \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots, & \text{all } x, \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, & \text{all } x, \\ \ln(1 + x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots, & -1 < x \leq 1. \end{aligned} \quad (1.16)$$

A reader might expect the series for $\ln(1 + x)$ to fail only near $x = -1$, where the logarithm becomes infinite, and to work for large positive x , where the logarithm rises smoothly. Figure 3 shows otherwise. A power series converges in an interval centred on the origin, and since the point $x = -1$ limits the radius of convergence to 1, the series diverges at $x = 2$ just as it does at $x = -2$.

Taylor's Theorem

The ratio test tells us where a Taylor series converges, but not whether its sum is equal to the function it came from, and for that we need the error made in stopping the series after a finite number

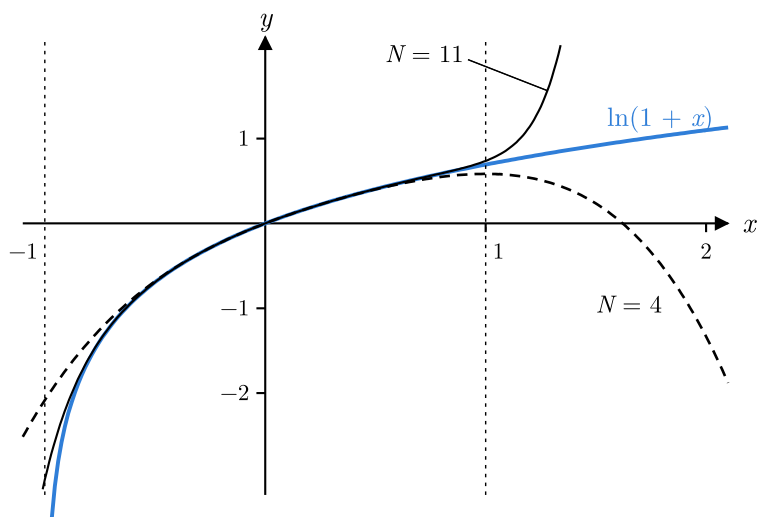


Figure 3. The function $\ln(1+x)$ and the sums of the first N terms of its Maclaurin series, for $N=4$ (dashed) and $N=11$. Inside the interval $-1 < x < 1$ the sum with more terms lies closer to the function. Beyond $x=1$ the sums move away from the function, and the sum with more terms moves away faster.

of terms. Taylor's theorem states that if f has $N + 1$ derivatives on an interval containing a and x , then

$$\begin{aligned} f(x) &= \sum_{n=0}^N \frac{f^{(n)}(a)}{n!} (x-a)^n + E_N(x), & E_N(x) \\ &= \frac{f^{(N+1)}(\xi)}{(N+1)!} (x-a)^{N+1}, \end{aligned} \quad (1.17)$$

for some number ξ between a and x . The expression for E_N is called the Lagrange form of the remainder. It has the form of the next term of the series, with the derivative evaluated at an unknown point ξ in place of a , and so its modulus is at most the largest value of $|f^{(N+1)}|$ between a and x , multiplied by $|x-a|^{N+1}/(N+1)!$.

Consider, for instance, the exponential, about $a = 0$. Every derivative of e^x is e^x , and $e^\xi \leq e^{|x|}$ for ξ between 0 and x , so

$$|E_N(x)| \leq e^{|x|} \frac{|x|^{N+1}}{(N+1)!}. \quad (1.18)$$

As N increases by one, the bound is multiplied by $|x|/(N+2)$, which is eventually less than $\frac{1}{2}$, so the bound tends to zero, and the Maclaurin series of the exponential converges to e^x for every x . The same argument applies to the sine and the cosine, whose derivatives never exceed 1 in modulus. The theorem also bounds the error of an approximation. The approximation $\sin x \approx x$ is the Maclaurin series stopped at $N = 2$, since the term in x^2 is zero, and its remainder is $-\cos \xi x^3/3!$, whose modulus is at most $|x|^3/6$.

Limits of the Form 0/0

When $f(a) = g(a) = 0$, the ratio $f(x)/g(x)$ is not defined at $x = a$, but it may approach a limit as $x \rightarrow a$. Taylor series give such limits directly. For example,

$$\frac{\sin x}{x} = \frac{x - x^3/3! + \dots}{x} = 1 - \frac{x^2}{6} + \dots, \quad (1.19)$$

which tends to 1 as $x \rightarrow 0$. The same idea gives the following general result. If $f(a) = g(a) = 0$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}, \quad (1.20)$$

provided that the limit on the right exists, and this result is called l'Hôpital's rule. When $g'(a) \neq 0$ the reason can be seen from the Taylor series: near a we have $f(x) \approx f'(a)(x-a)$ and $g(x) \approx g'(a)(x-a)$, and the factors $x-a$ cancel in the ratio. If $f'(a)$ and $g'(a)$ are both zero, the rule is applied again to f'/g' . The limit of $(e^x - 1 - x)/x^2$ as $x \rightarrow 0$ provides an example. The numerator and the denominator vanish at $x = 0$, and so do their first derivatives, $e^x - 1$ and $2x$, so we apply the rule twice:

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}, \quad (1.21)$$

in agreement with the Maclaurin series $e^x - 1 - x = x^2/2 + x^3/6 + \dots$.

1.2 Complex Numbers and Phasors

Complex numbers first appeared as a device for solving polynomial equations, and physicists often use them, in part because they make oscillations simple to describe. We start by recalling the complex numbers of the school course, then define the exponential of a complex number with the power series of Section 1.1 and use it to find roots, logarithms, and powers, and finally show how a sum of oscillations of the same frequency can be found by adding complex numbers.

Complex Numbers from School

A complex number is written $z = x + iy$, where x and y are real and $i^2 = -1$. The real number x is called the real part of z , written $\operatorname{Re} z$, and y is called the imaginary part, $\operatorname{Im} z$. Complex numbers are added by adding their real and imaginary parts separately, and they are multiplied by multiplying out the brackets and using $i^2 = -1$, so that

$$(x_1 + iy_1)(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1). \quad (1.22)$$

The number $z^* = x - iy$ is called the complex conjugate of z , and the product $zz^* = x^2 + y^2$ is real and positive unless $z = 0$. Its positive square root $|z|$ is called the modulus of z . A quotient is found by multiplying the numerator and the denominator by the conjugate of the denominator, which gives $z_1/z_2 = z_1z_2^*/|z_2|^2$. A quadratic

equation with real coefficients and a negative discriminant has two complex roots, and these roots are conjugates of each other.

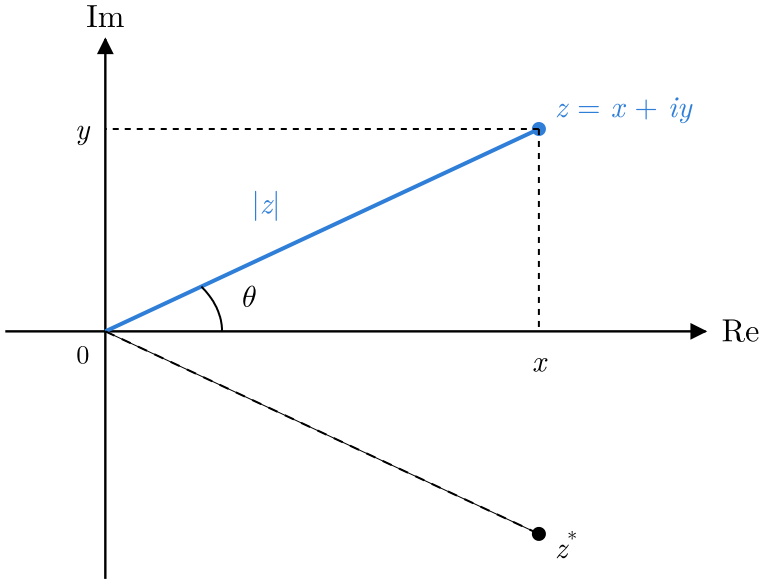


Figure 4. The Argand diagram. The number $z = x + iy$ is the point with coordinates (x, y) , at a distance $|z|$ from the origin and at an angle $\theta = \arg z$ to the positive real axis. Its conjugate z^* is the reflection of z in the real axis.

On the Argand diagram of Figure 4, the modulus $|z| = \sqrt{x^2 + y^2}$ is the distance of the point (x, y) from the origin, and the angle θ between the positive real axis and the line from the origin to z is called the argument of z , written $\arg z$. It satisfies $\tan \theta = y/x$, with the quadrant fixed by the signs of x and y , and the value in the range $-\pi < \theta \leq \pi$ is called its principal value. Since $x = |z| \cos \theta$ and $y = |z| \sin \theta$, the number can be written in the polar form

$$z = |z| (\cos \theta + i \sin \theta) . \quad (1.23)$$

Multiplying two numbers in polar form and using the formulas for the cosine and the sine of a sum of two angles shows that the modulus of a product is the product of the moduli, and its argument is the sum of the arguments.

The Complex Exponential

We define the exponential of a complex number z by the same power series as for a real number,

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots. \quad (1.24)$$

A series of complex terms is said to converge when the series of its real parts and the series of its imaginary parts both converge. The real and imaginary parts of a complex number are no larger in modulus than the number itself, so if $\sum |u_n|$ converges, the series of real parts and the series of imaginary parts both converge absolutely, by the comparison test of Section 1.1. For the exponential series the ratio of successive moduli is $|z|/(n+1)$, which tends to zero, so the series converges absolutely for every complex z .

The exponential of a sum is the product of the exponentials,

$$e^{z_1} e^{z_2} = e^{z_1 + z_2}. \quad (1.25)$$

To show this, we multiply the two series together and collect the terms in which the powers of z_1 and z_2 add up to n , a rearrangement that is allowed for absolutely convergent series. These terms are

$$\sum_{k=0}^n \frac{z_1^k}{k!} \frac{z_2^{n-k}}{(n-k)!} = \frac{1}{n!} \sum_{k=0}^n \binom{n}{k} z_1^k z_2^{n-k} = \frac{(z_1 + z_2)^n}{n!}, \quad (1.26)$$

by the binomial expansion, and summing over n gives the series for $e^{z_1 + z_2}$.

Now let $z = i\theta$, with θ real. The powers of i repeat in the pattern $i, -1, -i, 1$, and so

$$\begin{aligned} e^{i\theta} &= 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} + \cdots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \cdots\right), \end{aligned} \quad (1.27)$$

where the terms have been regrouped, as absolute convergence allows. The brackets are the Maclaurin series of the cosine and the sine, and therefore

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad (1.28)$$

which is called Euler's formula. By $\cos^2 \theta + \sin^2 \theta = 1$, the point $e^{i\theta}$ lies on the circle of unit radius centred on the origin, at argument θ . Any complex number can therefore be written as $z = |z|e^{i\theta}$, and the rule for multiplying numbers in polar form becomes the relation $e^{z_1}e^{z_2} = e^{z_1+z_2}$ for exponentials. Setting $\theta = \pi$ gives $e^{i\pi} = -1$, and setting $\theta = 2\pi$ gives $e^{2\pi i} = 1$, so that $e^{i(\theta+2\pi)} = e^{i\theta}$. Adding and subtracting Euler's formula and its conjugate, $e^{-i\theta} = \cos \theta - i \sin \theta$, gives

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (1.29)$$

De Moivre's Theorem

For a positive integer n , applying the relation $e^{z_1}e^{z_2} = e^{z_1+z_2}$ repeatedly gives $(e^{i\theta})^n = e^{in\theta}$, and since the same relation gives $e^{i\theta}e^{-i\theta} = e^0 = 1$, so that $e^{-i\theta} = 1/e^{i\theta}$, the result holds for any integer n . In terms of the cosine and the sine it is

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta, \quad (1.30)$$

which is called de Moivre's theorem. It turns the cosine and the sine of a multiple of an angle θ into powers of $\cos \theta$ and $\sin \theta$. As an example, consider the case $n = 3$. The binomial expansion of the left-hand side is $\cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$, and equating its real and imaginary parts to $\cos 3\theta$ and $\sin 3\theta$, and using $\cos^2 \theta + \sin^2 \theta = 1$, gives

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta, \quad \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta. \quad (1.31)$$

The exponential forms of the cosine and the sine work in the opposite direction, turning powers into multiple angles, which is the form needed to integrate a power of a cosine. Cubing the expression for $\cos \theta$ gives

$$\cos^3 \theta = \frac{e^{3i\theta} + 3e^{i\theta} + 3e^{-i\theta} + e^{-3i\theta}}{8} = \frac{\cos 3\theta + 3 \cos \theta}{4}, \quad (1.32)$$

in agreement with the formula for $\cos 3\theta$.

Roots of Unity

The complex numbers z that satisfy $z^n = 1$, for a positive integer n , are called the n th roots of unity. With $z = |z|e^{i\theta}$, the equation

becomes $|z|^n e^{in\theta} = 1 = e^{2\pi i k}$ for any integer k , so $|z| = 1$ and $n\theta = 2\pi k$. Let

$$\zeta = e^{2\pi i/n}. \quad (1.33)$$

The roots are then $1, \zeta, \zeta^2, \dots, \zeta^{n-1}$, which correspond to $k = 0, 1, \dots, n-1$. Other values of k give no new roots, because $\zeta^n = 1$, so that $\zeta^{k+n} = \zeta^k$.

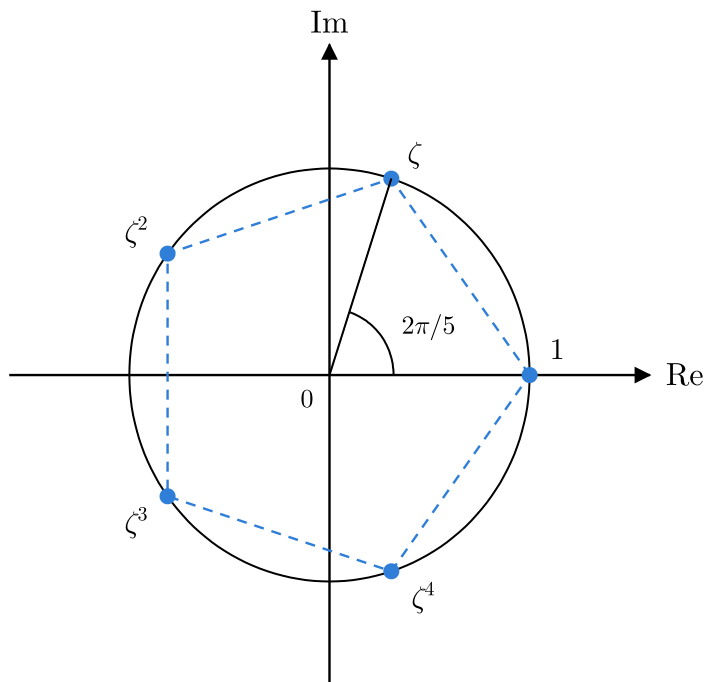


Figure 5. The fifth roots of unity, $1, \zeta, \zeta^2, \zeta^3, \zeta^4$, with $\zeta = e^{2\pi i/5}$. They lie on the unit circle at the corners of a regular pentagon, separated by angles of $2\pi/5$.

The roots lie on the unit circle, separated by angles of $2\pi/n$, at the corners of a regular polygon with one corner at 1, as Figure 5 shows for $n = 5$. For $n = 3$ the roots are 1 and $e^{\pm 2\pi i/3} = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$. For $n \geq 2$ the roots add up to zero, because their sum is a geometric series with ratio $\zeta \neq 1$, and the formula of Section 1.1, whose

derivation holds equally for a complex ratio, gives

$$1 + \zeta + \zeta^2 + \dots + \zeta^{n-1} = \frac{1 - \zeta^n}{1 - \zeta} = 0, \quad (1.34)$$

since $\zeta^n = 1$. The same method finds the n th roots of any complex number, and they too lie at the corners of a regular polygon centred on the origin.

Complex Logarithms and Powers

The logarithm of a complex number z is defined as a complex number w such that $e^w = z$. Let $w = X + iY$, with X and Y real, and let $z = |z|e^{i\theta}$. The equation $e^X e^{iY} = |z|e^{i\theta}$ requires the moduli to agree, which gives $e^X = |z|$, and the arguments to agree up to a multiple of 2π , which gives $Y = \theta + 2\pi k$. Therefore

$$\ln z = \ln |z| + i(\arg z + 2\pi k), \quad k = 0, \pm 1, \pm 2, \dots, \quad (1.35)$$

in which $\ln |z|$ is the ordinary logarithm of a positive number. The logarithm has infinitely many values, differing by multiples of $2\pi i$, because adding $2\pi i$ to w leaves e^w unchanged, and the value with $k = 0$ and $-\pi < \arg z \leq \pi$ is called the principal value. Negative numbers now have logarithms. The principal value of $\ln(-1)$ is $i\pi$, and that of $\ln i$ is $i\pi/2$. The relation $\ln(z_1 z_2) = \ln z_1 + \ln z_2$ holds only up to a multiple of $2\pi i$. With principal values and $z_1 = z_2 = -1$, the left-hand side is $\ln 1 = 0$, while the right-hand side is $2\pi i$.

A complex power is defined through the logarithm,

$$z^\gamma = e^{\gamma \ln z}, \quad (1.36)$$

and substituting the logarithm gives $z^\gamma = e^{\gamma \ln |z| + i\gamma \arg z} e^{2\pi i k \gamma}$, one value for each value of the logarithm, although these values need not all be different. When γ is an integer, the factor $e^{2\pi i k \gamma}$ is 1 and the power has the single value of the ordinary integer power. When $\gamma = 1/n$, the factor $e^{2\pi i k/n}$ is ζ^k , and the power has the n values of the n th roots. As an example, consider i^i . Using $\ln i = i(\pi/2 + 2\pi k)$, we find

$$i^i = e^{i \cdot i(\pi/2 + 2\pi k)} = e^{-\pi/2 - 2\pi k}, \quad (1.37)$$

so every value of i^i is real, and the principal value is $e^{-\pi/2}$.

Hyperbolic Functions

The hyperbolic cosine and the hyperbolic sine are defined as

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad \sinh x = \frac{e^x - e^{-x}}{2}, \quad (1.38)$$

and the function $\tanh x = \sinh x / \cosh x$ is called the hyperbolic tangent. They are the parts of e^x that are even and odd in x , and their Maclaurin series are the even and odd terms of the series for e^x , $\cosh x = 1 + x^2/2! + x^4/4! + \cdots$ and $\sinh x = x + x^3/3! + \cdots$.

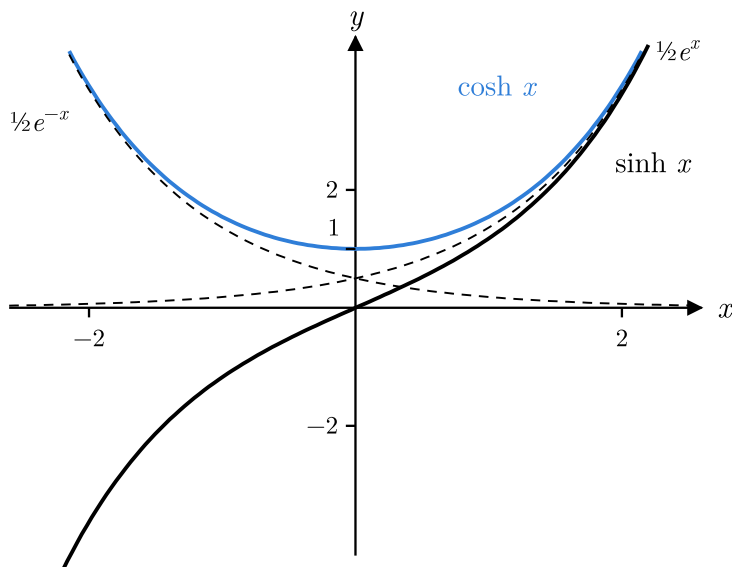


Figure 6. The hyperbolic functions $\cosh x$ and $\sinh x$. For large positive x both functions approach the dashed curve $\frac{1}{2}e^x$. For large negative x , $\cosh x$ approaches the dashed curve $\frac{1}{2}e^{-x}$, and $\sinh x$ approaches $-\frac{1}{2}e^{-x}$, the reflection of that curve in the x -axis.

In Figure 6 the hyperbolic cosine is even, with a minimum value of 1 at $x = 0$, and the hyperbolic sine is odd, passing through the origin with slope 1. For large positive x the term e^{-x} becomes negligible, and both functions approach $\frac{1}{2}e^x$. Squaring the definitions and

subtracting gives

$$\cosh^2 x - \sinh^2 x = \frac{(e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x})}{4} = 1, \quad (1.39)$$

so the point $(\cosh \tau, \sinh \tau)$ moves along the branch with $x \geq 1$ of the hyperbola $x^2 - y^2 = 1$ as τ varies, in the same way that $(\cos \tau, \sin \tau)$ moves round the circle $x^2 + y^2 = 1$, and this is the origin of the name. Differentiating the definitions gives $d(\cosh x)/dx = \sinh x$ and $d(\sinh x)/dx = \cosh x$, without the change of sign that appears in the derivative of the cosine.

The definitions make sense for complex arguments, and so do the exponential forms of the cosine and the sine, which define the cosine and the sine of a complex number. Comparing the two pairs of definitions gives

$$\begin{aligned} \cosh ix &= \cos x, & \sinh ix &= i \sin x, \\ \cos ix &= \cosh x, & \sin ix &= i \sinh x. \end{aligned} \quad (1.40)$$

These relations turn each trigonometric identity into a hyperbolic one. Replacing x by ix in $\cos^2 x + \sin^2 x = 1$, for example, gives $\cosh^2 x + (i \sinh x)^2 = 1$, which is the identity derived above.

Phasors

A mass on a spring, which we encountered when studying simple harmonic motion at school, oscillates with a displacement

$$x(t) = A \cos(\omega t + \alpha), \quad (1.41)$$

where A is the amplitude, ω is the angular frequency, and α is the phase constant. By Euler's formula, $A \cos(\omega t + \alpha)$ is the real part of $Ae^{i(\omega t + \alpha)}$, and so

$$x(t) = \operatorname{Re}(Ce^{i\omega t}), \quad C = Ae^{i\alpha}. \quad (1.42)$$

The constant C is called the complex amplitude, and its modulus is the amplitude A and its argument is the phase constant α . As time passes, the point $Ce^{i\omega t}$ moves round a circle of radius A on the Argand diagram with angular velocity ω , and the displacement is

the projection of this point on the real axis. The rotating complex number $Ce^{i\omega t}$, or often the complex amplitude C alone, is called a phasor.

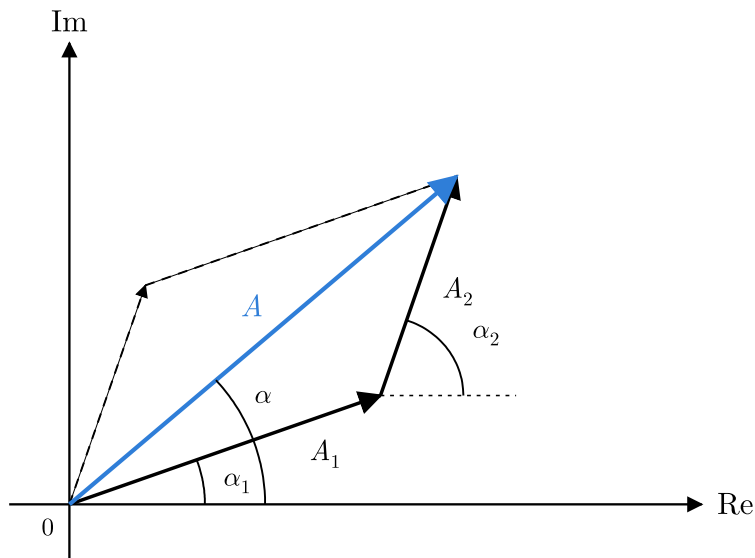


Figure 7. Adding two phasors of the same frequency. The complex amplitudes $A_1e^{i\alpha_1}$ and $A_2e^{i\alpha_2}$ are added as vectors, head to tail, and the resultant has modulus A and argument α . The dashed lines complete the parallelogram.

Consider now two oscillations of the same angular frequency, $x_1 = A_1 \cos(\omega t + \alpha_1)$ and $x_2 = A_2 \cos(\omega t + \alpha_2)$, which could, for example, be the displacements produced at one point by waves from two sources. Their complex amplitudes are $C_1 = A_1e^{i\alpha_1}$ and $C_2 = A_2e^{i\alpha_2}$, and since the real part of a sum is the sum of the real parts,

$$x_1 + x_2 = \text{Re} [(C_1 + C_2)e^{i\omega t}] . \quad (1.43)$$

The sum is an oscillation of the same angular frequency, whose complex amplitude $C = C_1 + C_2$ is found by adding C_1 and C_2 as vectors, as in Figure 7. The phasors $C_1e^{i\omega t}$ and $C_2e^{i\omega t}$ rotate at the same rate, so the diagram drawn at $t = 0$ keeps its shape as it turns. The amplitude $A = |C|$ follows from $|C|^2 = (C_1 + C_2)(C_1^* + C_2^*)$,

which gives

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos(\alpha_1 - \alpha_2), \quad (1.44)$$

where the cross terms $C_1C_2^* + C_1^*C_2$ have been combined using the exponential form of the cosine; this is the cosine rule for the triangle in Figure 7, in which the angle between the sides of lengths A_1 and A_2 is $\pi - (\alpha_2 - \alpha_1)$, whose cosine is $-\cos(\alpha_1 - \alpha_2)$. The phase constant of the sum is given by $\tan \alpha = \text{Im } C / \text{Re } C$, that is

$$\tan \alpha = \frac{A_1 \sin \alpha_1 + A_2 \sin \alpha_2}{A_1 \cos \alpha_1 + A_2 \cos \alpha_2}, \quad (1.45)$$

with the quadrant fixed by the signs of the numerator and the denominator. Oscillations in phase give $A = A_1 + A_2$, and oscillations that differ in phase by π give $|A_1 - A_2|$. For two oscillations of equal amplitude A_1 and phase difference δ , the identity $1 + \cos \delta = 2 \cos^2(\delta/2)$ gives $A = 2A_1 |\cos(\delta/2)|$.

Setting $\theta = \omega t$ in Euler's formula and differentiating with respect to t gives $de^{i\omega t}/dt = -\omega \sin \omega t + i\omega \cos \omega t = i\omega e^{i\omega t}$, so the velocity of the mass is

$$\frac{dx}{dt} = \text{Re}(i\omega C e^{i\omega t}). \quad (1.46)$$

The complex amplitude of the velocity is $i\omega C$, and since multiplying by $i = e^{i\pi/2}$ turns a phasor through a right angle, the velocity is ahead of the displacement by a quarter of a period. Differentiating the velocity with respect to t multiplies the complex amplitude by $i\omega$ once more, so the acceleration is $-\omega^2 x$, and therefore

$$\frac{d^2x}{dt^2} = -\omega^2 x, \quad (1.47)$$

which is the equation of simple harmonic motion.

The complex form can be used for linear operations, such as adding oscillations, multiplying them by real constants, and differentiating or integrating them, with the real part taken at the end. It cannot be used directly for a product, such as the square of the velocity in the kinetic energy, because the product of two real parts is not the real part of the product. You will find the time average of a product of two oscillations in Problem 7.

Problems

§1.1 Series, Limits, and Convergence

1. Use the ratio test to find the values of x for which $\sum_{n=1}^{\infty} nx^n$ converges, and show that the series diverges at both ends of this range.
2. Find the radius of convergence of $\sum_{n=0}^{\infty} n!x^n$ and of $\sum_{n=1}^{\infty} x^n/n^2$. Decide whether the second series converges at $x = R$ and at $x = -R$.
3. Use the comparison test to show that $\sum 1/n^p$ converges for every $p \geq 2$ and diverges for every $p \leq 1$.
4. Use Taylor's theorem with the Lagrange form of the remainder to show that the Maclaurin series of $\ln(1+x)$ converges to $\ln(1+x)$ at $x = 1$, and hence that $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \ln 2$.
5. Show that the sum S of an alternating series that satisfies the conditions of the alternating-series test lies between any two successive partial sums, and hence that stopping after N terms gives an error of at most b_{N+1} .
6. A quantity that decays at a rate λ often appears in the combination $(1 - e^{-\lambda t})/\lambda$. Show that this combination tends to t as $\lambda \rightarrow 0$, first with the Maclaurin series of the exponential and then with l'Hôpital's rule, treating λ as the variable.
7. Use Taylor's theorem to show that, for $x > 0$, the approximation $\sqrt{1+x} \approx 1 + x/2$ has an error of at most $x^2/8$.

§1.2 Complex Numbers and Phasors

1. Use de Moivre's theorem to express $\cos 4\theta$ in powers of $\cos \theta$. Express $\sin^4 \theta$ in terms of $\cos 2\theta$ and $\cos 4\theta$.
2. Find the four solutions of $z^4 = -1$ and draw them on an Argand diagram. By pairing each solution with its conjugate, write $z^4 + 1$ as a product of two quadratics with real coefficients.
3. Show that the product of the n th roots of unity is $(-1)^{n+1}$.

4. Find all the values of $\ln(-i)$ and of $(1+i)^i$, and give their principal values.
5. Starting from the definitions of the hyperbolic functions, show that $d(\tanh x)/dx = 1/\cosh^2 x$. Show that the inverse of the hyperbolic sine is $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$.
6. Waves from N equally spaced sources, which could, for example, be the slits of a diffraction grating, arrive at a point with equal amplitudes A and with phases that increase by δ from one source to the next, so that the displacement there is $\sum_{n=0}^{N-1} A \cos(\omega t + n\delta)$. Sum the complex amplitudes as a geometric series and show that the amplitude of the resultant oscillation is $A |\sin(N\delta/2)/\sin(\delta/2)|$.
7. Let $x_1 = \operatorname{Re}(C_1 e^{i\omega t})$ and $x_2 = \operatorname{Re}(C_2 e^{i\omega t})$. Show that the mean value of the product $x_1 x_2$ over one period $2\pi/\omega$ is $\frac{1}{2} \operatorname{Re}(C_1 C_2^*)$, and hence that the mean value of x_1^2 is $\frac{1}{2} |C_1|^2$.
8. A mass on a spring oscillates with displacement $x = \operatorname{Re}(C e^{i\omega t})$. At $t = 0$ its displacement is x_0 and its velocity is v_0 . Find C , and hence the amplitude of the oscillation.

Calculus of Change and Accumulation

2.1 Differentiation and Integration

The calculus taught at school rests on two operations, differentiation and integration, and on the theorem that establishes one as the inverse of the other. We begin by stating the school results, and then extend them to curves given by an equation between x and y , to derivatives of high order, to integrands that the school methods do not reach, and finally to integrals over an infinite range or of functions that become infinite. The two sections that follow this one carry the same ideas over to functions of more than one variable.

Calculus from School

The derivative of a function $f(x)$ is defined as the limit

$$f'(x) = \frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad (2.1)$$

which is the slope of the graph of f at the point x . The derivatives of the functions used most often are listed in the following table.

Function $f(x)$	Derivative $f'(x)$
x^n	nx^{n-1}
e^x	e^x
$\ln x$	$1/x$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$

Products, quotients, and composite functions are differentiated with the following rules, in which f and g are functions of x :

1. the product rule, $(fg)' = f'g + fg'$;
2. the quotient rule, $(f/g)' = (f'g - fg')/g^2$;
3. the chain rule, $df(g(x))/dx = f'(g(x))g'(x)$.

A point at which $f'(x) = 0$ is called a stationary point. It is a minimum if $f'' > 0$ there and a maximum if $f'' < 0$, and the reason can be seen from the Taylor series of Section 1.1: near a stationary point a , the function is $f(a) + \frac{1}{2}f''(a)(x-a)^2$ plus terms of higher order, and the sign of $f''(a)$ decides whether the function rises or falls on both sides. When $f''(a) = 0$ the test is inconclusive, and the higher terms have to be examined.

Integration and differentiation are inverse operations. If $F'(x) = f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a), \quad (2.2)$$

and the integral on the left is the area under the graph of f between $x = a$ and $x = b$, with area below the axis counted as negative. This result is called the fundamental theorem of calculus. Replacing b by a variable upper limit x and differentiating with respect to x gives

$$\frac{d}{dx} \int_a^x f(t) dt = f(x). \quad (2.3)$$

Integrals whose integrands cannot be recognised as the derivatives of standard functions are often found by substitution, in which a new variable $w = g(x)$ turns $\int f(g(x))g'(x) dx$ into $\int f(w) dw$, or

by integrating by parts,

$$\int_a^b f g' \, dx = [fg]_a^b - \int_a^b f' g \, dx, \quad (2.4)$$

which is the product rule integrated between a and b .

Implicit Differentiation

A curve is often given by an equation between x and y rather than by a formula for y . The circle $x^2 + y^2 = a^2$ is an example. We can find the slope without solving for y , by differentiating both sides of the equation with respect to x and treating y as a function of x . By the chain rule, the derivative of y^2 is $2y \, dy/dx$, and so

$$2x + 2y \frac{dy}{dx} = 0, \quad \frac{dy}{dx} = -\frac{x}{y}. \quad (2.5)$$

On the upper half of the circle, where $y = \sqrt{a^2 - x^2}$, differentiating directly gives $-x/\sqrt{a^2 - x^2}$, which agrees with $-x/y$. The result also says that the tangent at (x, y) is perpendicular to the radius, whose slope is y/x . This method is called implicit differentiation.

The same method gives the derivatives of inverse functions. Let $y = \arcsin x$, so that $x = \sin y$, with y between $-\pi/2$ and $\pi/2$. Differentiating $x = \sin y$ with respect to x gives $1 = \cos y \, dy/dx$, and therefore

$$\frac{d}{dx} \arcsin x = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - x^2}}, \quad (2.6)$$

where the positive square root is taken because $\cos y \geq 0$ in the range of y . Similarly, if $y = \sinh^{-1} x$, then $1 = \cosh y \, dy/dx$, and the identity $\cosh^2 y - \sinh^2 y = 1$ of Section 1.2 gives $dy/dx = 1/\sqrt{1 + x^2}$.

A product of several factors, or a function raised to a power that depends on x , is often easier to differentiate after taking its logarithm, a method called logarithmic differentiation. As an example, consider $y = x^x$, for $x > 0$. Then $\ln y = x \ln x$, and differentiating both sides with respect to x gives

$$\frac{1}{y} \frac{dy}{dx} = \ln x + 1, \quad \frac{dy}{dx} = x^x (\ln x + 1). \quad (2.7)$$

For a product $y = f_1 f_2 \cdots f_n$, the logarithm $\ln y = \ln f_1 + \ln f_2 + \cdots + \ln f_n$ is a sum, and differentiating both sides with respect to x gives $y'/y = f'_1/f_1 + f'_2/f_2 + \cdots + f'_n/f_n$.

Leibniz's Rule

Differentiating a product twice with the product rule gives $(fg)'' = f''g + 2f'g' + fg''$, and the coefficients 1, 2, 1 are those of the binomial expansion of $(1+x)^2$. The pattern continues, and the n th derivative of a product is

$$\frac{d^n}{dx^n}(fg) = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}, \quad (2.8)$$

which is called Leibniz's rule. We prove it by induction on n . For $n = 1$ it is the product rule. Suppose that it holds for some value of n . We differentiate both sides of the rule once more with respect to x . By the product rule, each term $f^{(k)} g^{(n-k)}$ gives $f^{(k+1)} g^{(n-k)} + f^{(k)} g^{(n-k+1)}$, and the term $f^{(k)} g^{(n+1-k)}$ in the result therefore arrives with the coefficient

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}, \quad (2.9)$$

where the equality, which is the rule for building Pascal's triangle, follows from writing the binomial coefficients on the left as factorials and putting them over a common denominator. This is Leibniz's rule with $n+1$ in place of n , and so the rule holds for every n .

Leibniz's rule is most useful when one of the factors is a polynomial, since only a few of its derivatives are non-zero. Consider, for instance, $x^2 e^x$. With $f = x^2$ and $g = e^x$, only the terms with $k = 0$, 1, and 2 are non-zero, and every derivative of g is e^x , so

$$\frac{d^n}{dx^n}(x^2 e^x) = e^x [x^2 + 2nx + n(n-1)]. \quad (2.10)$$

Partial Fractions

Let $N(x)/D(x)$ be a ratio of two polynomials, with N of lower degree than D ; if it is not, dividing D into N first leaves a polynomial plus a ratio of this kind. A polynomial of degree n has n complex roots, counted with multiplicity, a result called the fundamental theorem of algebra, whose proof is beyond the scope of this course.

When the coefficients of D are real, taking the complex conjugate of $D(z) = 0$ gives $D(z^*) = 0$, so the complex roots come in conjugate pairs, and each pair gives a real quadratic factor. The polynomial D can therefore be factorised into real linear factors and real quadratic factors with no real roots, and the ratio can be written as a sum of simpler fractions of the following kinds:

1. a linear factor $x - a$ gives a term $A/(x - a)$;
2. a repeated factor $(x - a)^m$ gives terms $A_1/(x - a) + A_2/(x - a)^2 + \dots + A_m/(x - a)^m$;
3. a quadratic factor $x^2 + \beta x + \gamma$ with no real roots gives a term $(Bx + C)/(x^2 + \beta x + \gamma)$, and a repeated quadratic factor $(x^2 + \beta x + \gamma)^m$ gives m such terms, with the powers $1, 2, \dots, m$ of the quadratic in their denominators, in the same pattern as the second kind.

The constants are found by multiplying through by D and comparing the coefficients of each power of x , or by substituting values of x at which all but one of the terms vanish, and the sum is called the partial-fraction expansion of the ratio.

As an example, consider the integral of $1/(x^2 - a^2)$. The denominator is $(x - a)(x + a)$, and we write

$$\frac{1}{x^2 - a^2} = \frac{A}{x - a} + \frac{B}{x + a}, \quad 1 = A(x + a) + B(x - a). \quad (2.11)$$

The second equation holds for every x . Setting $x = a$ gives $A = 1/(2a)$, and setting $x = -a$ gives $B = -1/(2a)$. Each term integrates to a logarithm, and therefore

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right| + \text{constant}. \quad (2.12)$$

As a second example, consider $1/[x(x^2 + a^2)]$, whose denominator has a quadratic factor with no real roots. We write it as $A/x + (Bx + C)/(x^2 + a^2)$, and multiplying through gives $1 = A(x^2 + a^2) + (Bx + C)x$. Setting $x = 0$ gives $A = 1/a^2$, the coefficient of x^2 gives $A + B = 0$, and the coefficient of x gives $C = 0$. The substitution $w = x^2 + a^2$ turns $\int x dx/(x^2 + a^2)$ into $\frac{1}{2} \ln(x^2 + a^2)$, and so

$$\int \frac{dx}{x(x^2 + a^2)} = \frac{1}{a^2} \left[\ln |x| - \frac{1}{2} \ln(x^2 + a^2) \right] + \text{constant}. \quad (2.13)$$

Had the constant C been non-zero, its term would have given $(C/a) \arctan(x/a)$, which follows from the substitution $x = a \tan w$.

Reduction Formulas

Some integrals depend on a whole number n , and integrating by parts relates the integral for n to the integral for a smaller value of n . A relation of this kind is called a reduction formula. As an example, consider

$$K_n = \int_0^{\pi/2} \sin^n x \, dx. \quad (2.14)$$

Let us integrate by parts, differentiating $\sin^{n-1} x$ and integrating $\sin x$ to $-\cos x$. For $n \geq 2$ the boundary term $\left[-\sin^{n-1} x \cos x\right]_0^{\pi/2}$ vanishes at both limits, and

$$K_n = (n-1) \int_0^{\pi/2} \sin^{n-2} x \cos^2 x \, dx = (n-1) (K_{n-2} - K_n), \quad (2.15)$$

where we have used $\cos^2 x = 1 - \sin^2 x$. Solving for K_n gives

$$K_n = \frac{n-1}{n} K_{n-2}. \quad (2.16)$$

Starting from $K_0 = \pi/2$ and $K_1 = 1$, the reduction formula gives $K_2 = \pi/4$, $K_3 = 2/3$, $K_4 = 3\pi/16$, and so on.

Improper Integrals

The fundamental theorem of calculus applies to a function that is finite on a finite range. An integral over an infinite range, or of a function that becomes infinite somewhere in the range, is called an improper integral, and it is defined as a limit of ordinary integrals. For an infinite range,

$$\int_a^\infty f(x) \, dx = \lim_{b \rightarrow \infty} \int_a^b f(x) \, dx, \quad (2.17)$$

and for a function that becomes infinite at the lower limit, the integral from a to b is defined as the limit of the integral from $a + \delta$ to b as the positive number δ tends to zero. The integral is said to converge if the limit exists and to diverge otherwise. A

function that becomes infinite at a point inside the range is treated by splitting the range at that point.

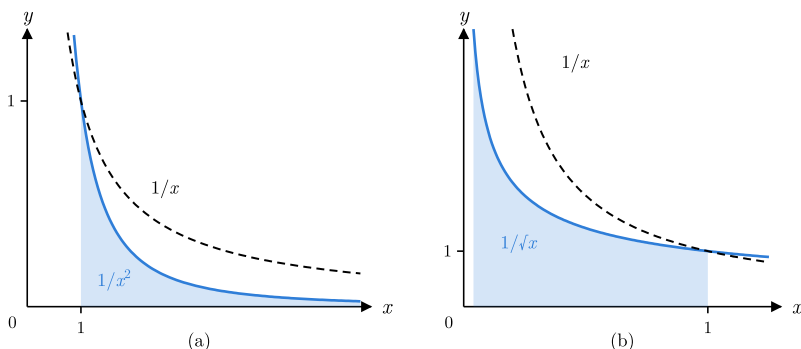


Figure 8. (a) The area under $1/x^2$ from $x = 1$ to infinity is finite, while the area under $1/x$ is infinite. (b) The area under $1/\sqrt{x}$ from $x = 0$ to $x = 1$ is finite, while the area under $1/x$ is infinite.

The functions x^{-p} for $p > 0$, some of which are drawn in Figure 8, provide an example. For $p \neq 1$,

$$\int_1^b x^{-p} dx = \frac{b^{1-p} - 1}{1-p}, \quad (2.18)$$

and for $p = 1$ the integral is $\ln b$. As $b \rightarrow \infty$, the power b^{1-p} tends to zero when $p > 1$ and grows without limit when $p < 1$, and $\ln b$ grows without limit. The integral of x^{-p} from 1 to infinity therefore converges, to $1/(p-1)$, when $p > 1$, and diverges when $p \leq 1$. Near the origin the situation is reversed. The integral of x^{-p} from δ to 1 is $(1 - \delta^{1-p})/(1-p)$, which tends to $1/(1-p)$ as $\delta \rightarrow 0$ when $p < 1$, while for $p \geq 1$ the integral from 0 to 1 diverges. The function $1/x$ lies on the borderline for the infinite range and for the range that starts at the origin, and its area is infinite in each case.

An improper integral of this kind arises in school physics. The work done by an external force in bringing a charge q from infinity to a distance r from a point charge Q , against the repulsion given by Coulomb's law, is

$$-\int_{\infty}^r \frac{Qq}{4\pi\epsilon_0 x^2} dx = \frac{Qq}{4\pi\epsilon_0 r}, \quad (2.19)$$

which is finite because the force goes as $1/x^2$, and this is why the

zero of the electric potential energy of the two charges can be placed at infinity.

The comparison test of Section 1.1 has a counterpart for integrals. If $0 \leq f(x) \leq g(x)$ for $x \geq a$, and the integral of g from a to infinity converges, then so does the integral of f . The reason is the same as for series: the integral of f from a to b increases with b and never exceeds the integral of g , so it has a limit. Consider, for instance, the integral of e^{-x^2} from 0 to infinity, which cannot be found from an indefinite integral in terms of standard functions. For $x \geq 1$ we have $x^2 \geq x$, and so $e^{-x^2} \leq e^{-x}$, whose integral from 1 to infinity is e^{-1} . The integral of e^{-x^2} from 1 to infinity therefore converges, and since the integral from 0 to 1 is an ordinary integral of a finite function, so does the integral from 0 to infinity.

As a second example, consider

$$J_n = \int_0^\infty x^n e^{-x} dx, \quad (2.20)$$

for a whole number n . Every term of the Maclaurin series of e^x is positive for $x > 0$, so e^x is greater than any one of those terms, and in particular $e^x > x^{n+3}/(n+3)!$. Therefore $x^n e^{-x} < (n+3)!/x^3$, which is less than $1/x^2$ once $x > (n+3)!$. The integral from $(n+3)!$ to infinity therefore converges by comparison with the integral of $1/x^2$, and since the integral from 0 to $(n+3)!$ is an ordinary integral of a finite function, so does the integral from 0 to infinity. Integrating by parts, with x^n differentiated and e^{-x} integrated, gives

$$J_n = \left[-x^n e^{-x} \right]_0^\infty + n \int_0^\infty x^{n-1} e^{-x} dx = n J_{n-1}, \quad (2.21)$$

since the boundary term vanishes at both limits for $n \geq 1$. With $J_0 = 1$, this reduction formula gives $J_n = n!$.

An improper integral has to be recognised as one before the fundamental theorem is used. The function $1/x^2$ is positive everywhere, and yet applying the fundamental theorem to it between -1 and 1 gives $[-1/x]_{-1}^1 = -2$. The error is that $1/x^2$ becomes infinite at $x = 0$, inside the range. Splitting the range there, the integral from δ to 1 is $1/\delta - 1$, which grows without limit as $\delta \rightarrow 0$, and the integral diverges.

2.2 Partial Differentiation

Many quantities in physics depend on more than one variable, such as the pressure of a fixed amount of gas, which depends on its volume and its temperature. To extend the derivative of Section 2.1 to such functions, we first define partial derivatives and the total differential, then use them to change variables and to find and classify stationary points, and finally differentiate an integral with respect to a parameter that appears inside it.

Partial Derivatives

A function $f(x, y)$ assigns a number to each point of a region of the plane. Its graph is the surface $z = f(x, y)$, which is often drawn instead as a set of contours, the curves on which f is constant, as a map shows a hill.

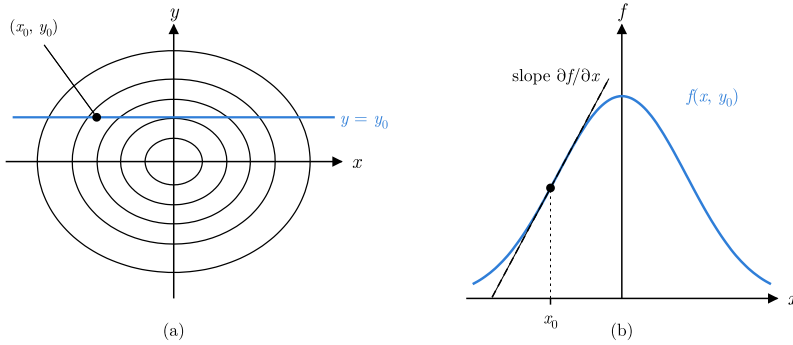


Figure 9. (a) Contours of a function $f(x, y)$ that has a single peak at the origin, with the line $y = y_0$. (b) The values of f along that line, plotted against x . The partial derivative $\partial f / \partial x$ at (x_0, y_0) is the slope of the tangent to this curve at $x = x_0$.

The partial derivative of f with respect to x is defined as

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}, \quad (2.22)$$

in which y is held constant. It is the slope of the curve cut from the surface by the plane $y = y_0$, as Figure 9 shows, and it is found by differentiating with respect to x while treating y as a constant. The partial derivative $\partial f / \partial y$ is defined in the same way. Where it

matters which variable is held constant, it is shown as a subscript, $(\partial f/\partial x)_y$, and f_x is sometimes written for $\partial f/\partial x$. As an example, consider the pressure of a fixed amount of an ideal gas, $P = nRT/V$, where n is the amount of gas and R is the gas constant. Its partial derivatives are

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{nR}{V}, \quad \left(\frac{\partial P}{\partial V}\right)_T = -\frac{nRT}{V^2}. \quad (2.23)$$

Differentiating the partial derivatives again gives four second derivatives, $\partial^2 f/\partial x^2$, $\partial^2 f/\partial y^2$, and the mixed derivatives

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right), \quad \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right). \quad (2.24)$$

For $f = x^2y + \sin xy$, for example, both mixed derivatives are $2x + \cos xy - xy \sin xy$. This equality is a general result: whenever the second derivatives of f are continuous,

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}, \quad (2.25)$$

so the order in which partial derivatives are taken does not matter. To see why, consider the combination

$$\Delta = f(x+h, y+k) - f(x+h, y) - f(x, y+k) + f(x, y), \quad (2.26)$$

where h and k are small. Grouping the first and third terms, and the second and fourth, shows that Δ is the change in the difference $f(x+h, y) - f(x, y)$ when y increases by k . That difference is approximately $h \partial f/\partial x$, so $\Delta \approx hk \partial^2 f/\partial y \partial x$. Grouping the first and second terms, and the third and fourth, gives in the same way $\Delta \approx hk \partial^2 f/\partial x \partial y$. Dividing by hk and letting h and k tend to zero gives the result, and the continuity of the second derivatives is what makes the approximations valid.

The Total Differential

When x changes by δx and y by δy , the change in f can be split into a change in x at fixed $y + \delta y$ and a change in y at fixed x :

$$\begin{aligned} \delta f &= [f(x+\delta x, y+\delta y) - f(x, y+\delta y)] + [f(x, y+\delta y) - f(x, y)] \\ &\approx \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y. \end{aligned} \quad (2.27)$$

The error is small compared with δx and δy when these are small, and in the limit we write

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy, \quad (2.28)$$

which is called the total differential of f . For the ideal gas, dividing $dP = (nR/V) dT - (nRT/V^2) dV$ by P gives $dP/P = dT/T - dV/V$.

An expression $A(x, y) dx + B(x, y) dy$ is called an exact differential if it is the total differential of some function f , that is, if $A = \partial f / \partial x$ and $B = \partial f / \partial y$. In that case the equality of mixed derivatives gives

$$\frac{\partial A}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial B}{\partial x}, \quad (2.29)$$

and this condition is the test for an exact differential. Conversely, if $\partial A / \partial y = \partial B / \partial x$ throughout a rectangle, the differential is exact there. To construct f , let $\mathcal{A}(x, y)$ be an integral of A with respect to x at fixed y , and we look for f in the form $\mathcal{A} + \chi(y)$. The requirement $\partial f / \partial y = B$ gives $\chi'(y) = B - \partial \mathcal{A} / \partial y$, whose derivative with respect to x is $\partial B / \partial x - \partial A / \partial y = 0$ by the equality of mixed derivatives of $\mathcal{A}$, so the right-hand side depends on y alone and can be integrated.

As an example, consider $(2xy + 1) dx + x^2 dy$. Here $\partial A / \partial y = 2x$ and $\partial B / \partial x = 2x$, so the differential is exact. Integrating A with respect to x gives $\mathcal{A} = x^2 y + x$, and then $\chi'(y) = x^2 - x^2 = 0$, so $f = x^2 y + x$, up to a constant. By contrast, $y dx - x dy$ is not exact, since $\partial A / \partial y = 1$ while $\partial B / \partial x = -1$.

The Chain Rule

If x and y both depend on a variable t , then $f(x(t), y(t))$ is a function of t alone. The coordinates of a thermometer carried across a heated plate, for example, depend on the time t ; for the thermometer, f is the temperature of the plate, and $f(x(t), y(t))$ is the temperature recorded at time t . Dividing the total differential by dt gives the chain rule for several variables,

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}. \quad (2.30)$$

If x and y depend on two variables, the chain rule applies to the

derivative with respect to each of these variables, with the other variable held constant. Plane polar coordinates provide an example, with $x = s \cos \phi$ and $y = s \sin \phi$, where s is the distance from the origin and ϕ is the angle from the x -axis. The partial derivatives of f with respect to s and ϕ are

$$\begin{aligned} \left(\frac{\partial f}{\partial s}\right)_\phi &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial s} = \cos \phi \frac{\partial f}{\partial x} + \sin \phi \frac{\partial f}{\partial y}, \\ \left(\frac{\partial f}{\partial \phi}\right)_s &= \frac{\partial f}{\partial x} \frac{\partial x}{\partial \phi} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \phi} = -s \sin \phi \frac{\partial f}{\partial x} + s \cos \phi \frac{\partial f}{\partial y}. \end{aligned} \quad (2.31)$$

Solving these two equations for the Cartesian derivatives gives

$$\frac{\partial f}{\partial x} = \cos \phi \frac{\partial f}{\partial s} - \frac{\sin \phi}{s} \frac{\partial f}{\partial \phi}, \quad \frac{\partial f}{\partial y} = \sin \phi \frac{\partial f}{\partial s} + \frac{\cos \phi}{s} \frac{\partial f}{\partial \phi}. \quad (2.32)$$

As a check, $f = x = s \cos \phi$ has $\partial f / \partial s = \cos \phi$ and $\partial f / \partial \phi = -s \sin \phi$, and the formulas give $\partial f / \partial x = \cos^2 \phi + \sin^2 \phi = 1$ and $\partial f / \partial y = \sin \phi \cos \phi - \cos \phi \sin \phi = 0$, as they should.

The subscripts matter here. From $s = \sqrt{x^2 + y^2}$ we have $(\partial s / \partial x)_y = x/s = \cos \phi$, and from $x = s \cos \phi$ we have $(\partial x / \partial s)_\phi = \cos \phi$. One might expect these derivatives to be reciprocals. Their product is $\cos^2 \phi$ rather than 1, because the first holds y constant and the second holds ϕ constant. With y held constant in both derivatives, they are reciprocals, since $x = \sqrt{s^2 - y^2}$ gives $(\partial x / \partial s)_y = s/x = 1/\cos \phi$.

Reciprocity and Cyclic Relations

Let x , y , and z be related by a single equation, so that any two of them fix the third; the pressure, volume, and temperature of a fixed amount of gas, for example, are related in this way. Regarding z as a function of x and y ,

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy. \quad (2.33)$$

When z is held constant, y is an ordinary function of x , and x is in turn a function of y . Differentiating the identity $x = x(y(x))$ with respect to x by the chain rule gives $1 = (dx/dy)(dy/dx)$, so the derivative of a function and the derivative of its inverse are

reciprocals, and this gives the reciprocity relation

$$\left(\frac{\partial x}{\partial y}\right)_z = \frac{1}{(\partial y/\partial x)_z}. \quad (2.34)$$

Setting $dz = 0$ in the total differential gives $(\partial y/\partial x)_z = -(\partial z/\partial x)_y/(\partial z/\partial y)_x$. Using the reciprocity relation to replace $1/(\partial z/\partial y)_x$ by $(\partial y/\partial z)_x$, and multiplying both sides by $(\partial x/\partial y)_z$, we find

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1, \quad (2.35)$$

which is called the cyclic relation. The minus sign runs against the expectation that the derivatives, written as fractions, would cancel to give $+1$; they do not cancel, because each holds a different variable constant. For the ideal gas, $(\partial P/\partial V)_T = -nRT/V^2$, $(\partial V/\partial T)_P = nR/P$, and $(\partial T/\partial P)_V = V/nR$, whose product is $-nRT/PV = -1$.

Taylor's Theorem in Two Variables

To expand $f(x+h, y+k)$ in powers of h and k , let $\psi(t) = f(x+th, y+tk)$. By the chain rule, $\psi'(t) = hf_x + kf_y$, with the partial derivatives evaluated at $(x+th, y+tk)$. Differentiating $\psi'(t)$ with respect to t by the chain rule, and using the equality of mixed derivatives,

$$\psi''(t) = h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}, \quad (2.36)$$

where f_{xx} stands for $\partial^2 f/\partial x^2$, and so on. The Maclaurin series of ψ , evaluated at $t = 1$, then gives

$$f(x+h, y+k) = f + (hf_x + kf_y) + \frac{1}{2!} (h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) + \cdots, \quad (2.37)$$

with f and its derivatives on the right evaluated at (x, y) .

Stationary Points

A point at which $f_x = f_y = 0$ is called a stationary point. Elsewhere the first-order change $hf_x + kf_y$ takes either sign, depending on the signs of h and k , so a maximum or a minimum can occur only at a stationary point. Near a stationary point, the Taylor expansion

gives

$$\begin{aligned}
 f(x+h, y+k) - f(x, y) & \\
 & \approx \frac{1}{2} (h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) \\
 & = \frac{1}{2f_{xx}} \left[(hf_{xx} + kf_{xy})^2 + (f_{xx}f_{yy} - f_{xy}^2) k^2 \right], \tag{2.38}
 \end{aligned}$$

where the second form, found by completing the square, assumes $f_{xx} \neq 0$. The sign of this change decides the nature of the point, as follows:

1. if $f_{xx}f_{yy} - f_{xy}^2 > 0$ and $f_{xx} > 0$, the bracket and the factor before it are positive, and the point is a minimum;
2. if $f_{xx}f_{yy} - f_{xy}^2 > 0$ and $f_{xx} < 0$, the bracket is positive and the factor negative, and the point is a maximum;
3. if $f_{xx}f_{yy} - f_{xy}^2 < 0$, the bracket is positive for $k = 0$ and negative where $hf_{xx} + kf_{xy} = 0$, so f rises in some directions and falls in others;
4. if $f_{xx}f_{yy} - f_{xy}^2 = 0$, the test is inconclusive.

A point of the third kind is called a saddle point, since the surface near it has the shape of a saddle. The third case also covers $f_{xx} = 0$, where the second form cannot be used. The condition $f_{xx}f_{yy} - f_{xy}^2 < 0$ then requires $f_{xy} \neq 0$, and the change $\frac{1}{2}(2hkf_{xy} + k^2f_{yy})$ takes both signs, since for $|h|$ large compared with $|k|$ its sign is that of hkf_{xy} , which reverses when h does.

Consider, for example, the function $f = x^3 + y^3 - 3axy$ with $a > 0$. The conditions $3x^2 - 3ay = 0$ and $3y^2 - 3ax = 0$ give $x(x^3 - a^3) = 0$ when $y = x^2/a$ from the first condition is substituted into the second, so the stationary points are $(0, 0)$ and (a, a) . The second derivatives are $f_{xx} = 6x$, $f_{yy} = 6y$, and $f_{xy} = -3a$. At the origin, $f_{xx}f_{yy} - f_{xy}^2 = -9a^2 < 0$, and the point is a saddle. At (a, a) , $f_{xx}f_{yy} - f_{xy}^2 = 27a^2 > 0$ with $f_{xx} = 6a > 0$, and the point is a minimum, where $f = -a^3$, as the contours of Figure 10 show.

Lagrange Multipliers

We often need the stationary values of a function $f(x, y)$ when x and y are related by a constraint $g(x, y) = 0$. A small displacement

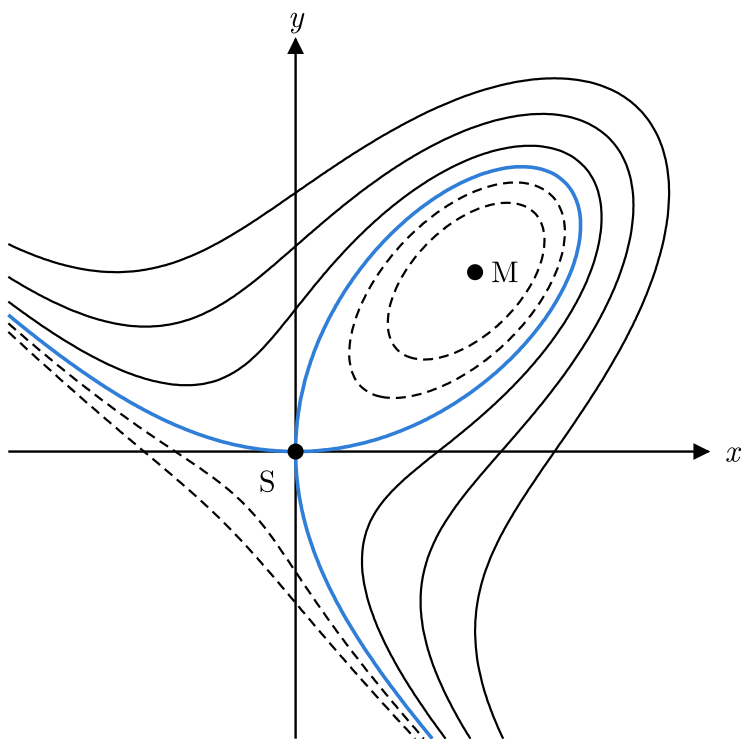


Figure 10. Contours of $f = x^3 + y^3 - 3axy$. The blue contour is $f = 0$, which crosses itself at the saddle point S at the origin. The dashed contours have $f < 0$, and those inside the loop of the blue contour surround the minimum M at (a, a) .

along the curve $g = 0$ satisfies $dg = g_x dx + g_y dy = 0$, and at a stationary point of f on the curve it also satisfies $df = f_x dx + f_y dy = 0$. The conditions $dg = 0$ and $df = 0$ must give the same ratio dy/dx , so $f_x g_y = f_y g_x$, that is, $f_x/g_x = f_y/g_y$ at a point where neither g_x nor g_y vanishes. Let λ be the common value of these ratios. Then

$$f_x - \lambda g_x = 0, \quad f_y - \lambda g_y = 0, \quad g = 0, \quad (2.39)$$

which are three equations for x , y , and λ . The constant λ is called a Lagrange multiplier. The first two equations are the conditions for $f - \lambda g$ to be stationary with λ held constant. The method extends to more variables, with one equation for each, and to several constraints, with one multiplier for each.

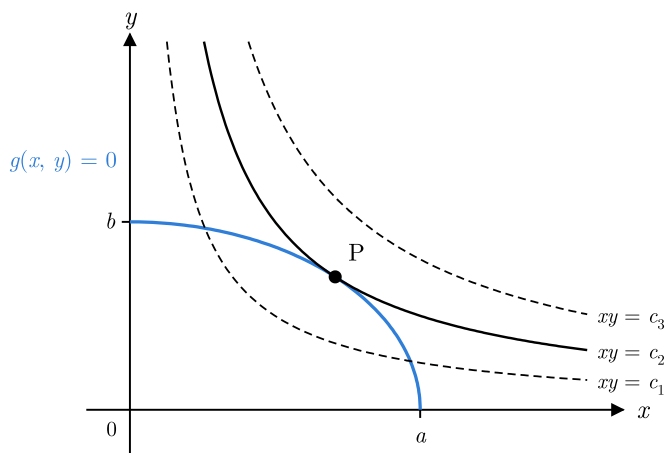


Figure 11. Stationary values of $f = xy$ on the part of the ellipse $g(x, y) = x^2/a^2 + y^2/b^2 - 1 = 0$ in the first quadrant. The contour $xy = c_1$ crosses the ellipse, the contour $xy = c_3$ does not meet it, and the contour $xy = c_2$ touches it at the point P , where f takes its largest value on this part of the ellipse.

The slope of a contour of f is $-f_x/f_y$ and that of the constraint curve is $-g_x/g_y$, so at a stationary point the contour of f touches the constraint curve, as in Figure 11. Where a contour crosses the curve, moving along the curve carries us onto contours with larger or smaller values of f .

As an example, consider the question of finding the rectangle of largest area, with sides parallel to the axes, that fits inside the ellipse $x^2/a^2 + y^2/b^2 = 1$. With one corner at (x, y) in the first quadrant, the area is $4xy$, and we take $f = xy$ with $g = x^2/a^2 + y^2/b^2 - 1$. The conditions are $y = 2\lambda x/a^2$ and $x = 2\lambda y/b^2$. Multiplying the first condition by x and the second by y shows that $xy = 2\lambda x^2/a^2 = 2\lambda y^2/b^2$, so $x^2/a^2 = y^2/b^2$, and the constraint makes each of these ratios $\frac{1}{2}$. The corner is at $(a/\sqrt{2}, b/\sqrt{2})$, and the largest area is $2ab$.

Differentiating an Integral with Respect to a Parameter

Let

$$F(\alpha) = \int_a^b f(x, \alpha) \, dx, \quad (2.40)$$

where a and b are constants and α is a parameter. Then

$$\frac{F(\alpha + \delta\alpha) - F(\alpha)}{\delta\alpha} = \int_a^b \frac{f(x, \alpha + \delta\alpha) - f(x, \alpha)}{\delta\alpha} \, dx. \quad (2.41)$$

As $\delta\alpha \rightarrow 0$, the integrand on the right tends to $\partial f/\partial\alpha$, and so

$$\frac{dF}{d\alpha} = \int_a^b \frac{\partial f}{\partial\alpha} \, dx. \quad (2.42)$$

This result is called differentiation under the integral sign. It holds when f and $\partial f/\partial\alpha$ are continuous on the range, which ensures that the difference quotient approaches $\partial f/\partial\alpha$ at the same rate at every point of the range, so that the limit can be taken inside the integral. For an infinite range it also holds, provided that, for all the values of α considered, $|\partial f/\partial\alpha|$ is less than some function of x alone whose integral over the range converges.

When the limits also depend on α , the area under the curve changes in a second way, as Figure 12 shows. Let $\Lambda(\alpha, a, b)$ be the integral of $f(x, \alpha)$ from a to b , regarded as a function of three variables. By the fundamental theorem of calculus, $\partial\Lambda/\partial b = f(b, \alpha)$ and $\partial\Lambda/\partial a = -f(a, \alpha)$, and the chain rule applied to $F(\alpha) = \Lambda(\alpha, a(\alpha), b(\alpha))$ gives

$$\frac{dF}{d\alpha} = \int_a^b \frac{\partial f}{\partial\alpha} \, dx + f(b, \alpha) \frac{db}{d\alpha} - f(a, \alpha) \frac{da}{d\alpha}, \quad (2.43)$$

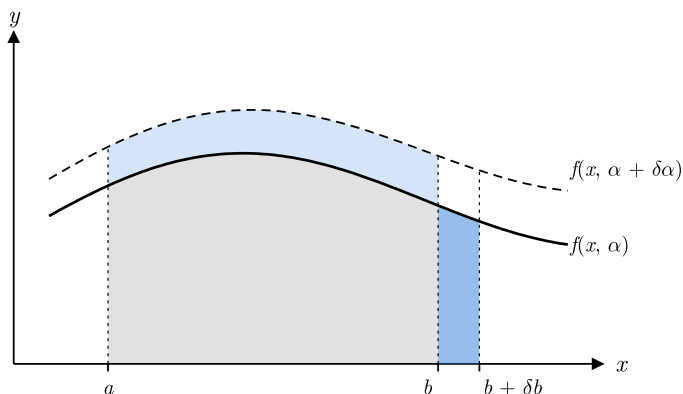


Figure 12. The change in $F(\alpha) = \int_a^{b(\alpha)} f(x, \alpha) dx$ when α changes by $\delta\alpha$. The pale band between the curves $f(x, \alpha)$ and $f(x, \alpha + \delta\alpha)$ comes from the change in the integrand, and has area close to $\delta\alpha \int_a^b (\partial f / \partial \alpha) dx$. The dark strip comes from the change δb in the upper limit, and has area close to $f(b, \alpha) \delta b$.

which is called Leibniz's rule for differentiating an integral.

As an example, consider

$$\int_0^\infty e^{-\alpha x} dx = \frac{1}{\alpha}, \quad \alpha > 0. \quad (2.44)$$

For α greater than some fixed $\alpha_0 > 0$, the derivative of the integrand, $-xe^{-\alpha x}$, has modulus less than $xe^{-\alpha_0 x}$, whose integral converges, and the same holds for higher derivatives. Differentiating both sides n times with respect to α , each differentiation bringing down a factor $-x$, gives

$$\int_0^\infty x^n e^{-\alpha x} dx = \frac{n!}{\alpha^{n+1}}. \quad (2.45)$$

For $\alpha = 1$ this is the integral $J_n = n!$ found in Section 2.1 with a reduction formula.

2.3 Multiple Integrals and Change of Variables

An ordinary integral adds up a quantity along a line, which is the subject of the calculus of a single variable. We often encounter situations in which we have to integrate over more than one variable, for example when computing the mass of a plate whose density varies from point to point. The section first defines such integrals and evaluates them as repeated single integrals, then finds how they change when new variables are used, which gives the area and volume elements of polar, cylindrical, and spherical coordinates, and finally applies the results to the Gaussian integral and to the masses, centres of mass, and moments of inertia of bodies.

Double Integrals

Let $\mathcal{R}$ be a region of the xy -plane, and let $f(x, y)$ be a function defined on it. We divide $\mathcal{R}$ into small pieces, of areas $\delta A_1, \delta A_2, \dots$, choose a point (x_i, y_i) in each piece, and form the sum $\sum_i f(x_i, y_i) \delta A_i$. As the pieces are made smaller and more numerous, the sum tends to a limit, which is called the double integral of f over $\mathcal{R}$ and written

$$\iint_{\mathcal{R}} f(x, y) \, dA. \quad (2.46)$$

With $f = 1$ the double integral is the area of $\mathcal{R}$, and with f equal to the mass per unit area σ of a thin plate, it is the mass of the plate.

A double integral is evaluated as two single integrals, one inside the other, which is called a repeated integral. Suppose that the region lies between the lines $x = a$ and $x = b$, and that for each x it runs from a lower curve $y = y_1(x)$ to an upper curve $y = y_2(x)$. We take the pieces to be small rectangles of area $dx \, dy$, add first the pieces in a thin vertical strip at a fixed x , and then add the strips, so that

$$\iint_{\mathcal{R}} f \, dA = \int_a^b \left[\int_{y_1(x)}^{y_2(x)} f(x, y) \, dy \right] dx. \quad (2.47)$$

If the region is instead described by horizontal strips, with x running from $x_1(y)$ to $x_2(y)$ for each y between c and d , the pieces can

be added along each strip first, and then

$$\iint_{\mathcal{R}} f \, dA = \int_c^d \left[\int_{x_1(y)}^{x_2(y)} f(x, y) \, dx \right] dy. \quad (2.48)$$

The two repeated integrals are equal, because each adds the same quantities $f \, \delta A$, grouped in different ways. The inner integral is often written with its differential next to its integral sign, as in $\int_a^b dx \int_{y_1}^{y_2} f \, dy$. When the region is a rectangle and the integrand is a product $X(x)Y(y)$, the inner integral over y is the same for every x , and the double integral is the product of two single integrals,

$$\int_a^b dx \int_c^d X(x)Y(y) \, dy = \int_a^b X(x) \, dx \int_c^d Y(y) \, dy. \quad (2.49)$$

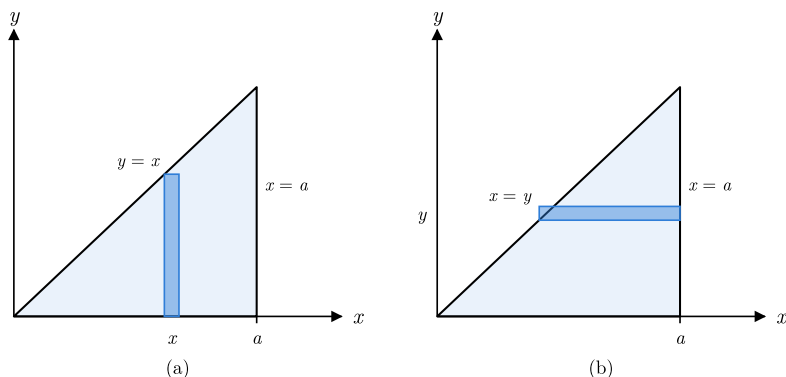


Figure 13. The triangle $0 \leq y \leq x \leq a$ described in two ways. (a) With vertical strips, x runs from 0 to a , and on each strip y runs from 0 to x . (b) With horizontal strips, y runs from 0 to a , and on each strip x runs from y to a .

Changing the order of integration can turn a repeated integral that cannot be done into one that can. Consider, for example, the repeated integral

$$\int_0^a dy \int_y^a \frac{\sin x}{x} \, dx. \quad (2.50)$$

The inner integral cannot be expressed in terms of standard functions. The region is the triangle of Figure 13, described here by

horizontal strips, as in panel (b). Describing it by vertical strips, as in panel (a), the integral becomes

$$\int_0^a \frac{\sin x}{x} dx \int_0^x dy = \int_0^a \sin x dx = 1 - \cos a. \quad (2.51)$$

A triple integral $\iiint_{\mathcal{V}} f dV$ over a region $\mathcal{V}$ of space is defined in the same way, and with f equal to the density ρ it gives the mass of the body that occupies $\mathcal{V}$. With $dV = dx dy dz$ it is evaluated as three nested single integrals, as you will do for a tetrahedron in Problem 6.

The Jacobian

Many regions are described more simply in other coordinates, such as a disc in plane polar coordinates. Let

$$x = x(u, v), \quad y = y(u, v) \quad (2.52)$$

be new coordinates u and v in the plane. The curves of constant u and of constant v form a grid, as in Figure 14, and we divide the region into the small cells of this grid.

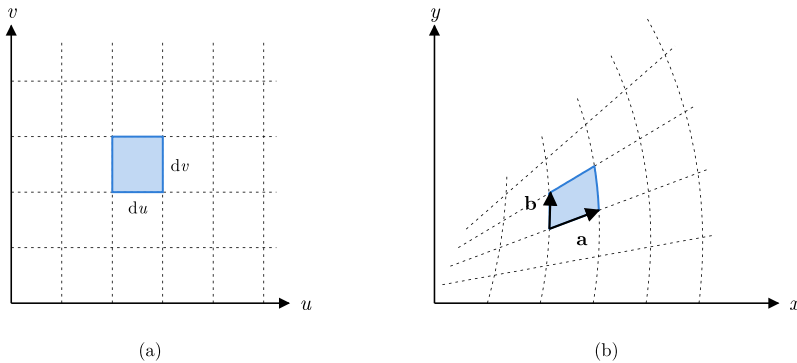


Figure 14. (a) A small rectangle of sides du and dv in the uv -plane. (b) Its image in the xy -plane, a small cell of the grid formed by curves of constant u and of constant v . When du and dv are small, the cell is close to the parallelogram with edges $\mathbf{a}$ and $\mathbf{b}$.

When u increases by du with v held constant, the point (x, y) moves

by the displacement

$$\mathbf{a} = \left(\frac{\partial x}{\partial u} \hat{\mathbf{x}} + \frac{\partial y}{\partial u} \hat{\mathbf{y}} \right) du, \quad (2.53)$$

and when v increases by dv with u held constant, it moves by $\mathbf{b} = (\partial x/\partial v \hat{\mathbf{x}} + \partial y/\partial v \hat{\mathbf{y}}) dv$. For small du and dv the cell is very nearly a parallelogram with edges $\mathbf{a}$ and $\mathbf{b}$, and the area of a parallelogram is the magnitude of the vector product of its edges. The vector product $\mathbf{a} \times \mathbf{b}$ points along $\hat{\mathbf{z}}$, and its magnitude gives the area of the cell as

$$dA = \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv, \quad \frac{\partial(x, y)}{\partial(u, v)} = \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u}. \quad (2.54)$$

The quantity $\partial(x, y)/\partial(u, v)$ is called the Jacobian of the change of variables. Its modulus is the ratio of the area of a small cell in the xy -plane to the area $du dv$ of the rectangle it comes from, and a double integral in the new variables is therefore

$$\iint_{\mathcal{R}} f dx dy = \iint_{\mathcal{R}'} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv, \quad (2.55)$$

where $\mathcal{R}'$ is the region of the uv -plane that corresponds to $\mathcal{R}$.

In three dimensions, let $\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$ depend on new coordinates u , v , and w . A small box of sides du , dv , and dw becomes a parallelepiped with edges $\mathbf{a} = (\partial \mathbf{r}/\partial u) du$, $\mathbf{b} = (\partial \mathbf{r}/\partial v) dv$, and $\mathbf{c} = (\partial \mathbf{r}/\partial w) dw$, where $\partial \mathbf{r}/\partial u$ is the vector whose components are the partial derivatives of x , y , and z with respect to u . The volume of a parallelepiped is $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$, since $|\mathbf{b} \times \mathbf{c}|$ is the area of the face with edges $\mathbf{b}$ and $\mathbf{c}$, and the component of $\mathbf{a}$ along $\mathbf{b} \times \mathbf{c}$, which is perpendicular to that face, is the height. The volume element is therefore

$$dV = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw, \quad \frac{\partial(x, y, z)}{\partial(u, v, w)} = \frac{\partial \mathbf{r}}{\partial u} \cdot \left(\frac{\partial \mathbf{r}}{\partial v} \times \frac{\partial \mathbf{r}}{\partial w} \right). \quad (2.56)$$

Polar, Cylindrical, and Spherical Coordinates

For plane polar coordinates, $x = s \cos \phi$ and $y = s \sin \phi$, the partial derivatives are $\partial x/\partial s = \cos \phi$, $\partial y/\partial s = \sin \phi$, $\partial x/\partial \phi = -s \sin \phi$,

and $\partial y / \partial \phi = s \cos \phi$, and the Jacobian is

$$\frac{\partial(x, y)}{\partial(s, \phi)} = s \cos^2 \phi + s \sin^2 \phi = s. \quad (2.57)$$

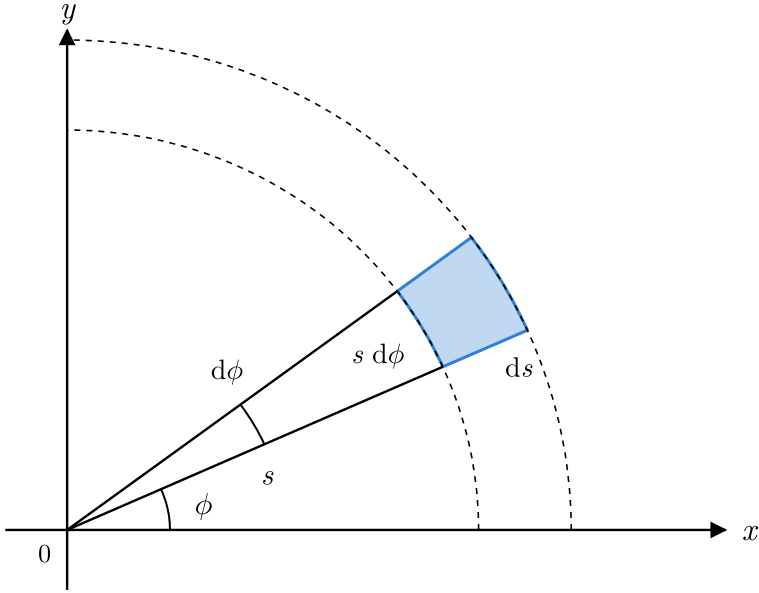


Figure 15. The area element in plane polar coordinates, bounded by the circles of radius s and $s + ds$ and by the lines at angles ϕ and $\phi + d\phi$. Its sides are ds and $s d\phi$.

The area element is therefore $dA = s ds d\phi$, and Figure 15 shows the reason directly. The cell between the circles of radius s and $s + ds$ and the lines at angles ϕ and $\phi + d\phi$ has one side of length ds and another that is an arc of radius s through the angle $d\phi$, of length $s d\phi$, and these sides are perpendicular. Cylindrical coordinates add the height z to the plane polar coordinates, and the cell is extended through a height dz , so that $dV = s ds d\phi dz$.

Spherical coordinates are defined by

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta, \quad (2.58)$$

where r is the distance from the origin, θ is the angle from the positive z -axis, running from 0 to π , and ϕ is the angle around the

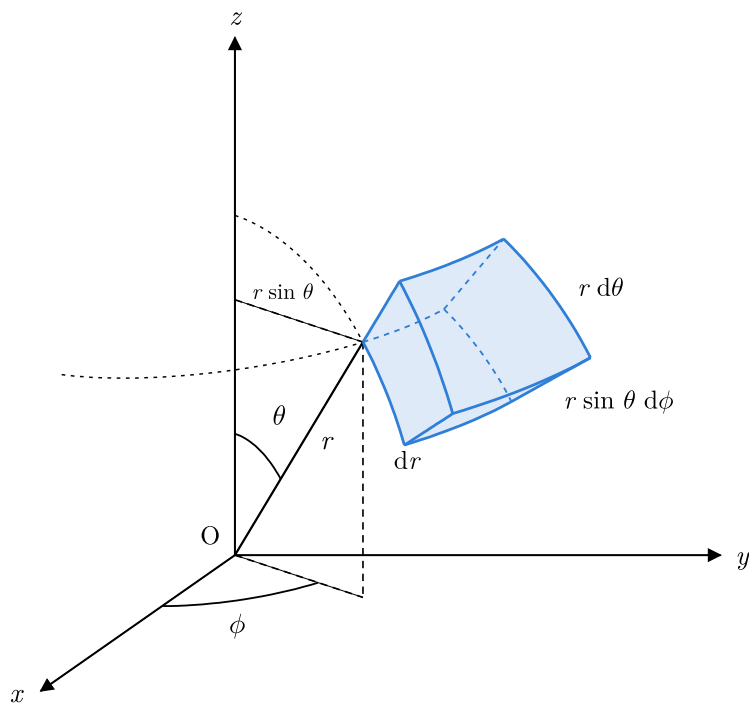


Figure 16. The volume element in spherical coordinates. The point at distance r from the origin O has polar angle θ and azimuthal angle ϕ , and lies at a distance $r \sin \theta$ from the z -axis. The element has perpendicular edges dr , $r d\theta$, and $r \sin \theta d\phi$.

z -axis, running from 0 to 2π . Figure 16 shows the cell between r and $r + dr$, θ and $\theta + d\theta$, and ϕ and $\phi + d\phi$. A change in r alone moves the point a distance dr along the radius. A change in θ alone moves it a distance $r d\theta$ along a circle of radius r centred on the origin, in the plane that contains the z -axis and the point, and the tangent to this circle is perpendicular to the radius. A change in ϕ alone moves the point a distance $r \sin \theta d\phi$ round a horizontal circle, whose radius $r \sin \theta$ is the distance of the point from the z -axis. This displacement is perpendicular to the plane that contains the z -axis and the point, and so it is perpendicular to the other two displacements, which lie in that plane. The three edges of the cell are therefore perpendicular, and

$$dV = r^2 \sin \theta dr d\theta d\phi, \quad (2.59)$$

a result you will check against the Jacobian in Problem 3. As a check, the volume of a sphere of radius a is

$$\int_0^a r^2 dr \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi = \frac{a^3}{3} \cdot 2 \cdot 2\pi = \frac{4}{3}\pi a^3, \quad (2.60)$$

which is the value quoted in your school textbooks.

The Gaussian Integral

In Section 2.1 we showed that the integral of e^{-x^2} from 0 to infinity converges, and since e^{-x^2} is even, so does the integral over the whole line,

$$G = \int_{-\infty}^{\infty} e^{-x^2} dx. \quad (2.61)$$

The function e^{-x^2} has no indefinite integral in terms of standard functions, but G can be found by squaring it. The same integral with y as the variable of integration is also G , and by the result for a product $X(x)Y(y)$ above, extended to the whole plane,

$$G^2 = \int_{-\infty}^{\infty} e^{-x^2} dx \int_{-\infty}^{\infty} e^{-y^2} dy = \iint e^{-(x^2+y^2)} dx dy, \quad (2.62)$$

where the double integral is over the whole plane. The integrand depends only on $x^2 + y^2 = s^2$, and in plane polar coordinates the

double integral becomes

$$G^2 = \int_0^{2\pi} d\phi \int_0^\infty e^{-s^2} s ds = 2\pi \left[-\frac{1}{2} e^{-s^2} \right]_0^\infty = \pi. \quad (2.63)$$

The factor s from the area element is, apart from a factor -2 , the derivative of the exponent $-s^2$, and this is what allows the integral over s to be done. Since G is positive,

$$\int_{-\infty}^\infty e^{-x^2} dx = \sqrt{\pi}. \quad (2.64)$$

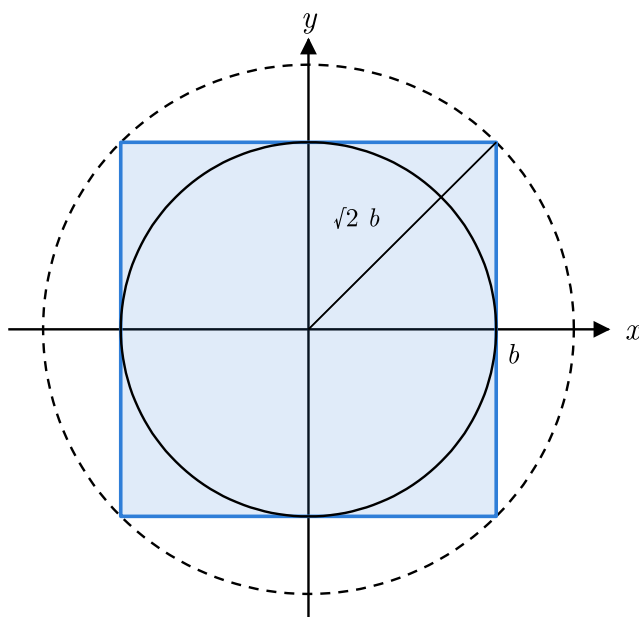


Figure 17. The square $|x| \leq b$, $|y| \leq b$ lies inside the circle of radius $\sqrt{2}b$ and contains the circle of radius b .

The product of two integrals from $-b$ to b is an integral over the square $|x| \leq b$, $|y| \leq b$, while the calculation in polar coordinates takes a limit over discs, and the limit over squares agrees with the limit over discs for the following reason. The square contains the disc of radius b and lies inside the disc of radius $\sqrt{2}b$, as Figure 17 shows, and since the integrand is positive, the integral over the

square lies between the integrals over the discs of radius b and $\sqrt{2}b$. The calculation above, with the upper limit on s finite, gives these as $\pi(1 - e^{-b^2})$ and $\pi(1 - e^{-2b^2})$. Each of these integrals tends to π as $b \rightarrow \infty$, and so does the integral over the square, which confirms the result.

For a positive constant α , the substitution $t = \sqrt{\alpha}x$ in the Gaussian integral gives

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}. \quad (2.65)$$

Integrals of $e^{-\alpha x^2}$ multiplied by even powers of x follow by differentiating this result with respect to α , as the integral of $e^{-\alpha x^2}$ was differentiated in Section 2.2. Differentiation under the integral sign is allowed here, because for α greater than some fixed $\alpha_0 > 0$ the modulus of the derivative of the integrand is less than $x^2 e^{-\alpha_0 x^2}$, whose integral converges. Differentiating both sides once with respect to α gives

$$\int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} dx = \frac{\sqrt{\pi}}{2} \alpha^{-3/2}. \quad (2.66)$$

Masses, Centres of Mass, and Moments of Inertia

The mass of a body of density ρ occupying a region $\mathcal{V}$ is $M = \iiint_{\mathcal{V}} \rho dV$. The position of its centre of mass is

$$\mathbf{R} = \frac{1}{M} \iiint_{\mathcal{V}} \mathbf{r} \rho dV, \quad (2.67)$$

which replaces the sum $\sum_i m_i \mathbf{r}_i / \sum_i m_i$ for a set of point masses by an integral over small pieces of mass ρdV . The moment of inertia of the body about an axis is defined as

$$I = \iiint_{\mathcal{V}} d_{\perp}^2 \rho dV, \quad (2.68)$$

where $d_{\perp}$ is the perpendicular distance of the volume element from the axis. The moment of inertia measures how far the mass lies from the axis, and a body with a larger moment of inertia needs a larger moment of force to give it the same angular acceleration.

As an example, consider a uniform solid hemisphere of radius a , with its flat face on the xy -plane and its centre at the origin. Its

mass is $M = \frac{2}{3}\pi a^3 \rho$. By symmetry the centre of mass lies on the z -axis, and in spherical coordinates, with $z = r \cos \theta$ and θ running from 0 to $\pi/2$,

$$MZ = \rho \int_0^a r^3 dr \int_0^{\pi/2} \cos \theta \sin \theta d\theta \int_0^{2\pi} d\phi = \rho \cdot \frac{a^4}{4} \cdot \frac{1}{2} \cdot 2\pi = \frac{\pi a^4 \rho}{4}, \quad (2.69)$$

where Z is the z -component of $\mathbf{R}$. Dividing by the mass gives $Z = 3a/8$.

As a second example, consider the moment of inertia of a uniform solid sphere of radius a and mass $M = \frac{4}{3}\pi a^3 \rho$ about the z -axis. The perpendicular distance from the axis is $d_\perp = r \sin \theta$, and so

$$I = \rho \int_0^a r^4 dr \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} d\phi. \quad (2.70)$$

With the substitution $\mu = \cos \theta$, for which $d\mu = -\sin \theta d\theta$ and $\sin^2 \theta = 1 - \mu^2$, the integral over θ becomes $\int_{-1}^1 (1 - \mu^2) d\mu = \frac{4}{3}$. Therefore $I = \rho \cdot (a^5/5) \cdot (4/3) \cdot 2\pi = 8\pi \rho a^5/15$, and in terms of the mass,

$$I = \frac{2}{5}Ma^2. \quad (2.71)$$

Problems

§2.1 Differentiation and Integration

1. Use implicit differentiation to find the slope of the ellipse $x^2/a^2 + y^2/b^2 = 1$ at the point (x_0, y_0) , and show that the tangent there is the line $xx_0/a^2 + yy_0/b^2 = 1$.
2. Show that $d(\arctan x)/dx = 1/(1 + x^2)$, and that $d(\tanh^{-1} x)/dx = 1/(1 - x^2)$ for $|x| < 1$.
3. Use Leibniz's rule to show that the n th derivative of $x \sin x$ is $x \sin(x + n\pi/2) + n \sin(x + (n-1)\pi/2)$.
4. Express $1/[x(x+a)^2]$ in partial fractions, and hence find its indefinite integral.
5. Use the reduction formula for K_n to show that $nK_n K_{n-1}$ does not depend on n , and hence that $K_n K_{n-1} = \pi/(2n)$.

6. Show that $\int_0^\infty dx/(1+x^p)$ converges for $p > 1$ and diverges for $0 < p \leq 1$. Show that $\int_0^1 \ln x \, dx = -1$.

§2.2 Partial Differentiation

1. Let $f = x^y$, for $x > 0$. Find $\partial f/\partial x$ and $\partial f/\partial y$, and check that the two mixed second derivatives are equal.
2. Show that $(y \cos x + 2x) dx + (\sin x + 3y^2) dy$ is an exact differential, and find the function of which it is the total differential.
3. The function $f = 2xy$ can be written in plane polar coordinates as $f = s^2 \sin 2\phi$. Use the expressions for $\partial f/\partial x$ and $\partial f/\partial y$ in terms of derivatives with respect to s and ϕ to find them, and check your answers against the Cartesian form.
4. The variables x , y , and z are related by $z = x^2y$, with x and y positive. Find $(\partial x/\partial y)_z$, $(\partial y/\partial z)_x$, and $(\partial z/\partial x)_y$, and verify the cyclic relation.
5. Find and classify the stationary points of $f = xy(a - x - y)$, where $a > 0$.
6. A closed rectangular box has edges x , y , and z , and a fixed total surface area A_0 . Use a Lagrange multiplier to show that its volume is stationary when the box is a cube.
7. Let $\kappa(x)$ be a continuous function, and let $F(t) = \int_0^t (t - x) \kappa(x) \, dx$. Use Leibniz's rule for differentiating an integral to show that $d^2F/dt^2 = \kappa(t)$.
8. Let $F(\alpha) = \int_0^\infty e^{-x} (\sin \alpha x)/x \, dx$. By differentiating under the integral sign, and writing the cosine that appears as the real part of a complex exponential, show that $dF/d\alpha = 1/(1 + \alpha^2)$, and hence that $F(\alpha) = \arctan \alpha$.

§2.3 Multiple Integrals and Change of Variables

1. Sketch the region of integration of $\int_0^a dx \int_x^a e^{-y^2} dy$, change the order of integration, and show that the integral is $\frac{1}{2}(1 - e^{-a^2})$.
2. Use the change of variables $x = a\xi$, $y = b\eta$ to show that the area of the ellipse $x^2/a^2 + y^2/b^2 \leq 1$ is πab .

3. Write down the vectors $\partial \mathbf{r} / \partial r$, $\partial \mathbf{r} / \partial \theta$, and $\partial \mathbf{r} / \partial \phi$ for spherical coordinates, and show that $\partial(x, y, z) / \partial(r, \theta, \phi) = r^2 \sin \theta$.
4. Show that the moment of inertia of a uniform solid cylinder of radius a , height H , and mass M about its axis is $\frac{1}{2}Ma^2$.
5. A sphere of radius a has density $\rho = \rho_0(1 - r/a)$. Find its mass M and its moment of inertia about a diameter, and show that the moment of inertia is $\frac{4}{15}Ma^2$. Explain why this is less than the value $\frac{2}{5}Ma^2$ for a uniform sphere.
6. Show that the volume of the tetrahedron bounded by the coordinate planes and the plane $x/a + y/b + z/c = 1$, with a , b , and c positive, is $abc/6$.
7. By completing the square in the exponent, show that $\int_{-\infty}^{\infty} e^{-\alpha x^2 + \beta x} dx = \sqrt{\pi/\alpha} e^{\beta^2/4\alpha}$ for $\alpha > 0$. By differentiating the Gaussian integral twice with respect to α , show that $\int_{-\infty}^{\infty} x^4 e^{-\alpha x^2} dx = \frac{3}{4}\sqrt{\pi} \alpha^{-5/2}$.

Vector Calculus and Field Geometry

3.1 Gradient, Divergence, and Curl

Many quantities in physics take a value at every point of a region of space, such as the temperature in a room, the velocity of the water in a river, and the electric field around a charged body. In this chapter we apply the calculus of functions of several variables, developed in Chapter 2, to quantities of this kind. This section introduces the three derivatives that act on them, which are called the gradient, the divergence, and the curl. In the sections that follow we first integrate such quantities along curves, over surfaces, and through volumes, then prove the theorems that connect those integrals, and finally write the three derivatives in cylindrical and spherical coordinates.

Vectors and Their Derivatives

We shall need the two products of vectors that you met at school. For vectors $\mathbf{A} = A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}$ and $\mathbf{B} = B_x\hat{\mathbf{x}} + B_y\hat{\mathbf{y}} + B_z\hat{\mathbf{z}}$, the scalar product is

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z = AB \cos \gamma, \quad (3.1)$$

where A and B are the magnitudes of the vectors and γ is the angle between them. The vector product is

$$\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y) \hat{\mathbf{x}} + (A_z B_x - A_x B_z) \hat{\mathbf{y}} + (A_x B_y - A_y B_x) \hat{\mathbf{z}}, \quad (3.2)$$

a vector of magnitude $AB \sin \gamma$, perpendicular to $\mathbf{A}$ and $\mathbf{B}$, in the direction given by the right-hand rule. Reversing the order of the factors reverses its sign, so $\mathbf{A} \times \mathbf{A} = \mathbf{0}$. The y - and z -components are obtained from the x -component by replacing each letter by the next one in the cycle $x \rightarrow y \rightarrow z \rightarrow x$, and several identities in this section are proved by working out one component and changing the letters in this way.

The derivative of a vector that depends on a parameter, such as the position $\mathbf{r}(t)$ of a moving particle at time t , is defined as for an ordinary function,

$$\frac{d\mathbf{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{dx}{dt} \hat{\mathbf{x}} + \frac{dy}{dt} \hat{\mathbf{y}} + \frac{dz}{dt} \hat{\mathbf{z}}, \quad (3.3)$$

where the second form holds because the unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ do not change. For a projectile with position $\mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{g} t^2$, for example, the velocity is $\mathbf{v}_0 + \mathbf{g} t$. Applying the product rule to each component gives

$$\frac{d}{dt}(\mathbf{A} \cdot \mathbf{B}) = \frac{d\mathbf{A}}{dt} \cdot \mathbf{B} + \mathbf{A} \cdot \frac{d\mathbf{B}}{dt}, \quad \frac{d}{dt}(\mathbf{A} \times \mathbf{B}) = \frac{d\mathbf{A}}{dt} \times \mathbf{B} + \mathbf{A} \times \frac{d\mathbf{B}}{dt}, \quad (3.4)$$

where the order of the factors in the vector products must be maintained. If $\mathbf{A}(t)$ has constant magnitude, the first rule applied to $\mathbf{A} \cdot \mathbf{A}$ gives $2\mathbf{A} \cdot d\mathbf{A}/dt = 0$, so the derivative of a vector of constant length is perpendicular to it; for example, the velocity of a particle moving on a circle is perpendicular to its position vector from the centre.

Scalar and Vector Fields

A function $f(\mathbf{r}) = f(x, y, z)$ that assigns a number to each point is called a scalar field. The temperature $T(\mathbf{r})$ in a room is one, and the height $f(x, y)$ of the ground, plotted against position on a map, is a scalar field in two dimensions. A function $\mathbf{F}(\mathbf{r})$ that assigns a vector to each point is called a vector field. Examples are the velocity $\mathbf{v}(\mathbf{r})$ of the water in a river and the electric field of a point

charge q at the origin,

$$\mathbf{E}(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3}, \quad (3.5)$$

where $r = |\mathbf{r}|$. Since $\mathbf{r}/r^3 = \hat{\mathbf{r}}/r^2$, where $\hat{\mathbf{r}} = \mathbf{r}/r$ is the unit vector pointing away from the charge, this field is Coulomb's law in vector form. A field may also depend on time, and its partial derivatives with respect to x , y , and z are then taken at fixed time.

The surfaces on which a scalar field is constant are called level surfaces, such as isotherms in a temperature field or the equipotential surfaces around a charge, and in two dimensions they are curves, such as the contour lines on a map, which indicate points of equal altitude. A vector field is pictured by its field lines, which are curves whose tangent at every point is parallel to the field. A small step $d\mathbf{r}$ along a field line is a multiple of $\mathbf{F}$, so the field lines satisfy

$$\frac{dx}{F_x} = \frac{dy}{F_y} = \frac{dz}{F_z}. \quad (3.6)$$

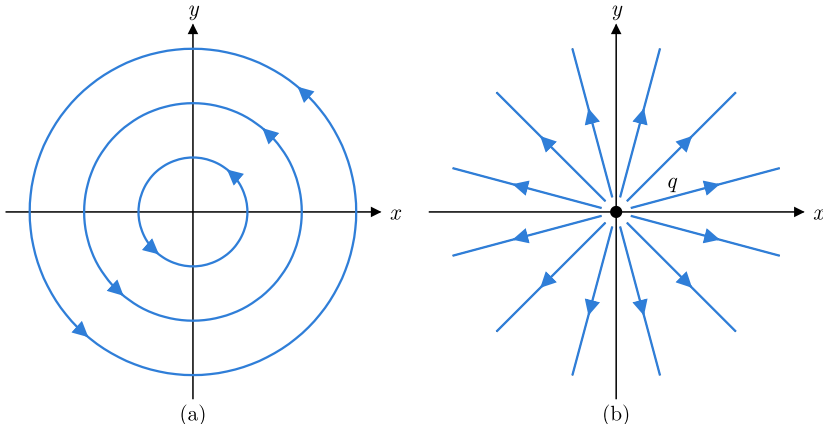


Figure 18. (a) Field lines of $\mathbf{F} = -y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$ in the plane $z = 0$, which are circles about the origin traversed anticlockwise. (b) Field lines of the electric field of a positive point charge q at the origin, which run radially outwards.

Consider the following example. The field lines of $\mathbf{F} = -y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$ lie in planes of constant z and satisfy $dx/(-y) = dy/x$, which can be rewritten as $x dx + y dy = 0$. Integrating this equation gives $x^2 + y^2 = \text{constant}$, so the field lines are circles about the z -axis, as

Figure 18(a) shows. The field lines of a positive point charge are straight lines running radially outwards, as in Figure 18(b).

The Directional Derivative and the Gradient

When we move from $\mathbf{r}$ to $\mathbf{r} + d\mathbf{r}$, a scalar field changes by the total differential of Section 2.2, $df = (\partial f/\partial x) dx + (\partial f/\partial y) dy + (\partial f/\partial z) dz$, which is the scalar product of $d\mathbf{r}$ with the vector

$$\nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{x}} + \frac{\partial f}{\partial y} \hat{\mathbf{y}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}}. \quad (3.7)$$

This vector is called the gradient of f , and $df = \nabla f \cdot d\mathbf{r}$. If we move a distance $d\ell$ in the direction of a unit vector $\hat{\mathbf{u}}$, so that $d\mathbf{r} = \hat{\mathbf{u}} d\ell$, the rate of change of f with distance is

$$\frac{df}{d\ell} = \hat{\mathbf{u}} \cdot \nabla f = |\nabla f| \cos \gamma, \quad (3.8)$$

where γ is the angle between $\hat{\mathbf{u}}$ and ∇f . This rate is called the directional derivative of f in the direction $\hat{\mathbf{u}}$, and $\partial f/\partial x$ is the directional derivative in the direction $\hat{\mathbf{x}}$. The directional derivative is largest when $\cos \gamma = 1$, so the gradient points in the direction in which f increases fastest, and its magnitude is that fastest rate. If instead $d\mathbf{r}$ lies in the level surface through the point, f does not change, so $\nabla f \cdot d\mathbf{r} = 0$, and the gradient is perpendicular to the level surfaces.

Figure 19 shows these results for the height of a hill. The gradient points straight uphill, across the contours, and it is largest where the contours are closest together, since the height then changes by a given amount over the shortest horizontal distance.

Let us find the gradient of the distance $r = (x^2 + y^2 + z^2)^{1/2}$ from the origin. By the chain rule $\partial r/\partial x = x/r$, and similarly for y and z , so $\nabla r = \mathbf{r}/r = \hat{\mathbf{r}}$. For a function $G(r)$ of the distance alone, the chain rule gives $\partial G/\partial x = G'(r) \partial r/\partial x$, so $\nabla G = G'(r) \hat{\mathbf{r}}$. Taking $G = 1/r$ gives $\nabla(1/r) = -\hat{\mathbf{r}}/r^2$, and therefore

$$\mathbf{E} = -\nabla \left(\frac{q}{4\pi\epsilon_0 r} \right), \quad (3.9)$$

so the field of the point charge is minus the gradient of its potential, which is the three-dimensional form of the relation between field

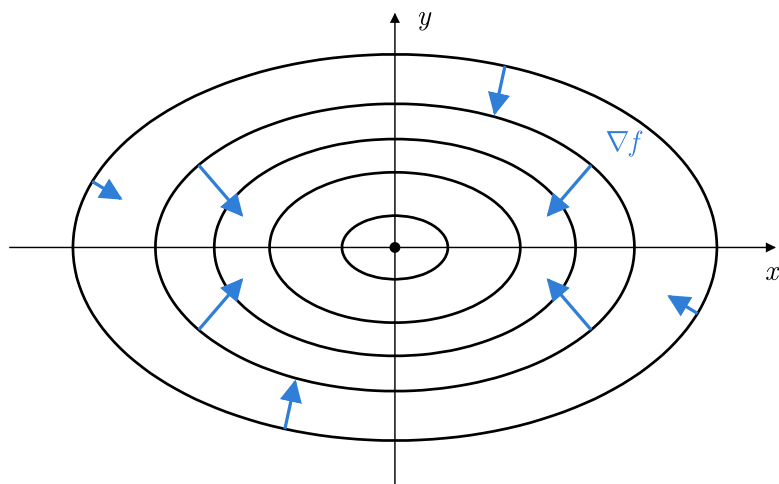


Figure 19. Contours of a hill of height $f(x, y) = f_0 \exp(-x^2/a^2 - y^2/b^2)$, drawn at equal intervals of height, with the gradient ∇f drawn at several points. Each arrow is perpendicular to the contour through its foot and points uphill, and the arrows are longest where the contours are closest together.

and potential quoted at school. The field is perpendicular to the equipotential spheres centred on the charge.

The Del Operator, Divergence, and Curl

The gradient is the result of applying the vector operator

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}, \quad (3.10)$$

which is called del, to f . Treating ∇ as a vector whose components are the three partial derivatives, its scalar product with a vector field $\mathbf{F}$ gives a scalar field,

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}, \quad (3.11)$$

which is called the divergence of $\mathbf{F}$, and its vector product gives a vector field,

$$\nabla \times \mathbf{F} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \hat{\mathbf{z}}, \quad (3.12)$$

which is called the curl of $\mathbf{F}$. The derivatives in ∇ act on the quantity to their right, so $\nabla \cdot \mathbf{F}$ must not be confused with the operator $\mathbf{F} \cdot \nabla = F_x \partial/\partial x + F_y \partial/\partial y + F_z \partial/\partial z$, which is a scalar operator that still waits for a field to its right to act on. The position vector $\mathbf{r}$, for example, has divergence $1 + 1 + 1 = 3$, and its curl is zero, since x does not depend on y or z , and similarly for the other components.

Divergence and Curl in a Flowing Fluid

The divergence and the curl are easiest to picture for the velocity field $\mathbf{v}(\mathbf{r})$ of a fluid. Figure 20 shows the following flows in the xy -plane, with k and ω constants:

1. the source flow $\mathbf{v} = k(x \hat{\mathbf{x}} + y \hat{\mathbf{y}})$, which has $\nabla \cdot \mathbf{v} = 2k$ and $\nabla \times \mathbf{v} = \mathbf{0}$;
2. the rotation $\mathbf{v} = \omega(-y \hat{\mathbf{x}} + x \hat{\mathbf{y}})$, which has $\nabla \cdot \mathbf{v} = 0$ and $\nabla \times \mathbf{v} = 2\omega \hat{\mathbf{z}}$;
3. the shear flow $\mathbf{v} = ky \hat{\mathbf{x}}$, which has $\nabla \cdot \mathbf{v} = 0$ and $\nabla \times \mathbf{v} = -k \hat{\mathbf{z}}$.

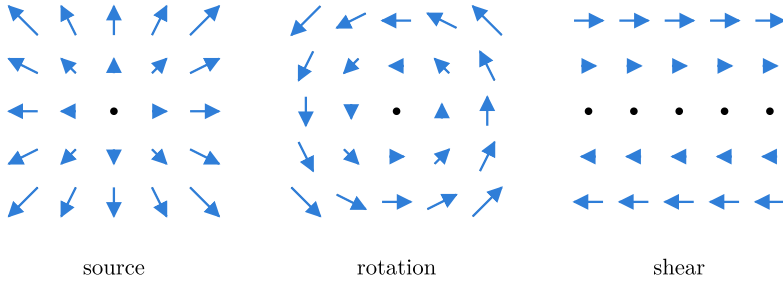


Figure 20. The velocity fields of three flows in the xy -plane, with an arrow centred on each point of a square grid: from left to right, the source flow, the rotation, and the shear flow, each with a positive constant. The dots mark points where the velocity is zero.

In the source flow the fluid moves directly away from the origin, faster the further it is from the origin. The rotation is the motion of a body turning at angular velocity ω about the z -axis, and in the shear flow layers of fluid slide over one another, faster the higher they are.

To see what the divergence measures, we follow a small box of fluid with sides Δx , Δy , and Δz parallel to the axes, whose faces move with the fluid. Its faces at $x + \Delta x$ and at x move in the x -direction with velocities $v_x(x + \Delta x, y, z)$ and $v_x(x, y, z)$, so by Taylor's theorem the length of the box grows at the rate

$$\frac{d(\Delta x)}{dt} = v_x(x + \Delta x, y, z) - v_x(x, y, z) \approx \frac{\partial v_x}{\partial x} \Delta x, \quad (3.13)$$

and the widths Δy and Δz grow in the same way. By the product rule, the volume $\Delta V = \Delta x \Delta y \Delta z$ therefore changes at the rate

$$\frac{1}{\Delta V} \frac{d(\Delta V)}{dt} = \frac{1}{\Delta x} \frac{d(\Delta x)}{dt} + \frac{1}{\Delta y} \frac{d(\Delta y)}{dt} + \frac{1}{\Delta z} \frac{d(\Delta z)}{dt} \approx \nabla \cdot \mathbf{v}. \quad (3.14)$$

The divergence of the velocity is therefore the rate at which a small volume of fluid expands, per unit volume, as the left-hand panel of Figure 21 shows for the source flow. In a gas, a positive divergence means that the gas is expanding. For a liquid such as water, whose density hardly changes, the divergence of the velocity is zero wherever no liquid is supplied or removed.

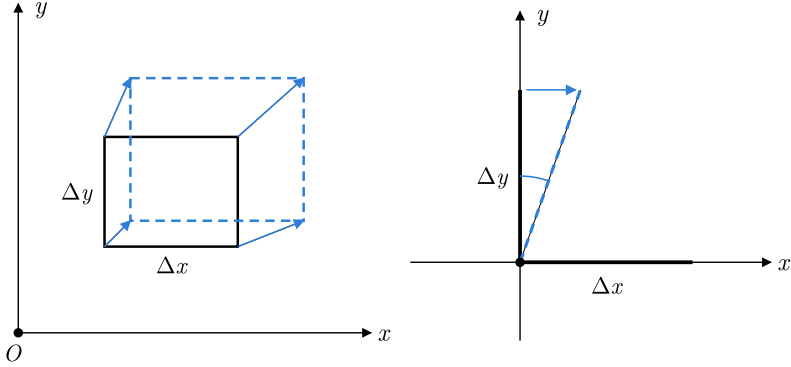


Figure 21. Small elements of fluid at a time t (black) and a short time later (blue, dashed), with the change exaggerated. Left: in the source flow every point moves directly away from O , and the rectangle grows in both directions. Right: in the shear flow the element along the x -axis does not move, while the element along the y -axis turns clockwise.

To see what the curl measures, we follow two short lines of fluid that start at the same point, one of length Δx along the x -direction and one of length Δy along the y -direction. The far end of the first line moves in the y -direction faster than the near end by $(\partial v_y / \partial x) \Delta x$, so that line turns anticlockwise with angular velocity $\partial v_y / \partial x$, as long as the angle it has turned through remains small. The far end of the second line moves in the x -direction faster than the near end by $(\partial v_x / \partial y) \Delta y$, which turns it clockwise, so its anticlockwise angular velocity is $-\partial v_x / \partial y$. The mean of the two angular velocities is

$$\frac{1}{2} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) = \frac{1}{2} (\nabla \times \mathbf{v})_z, \quad (3.15)$$

so the z -component of the curl is twice the mean angular velocity of the fluid about the z -axis near the point, and a small paddle wheel with its axle along z would turn at that mean angular velocity, which is half the z -component of the curl. For the rotation both lines turn with the fluid at angular velocity ω . The result for the shear flow is less expected, since one might think that fluid moving along straight lines does not rotate. As the right-hand panel of Figure 21 shows, however, the line along y turns clockwise at the rate k , because its upper end is carried faster than its lower end, while the line along x does not turn, and the mean angular velocity

is $-k/2$.

Identities

Of the identities that these derivatives satisfy, two follow from the equality of mixed partial derivatives, which we met in Section 2.2 and which holds for functions whose second derivatives are continuous. The first is that the curl of a gradient is zero,

$$\nabla \times (\nabla f) = \mathbf{0}, \quad (3.16)$$

since its x -component is $\partial^2 f / \partial y \partial z - \partial^2 f / \partial z \partial y = 0$, and the other components vanish in the same way. The field of a point charge, being a gradient, therefore has zero curl wherever it is defined. The second is that the divergence of a curl is zero,

$$\begin{aligned} \nabla \cdot (\nabla \times \mathbf{A}) &= \frac{\partial}{\partial x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \\ &\quad + \frac{\partial}{\partial z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) = 0, \end{aligned} \quad (3.17)$$

since the six terms cancel in pairs, the term in $\partial^2 A_z / \partial x \partial y$ with the term in $\partial^2 A_z / \partial y \partial x$, and so on.

The divergence of a gradient is

$$\nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \nabla^2 f, \quad (3.18)$$

and the operator ∇^2 is called the Laplacian. In Cartesian coordinates the Laplacian of a vector field acts on each component, $\nabla^2 \mathbf{A} = \nabla^2 A_x \hat{\mathbf{x}} + \nabla^2 A_y \hat{\mathbf{y}} + \nabla^2 A_z \hat{\mathbf{z}}$.

Finally, the three derivatives obey the following product rules:

$$\begin{aligned} \nabla(f\psi) &= f \nabla \psi + \psi \nabla f, \\ \nabla \cdot (f\mathbf{A}) &= f \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla f, \\ \nabla \times (f\mathbf{A}) &= f \nabla \times \mathbf{A} + \nabla f \times \mathbf{A}, \\ \nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}). \end{aligned} \quad (3.19)$$

Each rule is proved by writing out the components and applying the ordinary product rule. For the second rule, $\partial(fA_x)/\partial x = f \partial A_x / \partial x + A_x \partial f / \partial x$, and adding the corresponding terms for y and z gives the result. You will prove the third and the fourth in Problems 4 and 5.

As an example of the second rule, consider the field $r^n \mathbf{r}$, where n is a constant. With $\nabla \cdot \mathbf{r} = 3$ and $\nabla(r^n) = nr^{n-1} \hat{\mathbf{r}}$, the second rule gives

$$\nabla \cdot (r^n \mathbf{r}) = 3r^n + nr^{n-1} \hat{\mathbf{r}} \cdot \mathbf{r} = (n+3)r^n. \quad (3.20)$$

For $n = -3$ the divergence vanishes, so the field $\mathbf{r}/r^3$ of a point charge has zero divergence wherever it is defined. Since $\nabla(1/r) = -\mathbf{r}/r^3$, the same result states that

$$\nabla^2 \left(\frac{1}{r} \right) = 0, \quad r \neq 0. \quad (3.21)$$

3.2 Line, Surface, and Volume Integrals

In physics we often need integrals along curves, such as the work done by a force on a particle that moves along a path, and integrals over surfaces, such as the rate at which water flows through a net. This section first evaluates both kinds by describing the curve or the surface with parameters, then finds which vector fields have line integrals that depend only on the end points of the path, and finally treats integrals of vector fields over volumes.

Parametrised Curves

A curve C can be described by giving the position vector $\mathbf{r}(t)$ of its points as a function of a parameter t , which runs from t_A at one end, A , to t_B at the other, B . The parameter could, for example, be the time at which a moving particle passes each point. As t increases by dt , the point moves along the curve by

$$d\mathbf{r} = \frac{d\mathbf{r}}{dt} dt, \quad (3.22)$$

which is called the line element. As Figure 22 shows, $d\mathbf{r}$ is the chord between neighbouring points, and as $dt \rightarrow 0$ it lies along the tangent. Its length is $d\ell = |d\mathbf{r}/dt| dt$, where ℓ is the arc length, the distance measured along the curve from A , so the length of the curve is

$$\int_C d\ell = \int_{t_A}^{t_B} \left| \frac{d\mathbf{r}}{dt} \right| dt. \quad (3.23)$$

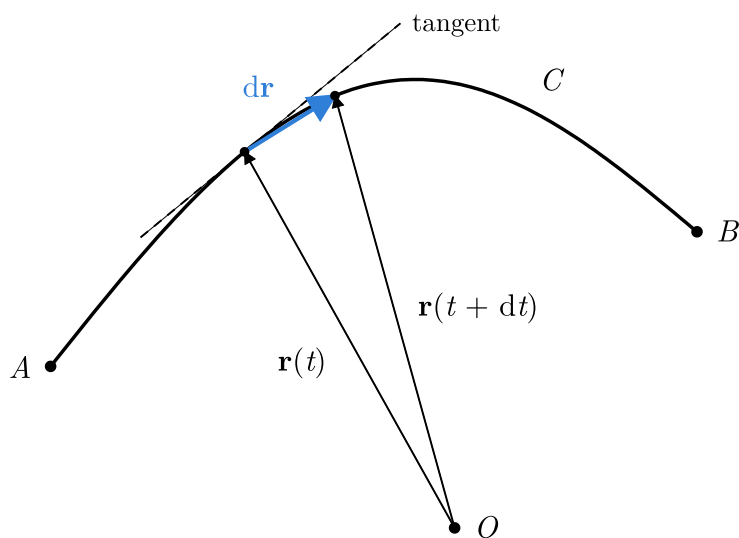


Figure 22. A curve C from A to B , described by the position vector $\mathbf{r}(t)$ from the origin O . The line element $d\mathbf{r} = \mathbf{r}(t+dt) - \mathbf{r}(t)$ joins neighbouring points of the curve, and as $dt \rightarrow 0$ it lies along the tangent.

A curve has many parametrisations, but since the chain rule gives $(d\mathbf{r}/dt) dt = (d\mathbf{r}/dt') dt'$ for any other parameter t' , the integrals of this section do not depend on which one is used, provided the curve is traversed in the same direction.

Line Integrals

Let $\mathbf{F}$ be a vector field and C a curve from A to B . Dividing the curve into short steps $d\mathbf{r}$ and adding the scalar products $\mathbf{F} \cdot d\mathbf{r}$ gives, in the limit of short steps, the line integral of $\mathbf{F}$ along C , which is evaluated as the ordinary integral

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{t_A}^{t_B} \mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt} dt. \quad (3.24)$$

When $\mathbf{F}$ is a force on a particle moving along C , the line integral is the work done by the force, since the work done in a short step is the component of the force along the step multiplied by its length. Traversing the curve in the opposite direction changes the sign of the integral. For a closed curve the integral is written $\oint_C \mathbf{F} \cdot d\mathbf{r}$ and is called the circulation of $\mathbf{F}$ round C .

The weight $\mathbf{F} = -mg\hat{\mathbf{z}}$ of a body of mass m near the surface of the Earth, with z the height, gives a simple example. On any path $\mathbf{F} \cdot d\mathbf{r} = -mg dz$, and the work done is $-mg(z_B - z_A)$, which depends only on the heights of the ends of the path.

As another example, consider the field $\mathbf{F} = -y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$ of Section 3.1, integrated from $(R, 0, 0)$ to $(0, R, 0)$ along the two paths of Figure 23, a quarter circle and a straight line. Along the quarter circle $\mathbf{r} = R \cos t \hat{\mathbf{x}} + R \sin t \hat{\mathbf{y}}$, with $0 \leq t \leq \pi/2$, the field equals $d\mathbf{r}/dt$, and so

$$\int \mathbf{F} \cdot d\mathbf{r} = \int_0^{\pi/2} (R^2 \sin^2 t + R^2 \cos^2 t) dt = \frac{\pi R^2}{2}. \quad (3.25)$$

Along the straight line $\mathbf{r} = (1-t)R\hat{\mathbf{x}} + tR\hat{\mathbf{y}}$, with $0 \leq t \leq 1$, the integrand is $(-tR)(-R) + (1-t)R \cdot R = R^2$, and the integral is R^2 , so the two paths give different results. Round the whole circle the circulation is $2\pi R^2$, and a force of this form would do work on a particle carried once round the circle.

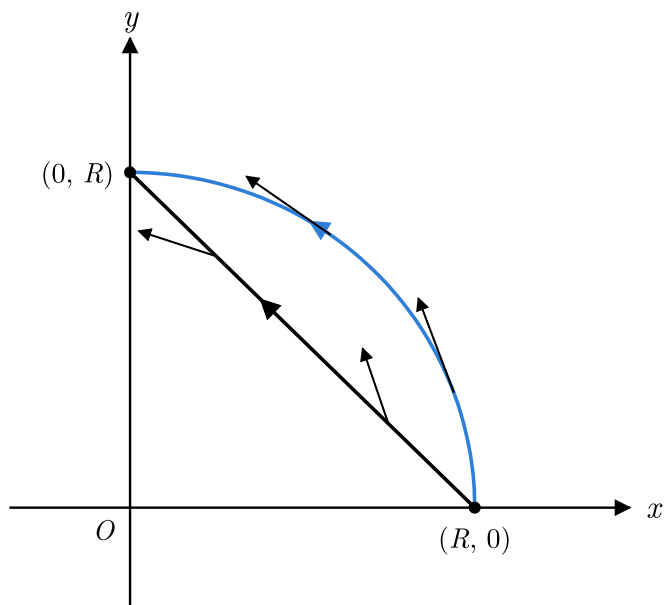


Figure 23. The quarter circle (blue) and the straight line from $(R, 0, 0)$ to $(0, R, 0)$ in the plane $z = 0$, with arrows of the field $\mathbf{F} = -y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$ at points on each path. On the circle the field points along the path, and on the line it points along the path only at the midpoint.

Conservative Fields and Potentials

The question that arises from these examples is which vector fields have line integrals that depend only on the end points, such as the line integral of the weight. For a vector field $\mathbf{F}$ defined throughout a connected region, one in which any two points can be joined by a path that stays inside it, the following conditions are equivalent:

1. the integral $\int_A^B \mathbf{F} \cdot d\mathbf{r}$ takes the same value on every path from A to B in the region, for every pair of points A and B ;
2. the circulation $\oint \mathbf{F} \cdot d\mathbf{r}$ is zero round every closed curve in the region;
3. there is a scalar field f with $\mathbf{F} = \nabla f$ throughout the region.

A field with these properties is called conservative, and f is called a scalar potential for $\mathbf{F}$. In mechanics a conservative force is often written as $\mathbf{F} = -\nabla U$, where U is the potential energy.

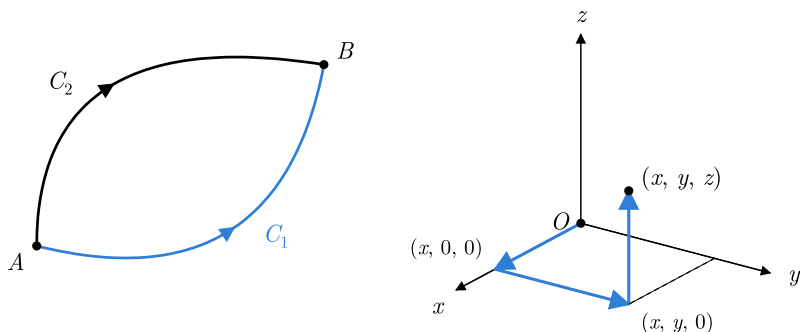


Figure 24. Left: two paths C_1 and C_2 from A to B , which together make a closed curve. Right: the path from the origin O to the point (x, y, z) in three steps parallel to the axes, used to construct a potential.

For the first two conditions, let C_1 and C_2 be two paths from A to B , as in the left-hand panel of Figure 24. Going out along C_1 and back along C_2 makes a closed curve, and since reversing C_2 changes the sign of its integral, the circulation round that curve is $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} - \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$. The circulation vanishes exactly when the integrals along C_1 and C_2 are equal, and since every closed

curve can be split into two such paths, the first two conditions are equivalent.

If $\mathbf{F} = \nabla f$, the chain rule gives $df(\mathbf{r}(t))/dt = \nabla f \cdot d\mathbf{r}/dt$ along any path, so that

$$\int_A^B \mathbf{F} \cdot d\mathbf{r} = \int_{t_A}^{t_B} \frac{df}{dt} dt = f(B) - f(A), \quad (3.26)$$

which depends only on the end points. Conversely, if the first condition holds, we choose a fixed point O and let $f(\mathbf{r})$ be the integral of $\mathbf{F} \cdot d\mathbf{r}$ from O to $\mathbf{r}$, along any path. Reaching $(x + \Delta x, y, z)$ by way of (x, y, z) and a short step in the x -direction gives

$$\begin{aligned} f(x + \Delta x, y, z) - f(x, y, z) &= \int_x^{x+\Delta x} F_x(x', y, z) dx' \\ &\approx F_x(x, y, z) \Delta x, \end{aligned} \quad (3.27)$$

so $\partial f/\partial x = F_x$, and the other components follow in the same way. The third condition is therefore equivalent to the first.

Since the curl of a gradient is zero, by Section 3.1, a conservative field has zero curl. The converse gives a practical test, and also a way of finding the potential. Suppose that $\nabla \times \mathbf{F} = \mathbf{0}$ throughout space, or at least throughout a box with sides parallel to the axes that contains the origin, and define f by integrating along the path of the right-hand panel of Figure 24:

$$\begin{aligned} f(x, y, z) &= \int_0^x F_x(x', 0, 0) dx' + \int_0^y F_y(x, y', 0) dy' \\ &\quad + \int_0^z F_z(x, y, z') dz'. \end{aligned} \quad (3.28)$$

We check that $\nabla f = \mathbf{F}$, using the fundamental theorem of calculus and differentiation under the integral sign from Section 2.2. Only the last integral depends on z , so $\partial f/\partial z = F_z(x, y, z)$. On differentiating with respect to y , the first integral gives nothing, the second gives $F_y(x, y, 0)$, and in the third we use the x -component of $\nabla \times \mathbf{F} = \mathbf{0}$, which states that $\partial F_z/\partial y = \partial F_y/\partial z$:

$$\frac{\partial f}{\partial y} = F_y(x, y, 0) + \int_0^z \frac{\partial F_y}{\partial z'}(x, y, z') dz' = F_y(x, y, z). \quad (3.29)$$

On differentiating with respect to x , the other two components of $\nabla \times \mathbf{F} = \mathbf{0}$, $\partial F_y / \partial x = \partial F_x / \partial y$ and $\partial F_z / \partial x = \partial F_x / \partial z$, give in the same way

$$\begin{aligned} \frac{\partial f}{\partial x} &= F_x(x, 0, 0) + [F_x(x, y, 0) - F_x(x, 0, 0)] \\ &\quad + [F_x(x, y, z) - F_x(x, y, 0)] \\ &= F_x(x, y, z). \end{aligned} \quad (3.30)$$

A field whose curl vanishes throughout such a region is therefore conservative. This result is the three-dimensional form of the test for an exact differential in Section 2.2, since the three conditions on mixed partial derivatives for $F_x dx + F_y dy + F_z dz$ are the three components of $\nabla \times \mathbf{F} = \mathbf{0}$.

Consider, for example, the field $\mathbf{F} = (2xy + z^3)\hat{\mathbf{x}} + x^2\hat{\mathbf{y}} + 3xz^2\hat{\mathbf{z}}$. The x -, y -, and z -components of its curl are $0 - 0$, $3z^2 - 3z^2$, and $2x - 2x$, which are all zero. Along the path of Figure 24 the first integral vanishes, since $F_x(x', 0, 0) = 0$, the second is x^2y , and the third is $\int_0^z 3xz'^2 dz' = xz^3$, which gives the potential

$$f = x^2y + xz^3, \quad (3.31)$$

whose partial derivatives are the components of $\mathbf{F}$.

The requirement that the curl vanish throughout such a region cannot be dropped. You will show in Problem 3 that the field $(-y\hat{\mathbf{x}} + x\hat{\mathbf{y}})/(x^2 + y^2)$, which is not defined on the z -axis, has zero curl everywhere else and yet a circulation of 2π round any circle centred on the z -axis and lying in a plane perpendicular to it. No box that contains such a circle avoids the z -axis, so the argument above does not apply.

Parametrised Surfaces

A surface S can be described by a position vector $\mathbf{r}(u, v)$ that depends on two parameters, and the curves of constant u and of constant v form a grid on it, as in Figure 25. Changing u by du moves the point by $(\partial \mathbf{r} / \partial u) du$, and changing v by dv moves it by $(\partial \mathbf{r} / \partial v) dv$, so the small patch between neighbouring grid lines is very nearly a parallelogram with these sides, and its area is the

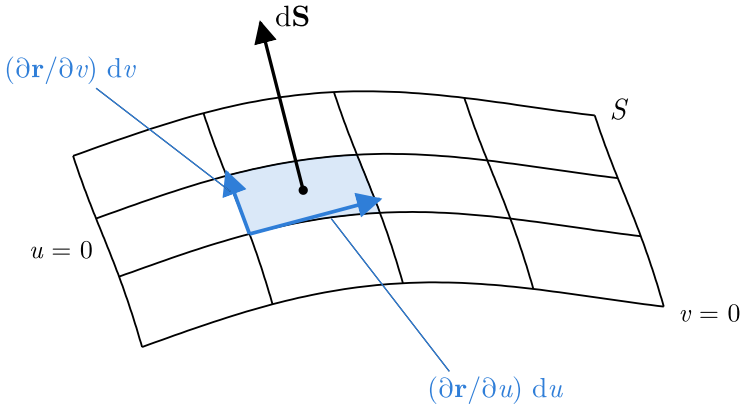


Figure 25. A surface S described by two parameters u and v , with the grid of lines on which one of them is constant. The shaded patch is very nearly a parallelogram with sides $(\partial \mathbf{r} / \partial u) du$ and $(\partial \mathbf{r} / \partial v) dv$, and its vector area $d\mathbf{S}$ is perpendicular to the surface.

magnitude of their vector product. The vector

$$d\mathbf{S} = \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} du dv = \hat{\mathbf{n}} dS \quad (3.32)$$

is called the vector area element. Its magnitude dS is the area of the patch, and its direction is that of the unit normal $\hat{\mathbf{n}}$, since the vector product is perpendicular to both vectors that lie along the surface. Exchanging u and v reverses $d\mathbf{S}$, and for a closed surface the convention is to choose the normal that points outwards.

As an example, consider the sphere of radius R centred on the origin, described by the angles θ and ϕ of Section 2.3, with $\mathbf{r} = R \sin \theta \cos \phi \hat{\mathbf{x}} + R \sin \theta \sin \phi \hat{\mathbf{y}} + R \cos \theta \hat{\mathbf{z}}$. The tangent vectors are

$$\begin{aligned} \frac{\partial \mathbf{r}}{\partial \theta} &= R (\cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}}), & \frac{\partial \mathbf{r}}{\partial \phi} &= R (-\sin \theta \sin \phi \hat{\mathbf{x}} + \sin \theta \cos \phi \hat{\mathbf{y}}), \end{aligned} \quad (3.33)$$

and the formula for the vector product, with $\cos^2 \phi + \sin^2 \phi = 1$ and $\cos^2 \theta + \sin^2 \theta = 1$, gives, as you found in Problem 3 of Section

2.3 with r in place of R ,

$$\begin{aligned}\frac{\partial \mathbf{r}}{\partial \theta} \times \frac{\partial \mathbf{r}}{\partial \phi} &= R^2 \sin \theta (\sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}) \\ &= R^2 \sin \theta \hat{\mathbf{r}}.\end{aligned}\quad (3.34)$$

The vector area element of the sphere is therefore $d\mathbf{S} = R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}$, pointing outwards, and integrating its magnitude over the angles gives the area $4\pi R^2$.

Flux

The integral of a vector field over a surface,

$$\int_S \mathbf{F} \cdot d\mathbf{S} = \iint \mathbf{F}(\mathbf{r}(u, v)) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du dv, \quad (3.35)$$

is called the flux of $\mathbf{F}$ through S , written $\oint_S \mathbf{F} \cdot d\mathbf{S}$ for a closed surface. Only the component of $\mathbf{F}$ along the normal contributes.

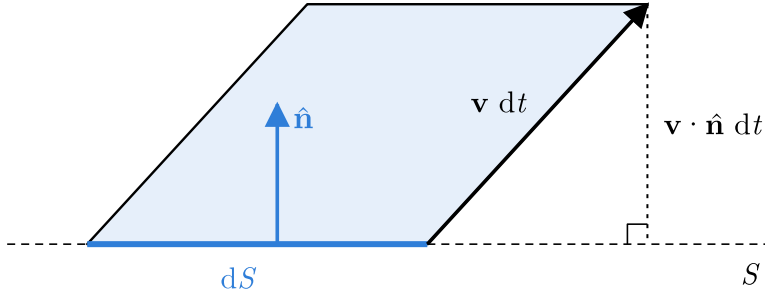


Figure 26. A patch dS of a surface S , seen edge-on. In a time dt the fluid that crosses the patch fills a slanted prism with edges $\mathbf{v} dt$ and height $\mathbf{v} \cdot \hat{\mathbf{n}} dt$.

The name comes from the flow of a fluid of velocity $\mathbf{v}$. In a short time dt each particle of fluid moves by $\mathbf{v} dt$, so the fluid that crosses a patch dS fills the slanted prism of Figure 26. Its volume is the base area multiplied by the perpendicular height $\mathbf{v} \cdot \hat{\mathbf{n}} dt$, which is $\mathbf{v} \cdot d\mathbf{S} dt$. The flux $\int_S \mathbf{v} \cdot d\mathbf{S}$ is therefore the volume of fluid crossing S per unit time, counted as positive in the direction of the normal.

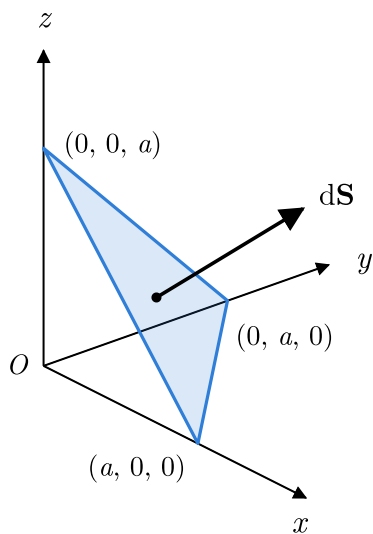


Figure 27. The triangle cut from the plane $x+y+z=a$ by the coordinate planes, with the element $d\mathbf{S}$ drawn from its centroid along the normal $\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}$, away from the origin.

Let us consider three examples, beginning with a plane. Let S be the triangle cut from the plane $x+y+z = a$ by the coordinate planes, shown in Figure 27, and let $\mathbf{F} = \mathbf{r}$. With $\mathbf{r} = u\hat{\mathbf{x}} + v\hat{\mathbf{y}} + (a-u-v)\hat{\mathbf{z}}$, where $u \geq 0$, $v \geq 0$, and $u+v \leq a$, the tangent vectors are $\hat{\mathbf{x}} - \hat{\mathbf{z}}$ and $\hat{\mathbf{y}} - \hat{\mathbf{z}}$, whose vector product is $\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}$. The integrand is $u+v+(a-u-v) = a$, and multiplying by the area $a^2/2$ of the triangle in the uv -plane gives

$$\int_S \mathbf{r} \cdot d\mathbf{S} = \frac{a^3}{2}. \quad (3.36)$$

Next, let S be the closed cylinder of radius R about the z -axis, from $z = 0$ to $z = H$, with the same field. On the curved surface, with $\mathbf{r} = R \cos \phi \hat{\mathbf{x}} + R \sin \phi \hat{\mathbf{y}} + z \hat{\mathbf{z}}$, the tangent vectors are $-R \sin \phi \hat{\mathbf{x}} + R \cos \phi \hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$, and their vector product, $R \cos \phi \hat{\mathbf{x}} + R \sin \phi \hat{\mathbf{y}}$, points away from the axis. The integrand is R^2 , and the flux through the curved surface is $2\pi R^2 H$. On the top the outward normal is $\hat{\mathbf{z}}$ and $\mathbf{r} \cdot \hat{\mathbf{z}} = H$, which gives $\pi R^2 H$, and on the bottom $\mathbf{r} \cdot (-\hat{\mathbf{z}}) = 0$. The total flux out of the cylinder is therefore

$$\oint_S \mathbf{r} \cdot d\mathbf{S} = 3\pi R^2 H. \quad (3.37)$$

Finally, let S be the sphere of radius R centred on the origin, with the field $\hat{\mathbf{r}}/r^2$, which has the form of the field of a point charge. With the vector area element found above,

$$\oint_S \frac{\hat{\mathbf{r}}}{r^2} \cdot d\mathbf{S} = \int_0^{2\pi} \int_0^\pi \frac{1}{R^2} R^2 \sin \theta \, d\theta \, d\phi = 4\pi, \quad (3.38)$$

which does not depend on R , because the area grows as R^2 while the field goes as $1/R^2$.

Volume Integrals

Integrals of scalar fields over volumes were treated in Section 2.3. A vector field is integrated over a volume by integrating each Cartesian component, since $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ do not change from point to point:

$$\int_V \mathbf{F} \, dV = \hat{\mathbf{x}} \int_V F_x \, dV + \hat{\mathbf{y}} \int_V F_y \, dV + \hat{\mathbf{z}} \int_V F_z \, dV. \quad (3.39)$$

A unit vector such as $\hat{\mathbf{r}}$ points in different directions at different points, so it must be written in Cartesian components before it is integrated. Consider, as an illustration, the integral of $\hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}$ over the solid hemisphere $r \leq R$, $z \geq 0$. The x - and y -components vanish, because $\cos \phi$ and $\sin \phi$ integrate to zero over $0 \leq \phi \leq 2\pi$, and with the volume element of Section 2.3 the z -component is

$$\int_0^{2\pi} \int_0^{\pi/2} \int_0^R \cos \theta r^2 \sin \theta dr d\theta d\phi = 2\pi \cdot \frac{1}{2} \cdot \frac{R^3}{3} = \frac{\pi R^3}{3}, \quad (3.40)$$

so the integral is $(\pi R^3/3) \hat{\mathbf{z}}$.

3.3 The Integral Theorems of Gauss, Stokes, and Green

The fundamental theorem of calculus relates the integral of a derivative over an interval to the values of the function at the ends. This section proves three theorems of the same kind for the derivatives of Section 3.1, each relating an integral over a region to an integral over its boundary, and then applies them.

Green's Theorem in the Plane

Let Ω be a region of the xy -plane bounded by a closed curve C , and let $P(x, y)$ and $Q(x, y)$ be functions whose partial derivatives are continuous in Ω and on C . Green's theorem in the plane states that

$$\oint_C (P dx + Q dy) = \iint_{\Omega} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy, \quad (3.41)$$

where C is traversed anticlockwise, so that the region lies on the left of a point moving round it.

Suppose first that every line parallel to the y -axis crosses C at most twice, so that Ω lies between a lower curve $y = y_1(x)$ and an upper curve $y = y_2(x)$ for $x_{\min} \leq x \leq x_{\max}$, as in Figure 28. Integrating the term in P first with respect to y , as in Section 2.3, the fundamental theorem of calculus gives

$$-\iint_{\Omega} \frac{\partial P}{\partial y} dx dy = \int_{x_{\min}}^{x_{\max}} P(x, y_1(x)) dx - \int_{x_{\min}}^{x_{\max}} P(x, y_2(x)) dx. \quad (3.42)$$

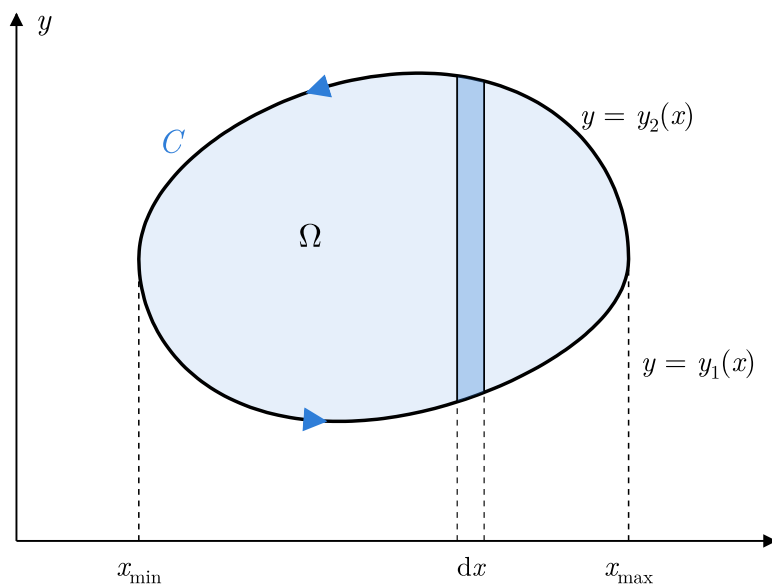


Figure 28. A region Ω between the curves $y = y_1(x)$ and $y = y_2(x)$, whose boundary C is traversed anticlockwise. The integral with respect to y is taken first, over strips such as the shaded one.

The right-hand side is $\oint_C P dx$, since the lower curve is traversed from $x_{\min}$ to $x_{\max}$ and the upper curve in the opposite direction, while vertical segments of C contribute nothing, because $dx = 0$ along them. If lines parallel to the x -axis also cross C at most twice, integrating first with respect to x gives in the same way $\oint_C Q dy = \iint_{\Omega} (\partial Q / \partial x) dx dy$, and adding the two results proves the theorem. A region of any other shape can be cut into pieces of this shape, and each cut is then traversed once in each direction, so its contributions to the line integrals cancel.

With $P = -y/2$ and $Q = x/2$ the integrand on the right is 1, and Green's theorem gives the area of Ω as

$$\iint_{\Omega} dx dy = \frac{1}{2} \oint_C (x dy - y dx) . \quad (3.43)$$

For the ellipse $x = a \cos t$, $y = b \sin t$, we have $x dy - y dx = ab dt$, using $\cos^2 t + \sin^2 t = 1$, and the area is $\frac{1}{2} \cdot 2\pi ab = \pi ab$.

The Divergence Theorem

Let V be a volume enclosed by a closed surface S with outward normal, and let $\mathbf{F}$ be a vector field whose partial derivatives are continuous in V and on S . The divergence theorem, which is also called Gauss's theorem, states that

$$\int_V \nabla \cdot \mathbf{F} dV = \oint_S \mathbf{F} \cdot d\mathbf{S} . \quad (3.44)$$

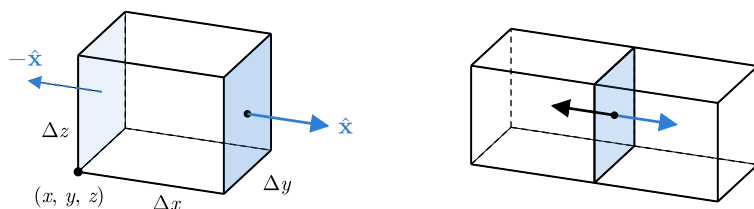


Figure 29. Left: a small box, whose two shaded faces have outward normals $\hat{\mathbf{x}}$ and $-\hat{\mathbf{x}}$. Right: on a face shared by two boxes, their outward normals point in opposite directions.

We prove the theorem first for the small box of Figure 29, with one corner at (x, y, z) and sides Δx , Δy , and Δz . The face at $x + \Delta x$ has outward normal $\hat{\mathbf{x}}$ and area $\Delta y \Delta z$, and the face at x has outward normal $-\hat{\mathbf{x}}$. Taking F_x at the centre of each face, since its variation across a face matters only at higher order, Taylor's theorem gives the flux through the two faces as

$$[F_x(x + \Delta x, y', z') - F_x(x, y', z')] \Delta y \Delta z \approx \frac{\partial F_x}{\partial x} \Delta x \Delta y \Delta z, \quad (3.45)$$

where (y', z') is the centre of each face. The other two pairs of faces give $(\partial F_y / \partial y) \Delta V$ and $(\partial F_z / \partial z) \Delta V$, where $\Delta V = \Delta x \Delta y \Delta z$, so the flux out of the box is $\nabla \cdot \mathbf{F} \Delta V$, with an error that becomes negligible compared with ΔV as the box shrinks.

We now divide V into many small boxes and add the fluxes out of all of them. A face shared by two neighbouring boxes appears twice in the sum, with outward normals in opposite directions, as the right-hand panel of Figure 29 shows, so its two contributions cancel. The faces that are left make up S , and the sum of $\nabla \cdot \mathbf{F} \Delta V$ becomes the integral over V as the boxes shrink, which proves the theorem. The boxes that meet S are cut by it into pieces that are not boxes, and the careful handling of these pieces is beyond the scope of this course.

Applying the theorem to a small region of volume ΔV round a point, and dividing by ΔV , gives a definition of the divergence that refers to no coordinates,

$$\nabla \cdot \mathbf{F} = \lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \oint \mathbf{F} \cdot d\mathbf{S}, \quad (3.46)$$

where the small region may have any shape. For the velocity of a fluid, this flux per unit volume agrees with the rate of expansion found in Section 3.1.

Stokes's Theorem

Let S be a surface bounded by a closed curve C , with $\mathbf{F}$ as before. The direction round C and the normal to S are related by the right-hand rule: when the thumb of the right hand points along $\hat{\mathbf{n}}$, the fingers curl in the direction in which C is traversed. Stokes's

theorem states that

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{r}. \quad (3.47)$$

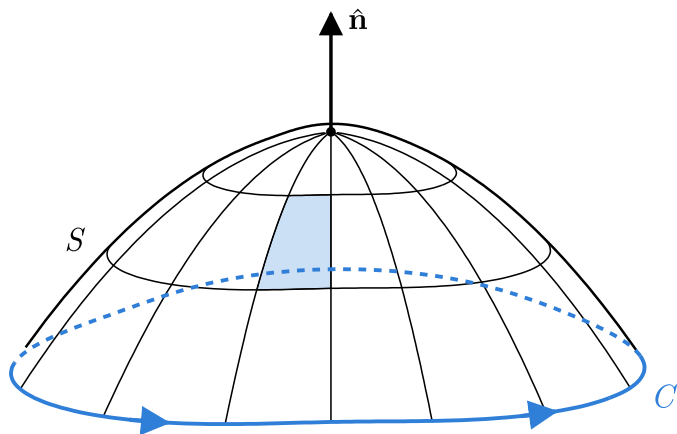


Figure 30. A surface S bounded by a curve C , which is traversed anti-clockwise when seen from above, as the right-hand rule requires for the upward normal $\hat{\mathbf{n}}$. The grid divides S into small patches, one of which is shaded.

The proof has three steps, which treat first a small square in the xy -plane, then a small flat patch of any orientation, and finally a surface built from such patches, as in Figure 30.

For the first step, consider the square of Figure 31, with sides Δx and Δy , traversed anticlockwise when seen from the positive z -axis, so that its normal is $\hat{\mathbf{z}}$. With $\mathbf{F}$ taken at the midpoint of each side, the bottom and top sides contribute $F_x \Delta x$ at y and $-F_x \Delta x$ at $y + \Delta y$, and the right-hand and left-hand sides contribute $F_y \Delta y$ at $x + \Delta x$ and $-F_y \Delta y$ at x . By Taylor's theorem the circulation is

$$\oint \mathbf{F} \cdot d\mathbf{r} \approx \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \Delta x \Delta y = (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{z}} \Delta x \Delta y, \quad (3.48)$$

which is the flux of $\nabla \times \mathbf{F}$ through the square.

For the second step, let the patch be the parallelogram of Figure 32, with one corner at $\mathbf{r}_0$ and sides $\mathbf{L}$ and $\mathbf{M}$, traversed from $\mathbf{r}_0$ along $\mathbf{L}$, then along $\mathbf{M}$, then back along $-\mathbf{L}$ and $-\mathbf{M}$, so that its

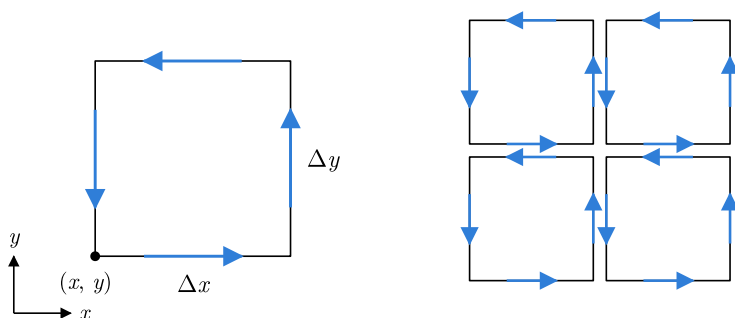


Figure 31. Left: a small square traversed anticlockwise. Right: four neighbouring squares, drawn slightly apart, in which each shared edge is traversed once in each direction.

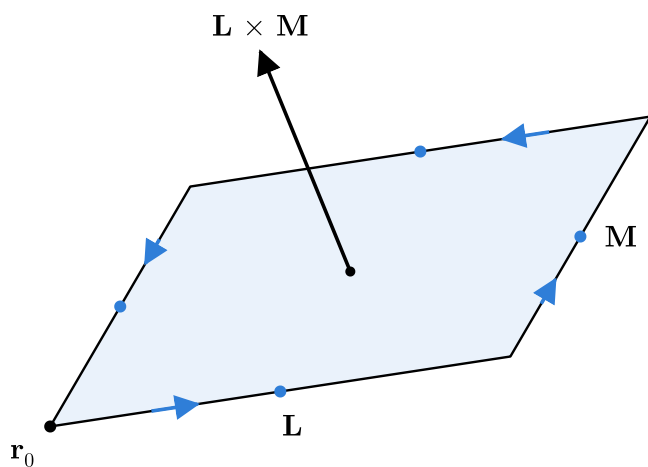


Figure 32. The parallelogram with one corner at $\mathbf{r}_0$ and sides $\mathbf{L}$ and $\mathbf{M}$, traversed in the direction of the arrows. The dots mark the midpoints of the four sides, at which $\mathbf{F}$ is evaluated, and the normal $\mathbf{L} \times \mathbf{M}$ is drawn from the centre.

vector area is $\Delta \mathbf{S} = \mathbf{L} \times \mathbf{M}$. With $\mathbf{F}$ taken at the midpoint of each side, the circulation is

$$\begin{aligned} & \mathbf{F}(\mathbf{r}_0 + \tfrac{1}{2}\mathbf{L}) \cdot \mathbf{L} + \mathbf{F}(\mathbf{r}_0 + \mathbf{L} + \tfrac{1}{2}\mathbf{M}) \cdot \mathbf{M} \\ & - \mathbf{F}(\mathbf{r}_0 + \mathbf{M} + \tfrac{1}{2}\mathbf{L}) \cdot \mathbf{L} - \mathbf{F}(\mathbf{r}_0 + \tfrac{1}{2}\mathbf{M}) \cdot \mathbf{M}. \end{aligned} \quad (3.49)$$

We first pair the opposite sides. In the first and third terms, the points at which $\mathbf{F}$ is evaluated differ by $\mathbf{M}$, and the first-order term of Taylor's theorem, which Section 2.2 gave for a function of two variables and which extends in the same way to three, gives $\mathbf{F}(\mathbf{r} + \mathbf{M}) - \mathbf{F}(\mathbf{r}) \approx M_x \partial \mathbf{F} / \partial x + M_y \partial \mathbf{F} / \partial y + M_z \partial \mathbf{F} / \partial z$. The second and fourth terms are treated in the same way with $\mathbf{L}$, and the circulation becomes

$$\begin{aligned} \oint \mathbf{F} \cdot d\mathbf{r} & \approx \left(L_x \frac{\partial \mathbf{F}}{\partial x} + L_y \frac{\partial \mathbf{F}}{\partial y} + L_z \frac{\partial \mathbf{F}}{\partial z} \right) \cdot \mathbf{M} \\ & - \left(M_x \frac{\partial \mathbf{F}}{\partial x} + M_y \frac{\partial \mathbf{F}}{\partial y} + M_z \frac{\partial \mathbf{F}}{\partial z} \right) \cdot \mathbf{L}. \end{aligned} \quad (3.50)$$

Next we collect the terms according to the components of $\mathbf{L}$ and $\mathbf{M}$ that they contain. The term in $L_x M_x$ appears in each bracket as $L_x M_x \partial F_x / \partial x$, and the two terms cancel, as do the terms in $L_y M_y$ and $L_z M_z$. The terms in $L_x M_y$ and $L_y M_x$ are

$$\begin{aligned} & L_x M_y \frac{\partial F_y}{\partial x} + L_y M_x \frac{\partial F_x}{\partial y} - M_y L_x \frac{\partial F_x}{\partial y} - M_x L_y \frac{\partial F_y}{\partial x} \\ & = (L_x M_y - L_y M_x) \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right), \end{aligned} \quad (3.51)$$

which is the product of the z -components of $\mathbf{L} \times \mathbf{M}$ and $\nabla \times \mathbf{F}$. Finally, the terms in the other pairs of components give the products of the x -components and of the y -components, with the letters changed cyclically, so that

$$\oint \mathbf{F} \cdot d\mathbf{r} \approx (\nabla \times \mathbf{F}) \cdot (\mathbf{L} \times \mathbf{M}) = (\nabla \times \mathbf{F}) \cdot \Delta \mathbf{S}. \quad (3.52)$$

For the third step, we divide S into small patches with the grid of a parametrisation, as in Section 3.2, so that each patch is very nearly a parallelogram, and traverse each patch in the direction set by the right-hand rule. When the circulations are added, each edge shared by two patches is traversed once in each direction, as the right-hand panel of Figure 31 shows, and its contributions cancel.

The edges that are left make up C , while the sum of $(\nabla \times \mathbf{F}) \cdot \Delta \mathbf{S}$ becomes the flux of the curl through S , which proves the theorem.

Applying the theorem to a small flat patch of area ΔS and normal $\hat{\mathbf{n}}$ gives a definition of the curl that refers to no coordinates,

$$\hat{\mathbf{n}} \cdot (\nabla \times \mathbf{F}) = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint \mathbf{F} \cdot d\mathbf{r}, \quad (3.53)$$

so the component of the curl in any direction is the circulation per unit area round a small loop perpendicular to that direction. Green's theorem is the special case in which S is a region of the xy -plane and $\mathbf{F} = P \hat{\mathbf{x}} + Q \hat{\mathbf{y}}$.

The Continuity Equation

In each of the following three applications, the divergence theorem turns a statement about a finite volume into an equation that holds at each point, and the step from the integral to the equation rests on the argument below. Suppose that a continuous function χ has $\int_V \chi dV = 0$ for every volume V . If χ were positive at some point, it would, being continuous, be positive throughout a small sphere round that point, and its integral over that sphere would be positive. A similar argument excludes negative values, so $\chi = 0$ everywhere.

Consider a fluid of density $\rho(\mathbf{r}, t)$ and velocity $\mathbf{v}(\mathbf{r}, t)$, and a fixed volume V enclosed by a surface S . By Section 3.2, the mass leaving V per unit time is $\oint_S \rho \mathbf{v} \cdot d\mathbf{S}$, and since mass is neither created nor destroyed, the mass inside V decreases at that rate:

$$\frac{d}{dt} \int_V \rho dV = - \oint_S \rho \mathbf{v} \cdot d\mathbf{S}. \quad (3.54)$$

Since V does not move, the derivative on the left can be taken under the integral sign, as in Section 2.2, and the divergence theorem turns the right-hand side into $-\int_V \nabla \cdot (\rho \mathbf{v}) dV$. The integral of $\partial \rho / \partial t + \nabla \cdot (\rho \mathbf{v})$ therefore vanishes over every volume, and by the argument above

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad (3.55)$$

which is called the continuity equation. For a liquid of constant density it reduces to $\nabla \cdot \mathbf{v} = 0$, in agreement with Section 3.1.

The Diffusion Equation for Heat

Let us turn next to a solid body of density ρ and specific heat capacity c , both constant, with temperature $T(\mathbf{r}, t)$. At school you met the law of conduction, by which heat flows through a slab of area A and thickness Δx , whose faces differ in temperature by ΔT , at the rate $\kappa A \Delta T / \Delta x$, where κ is the thermal conductivity. In three dimensions the flow is described by the heat flux $\mathbf{j}$, whose component along a unit normal $\hat{\mathbf{n}}$ is the rate of flow of heat per unit area across a small surface with that normal, and the law of conduction becomes

$$\mathbf{j} = -\kappa \nabla T, \quad (3.56)$$

which is called Fourier's law, and which reduces to the school form for the slab.

The heat energy in a fixed volume V , measured from a reference temperature, is $\int_V \rho c T \, dV$. If no heat is produced inside the body, the conservation of energy requires

$$\frac{d}{dt} \int_V \rho c T \, dV = - \oint_S \mathbf{j} \cdot d\mathbf{S} = \oint_S \kappa \nabla T \cdot d\mathbf{S} = \int_V \kappa \nabla^2 T \, dV, \quad (3.57)$$

where S is the surface of V , and the last step uses the divergence theorem and $\nabla \cdot (\nabla T) = \nabla^2 T$, with κ taken to be constant throughout the body, as ρ and c are. Taking the time derivative under the integral sign, the argument above gives $\rho c \partial T / \partial t = \kappa \nabla^2 T$, which can be rewritten as

$$\frac{\partial T}{\partial t} = D \nabla^2 T, \quad D = \frac{\kappa}{\rho c}. \quad (3.58)$$

This equation is called the diffusion equation, or the heat equation, and D is called the thermal diffusivity.

Gauss's Law

The last application concerns a point charge q at the origin, whose field is given by Coulomb's law as $\mathbf{E} = q \hat{\mathbf{r}} / (4\pi\epsilon_0 r^2)$. By the result of Section 3.2 for the flux of $\hat{\mathbf{r}}/r^2$, the flux of $\mathbf{E}$ out through any sphere centred on the charge is q/ϵ_0 . We now show that the same flux passes through any closed surface S that encloses the charge.

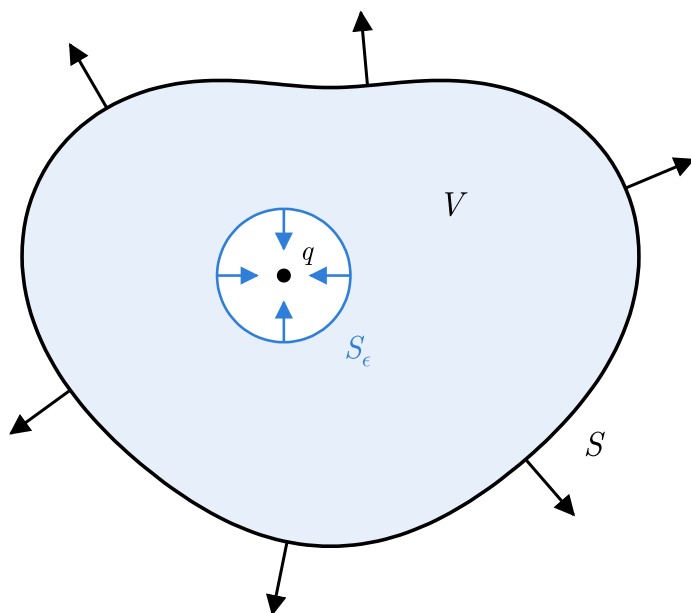


Figure 33. A closed surface S enclosing a point charge q , and a small sphere S_ϵ centred on the charge. The normal pointing out of the shaded region V points towards the charge on S_ϵ .

Since $\mathbf{E}$ is not defined at the charge, we apply the divergence theorem to the region V between S and a small sphere S_ε of radius ε centred on the charge, as in Figure 33. The field is defined throughout V , and we showed in Section 3.1 that its divergence, a multiple of $\nabla \cdot (\mathbf{r}/r^3)$, is zero there. On S_ε the normal pointing out of V points towards the charge, so the flux out of V through S_ε is $-q/\varepsilon_0$, and the divergence theorem gives

$$0 = \int_V \nabla \cdot \mathbf{E} \, dV = \oint_S \mathbf{E} \cdot d\mathbf{S} - \frac{q}{\varepsilon_0}. \quad (3.59)$$

The flux through S is therefore q/ε_0 for a surface of any shape. If S does not enclose the charge, the divergence of $\mathbf{E}$ is zero throughout the volume inside S , and the flux is zero. The fields of several charges add, so for any number of point charges

$$\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{q_{\text{enc}}}{\varepsilon_0}, \quad (3.60)$$

where q_{enc} is the total charge enclosed by S . This is Gauss's law. For charge spread through space with charge per unit volume ρ , the enclosed charge is $\int_V \rho \, dV$, where V is now the volume inside S , and the divergence theorem turns Gauss's law into $\int_V (\nabla \cdot \mathbf{E} - \rho/\varepsilon_0) \, dV = 0$. Since this integral vanishes for every volume, the argument at the start of the subsection on the continuity equation gives

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad (3.61)$$

which is Gauss's law in differential form.

3.4 Curvilinear Coordinates

Cartesian coordinates suit problems with flat boundaries, but many problems of physics have the symmetry of a cylinder or of a sphere, such as the electric field round a charged wire or the temperature inside a ball that is cooling. Section 2.3 introduced cylindrical and spherical coordinates and found their volume elements. Here we write the gradient, the divergence, the curl, and the Laplacian in such coordinates. The results are derived for a general class of coordinates, called orthogonal curvilinear coordinates, and then written out for the cylindrical and spherical cases.

Orthogonal Curvilinear Coordinates

Let u_1 , u_2 , and u_3 be three coordinates that fix the position of a point, so that the position vector is a function $\mathbf{r}(u_1, u_2, u_3)$. The surfaces on which one coordinate is constant are called coordinate surfaces, and the curves along which one coordinate varies while the other two are fixed are called coordinate lines. In spherical coordinates, for example, the surfaces of constant r are spheres centred on the origin, and the coordinate lines along which only ϕ varies are circles about the z -axis. The vector $\partial\mathbf{r}/\partial u_1$ is tangent to the coordinate line along which u_1 varies, and similarly for the other two coordinates. Let

$$h_1 = \left| \frac{\partial\mathbf{r}}{\partial u_1} \right|, \quad \hat{\mathbf{e}}_1 = \frac{1}{h_1} \frac{\partial\mathbf{r}}{\partial u_1}, \quad (3.62)$$

with h_2 , $\hat{\mathbf{e}}_2$, h_3 , and $\hat{\mathbf{e}}_3$ defined in the same way. The quantities h_1 , h_2 , and h_3 are called scale factors, and $\hat{\mathbf{e}}_1$, $\hat{\mathbf{e}}_2$, and $\hat{\mathbf{e}}_3$ are unit vectors along the three coordinate lines through the point. The coordinates are called orthogonal if these three unit vectors are perpendicular to one another at every point, and we number the coordinates so that $\hat{\mathbf{e}}_1 \times \hat{\mathbf{e}}_2 = \hat{\mathbf{e}}_3$, as for $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$. All the coordinates in this section are orthogonal.

When the coordinates change by du_1 , du_2 , and du_3 , the chain rule gives the change in position as

$$\begin{aligned} d\mathbf{r} &= \frac{\partial\mathbf{r}}{\partial u_1} du_1 + \frac{\partial\mathbf{r}}{\partial u_2} du_2 + \frac{\partial\mathbf{r}}{\partial u_3} du_3 \\ &= h_1 du_1 \hat{\mathbf{e}}_1 + h_2 du_2 \hat{\mathbf{e}}_2 + h_3 du_3 \hat{\mathbf{e}}_3. \end{aligned} \quad (3.63)$$

A change du_1 in the first coordinate therefore moves the point a distance $h_1 du_1$, which is the reason for the name scale factor. Since the unit vectors are perpendicular, Pythagoras's theorem gives the square of the length of $d\mathbf{r}$ as

$$d\ell^2 = h_1^2 du_1^2 + h_2^2 du_2^2 + h_3^2 du_3^2. \quad (3.64)$$

A small box bounded by coordinate surfaces has perpendicular edges of lengths $h_1 du_1$, $h_2 du_2$, and $h_3 du_3$, so the volume element is $dV = h_1 h_2 h_3 du_1 du_2 du_3$, and the area element on a surface of constant u_3 is $h_1 h_2 du_1 du_2$. A vector field is written in terms of the unit vectors at each point as $\mathbf{F} = F_1 \hat{\mathbf{e}}_1 + F_2 \hat{\mathbf{e}}_2 + F_3 \hat{\mathbf{e}}_3$, with

$F_1 = \mathbf{F} \cdot \hat{\mathbf{e}}_1$ and so on. The unit vectors $\hat{\mathbf{e}}_1$, $\hat{\mathbf{e}}_2$, and $\hat{\mathbf{e}}_3$ point in different directions at different points, unlike $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$, and this variation from point to point is the source of most of the differences between the formulas of this section and the Cartesian ones.

Cylindrical and Spherical Coordinates

In cylindrical coordinates (s, ϕ, z) the position vector is $\mathbf{r} = s \cos \phi \hat{\mathbf{x}} + s \sin \phi \hat{\mathbf{y}} + z \hat{\mathbf{z}}$. Differentiating with respect to each coordinate gives

$$\begin{aligned} \frac{\partial \mathbf{r}}{\partial s} &= \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}, & \frac{\partial \mathbf{r}}{\partial \phi} &= -s \sin \phi \hat{\mathbf{x}} + s \cos \phi \hat{\mathbf{y}}, & \frac{\partial \mathbf{r}}{\partial z} &= \hat{\mathbf{z}}, \end{aligned} \quad (3.65)$$

so the scale factors are $h_1 = 1$, $h_2 = s$, and $h_3 = 1$, and the unit vectors are

$$\hat{\mathbf{s}} = \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}}, \quad \hat{\phi} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}, \quad \hat{\mathbf{z}}. \quad (3.66)$$

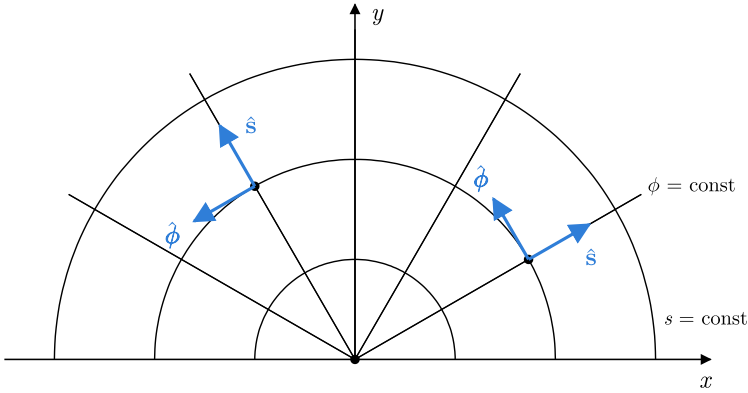


Figure 34. The coordinate lines of plane polar coordinates, which are circles of constant s and rays of constant ϕ , with the unit vectors $\hat{\mathbf{s}}$ and $\hat{\phi}$ drawn at two points. The unit vectors at the two points point in different directions.

The scalar product $\hat{\mathbf{s}} \cdot \hat{\phi}$ is zero, and using $\cos^2 \phi + \sin^2 \phi = 1$ the vector product $\hat{\mathbf{s}} \times \hat{\phi}$ is $\hat{\mathbf{z}}$, so the coordinates are orthogonal and numbered in the right order. The unit vector $\hat{\mathbf{s}}$ points directly away

from the z -axis and $\hat{\phi}$ points round it, in the direction of increasing ϕ , as Figure 34 shows for two points in the plane $z = 0$. The volume element $h_1 h_2 h_3 ds d\phi dz = s ds d\phi dz$ agrees with the one found from the Jacobian in Section 2.3.

The unit vectors $\hat{s}$ and $\hat{\phi}$ depend on ϕ , and differentiating the expressions above gives

$$\frac{\partial \hat{s}}{\partial \phi} = \hat{\phi}, \quad \frac{\partial \hat{\phi}}{\partial \phi} = -\hat{s}, \quad (3.67)$$

while neither of them depends on s or z . Consider, for example, a particle moving in the plane $z = 0$, whose position is $\mathbf{r} = s\hat{s}$, with s and ϕ both functions of time. By the chain rule, $d\hat{s}/dt = (\partial \hat{s}/\partial \phi)(d\phi/dt) = (d\phi/dt)\hat{\phi}$, and by the product rule the velocity is

$$\frac{d\mathbf{r}}{dt} = \frac{ds}{dt}\hat{s} + s\frac{d\phi}{dt}\hat{\phi}. \quad (3.68)$$

The first term is the rate at which the particle moves away from the origin, and the second is its motion round the origin. For motion on a circle s is constant, and the speed is the radius multiplied by the angular speed $d\phi/dt$, a result you will know from the study of circular motion at school.

In spherical coordinates (r, θ, ϕ) the position vector is $\mathbf{r} = r \sin \theta \cos \phi \hat{x} + r \sin \theta \sin \phi \hat{y} + r \cos \theta \hat{z}$. The derivative with respect to r is a unit vector, and the derivatives with respect to θ and ϕ were found in Section 3.2 for a sphere of radius R . With R replaced by r , these derivatives give the scale factors $h_1 = 1$, $h_2 = r$, and $h_3 = r \sin \theta$, and the unit vectors

$$\begin{aligned} \hat{\mathbf{r}} &= \sin \theta \cos \phi \hat{x} + \sin \theta \sin \phi \hat{y} + \cos \theta \hat{z}, \\ \hat{\theta} &= \cos \theta \cos \phi \hat{x} + \cos \theta \sin \phi \hat{y} - \sin \theta \hat{z}, \\ \hat{\phi} &= -\sin \phi \hat{x} + \cos \phi \hat{y}. \end{aligned} \quad (3.69)$$

The scalar product $\hat{\mathbf{r}} \cdot \hat{\theta}$ is $\sin \theta \cos \theta (\cos^2 \phi + \sin^2 \phi) - \cos \theta \sin \theta = 0$, and the other two scalar products vanish in the same way, so the coordinates are orthogonal. In Section 3.2 we found that $\partial \mathbf{r}/\partial \theta \times \partial \mathbf{r}/\partial \phi$ points along $\hat{\mathbf{r}}$, so $\hat{\theta} \times \hat{\phi} = \hat{\mathbf{r}}$, and by the cyclic pattern of the vector product $\hat{\mathbf{r}} \times \hat{\theta} = \hat{\phi}$, so the numbering is in the right order. The vector $\hat{\phi}$ is the same as in cylindrical coordinates. Figure

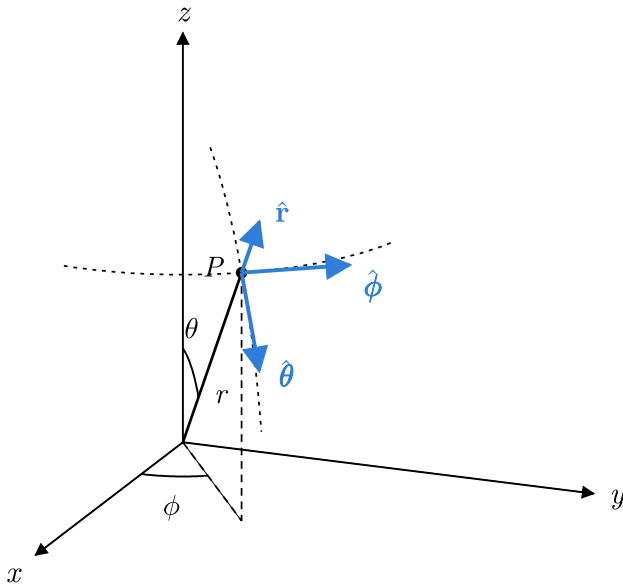


Figure 35. The spherical coordinates r , θ , and ϕ of a point P , and the unit vectors $\hat{r}$, $\hat{\theta}$, and $\hat{\phi}$ at P . The dotted curves are the coordinate lines through P along which only θ varies and only ϕ varies, and $\hat{\theta}$ and $\hat{\phi}$ are tangent to them.

35 shows the three unit vectors at a point P . If the positive z -axis is taken to point towards the north pole of a globe, $\hat{\mathbf{r}}$ points vertically upwards, $\hat{\boldsymbol{\theta}}$ points south along the meridian through P , and $\hat{\boldsymbol{\phi}}$ points east along the circle of latitude through P . The volume element is $h_1 h_2 h_3 \, dr \, d\theta \, d\phi = r^2 \sin \theta \, dr \, d\theta \, d\phi$, in agreement with Section 2.3.

The Gradient

The change in a scalar field f between neighbouring points can be written in two ways. By the chain rule it is $df = (\partial f / \partial u_1) du_1 + (\partial f / \partial u_2) du_2 + (\partial f / \partial u_3) du_3$, and by Section 3.1 it is $df = \nabla f \cdot d\mathbf{r}$. Let the components of the gradient along the unit vectors be $(\nabla f)_1$, $(\nabla f)_2$, and $(\nabla f)_3$. Since the unit vectors are perpendicular, the scalar product of ∇f with the expression for $d\mathbf{r}$ above is $(\nabla f)_1 h_1 du_1 + (\nabla f)_2 h_2 du_2 + (\nabla f)_3 h_3 du_3$. The two expressions for df must agree for every choice of the independent changes du_1 , du_2 , and du_3 , so their coefficients are equal, and

$$\nabla f = \frac{1}{h_1} \frac{\partial f}{\partial u_1} \hat{\mathbf{e}}_1 + \frac{1}{h_2} \frac{\partial f}{\partial u_2} \hat{\mathbf{e}}_2 + \frac{1}{h_3} \frac{\partial f}{\partial u_3} \hat{\mathbf{e}}_3. \quad (3.70)$$

Each component is the rate of change of f with distance along a coordinate line, since a change du_1 changes f by $(\partial f / \partial u_1) du_1$ over the distance $h_1 du_1$.

The Divergence

For the divergence we use the definition of Section 3.3, as the flux out of a small region divided by its volume, and take the region to be the small box bounded by the coordinate surfaces at u_1 and $u_1 + \Delta u_1$, at u_2 and $u_2 + \Delta u_2$, and at u_3 and $u_3 + \Delta u_3$. Figure 36 shows such a box for spherical coordinates.

The two faces on which u_1 is constant have outward normals $\hat{\mathbf{e}}_1$ at $u_1 + \Delta u_1$ and $-\hat{\mathbf{e}}_1$ at u_1 , and each has area $h_2 h_3 \Delta u_2 \Delta u_3$, evaluated on that face. The flux through the pair is therefore

$$\begin{aligned} & \left[(F_1 h_2 h_3) \Big|_{u_1 + \Delta u_1} - (F_1 h_2 h_3) \Big|_{u_1} \right] \Delta u_2 \Delta u_3 \\ & \approx \frac{\partial (h_2 h_3 F_1)}{\partial u_1} \Delta u_1 \Delta u_2 \Delta u_3, \end{aligned} \quad (3.71)$$

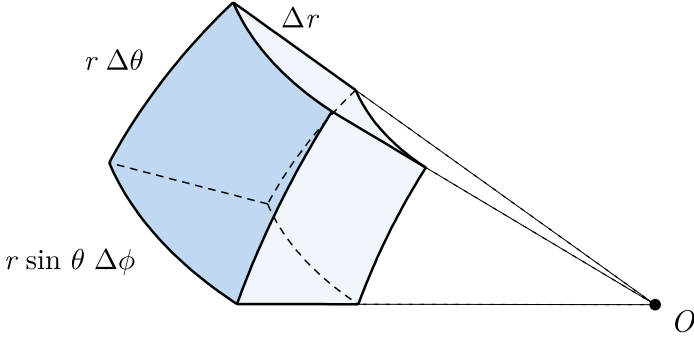


Figure 36. A small box bounded by coordinate surfaces, drawn for spherical coordinates, whose edges are $h_1 \Delta u_1 = \Delta r$, $h_2 \Delta u_2 = r \Delta \theta$, and $h_3 \Delta u_3 = r \sin \theta \Delta \phi$. The inner and outer faces, on which r is constant, are shaded, and the outer face has the larger area.

by Taylor's theorem. The scale factors must be kept inside the derivative, because the area of a face depends on where the face is. In spherical coordinates the outer face of the box has the area $r^2 \sin \theta \Delta \theta \Delta \phi$ evaluated at the larger radius, as Figure 36 shows. The other two pairs of faces give similar terms, and dividing the total flux by the volume $h_1 h_2 h_3 \Delta u_1 \Delta u_2 \Delta u_3$ gives

$$\nabla \cdot \mathbf{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial(h_2 h_3 F_1)}{\partial u_1} + \frac{\partial(h_3 h_1 F_2)}{\partial u_2} + \frac{\partial(h_1 h_2 F_3)}{\partial u_3} \right]. \quad (3.72)$$

By analogy with the gradient, one might expect the divergence to be $(1/h_1) \partial F_1 / \partial u_1$ plus two similar terms. The field $\hat{\mathbf{r}}$ shows why this expectation fails. Its component along $\hat{\mathbf{r}}$ is the constant 1, and the expected formula would give a divergence of zero, whereas in Problem 3 of Section 3.1 you found $\nabla \cdot \hat{\mathbf{r}} = 2/r$. A fluid moving outwards with unit speed in every direction spreads over larger and larger spheres, so more fluid leaves the outer face of the box than enters the inner face. The formula above, with $h_2 h_3 = r^2 \sin \theta$ and $h_1 h_2 h_3 = r^2 \sin \theta$, gives $(1/r^2) \partial(r^2) / \partial r = 2/r$, as it should.

The Curl

For the curl we use the definition of Section 3.3, as the circulation round a small loop divided by its area. The component along $\hat{\mathbf{e}}_3$ is found from a loop on a surface of constant u_3 , bounded by the coordinate lines at u_1 and $u_1 + \Delta u_1$ and at u_2 and $u_2 + \Delta u_2$, which is one face of a box like that of Figure 36. The loop is traversed along $\hat{\mathbf{e}}_1$, then along $\hat{\mathbf{e}}_2$, then back along $-\hat{\mathbf{e}}_1$ and $-\hat{\mathbf{e}}_2$, which is the direction set by the right-hand rule, since $\hat{\mathbf{e}}_1 \times \hat{\mathbf{e}}_2 = \hat{\mathbf{e}}_3$. The two sides along which u_1 varies have length $h_1 \Delta u_1$ and contribute $(F_1 h_1) \Delta u_1$ evaluated at u_2 and minus the same quantity evaluated at $u_2 + \Delta u_2$. The two sides along which u_2 varies contribute $(F_2 h_2) \Delta u_2$ evaluated at $u_1 + \Delta u_1$ and minus the same quantity evaluated at u_1 . By Taylor's theorem the circulation is

$$\oint \mathbf{F} \cdot d\mathbf{r} \approx \left[\frac{\partial(h_2 F_2)}{\partial u_1} - \frac{\partial(h_1 F_1)}{\partial u_2} \right] \Delta u_1 \Delta u_2, \quad (3.73)$$

and dividing by the area $h_1 h_2 \Delta u_1 \Delta u_2$ gives the component of the curl along $\hat{\mathbf{e}}_3$. The other components follow by changing the numbers cyclically, and

$$\begin{aligned} \nabla \times \mathbf{F} = & \frac{\hat{\mathbf{e}}_1}{h_2 h_3} \left[\frac{\partial(h_3 F_3)}{\partial u_2} - \frac{\partial(h_2 F_2)}{\partial u_3} \right] \\ & + \frac{\hat{\mathbf{e}}_2}{h_3 h_1} \left[\frac{\partial(h_1 F_1)}{\partial u_3} - \frac{\partial(h_3 F_3)}{\partial u_1} \right] \\ & + \frac{\hat{\mathbf{e}}_3}{h_1 h_2} \left[\frac{\partial(h_2 F_2)}{\partial u_1} - \frac{\partial(h_1 F_1)}{\partial u_2} \right]. \end{aligned} \quad (3.74)$$

With all three scale factors equal to 1, the formulas for the divergence and the curl reduce to the Cartesian formulas of Section 3.1.

The Laplacian

The Laplacian of a scalar field is the divergence of its gradient. Substituting the components of the gradient, $F_1 = (1/h_1) \partial f / \partial u_1$ and so on, into the formula for the divergence gives

$$\begin{aligned} \nabla^2 f = & \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial u_1} \right) + \frac{\partial}{\partial u_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial u_2} \right) \right. \\ & \left. + \frac{\partial}{\partial u_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial u_3} \right) \right]. \end{aligned} \quad (3.75)$$

In Cartesian coordinates the Laplacian of a vector field acts on each component separately. In curvilinear coordinates the unit vectors change from point to point, and the Laplacian of a vector field takes a more complicated form. The Laplacian of a vector field in curvilinear coordinates is beyond the scope of this course.

The Formulas in Cylindrical and Spherical Coordinates

Substituting the scale factors $h_1 = 1$, $h_2 = s$, and $h_3 = 1$ into the general formulas gives, in cylindrical coordinates,

$$\begin{aligned}
 \nabla f &= \frac{\partial f}{\partial s} \hat{\mathbf{s}} + \frac{1}{s} \frac{\partial f}{\partial \phi} \hat{\boldsymbol{\phi}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}}, \\
 \nabla \cdot \mathbf{F} &= \frac{1}{s} \frac{\partial(sF_s)}{\partial s} + \frac{1}{s} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}, \\
 \nabla \times \mathbf{F} &= \left(\frac{1}{s} \frac{\partial F_z}{\partial \phi} - \frac{\partial F_\phi}{\partial z} \right) \hat{\mathbf{s}} + \left(\frac{\partial F_s}{\partial z} - \frac{\partial F_z}{\partial s} \right) \hat{\boldsymbol{\phi}} \\
 &\quad + \frac{1}{s} \left(\frac{\partial(sF_\phi)}{\partial s} - \frac{\partial F_s}{\partial \phi} \right) \hat{\mathbf{z}}, \\
 \nabla^2 f &= \frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial f}{\partial s} \right) + \frac{1}{s^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2},
 \end{aligned} \tag{3.76}$$

where F_s , F_ϕ , and F_z are the components of $\mathbf{F}$ along $\hat{\mathbf{s}}$, $\hat{\boldsymbol{\phi}}$, and $\hat{\mathbf{z}}$. The formulas for plane polar coordinates are the same, with the terms containing z and F_z left out. Substituting $h_1 = 1$, $h_2 = r$, and $h_3 = r \sin \theta$ gives, in spherical coordinates,

$$\begin{aligned}
 \nabla f &= \frac{\partial f}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\boldsymbol{\phi}}, \\
 \nabla \cdot \mathbf{F} &= \frac{1}{r^2} \frac{\partial(r^2 F_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta F_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}, \\
 \nabla \times \mathbf{F} &= \frac{1}{r \sin \theta} \left(\frac{\partial(\sin \theta F_\phi)}{\partial \theta} - \frac{\partial F_\theta}{\partial \phi} \right) \hat{\mathbf{r}} \\
 &\quad + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial F_r}{\partial \phi} - \frac{\partial(r F_\phi)}{\partial r} \right) \hat{\boldsymbol{\theta}} + \frac{1}{r} \left(\frac{\partial(r F_\theta)}{\partial r} - \frac{\partial F_r}{\partial \theta} \right) \hat{\boldsymbol{\phi}}, \\
 \nabla^2 f &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) \\
 &\quad + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2},
 \end{aligned} \tag{3.77}$$

where common factors of r and $\sin \theta$ have been cancelled from the numerators and the denominators.

These formulas shorten many calculations with fields that have the symmetry of the coordinates. The field of a point charge, which is a multiple of $\hat{\mathbf{r}}/r^2$, gives a first example. Its only component is $F_r = 1/r^2$, so the divergence is $(1/r^2) \partial(r^2 \cdot r^{-2})/\partial r = 0$ wherever $r \neq 0$, which we found in Section 3.1 with the product rule. In the same way, $f = 1/r$ has $\partial f/\partial r = -1/r^2$, and its Laplacian is $(1/r^2) \partial(-r^2/r^2)/\partial r = 0$, again wherever $r \neq 0$.

As another example, consider the rotation of Section 3.1, $\omega(-y\hat{\mathbf{x}} + x\hat{\mathbf{y}})$. The vector $-y\hat{\mathbf{x}} + x\hat{\mathbf{y}}$ is $s\hat{\phi}$, since $x = s \cos \phi$ and $y = s \sin \phi$, so the field is $\omega s \hat{\phi}$ and its only component is $F_\phi = \omega s$. The formula for the curl gives $(1/s) \partial(\omega s^2)/\partial s \hat{\mathbf{z}} = 2\omega \hat{\mathbf{z}}$, as before. The field of Problem 3 of Section 3.2, $(-y\hat{\mathbf{x}} + x\hat{\mathbf{y}})/(x^2 + y^2)$, is $\hat{\phi}/s$, with $F_\phi = 1/s$, and its curl is $(1/s) \partial(s \cdot s^{-1})/\partial s \hat{\mathbf{z}} = \mathbf{0}$ wherever $s \neq 0$, without the use of the quotient rule. This form also shows why its circulation round a circle of radius s does not depend on s , since the field is tangent to the circle and has magnitude $1/s$, and the circumference is $2\pi s$.

Problems

§3.1 Gradient, Divergence, and Curl

1. Show that the vector $(x/a_1^2)\hat{\mathbf{x}} + (y/a_2^2)\hat{\mathbf{y}} + (z/a_3^2)\hat{\mathbf{z}}$ is perpendicular to the ellipsoid $x^2/a_1^2 + y^2/a_2^2 + z^2/a_3^2 = 1$ at the point (x, y, z) on it, and check the result for a sphere, where $a_1 = a_2 = a_3$.
2. Find the directional derivative of $f = xyz$ in the direction of $\hat{\mathbf{r}}$, and show that it equals $3f/r$. Explain this result by noting what happens to f when x , y , and z are all multiplied by the same factor.
3. Show that $\nabla(r^n) = nr^{n-2}\mathbf{r}$, that $\nabla \cdot \hat{\mathbf{r}} = 2/r$, and that $\nabla \times (r^n \mathbf{r}) = \mathbf{0}$ for any constant n .
4. Prove that $\nabla \times (f\mathbf{A}) = f \nabla \times \mathbf{A} + \nabla f \times \mathbf{A}$.
5. Prove that $\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$.
6. Let $\boldsymbol{\omega}$ be a constant vector. Show that the velocity field $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$

of a body rotating with angular velocity $\boldsymbol{\omega}$ has $\nabla \cdot \mathbf{v} = 0$ and $\nabla \times \mathbf{v} = 2\boldsymbol{\omega}$.

7. Find the field lines of $\mathbf{F} = x\hat{\mathbf{x}} - y\hat{\mathbf{y}}$. Show that the divergence and the curl of $\mathbf{F}$ are both zero, and describe what happens to the small box and the two short lines of fluid of this section when $\mathbf{F}$ is the velocity of a fluid.
8. Show that $\nabla^2(f\psi) = f\nabla^2\psi + 2\nabla f \cdot \nabla\psi + \psi\nabla^2 f$.
9. Show that $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$, by working out the x -component of each side.
10. A particle moves so that its acceleration is always parallel to its position vector. Show that $\mathbf{r} \times d\mathbf{r}/dt$ does not change with time.

§3.2 Line, Surface, and Volume Integrals

1. Find the length of one arch of the cycloid $\mathbf{r}(t) = R(t - \sin t)\hat{\mathbf{x}} + R(1 - \cos t)\hat{\mathbf{y}}$, with $0 \leq t \leq 2\pi$, which is the curve traced by a point on the rim of a wheel of radius R rolling along the x -axis.
2. Show that $\mathbf{F} = y^2\hat{\mathbf{x}} + 2xy\hat{\mathbf{y}}$ is conservative and find a potential for it. Evaluate $\int \mathbf{F} \cdot d\mathbf{r}$ from the origin to $(a, b, 0)$ along the straight line, and again along the x -axis to $(a, 0, 0)$ and then parallel to the y -axis, and check both results against the potential.
3. Show that the field $\mathbf{F} = (-y\hat{\mathbf{x}} + x\hat{\mathbf{y}})/(x^2 + y^2)$ has zero curl everywhere except on the z -axis, and that its circulation round a circle of radius R centred on the z -axis in the plane $z = 0$, taken anticlockwise when seen from above, is 2π .
4. A force $\mathbf{F} = -k_s\mathbf{r}$, with k_s a positive constant, acts on a particle. Find a potential for the force, and hence show that the work it does when the particle moves from A to B is $\frac{1}{2}k_s(r_A^2 - r_B^2)$, where r_A and r_B are the distances of the two points from the origin.
5. Find the flux of $\mathbf{F} = z\hat{\mathbf{z}}$ outwards through the sphere of radius R centred on the origin.
6. Find the area of the part of the sphere of radius R centred on the origin that lies within an angle α of the positive z -axis, and show that it equals the area of the curved surface of a cylinder of radius R with the same height as the cap.

7. Show that a surface $z = w(x, y)$, described by the parameters x and y , has $dS = [1 + (\partial w/\partial x)^2 + (\partial w/\partial y)^2]^{1/2} dx dy$. Use this result to find the area of the part of the paraboloid $z = (x^2 + y^2)/2p$ that lies above the disc $x^2 + y^2 \leq a^2$, where p is a positive constant.
8. Water flows along a straight pipe of radius R whose axis is the z -axis, with velocity $\mathbf{v} = v_0(1 - s^2/R^2)\hat{\mathbf{z}}$, where s is the distance from the axis and v_0 is a constant. Show that the volume of water passing through a cross-section per unit time is the cross-sectional area multiplied by half the speed on the axis.

§3.3 The Integral Theorems of Gauss, Stokes, and Green

1. Verify the divergence theorem for the field $\mathbf{F} = \mathbf{r}$ and the sphere of radius R centred on the origin.
2. Let $\mathbf{F} = z\hat{\mathbf{x}} + x\hat{\mathbf{y}} + y\hat{\mathbf{z}}$, and let C be the circle of radius R in the xy -plane centred on the origin, traversed anticlockwise when seen from above. Find the circulation of $\mathbf{F}$ round C , and verify Stokes's theorem using first the flat disc bounded by C and then the hemisphere $r = R, z \geq 0$.
3. Use Green's theorem to find the area enclosed by the curve $x = a \cos^3 t, y = a \sin^3 t$, with $0 \leq t \leq 2\pi$.
4. Apply the divergence theorem to the field $f\nabla\psi$ to show that $\int_V (f\nabla^2\psi + \nabla f \cdot \nabla\psi) dV = \oint_S f\nabla\psi \cdot d\mathbf{S}$. Deduce that $\int_V (f\nabla^2\psi - \psi\nabla^2 f) dV = \oint_S (f\nabla\psi - \psi\nabla f) \cdot d\mathbf{S}$.
5. By applying the divergence theorem to a field $\mathbf{c}$ that is the same at every point, show that $\oint_S d\mathbf{S} = \mathbf{0}$ for any closed surface S .
6. Show that the volume enclosed by a closed surface S is $\frac{1}{3} \oint_S \mathbf{r} \cdot d\mathbf{S}$, and check the result for the tetrahedron cut off by the plane $x + y + z = a$ and the three coordinate planes, using the triangle of Section 3.2.
7. Charge is spread with constant charge per unit volume ρ through a ball of radius R centred on the origin. Assuming that the field has the form $\mathbf{E} = E(r)\hat{\mathbf{r}}$, apply Gauss's law to a sphere of radius r to find $E(r)$ for $r < R$ and for $r > R$. Check that $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ inside the ball and $\nabla \cdot \mathbf{E} = 0$ outside it.

8. A gas fills all of space and moves with velocity $\mathbf{v} = k\mathbf{r}$, where k is a constant, and its density $\rho(t)$ is the same at every point. Use the continuity equation to find how ρ changes with time, and compare the result with the rate of expansion of a small volume found in Section 3.1.
9. The surface of a solid body is insulated, so that no heat crosses it. Show that the total heat energy $\int_V \rho c T \, dV$ in the body does not change with time.

§3.4 Curvilinear Coordinates

1. Parabolic cylindrical coordinates (u, v, z) are defined by $x = \frac{1}{2}(u^2 - v^2)$, $y = uv$, and $z = z$. Show that they are orthogonal, find the scale factors and the volume element, and show that $\nabla^2 f = (u^2 + v^2)^{-1}(\partial^2 f / \partial u^2 + \partial^2 f / \partial v^2) + \partial^2 f / \partial z^2$.
2. Show that $\partial \hat{\mathbf{r}} / \partial \theta = \hat{\boldsymbol{\theta}}$ and $\partial \hat{\mathbf{r}} / \partial \phi = \sin \theta \hat{\boldsymbol{\phi}}$, and that $\hat{\mathbf{r}}$ does not depend on r . Hence show that a particle whose spherical coordinates change with time has velocity $(dr/dt) \hat{\mathbf{r}} + r(d\theta/dt) \hat{\boldsymbol{\theta}} + r \sin \theta (d\phi/dt) \hat{\boldsymbol{\phi}}$.
3. Show that $\nabla^2(r^n) = n(n+1)r^{n-2}$, and find the values of n for which $\nabla^2(r^n) = 0$ wherever $r \neq 0$.
4. In cylindrical coordinates, show that $\nabla \cdot \hat{\mathbf{s}} = 1/s$, $\nabla \cdot \hat{\boldsymbol{\phi}} = 0$, and $\nabla \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}/s$. Check the first result by writing $\hat{\mathbf{s}}$ in Cartesian components and using the Cartesian formula for the divergence.
5. Show that any field of the form $G(r) \hat{\mathbf{r}}$ has zero curl, and that its divergence is $r^{-2} d(r^2 G)/dr$. Find all the functions G for which the divergence has the same constant value K at every point with $r \neq 0$, and compare the result with the field inside the charged ball of Problem 7 of Section 3.3.
6. Use the formula for the gradient in spherical coordinates to find $\nabla(r \cos \theta)$, and show that the result is $\hat{\mathbf{z}}$.
7. The temperature in a long solid cylinder depends only on the distance s from the axis and on the time t . Write the diffusion equation of Section 3.3 for this temperature in cylindrical coordinates.

Linear Algebra and Vector Spaces

4.1 Matrices, Determinants, and Linear Equations

Many relations in physics are linear, in the sense that doubling a cause doubles its effect and the effects of two causes add. The extension of a spring, for example, is proportional to the force applied to it. Sets of such relations are conveniently written with matrices. In this chapter we first develop the algebra of matrices and use it to solve simultaneous linear equations, then set the same ideas in the more general language of vector spaces, and finally find the directions that a matrix only stretches, as opposed to rotating them in addition to stretching them. These directions describe, for example, the vibrations of coupled oscillators.

Matrices Acting on Vectors

As an example of a linear relation, consider a rotation of the plane about the origin through an angle θ in the anticlockwise sense, as in Figure 37. A point at a distance $|\mathbf{x}|$ from the origin, on a line at an angle ψ to the x_1 -axis, has coordinates $x_1 = |\mathbf{x}| \cos \psi$ and $x_2 = |\mathbf{x}| \sin \psi$. After the rotation the line is at an angle $\psi + \theta$, so

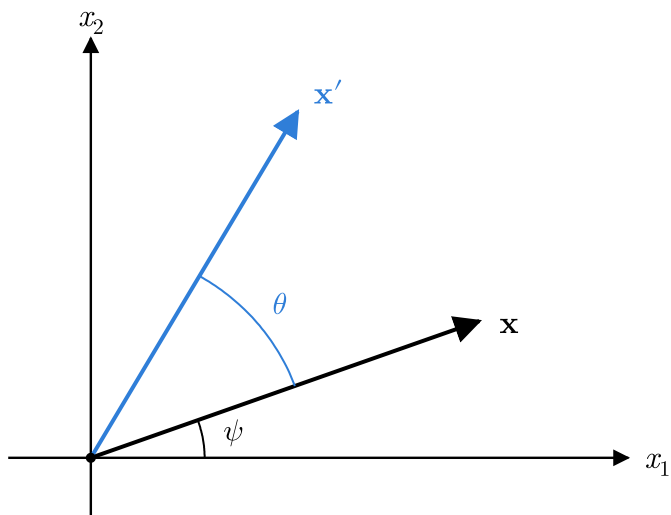


Figure 37. A rotation of the plane through an angle θ about the origin. The vector $\mathbf{x}$, at an angle ψ to the x_1 -axis, is carried into the vector $\mathbf{x}'$, which has the same length and lies at an angle $\psi + \theta$ to the axis.

by the compound-angle formulas the new coordinates are

$$\begin{aligned}x'_1 &= |\mathbf{x}| \cos(\psi + \theta) = x_1 \cos \theta - x_2 \sin \theta, \\x'_2 &= |\mathbf{x}| \sin(\psi + \theta) = x_1 \sin \theta + x_2 \cos \theta.\end{aligned}\tag{4.1}$$

Collecting the coefficients into an array, we write this pair of equations as

$$\begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \tag{4.2}$$

or $\mathbf{x}' = \mathbf{R}(\theta) \mathbf{x}$.

A rectangular array of numbers with m rows and n columns is called an $m \times n$ matrix, and the entry in row i and column j of a matrix $\mathbf{A}$ is written A_{ij} . A matrix with a single column is called a column vector, often written on one line as $\mathbf{x} = (x_1, \dots, x_n)$ to save space. An $m \times n$ matrix acts on a column vector with n components to give one with m components, according to the definition

$$(\mathbf{A}\mathbf{x})_i = \sum_{j=1}^n A_{ij} x_j, \tag{4.3}$$

so that the i th component is found by running along row i of the matrix and down the column vector, multiplying corresponding entries and adding the products. The map $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ is linear, since the definition gives $\mathbf{A}(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha\mathbf{A}\mathbf{x} + \beta\mathbf{A}\mathbf{y}$ for any numbers α and β .

Matrix Algebra

Matrices of the same size are added by adding corresponding entries, and a matrix is multiplied by a number by multiplying every entry by that number. The product of two matrices is defined to describe one map followed by another. If $\mathbf{B}$ acts on $\mathbf{x}$ and $\mathbf{A}$ then acts on the result, the definition above gives

$$(\mathbf{A}(\mathbf{B}\mathbf{x}))_i = \sum_j A_{ij} \sum_k B_{jk} x_k = \sum_k \left(\sum_j A_{ij} B_{jk} \right) x_k, \tag{4.4}$$

and we therefore define the product $\mathbf{AB}$ as the matrix with entries

$$(\mathbf{AB})_{ik} = \sum_j A_{ij} B_{jk}. \tag{4.5}$$

The product exists only when the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$. It is associative, $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$, because both sides describe the map $\mathbf{C}$, followed by $\mathbf{B}$, followed by $\mathbf{A}$. The square matrix $\mathbf{I}$ with ones on its leading diagonal and zeros elsewhere is called the identity matrix, and it satisfies $\mathbf{AI} = \mathbf{IA} = \mathbf{A}$.

A pair of rotations of the plane, performed in either order, turns it through the sum of their angles, as you will check in Problem 1. One might expect from this that the order of the factors in a matrix product never matters, as for ordinary numbers. This expectation turns out to be mistaken. As an example, consider the reflection in the x_1 -axis, which changes the sign of x_2 and is described by the matrix $\mathbf{P}$ with rows $(1, 0)$ and $(0, -1)$. Multiplying out gives

$$\mathbf{PR}(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{pmatrix}, \quad \mathbf{R}(\theta)\mathbf{P} = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}, \quad (4.6)$$

which differ unless $\sin \theta = 0$. A vector along the x_1 -axis, rotated and then reflected, ends at an angle $-\theta$ to the axis, while if it is reflected first the rotation carries it to an angle $+\theta$. Matrix multiplication is, in general, not commutative.

The transpose $\mathbf{A}^T$ of a matrix is found by exchanging its rows and columns, so that $(\mathbf{A}^T)_{ij} = A_{ji}$, and you will show in Problem 4 that $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$. A square matrix equal to its transpose is called symmetric. For two column vectors, the product $\mathbf{x}^T\mathbf{y} = \sum_i x_i y_i$ of a row vector with a column vector is the scalar product, written in components.

Determinants

For a two-by-two matrix the determinant is defined as

$$\det \mathbf{A} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc. \quad (4.7)$$

The matrix carries $(1, 0)$ and $(0, 1)$ into its columns $\mathbf{u}$ and $\mathbf{v}$, so it carries the unit square into the parallelogram of Figure 38. Of the pieces of the dashed rectangle that lie outside the parallelogram, two are triangles with sides a and c at the right angle, two are triangles with sides b and d , and two are rectangles with sides b

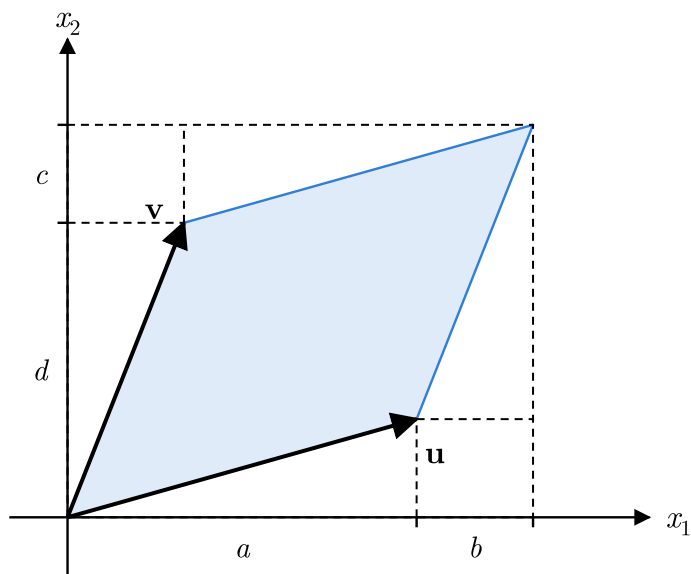


Figure 38. The parallelogram whose sides are the columns $\mathbf{u} = (a, c)$ and $\mathbf{v} = (b, d)$ of a two-by-two matrix. Its area is that of the dashed rectangle, $(a + b)(c + d)$, less the four right-angled triangles and the two small rectangles outside the parallelogram.

and c , so the area of the parallelogram is

$$(a+b)(c+d) - ac - bd - 2bc = ad - bc. \quad (4.8)$$

Because the map is linear, every small square is carried into a parallelogram with $|\det \mathbf{A}|$ times its area, so the area of any region is multiplied by $|\det \mathbf{A}|$. The sign of the determinant records whether the sense of rotation from $\mathbf{u}$ to $\mathbf{v}$ is the same as that from $(1, 0)$ to $(0, 1)$, so that the map preserves the sense of rotation, or the opposite, so that the map reverses it. The rotation has $\det \mathbf{R}(\theta) = \cos^2 \theta + \sin^2 \theta = 1$ and preserves areas and sense, while the reflection has $\det \mathbf{P} = -1$ and reverses the sense.

For a three-by-three matrix the determinant is defined by

$$\begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{vmatrix} = A_{11} \begin{vmatrix} A_{22} & A_{23} \\ A_{32} & A_{33} \end{vmatrix} - A_{12} \begin{vmatrix} A_{21} & A_{23} \\ A_{31} & A_{33} \end{vmatrix} \\ + A_{13} \begin{vmatrix} A_{21} & A_{22} \\ A_{31} & A_{32} \end{vmatrix}. \quad (4.9)$$

Writing out the vector product in components shows that the right-hand side is the scalar triple product $\mathbf{r}_1 \cdot (\mathbf{r}_2 \times \mathbf{r}_3)$ of the three rows, which is, up to its sign, the volume of the parallelepiped with those rows as edges. For larger matrices, the determinant M_{ij} of the matrix left by deleting row i and column j is called a minor, and $C_{ij} = (-1)^{i+j} M_{ij}$ is called the cofactor of the entry A_{ij} . The determinant of an $n \times n$ matrix is defined by the expansion along the first row,

$$\det \mathbf{A} = \sum_{j=1}^n A_{1j} C_{1j}, \quad (4.10)$$

of which the three-by-three definition is the case $n = 3$. Determinants have the following properties:

1. expanding along any row, or down any column, gives the same value, and in particular $\det \mathbf{A}^T = \det \mathbf{A}$;
2. multiplying one row by a number α multiplies the determinant by α , and if one row is the sum of two row vectors, the determinant is the sum of the two determinants with that row replaced by each of those vectors in turn;

3. exchanging two rows changes the sign of the determinant, so a matrix with two equal rows has a determinant equal to zero;
4. adding a multiple of one row to another row leaves the determinant unchanged.

The second property follows from the expansion along the row in question, since the cofactors of that row contain none of its entries. For three-by-three matrices, you will check the first property for the second row and for the transpose in Problem 3, and the third row and the columns are dealt with in the same way; the third property holds because exchanging two vectors in a scalar triple product changes its sign. The proofs of these two properties for larger matrices are beyond the scope of this course. The fourth property follows from the second and the third, since replacing row i by $\mathbf{r}_i + \alpha \mathbf{r}_k$ gives $\det \mathbf{A}$ plus α times the determinant of a matrix with two equal rows, which is zero. By the first property, the same statements hold for columns.

Since the map $\mathbf{B}$ multiplies every volume by $\det \mathbf{B}$, and the map $\mathbf{A}$ applied afterwards multiplies it by $\det \mathbf{A}$, with signs that combine as the senses of the two maps do, the determinant of the product satisfies the relation

$$\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}. \quad (4.11)$$

This argument covers two-by-two and three-by-three matrices, for which the determinant has been shown to be an area or a volume. The relation holds for square matrices of every size, but its proof for larger matrices is beyond the scope of this course.

The Inverse of a Matrix

The inverse of a square matrix $\mathbf{A}$ is the matrix $\mathbf{A}^{-1}$ with $\mathbf{A}^{-1}\mathbf{A} = \mathbf{AA}^{-1} = \mathbf{I}$, which describes the map that undoes the map $\mathbf{A}$. It can exist only when $\det \mathbf{A} \neq 0$, since the relation $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$ gives $\det \mathbf{A}^{-1} \det \mathbf{A} = 1$. A matrix whose determinant is zero is called singular. A singular two-by-two matrix squashes the plane onto a line or a point, and the vector that each image came from cannot be recovered.

The adjugate $\text{adj} \mathbf{A}$ is the transpose of the matrix of cofactors,

$(\text{adj } \mathbf{A})_{jk} = C_{kj}$, and

$$(\mathbf{A} \text{adj } \mathbf{A})_{ik} = \sum_j A_{ij} C_{kj}. \quad (4.12)$$

When $k = i$ the sum is the expansion of $\det \mathbf{A}$ along row i . When $k \neq i$ it is the expansion along row k of the matrix obtained from $\mathbf{A}$ by replacing row k with a copy of row i , since the cofactors of row k do not involve that row, and this matrix has two equal rows, so the sum is zero. Therefore $\mathbf{A} \text{adj } \mathbf{A} = (\det \mathbf{A}) \mathbf{I}$, and when $\det \mathbf{A} \neq 0$,

$$\mathbf{A}^{-1} = \frac{\text{adj } \mathbf{A}}{\det \mathbf{A}}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad (4.13)$$

the second formula being the two-by-two case, which gives $\mathbf{R}(\theta)^{-1} = \mathbf{R}(-\theta) = \mathbf{R}(\theta)^T$. Since $\mathbf{B}^{-1}\mathbf{A}^{-1}$ undoes first $\mathbf{A}$ and then $\mathbf{B}$, the inverse of a product is $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.

Simultaneous Linear Equations

A set of m linear equations in n unknowns can be written as $\mathbf{Ax} = \mathbf{b}$, with $\mathbf{A}$ the $m \times n$ matrix of coefficients and $\mathbf{b}$ the column of right-hand sides. When $\mathbf{A}$ is square and not singular, the single solution is $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$, but in practice the equations are usually solved by Gaussian elimination, which uses the following operations:

1. exchanging two equations;
2. multiplying an equation by a non-zero number;
3. adding a multiple of one equation to another.

Each operation can be undone by another of the same kind, so none changes the set of solutions. They are applied to the augmented matrix, $\mathbf{A}$ with the column $\mathbf{b}$ written alongside it, to clear the entries below the leading diagonal, column by column.

As an example, consider the equations

$$x_1 + x_2 + x_3 = b_1, \quad x_1 + 2x_2 + 3x_3 = b_2, \quad x_1 + 3x_2 + \eta x_3 = b_3, \quad (4.14)$$

in which b_1 , b_2 , b_3 , and η are constants. Subtracting the first row of the augmented matrix from the other two clears the first column, and subtracting twice the new second row from the third clears the

second column:

$$\begin{aligned} \left(\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 1 & 2 & 3 & b_2 \\ 1 & 3 & \eta & b_3 \end{array} \right) &\rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 1 & 2 & b_2 - b_1 \\ 0 & 2 & \eta - 1 & b_3 - b_1 \end{array} \right) \\ &\rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 1 & 2 & b_2 - b_1 \\ 0 & 0 & \eta - 5 & b_1 - 2b_2 + b_3 \end{array} \right). \end{aligned} \quad (4.15)$$

A matrix in which each row that is not entirely zero begins with more zeros than the row above it, and in which the rows that are entirely zero, if there are any, come last, is said to be in echelon form. The number of rows of a matrix in echelon form that are not entirely zero is called its rank, and the rank of any other matrix is the rank of the echelon form that elimination produces from it. If $\eta \neq 5$, the rows, taken from the bottom up, give x_3 , then x_2 , then x_1 , and there is exactly one solution. If $\eta = 5$, the last row becomes $0 = b_1 - 2b_2 + b_3$. When the right-hand side of this equation is not zero, there is no solution. When it is zero, x_3 may take any value τ , and the rows above give $x_2 = b_2 - b_1 - 2\tau$ and $x_1 = 2b_1 - b_2 + \tau$, so there are infinitely many solutions.

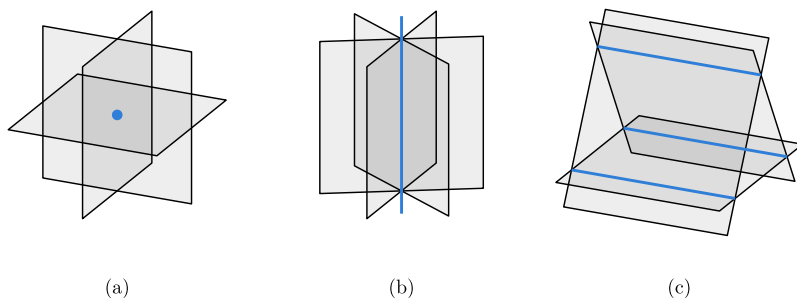


Figure 39. The planes of three linear equations in three unknowns. (a) The planes meet in a point, and there is one solution. (b) The planes share a line, and there are infinitely many solutions. (c) Each pair of planes meets in a line, the three lines are parallel, and there is no solution.

Each equation in three unknowns describes a plane, and Figure 39 illustrates the three cases with one arrangement of planes for each. Planes can fail to meet in other ways as well, for example when two

of them are parallel. In general, if the rank of the augmented matrix exceeds that of $\mathbf{A}$, elimination has produced an equation that sets zero equal to a non-zero number, and there is no solution. If the two ranks are equal to each other and to n , there is exactly one solution, and if they are equal but less than n , there are infinitely many, with $n - \text{rank } \mathbf{A}$ constants that can be chosen freely. In the example, the determinant of the coefficients is $\eta - 5$, which vanishes exactly when elimination leaves a row of zeros among the coefficients. The same is true in general, because row operations change a determinant only by non-zero factors, and the determinant of a square matrix in echelon form is the product of its diagonal entries, as you will show in Problem 5. A square matrix therefore has rank n exactly when its determinant is not zero. In particular, n equations $\mathbf{Ax} = \mathbf{0}$ with every right-hand side zero have a solution other than $\mathbf{x} = \mathbf{0}$ if and only if $\det \mathbf{A} = 0$.

Cramer's Rule

When $\det \mathbf{A} \neq 0$, let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be the columns of $\mathbf{A}$, so that the equations say $\mathbf{b} = \sum_j x_j \mathbf{a}_j$, and let $\mathbf{A}_i$ be $\mathbf{A}$ with its i th column replaced by $\mathbf{b}$. By the second property, applied to columns,

$$\det \mathbf{A}_i = \sum_j x_j \det(\mathbf{a}_1, \dots, \mathbf{a}_j, \dots, \mathbf{a}_n), \quad (4.16)$$

where $\mathbf{a}_j$ stands in the i th place of each determinant. Every term with $j \neq i$ has two equal columns and vanishes by the third property, leaving $x_i \det \mathbf{A}$, so

$$x_i = \frac{\det \mathbf{A}_i}{\det \mathbf{A}}, \quad (4.17)$$

which is called Cramer's rule.

4.2 Vector Spaces and Linear Maps

The column vectors of Section 4.1 are added and multiplied by numbers according to the same rules as the arrows used at school for displacements and forces. Polynomials and functions obey the same rules, and results proved from the rules alone hold for all of these objects together. This section states the rules as a definition, builds on them the ideas of a basis and of dimension, and shows

that a linear map is described by a matrix once a basis has been chosen.

The Axioms of a Vector Space

A vector space is a set V of objects, called vectors, with an operation that adds two vectors and an operation that multiplies a vector by a number, called a scalar, such that the following axioms hold for all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and all scalars α and β :

1. the sum $\mathbf{u} + \mathbf{v}$ and the multiple $\alpha\mathbf{u}$ are also in V ;
2. addition is commutative and associative, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ and $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$;
3. there is a zero vector $\mathbf{0}$ with $\mathbf{u} + \mathbf{0} = \mathbf{u}$, and every vector $\mathbf{u}$ has a negative $-\mathbf{u}$ with $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$;
4. $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$ and $(\alpha + \beta)\mathbf{u} = \alpha\mathbf{u} + \beta\mathbf{u}$;
5. $\alpha(\beta\mathbf{u}) = (\alpha\beta)\mathbf{u}$ and $1\mathbf{u} = \mathbf{u}$.

When the scalars are real, V is called a real vector space, and when they are complex, a complex vector space. The following sets are real vector spaces:

1. the arrows of school physics, added by the triangle rule;
2. the column vectors with n real components, added component by component, a space written $\mathbb{R}^n$;
3. the polynomials in x of degree at most n with real coefficients, added by adding coefficients;
4. the real functions on an interval $a \leq x \leq b$, added point by point, $(f + g)(x) = f(x) + g(x)$.

For the last three, the axioms are the rules of arithmetic applied to each component, coefficient, or value. A set of this kind can fail to be a vector space by breaking the first axiom. The polynomials of degree exactly n do not form one, because $x^n + (1 - x^n) = 1$, and nor do the functions with $f(a) = 1$, since the sum of two of them has the value 2 at $x = a$. The functions with $f(a) = f(b) = 0$, which could, for example, describe the shapes of a string clamped at both ends, do form a vector space.

A subset of a vector space that is itself a vector space under the same operations is called a subspace. A subset is a subspace if the sums and multiples of its members lie inside it, since the other axioms hold because they hold in the larger space. The vectors in $\mathbb{R}^3$ perpendicular to a fixed vector, for instance, form a subspace, a plane through the origin.

Linear Independence, Bases, and Dimension

A vector of the form $\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k$ is called a linear combination of the vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$. The set of all their linear combinations is called their span, and a pair of arrows that do not lie along one line, for instance, spans a plane. The vectors are called linearly independent if

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_k \mathbf{v}_k = \mathbf{0} \quad (4.18)$$

only when every α_i is zero, and linearly dependent otherwise. In a dependent set some coefficient, say α_1 , is not zero, and dividing by it expresses $\mathbf{v}_1$ as a combination of the others.

For n vectors in $\mathbb{R}^n$, let $\mathbf{A}$ be the matrix whose columns are the vectors. The equation above is then $\mathbf{A}\boldsymbol{\alpha} = \mathbf{0}$, with $\boldsymbol{\alpha}$ the column of coefficients, and by the result of Section 4.1 on equations whose right-hand sides are all zero, it has a solution other than $\boldsymbol{\alpha} = \mathbf{0}$ if and only if $\det \mathbf{A} = 0$. The vectors are therefore independent exactly when $\det \mathbf{A} \neq 0$. For the polynomials $1, x$, and x^2 on an interval, if $\alpha_0 + \alpha_1 x + \alpha_2 x^2 = 0$ for every x in the interval, the polynomial has infinitely many roots, whereas a non-zero polynomial of degree at most two has at most two, so the three coefficients vanish and the polynomials are independent.

A set of linearly independent vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ that spans V is called a basis of V . Every vector can be written as $\mathbf{v} = \sum_i v_i \mathbf{e}_i$, and in only one way, since subtracting a second expression $\sum_i v'_i \mathbf{e}_i$ gives $\sum_i (v_i - v'_i) \mathbf{e}_i = \mathbf{0}$, and independence makes every $v'_i = v_i$. The numbers v_i are called the components of $\mathbf{v}$ in the basis. The standard basis of $\mathbb{R}^n$ consists of $\mathbf{e}_1 = (1, 0, \dots, 0)$, $\mathbf{e}_2 = (0, 1, 0, \dots, 0)$, and so on, and for the polynomials of degree at most two, $1, x$, and x^2 form a basis in which the components of $c_0 + c_1 x + c_2 x^2$ are (c_0, c_1, c_2) .

A space has many bases, but they all contain the same number of vectors. To see why, suppose that $\mathbf{e}_1, \dots, \mathbf{e}_m$ span V , and let $\mathbf{f}_1, \dots, \mathbf{f}_n$ be vectors in V with $n > m$. Each of the vectors $\mathbf{f}_j$ can be written as $\mathbf{f}_j = \sum_i S_{ij} \mathbf{e}_i$, with an $m \times n$ matrix S . The m equations $\sum_j S_{ij} \alpha_j = 0$ in n unknowns have rank at most $m < n$, so by Section 4.1 they have solutions with the α_j not all zero, and for such a solution

$$\sum_j \alpha_j \mathbf{f}_j = \sum_i \left(\sum_j S_{ij} \alpha_j \right) \mathbf{e}_i = \mathbf{0}. \quad (4.19)$$

An independent set therefore has no more members than a spanning set, and two bases, each both independent and spanning, have the same number of members. This number is called the dimension of the space. The dimension of $\mathbb{R}^n$ is n , and that of the polynomials of degree at most n is $n + 1$. The functions on an interval include, for every N , the polynomials $1, x, \dots, x^N$, which are independent by the argument given above for $1, x$, and x^2 , since a non-zero polynomial of degree at most N has at most N roots, so no finite set spans them, and their space is called infinite-dimensional. The rest of this chapter concerns spaces of finite dimension, in which each vector is described by its column of components, so that the column vectors and matrices of Section 4.1 apply to them all.

Linear Maps and Their Matrices

A map T from a vector space V to a vector space W is called linear if

$$T(\alpha \mathbf{u} + \beta \mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}) \quad (4.20)$$

for all vectors and scalars. A rotation of arrows is linear, and so is the derivative $D = d/dx$ acting on polynomials, since the derivative of a sum is the sum of the derivatives and constant factors can be taken outside.

Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be a basis of V and $\mathbf{f}_1, \dots, \mathbf{f}_m$ a basis of W , and expand the image of each basis vector of V as

$$T(\mathbf{e}_j) = \sum_{i=1}^m A_{ij} \mathbf{f}_i. \quad (4.21)$$

The coefficients form an $m \times n$ matrix A , called the matrix of T in

these bases, whose j th column holds the components of the image of $\mathbf{e}_j$. By linearity, a vector $\mathbf{v} = \sum_j v_j \mathbf{e}_j$ is carried into

$$T(\mathbf{v}) = \sum_j v_j T(\mathbf{e}_j) = \sum_i \left(\sum_j A_{ij} v_j \right) \mathbf{f}_i, \quad (4.22)$$

so the components of $T(\mathbf{v})$ are $\mathbf{A}$ times the components of $\mathbf{v}$, and, by the argument of Section 4.1, one map followed by another is described by the product of their matrices.

As an example, consider the derivative acting on the polynomials of degree at most two, with the basis $1, x, x^2$. Since $D1 = 0$, $Dx = 1$, and $Dx^2 = 2x$, the columns of the matrix are the components of 0 , 1 , and $2x$, and

$$D = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}. \quad (4.23)$$

Acting on (c_0, c_1, c_2) , the matrix gives $(c_1, 2c_2, 0)$, the components of the derivative $c_1 + 2c_2x$. The cube of D is the zero matrix, because the third derivative of a quadratic vanishes, and D is singular, in agreement with the fact that the derivative sends every constant to zero, so that the constant term of a polynomial cannot be recovered from its derivative.

Change of Basis

The components of a vector depend on the basis, as Figure 40 shows. Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be an old basis and $\mathbf{e}'_1, \dots, \mathbf{e}'_n$ a new one, with

$$\mathbf{e}'_j = \sum_i S_{ij} \mathbf{e}_i, \quad (4.24)$$

so that the j th column of $\mathbf{S}$ holds the old components of $\mathbf{e}'_j$. A vector with new components x'_j is $\sum_j x'_j \mathbf{e}'_j = \sum_i \left(\sum_j S_{ij} x'_j \right) \mathbf{e}_i$, so its old and new columns of components are related by $\mathbf{x} = \mathbf{S}\mathbf{x}'$. The columns of $\mathbf{S}$ describe a basis and are independent, so $\mathbf{S}$ has an inverse, and $\mathbf{x}' = \mathbf{S}^{-1}\mathbf{x}$. For the basis of Figure 40,

$$\mathbf{S} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{S}^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}. \quad (4.25)$$

If a linear map of V into itself has the matrix $\mathbf{A}$ in the old basis, the old components of a vector and its image are related by $\mathbf{y} = \mathbf{A}\mathbf{x}$,

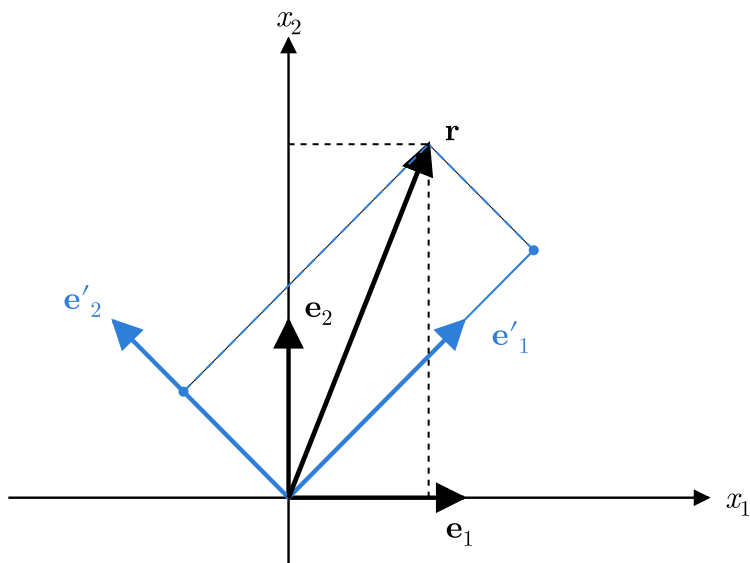


Figure 40. A vector $\mathbf{r}$ described in two bases. The black dashed lines, parallel to the standard basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$, give its components in the standard basis, and the blue dashed lines, parallel to the new basis vectors $\mathbf{e}'_1 = \mathbf{e}_1 + \mathbf{e}_2$ and $\mathbf{e}'_2 = -\mathbf{e}_1 + \mathbf{e}_2$, give its components in the new basis.

and the new components by $\mathbf{y}' = \mathbf{S}^{-1}\mathbf{y} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}\mathbf{x}'$. The matrix of the map in the new basis is therefore

$$\mathbf{A}' = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}. \quad (4.26)$$

This relation is called a similarity transformation, and matrices related by it are said to be similar. Similar matrices describe the same map, so any number that describes the map itself is the same for both matrices. The determinant is one such number, since the relation $\det(\mathbf{A}\mathbf{B}) = \det \mathbf{A} \det \mathbf{B}$ of Section 4.1 gives $\det \mathbf{A}' = \det \mathbf{A}$, as it must for a factor by which the map scales volumes. The trace, $\text{tr } \mathbf{A} = \sum_i A_{ii}$, the sum of the diagonal entries, is another, as you will show in Problem 4.

A well-chosen basis can simplify the matrix of a map. As an example, consider the reflection in the line $x_2 = x_1$, which exchanges the two components of a vector. In the standard basis and in the basis of Figure 40 its matrices are

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{A}' = \mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (4.27)$$

since $\mathbf{e}'_1$ lies along the mirror line and is unchanged, while $\mathbf{e}'_2$ is perpendicular to it and is reversed.

The Dual Space

A linear map from a vector space V to its scalars is called a linear functional. On $\mathbb{R}^n$, the map $\mathbf{x} \mapsto \mathbf{a}^T \mathbf{x}$ for a fixed $\mathbf{a}$ is one, and on the polynomials, so are the maps that send p to $p(0)$ and to $\int_a^b p(x) dx$. A physical instance is the change $d\Phi = \nabla \Phi \cdot d\mathbf{r}$ in a scalar field Φ , which, as we found in Section 3.1 when we wrote the total differential as the scalar product of the gradient with the displacement, depends linearly on the displacement $d\mathbf{r}$. With $(f + g)(\mathbf{v}) = f(\mathbf{v}) + g(\mathbf{v})$ and $(\alpha f)(\mathbf{v}) = \alpha f(\mathbf{v})$, the linear functionals on V form a vector space, called the dual space of V and denoted V^* .

In a basis $\mathbf{e}_1, \dots, \mathbf{e}_n$, a functional f is fixed by the numbers $f_j = f(\mathbf{e}_j)$, since linearity gives

$$f(\mathbf{v}) = \sum_j f_j v_j = \begin{pmatrix} f_1 & f_2 & \cdots & f_n \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}. \quad (4.28)$$

Vectors are therefore represented by columns and functionals by rows, and the value of a functional on a vector is the product of the row with the column. The functional $p \mapsto p(0)$ on the polynomials of degree at most two, for instance, is the row $(1, 0, 0)$. The functionals ε_i defined by $\varepsilon_i(\mathbf{e}_j) = \delta_{ij}$, where δ_{ij} , which is called the Kronecker delta, is 1 when $i = j$ and 0 otherwise, pick out the components, $\varepsilon_i(\mathbf{v}) = v_i$. Every functional is $f = \sum_i f_i \varepsilon_i$, and the ε_i are independent, so they form a basis of V^* , called the dual basis, and V^* has the same dimension as V .

Under a change of basis the new components of f are

$$f'_j = f(\mathbf{e}'_j) = \sum_i S_{ij} f(\mathbf{e}_i) = \sum_i f_i S_{ij}, \quad (4.29)$$

so the row is multiplied on the right by $\mathbf{S}$, while the column of a vector is multiplied on the left by $\mathbf{S}^{-1}$. The value $f'\mathbf{x}' = f\mathbf{S}\mathbf{S}^{-1}\mathbf{x} = f\mathbf{x}$ is then the same in both bases, as it must be, since it is fixed by the functional and the vector alone.

4.3 Eigenvalues, Diagonalisation, and Normal Modes

The reflection in the line $x_2 = x_1$, at the end of Section 4.2, has a diagonal matrix in a basis of two vectors that it carries into multiples of themselves. We now look for such vectors for a general square matrix, and use the basis they form first to simplify quadratic forms and then to find the vibrations of two masses coupled by springs.

Eigenvalues and Eigenvectors

Let $\mathbf{A}$ be an $n \times n$ matrix. A column vector $\mathbf{x} \neq \mathbf{0}$ that satisfies

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x} \quad (4.30)$$

for some number λ is called an eigenvector of $\mathbf{A}$, and λ is called the eigenvalue to which it belongs. The map leaves the line of an eigenvector unchanged and multiplies its length by $|\lambda|$, reversing it if λ is negative, and any non-zero multiple of an eigenvector is an eigenvector with the same eigenvalue. Figure P4-4a shows the contrast for the matrix of the first example below: a vector in a general direction is turned as well as stretched, while a vector along either eigenvector is only stretched.

Written as $(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = \mathbf{0}$, the equation is a set of n equations with every right-hand side zero, so by Section 4.1 it has a solution $\mathbf{x} \neq \mathbf{0}$ if and only if

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0, \quad (4.31)$$

which is called the characteristic equation of $\mathbf{A}$. Its left-hand side is a polynomial of degree n in λ , which has n complex roots when repeated roots are counted as often as they occur, by the fundamental theorem of algebra stated in Section 2.1. The eigenvalues are these roots, and for each of them the eigenvectors are found by elimination.

As an example, consider the symmetric matrix

$$\mathbf{A} = \begin{pmatrix} \mu & \nu \\ \nu & \mu \end{pmatrix}, \quad (4.32)$$

with $\nu \neq 0$. The characteristic equation $(\mu - \lambda)^2 - \nu^2 = 0$ has the roots $\mu + \nu$ and $\mu - \nu$. For $\lambda = \mu + \nu$ both equations reduce to $x_2 = x_1$, so the eigenvectors are the multiples of $(1, 1)$, and for $\lambda = \mu - \nu$ they reduce to $x_2 = -x_1$, with eigenvectors $(1, -1)$. The reflection of Section 4.2 is the case $\mu = 0$, $\nu = 1$.

As a second example, the rotation matrix $\mathbf{R}(\theta)$ of Section 4.1 has the characteristic equation $(\cos \theta - \lambda)^2 + \sin^2 \theta = 0$, whose roots are $\lambda = \cos \theta \pm i \sin \theta = e^{\pm i\theta}$, by Euler's formula from Section 1.2. The eigenvalues are complex unless θ is a whole multiple of π , as the geometry requires, since any other rotation turns every real vector to a new direction. You will find the complex eigenvectors in Problem 2.

The eigenvalues do not depend on the basis used to describe the map. If $\mathbf{A}' = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}$, then $\mathbf{A}' - \lambda \mathbf{I} = \mathbf{S}^{-1}(\mathbf{A} - \lambda \mathbf{I})\mathbf{S}$, and the relation $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$ of Section 4.1 shows that similar matrices have the same characteristic equation. You will show in Problem 1 that the sum of the eigenvalues of a two-by-two matrix is its trace, and that the product of the eigenvalues of any square matrix is its determinant, in keeping with the result of Section 4.2 that the trace and the determinant are unchanged by a change of basis.

Diagonalisation

Suppose that A has n linearly independent eigenvectors $\mathbf{x}_1, \dots, \mathbf{x}_n$, with eigenvalues $\lambda_1, \dots, \lambda_n$ that need not all differ. Let S be the matrix whose j th column is $\mathbf{x}_j$, and the diagonal matrix with entries $\lambda_1, \dots, \lambda_n$. The j th column of AS is $\lambda_j \mathbf{x}_j$, which is also the j th column of $S \Lambda$, and S has an inverse because its columns are independent, so

$$S^{-1}AS = \Lambda. \quad (4.33)$$

The matrix A is said to be diagonalised by S , and by Section 4.2, $S^{-1}AS$ is the matrix of the same map in the basis of eigenvectors. One use of this result is in calculating powers. Since $A = S \Lambda S^{-1}$, the factors $S^{-1}S$ inside A^k reduce to I , and $A^k = S \Lambda^k S^{-1}$, where Λ^k has the diagonal entries λ_j^k .

A matrix with n different eigenvalues always has n independent eigenvectors. To prove this, suppose that eigenvectors $\mathbf{x}_j$ belonging to different eigenvalues were dependent, and choose the relation $\sum_j \alpha_j \mathbf{x}_j = \mathbf{0}$ between them with the fewest non-zero coefficients. It has at least two terms, since an eigenvector is not zero. Let λ_k be the eigenvalue of one of its eigenvectors. Multiplying the relation by $A - \lambda_k I$ multiplies each $\mathbf{x}_j$ by $\lambda_j - \lambda_k$, which gives

$$\sum_j \alpha_j (\lambda_j - \lambda_k) \mathbf{x}_j = \mathbf{0}, \quad (4.34)$$

a relation in which the term with $j = k$ has disappeared and every other non-zero coefficient α_j has been multiplied by the non-zero number $\lambda_j - \lambda_k$. It therefore has fewer non-zero coefficients than the relation we chose, but at least one, which is a contradiction, so the eigenvectors are independent.

When eigenvalues are repeated, there may be too few eigenvectors. As an example, consider the matrix A with rows $(1, 1)$ and $(0, 1)$, which slides each point parallel to the x_1 -axis by a distance equal to its x_2 coordinate. Its characteristic equation $(1 - \lambda)^2 = 0$ has the single root 1, and $(A - I)\mathbf{x} = \mathbf{0}$ reduces to $x_2 = 0$, so every eigenvector is a multiple of $(1, 0)$, and the matrix cannot be diagonalised.

Real Symmetric Matrices

Real symmetric matrices, such as the first example above, arise often in physics, and they have two properties that simplify their diagonalisation.

The first property is that their eigenvalues are real. Let $\mathbf{A}$ be real and symmetric, with $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, where λ and $\mathbf{x}$ may for the moment be complex. Multiplying on the left by the row $\mathbf{x}^{*\mathrm{T}}$, the transpose of the complex conjugate of $\mathbf{x}$, gives $\mathbf{x}^{*\mathrm{T}}\mathbf{A}\mathbf{x} = \lambda\mathbf{x}^{*\mathrm{T}}\mathbf{x}$. Since $\mathbf{A}$ is real, the conjugate of the eigenvalue equation is $\mathbf{A}\mathbf{x}^* = \lambda^*\mathbf{x}^*$. Taking its transpose with the result $(\mathbf{A}\mathbf{x}^*)^{\mathrm{T}} = \mathbf{x}^{*\mathrm{T}}\mathbf{A}^{\mathrm{T}}$, which you showed for any product in Problem 4 of Section 4.1, using $\mathbf{A}^{\mathrm{T}} = \mathbf{A}$, and multiplying on the right by $\mathbf{x}$, gives $\mathbf{x}^{*\mathrm{T}}\mathbf{A}\mathbf{x} = \lambda^*\mathbf{x}^{*\mathrm{T}}\mathbf{x}$. Subtracting the two expressions for $\mathbf{x}^{*\mathrm{T}}\mathbf{A}\mathbf{x}$ gives

$$(\lambda - \lambda^*) \sum_i |x_i|^2 = 0, \quad (4.35)$$

and the sum is positive because $\mathbf{x} \neq \mathbf{0}$, so λ is real. The equations for the eigenvectors then have real coefficients, and the eigenvectors can be taken to be real.

The second property is that eigenvectors belonging to different eigenvalues are orthogonal. Let $\mathbf{A}\mathbf{x}_1 = \lambda_1\mathbf{x}_1$ and $\mathbf{A}\mathbf{x}_2 = \lambda_2\mathbf{x}_2$ with $\lambda_1 \neq \lambda_2$. Then $\mathbf{x}_2^{\mathrm{T}}\mathbf{A}\mathbf{x}_1 = \lambda_1\mathbf{x}_2^{\mathrm{T}}\mathbf{x}_1$, and by the same result on transposes and the symmetry of $\mathbf{A}$, the left-hand side is also $(\mathbf{A}\mathbf{x}_2)^{\mathrm{T}}\mathbf{x}_1 = \lambda_2\mathbf{x}_2^{\mathrm{T}}\mathbf{x}_1$. Subtracting the two expressions for $\mathbf{x}_2^{\mathrm{T}}\mathbf{A}\mathbf{x}_1$ gives $(\lambda_1 - \lambda_2)\mathbf{x}_2^{\mathrm{T}}\mathbf{x}_1 = 0$, so $\mathbf{x}_2^{\mathrm{T}}\mathbf{x}_1 = 0$, as for $(1, 1)$ and $(1, -1)$ in the first example. A real symmetric matrix also has n independent eigenvectors when eigenvalues are repeated, and those belonging to a repeated eigenvalue can be chosen orthogonal, but the proof of this result is beyond the scope of this course.

If each eigenvector is divided by its length, the matrix $\mathbf{S}$ with these normalised eigenvectors as columns satisfies $\mathbf{S}^{\mathrm{T}}\mathbf{S} = \mathbf{I}$, since the entries of $\mathbf{S}^{\mathrm{T}}\mathbf{S}$ are the scalar products $\mathbf{x}_i^{\mathrm{T}}\mathbf{x}_j$. A real matrix whose inverse is its transpose is called an orthogonal matrix. It preserves scalar products, since $(\mathbf{S}\mathbf{x})^{\mathrm{T}}(\mathbf{S}\mathbf{y}) = \mathbf{x}^{\mathrm{T}}\mathbf{S}^{\mathrm{T}}\mathbf{S}\mathbf{y} = \mathbf{x}^{\mathrm{T}}\mathbf{y}$, and therefore lengths and angles, and its determinant is ± 1 , because $(\det \mathbf{S})^2 = \det \mathbf{S}^{\mathrm{T}} \det \mathbf{S} = 1$. The rotation $\mathbf{R}(\theta)$ and the reflection $\mathbf{P}$ are orthogonal. Every real symmetric matrix can therefore be

diagonalised by an orthogonal matrix,

$$\mathbf{S}^T \mathbf{A} \mathbf{S} = \mathbf{D}. \quad (4.36)$$

For the first example, the normalised eigenvectors $(1, 1)/\sqrt{2}$ and $(-1, 1)/\sqrt{2}$, belonging to $\mu + \nu$ and $\mu - \nu$, are the columns of the rotation $\mathbf{R}(\pi/4)$.

Quadratic Forms and the Principal Axes of a Conic

A quadratic form in the variables $x_1, \dots, x_n$ is a sum of terms, each of which is a constant multiplied by the product of two of the variables, the two being allowed to be the same, so that squares such as x_1^2 are included. Any quadratic form can be written as

$$Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \sum_{i,j} A_{ij} x_i x_j \quad (4.37)$$

with $\mathbf{A}$ symmetric, by dividing the coefficient of each product $x_i x_j$ with $i \neq j$ equally between A_{ij} and A_{ji} , so that, for instance, $A_{11}x_1^2 + 2A_{12}x_1x_2 + A_{22}x_2^2$ belongs to the matrix with rows (A_{11}, A_{12}) and (A_{12}, A_{22}) . Let $\mathbf{S}$ be the orthogonal matrix that diagonalises $\mathbf{A}$, and let $\mathbf{x} = \mathbf{S}\mathbf{y}$. Then

$$Q = \mathbf{y}^T \mathbf{S}^T \mathbf{A} \mathbf{S} \mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2, \quad (4.38)$$

which contains no products of different variables. By Section 4.2, the y_i are the components of $\mathbf{x}$ along the normalised eigenvectors, which are perpendicular to one another. Their directions are called the principal axes of the quadratic form.

A conic centred on the origin is a curve $Q(\mathbf{x}) = 1$ in the plane, and in the new variables it is $\lambda_1 y_1^2 + \lambda_2 y_2^2 = 1$. If both eigenvalues are positive, the curve is an ellipse with semi-axes $1/\sqrt{\lambda_1}$ and $1/\sqrt{\lambda_2}$ along the principal axes, and if they have opposite signs it is a hyperbola. As an example, consider the curve

$$x_1^2 + 2\gamma x_1 x_2 + x_2^2 = 1, \quad (4.39)$$

with γ a constant and $0 < \gamma < 1$. Its matrix is the first example with $\mu = 1$ and $\nu = \gamma$, so in the new variables the curve is $(1 + \gamma)y_1^2 + (1 - \gamma)y_2^2 = 1$. It is the ellipse of Figure 41, with its shorter semi-axis, $1/\sqrt{1 + \gamma}$, along the line $x_2 = x_1$, and its longer semi-axis, $1/\sqrt{1 - \gamma}$, along $x_2 = -x_1$. When $\gamma = 0$ it becomes the unit circle.

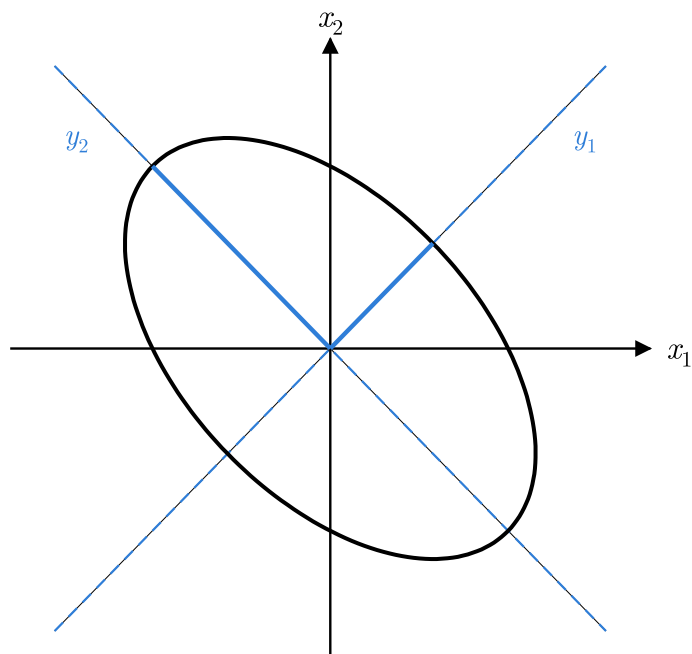


Figure 41. The ellipse $x_1^2 + 2\gamma x_1 x_2 + x_2^2 = 1$, drawn for $\gamma = \frac{1}{2}$. The principal axes, along the eigenvectors $(1, 1)$ and $(-1, 1)$, are the dashed lines, and the y_1 - and y_2 -axes lie along them. The semi-axes, drawn solid, have lengths $1/\sqrt{1 + \gamma}$ and $1/\sqrt{1 - \gamma}$.

Normal Modes

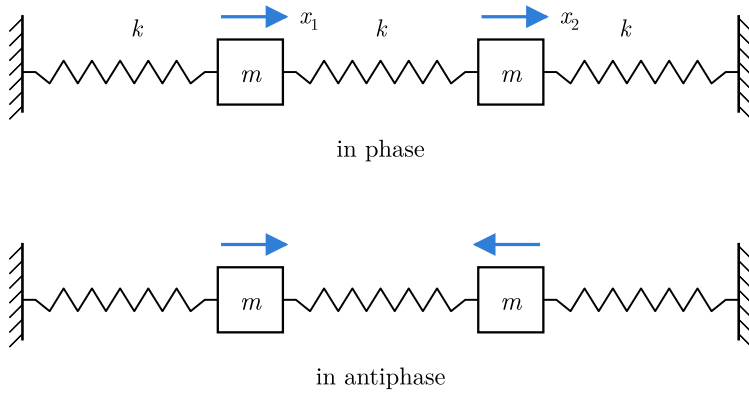


Figure 42. A pair of equal masses m , joined to each other and to two fixed walls by three identical springs of spring constant k . The displacements x_1 and x_2 are measured to the right from the equilibrium positions. In the first normal mode, drawn in the upper row, the masses move together, and in the second, drawn in the lower row, they move in opposite directions.

As an example of coupled motion, consider two equal masses m that slide without friction on a horizontal surface, joined to each other and to two fixed walls by three identical springs of spring constant k , as in Figure 42. The springs are unstretched at equilibrium, and x_1 and x_2 are the displacements of the masses to the right. By Hooke's law, the left-hand spring, stretched by x_1 , exerts a force $-kx_1$ on the first mass; the middle spring, stretched by $x_2 - x_1$, pulls the first mass to the right with a force $k(x_2 - x_1)$ and the second mass to the left with an equal force; and the right-hand spring, compressed by x_2 , exerts a force $-kx_2$ on the second mass. With a dot for a derivative with respect to time, Newton's second law for each mass gives

$$\begin{aligned} m\ddot{x}_1 &= -2kx_1 + kx_2, \\ m\ddot{x}_2 &= kx_1 - 2kx_2, \end{aligned} \tag{4.40}$$

or, as one matrix equation,

$$m\ddot{\mathbf{x}} = -\mathbf{K}\mathbf{x}, \quad \mathbf{K} = \begin{pmatrix} 2k & -k \\ -k & 2k \end{pmatrix}. \tag{4.41}$$

Each equation contains both displacements, so the motions of the masses are coupled. Let us look for motions in which both masses oscillate with the same angular frequency ω , with a fixed ratio of displacements, by trying

$$\mathbf{x} = \mathbf{a} \cos \omega t, \quad (4.42)$$

where $\mathbf{a}$ is a constant column vector. Since $\ddot{\mathbf{x}} = -\omega^2 \mathbf{a} \cos \omega t$, the matrix equation holds at every instant if

$$\mathbf{K}\mathbf{a} = m\omega^2 \mathbf{a}, \quad (4.43)$$

which is the eigenvalue equation for $\mathbf{K}$, with $\lambda = m\omega^2$. The matrix $\mathbf{K}$ is the first example with $\mu = 2k$ and $\nu = -k$, so its eigenvalues are k , with eigenvector $\mathbf{a}_1 = (1, 1)$, and $3k$, with eigenvector $\mathbf{a}_2 = (1, -1)$, and the trial solution works for the two frequencies

$$\omega_1 = \sqrt{\frac{k}{m}}, \quad \omega_2 = \sqrt{\frac{3k}{m}}, \quad (4.44)$$

the square roots of the eigenvalues of $\mathbf{K}/m$. Motions in which every part of a system oscillates at one frequency are called normal modes, and their frequencies are called normal frequencies.

Both frequencies can be understood from school physics. In the first mode the masses move together, the middle spring keeps its natural length, and each mass oscillates as if on its outer spring alone, with angular frequency $\sqrt{k/m}$. In the second mode the masses move oppositely and the midpoint of the middle spring stays still, so each mass is held by its outer spring and by half the middle spring, which is twice as stiff as the whole, and the total spring constant is $k + 2k = 3k$.

Since the second derivative of $\sin \omega t$ is $-\omega^2 \sin \omega t$, the trial solution $\mathbf{a} \sin \omega t$ leads to the same eigenvalue equation and is also a solution for each mode, and since the equations are linear, the general motion is

$$\mathbf{x} = \mathbf{a}_1 (\alpha_1 \cos \omega_1 t + \beta_1 \sin \omega_1 t) + \mathbf{a}_2 (\alpha_2 \cos \omega_2 t + \beta_2 \sin \omega_2 t), \quad (4.45)$$

with four constants fixed by the initial positions and velocities of the masses. The same result follows from the basis of eigenvectors. Adding and subtracting the equations of motion shows that $q_1 = \frac{1}{2}(x_1 + x_2)$ and $q_2 = \frac{1}{2}(x_1 - x_2)$ obey the equations of simple

harmonic motion

$$\frac{d^2 q_1}{dt^2} = -\omega_1^2 q_1, \quad \frac{d^2 q_2}{dt^2} = -\omega_2^2 q_2. \quad (4.46)$$

The variables q_1 and q_2 , which are called normal coordinates, are the components of $\mathbf{x}$ in the basis $\mathbf{a}_1, \mathbf{a}_2$, in which $\mathbf{K}$ is diagonal.

As an example, suppose that the first mass is pulled a distance d to the right, the second is held at equilibrium, and both masses are released from rest. Then q_1 and q_2 both start at $d/2$ with zero velocity, so $q_1 = \frac{1}{2}d \cos \omega_1 t$ and $q_2 = \frac{1}{2}d \cos \omega_2 t$, and

$$x_1 = \frac{d}{2} (\cos \omega_1 t + \cos \omega_2 t), \quad x_2 = \frac{d}{2} (\cos \omega_1 t - \cos \omega_2 t). \quad (4.47)$$

Each displacement is a sum of two oscillations whose frequencies are in the ratio $\sqrt{3}$, which is not a ratio of whole numbers, so the motion never repeats exactly.

Problems

§4.1 Matrices, Determinants, and Linear Equations

1. Using the compound-angle formulas, show that $\mathbf{R}(\theta_1)\mathbf{R}(\theta_2) = \mathbf{R}(\theta_1 + \theta_2)$. Deduce that $\mathbf{R}(\theta)^{-1} = \mathbf{R}(-\theta)$, and check that this is equal to $\mathbf{R}(\theta)^T$.
2. Show that $\mathbf{PR}(\theta) = \mathbf{R}(-\theta)\mathbf{P}$, where $\mathbf{P}$ is the reflection in the x_1 -axis. Hence show that $(\mathbf{PR}(\theta))^2 = \mathbf{I}$, and find the line of vectors that $\mathbf{PR}(\theta)$ leaves unchanged.
3. For a general three-by-three matrix, show that expanding the determinant along the second row gives the same result as expanding it along the first row, and that $\det \mathbf{A}^T = \det \mathbf{A}$.
4. Show from the definition of the product in components that $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$, and deduce that the transpose of the inverse of a matrix is the inverse of its transpose.
5. A square matrix whose entries below the leading diagonal are all zero is called upper triangular. Show that its determinant is the product of its diagonal entries, and explain how Gaussian elimination can be used to evaluate a determinant.

6. Solve $ax_1 + bx_2 = e$ and $cx_1 + dx_2 = f$ by Cramer's rule, and check the result with the inverse of the matrix of coefficients.
7. Find the values of the constant κ for which the equations $(1 - \kappa)x_1 + x_2 = 0$ and $x_1 + (1 - \kappa)x_2 = 0$ have a solution other than $x_1 = x_2 = 0$, and find the solutions for each such value.

§4.2 Vector Spaces and Linear Maps

1. Decide which of the following sets are vector spaces under the usual operations, and for each one that is, find its dimension: (a) the vectors in $\mathbb{R}^3$ with $x_1 + x_2 + x_3 = 0$; (b) the vectors in $\mathbb{R}^3$ with $x_1 + x_2 + x_3 = 1$; (c) the polynomials p of degree at most three with $p(1) = 0$; (d) the real functions f on $0 \leq x \leq L$ with $f(0) = f(L) = 0$.
2. Show that 1 , $1+x$, and $1+x+x^2$ form a basis of the polynomials of degree at most two, and find the components of x^2 in this basis.
3. Find the matrix D of d/dx on the space spanned by $\cos x$ and $\sin x$, in that basis, and show that $D^2 = -I$. Find D^{-1} , and show that it sends each function in the space to the one integral of it that lies in the space.
4. Show that $\text{tr}(AB) = \text{tr}(BA)$ for square matrices of the same size, and hence that similar matrices have the same trace.
5. On the polynomials of degree at most two, let $I(p) = \int_{-1}^1 p(x) dx$ and $E(p) = p(1)$. Find the rows that represent I and E in the basis $1, x, x^2$, and find all p with $I(p) = E(p) = 0$.
6. Show that e^x and e^{-x} are linearly independent on any interval. Find the matrices of d/dx on the space they span in the bases $\cosh x, \sinh x$ and e^x, e^{-x} , and check that they are related by a similarity transformation.

§4.3 Eigenvalues, Diagonalisation, and Normal Modes

1. Show that the characteristic equation of a two-by-two matrix is $\lambda^2 - (\text{tr } A)\lambda + \det A = 0$, and hence that the sum of its eigenvalues is its trace and their product is its determinant. For an

- $n \times n$ matrix, show that the product of the eigenvalues is the determinant.
- Find the eigenvectors of $R(\theta)$ that belong to the eigenvalues $e^{i\theta}$ and $e^{-i\theta}$.
 - Find the eigenvalues and normalised eigenvectors of the matrix with rows $(0, 1, 0)$, $(1, 0, 1)$, and $(0, 1, 0)$. Check that the eigenvectors are orthogonal, and write down the orthogonal matrix that diagonalises the matrix.
 - Using $A^k = S^k S^{-1}$, find A^k for the matrix A with rows (μ, ν) and (ν, μ) .
 - Describe the curve $x_1^2 + 2\gamma x_1 x_2 + x_2^2 = 1$ when $\gamma > 1$ and when $\gamma = 1$, giving the directions of its principal axes and the points where it crosses them.
 - In the system of Figure 42, the middle spring is replaced by one of spring constant k' . Show that the normal frequencies are $\omega_1 = \sqrt{k/m}$ and $\omega_2 = \sqrt{(k + 2k')/m}$, with the same normal modes. With the initial conditions of the example in the text, show that $x_1 = d \cos(\frac{1}{2}\Delta t) \cos \bar{\omega} t$ and $x_2 = d \sin(\frac{1}{2}\Delta t) \sin \bar{\omega} t$, where $\Delta = \omega_2 - \omega_1$ and $\bar{\omega} = \frac{1}{2}(\omega_1 + \omega_2)$. Describe the motion when $k' \ll k$.

Ordinary Differential Equations

5.1 First-Order Equations

An equation that relates a function to its derivatives is called a differential equation. When the function depends on a single variable, the equation is called an ordinary differential equation, and when it depends on several and the equation contains partial derivatives, a partial differential equation. The order of an equation is the order of its highest derivative, and the equation is called linear if the unknown function and its derivatives appear only to the first power and are not multiplied together. This section treats equations of the first order, beginning with the methods and the equation of simple harmonic motion met at school, and the next section treats linear equations of higher order.

Separation of Variables

The general solution of a first-order equation contains one arbitrary constant, which is fixed by an initial condition giving the value of the unknown function at one value of the independent variable. An equation of the form

$$\frac{dy}{dx} = X(x)Y(y), \quad (5.1)$$

whose right-hand side is a function of x times a function of y , is solved by separating the variables:

$$\int \frac{dy}{Y(y)} = \int X(x) dx + C. \quad (5.2)$$

Examples often met at school are those of exponential growth and decay. The number N of undecayed nuclei in a radioactive sample decreases at a rate proportional to N , so

$$\frac{dN}{dt} = -\frac{N}{\tau}, \quad N(t) = N_0 e^{-t/\tau}, \quad (5.3)$$

where N_0 is the number at $t = 0$, and τ , the reciprocal of the decay constant, is called the mean lifetime; the half-life is $\tau \ln 2$. A quantity whose rate of increase is proportional to itself grows in the same way, with the sign of the exponent reversed. Newton's law of cooling states that a body in surroundings at a constant temperature T_s cools at a rate proportional to the difference between its temperature T and T_s , so

$$\frac{dT}{dt} = -\kappa(T - T_s), \quad T(t) = T_s + (T(0) - T_s)e^{-\kappa t}, \quad (5.4)$$

where κ is a positive constant that depends on the body and its surroundings.

Simple Harmonic Motion

An equation of the second order that you have met at school is that of simple harmonic motion. Let a dot denote a derivative with respect to time. A mass m on a spring of spring constant k , displaced by x from equilibrium, feels the force $-kx$ by Hooke's law, so Newton's second law gives

$$\ddot{x} = -\omega_0^2 x, \quad \omega_0 = \sqrt{\frac{k}{m}}. \quad (5.5)$$

The general solution is

$$x(t) = A \cos \omega_0 t + B \sin \omega_0 t = R \cos(\omega_0 t - \delta), \quad (5.6)$$

where, by the compound-angle formula, the amplitude R and the phase δ satisfy $A = R \cos \delta$ and $B = R \sin \delta$. The period, $2\pi/\omega_0$, does not depend on the amplitude. A second-order equation needs two initial conditions, such as the initial position and velocity, to fix its two constants.

Direction Fields

A first-order equation $dy/dx = S(x, y)$ gives the slope of the solution curve through each point of the (x, y) -plane. Short segments drawn with these slopes at the points of a grid form a direction field, and every solution curve runs along the segments it meets. When S and $\partial S/\partial y$ are continuous, exactly one solution curve passes through each point, so solution curves never cross; the proof of this result is beyond the scope of this course.

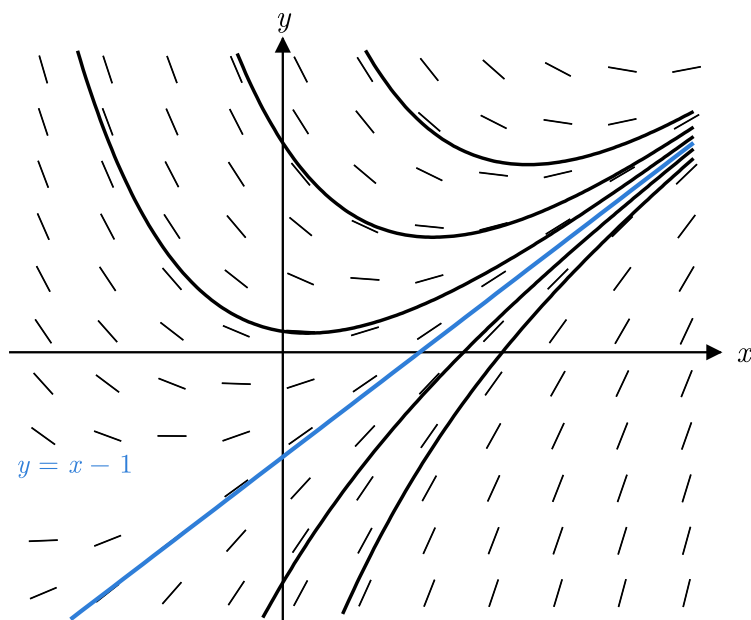


Figure 43. The direction field of $dy/dx = x - y$, with several solution curves. The straight line $y = x - 1$ is itself a solution, and every other solution curve approaches it as x increases.

The direction field of $dy/dx = x - y$ is shown in Figure 43. The segments are horizontal on the line $y = x$ and slope downwards above it. On the line $y = x - 1$ the equation gives the slope 1, which is the slope of the line, so the line is a solution. Above this line the slope $x - y$ is less than one, so a solution curve that starts there comes closer to the line as x increases, and below the line the

slope is greater than one, with the same effect. The figure suggests that every solution approaches the line, and the next subsection shows that every solution does so.

The Integrating Factor

The general linear equation of the first order is

$$\frac{dy}{dx} + p(x)y = f(x), \quad (5.7)$$

with p and f given functions. When f is not zero, the variables cannot in general be separated. Let us multiply the equation by a function $\mu(x)$, to give $\mu y' + \mu p y = \mu f$, where a prime denotes a derivative with respect to x . By the product rule, $(\mu y)' = \mu y' + \mu' y$, so the left-hand side is the derivative of μy if $\mu' = p\mu$, a separable equation whose solution is

$$\mu(x) = \exp \left(\int p(x) dx \right), \quad (5.8)$$

with the constant of integration dropped, since one such function is enough. The equation is then $(\mu y)' = \mu f$, and integrating gives

$$y(x) = \frac{1}{\mu(x)} \left[\int \mu(x) f(x) dx + C \right]. \quad (5.9)$$

The function μ is called an integrating factor.

As an example, consider the equation of Figure 43, $y' + y = x$. Here $\mu = e^x$, so $(e^x y)' = x e^x$, and integrating by parts gives $\int x e^x dx = (x - 1)e^x$, so that

$$y = x - 1 + C e^{-x}. \quad (5.10)$$

The choice $C = 0$ gives the straight line of the figure, and for every other C the term $C e^{-x}$ decays, so every solution approaches the line. The term $C e^{-x}$ is the general solution of the equation with $f = 0$, and $x - 1$ is one particular solution of the full equation.

Cooling in Surroundings Whose Temperature Changes

With surroundings whose temperature $T_s(t)$ changes, Newton's law of cooling is the linear equation

$$\frac{dT}{dt} + \kappa T = \kappa T_s(t). \quad (5.11)$$

As an example, suppose that the body and its surroundings are at the temperature T_0 at $t = 0$, and that the surroundings then warm steadily, $T_s = T_0 + \eta t$, at a constant rate η . With the integrating factor $e^{\kappa t}$ the equation becomes

$$\frac{d}{dt}(e^{\kappa t}T) = \kappa(T_0 + \eta t)e^{\kappa t}, \quad (5.12)$$

and integrating, the second term by parts, gives

$$T = T_0 + \eta\left(t - \frac{1}{\kappa}\right) + Ce^{-\kappa t}, \quad (5.13)$$

with $C = \eta/\kappa$ from the initial condition. After a time long compared with $1/\kappa$ the exponential has decayed, and the body follows the surroundings with a constant lag η/κ , the rise in their temperature in a time $1/\kappa$. At early times, when $\kappa t \ll 1$, the Taylor series of the exponential $e^{-\kappa t}$ from Section 1.1, taken to the second order, gives $T \approx T_0 + \frac{1}{2}\kappa\eta t^2$, so the body starts to warm slowly, because at first it is at the temperature of its surroundings.

A Falling Body with Linear Drag

As a second application, consider a body of mass m falling from rest through air that exerts a drag force bv opposing its motion, where v is the downward velocity and b is a drag coefficient to be measured. Newton's second law gives

$$m\frac{dv}{dt} = mg - bv, \quad (5.14)$$

which is linear, with the integrating factor $e^{bt/m}$, and the solution that starts from rest is

$$v = \frac{mg}{b}(1 - e^{-bt/m}). \quad (5.15)$$

The velocity approaches the terminal velocity mg/b , at which the drag balances the weight, over a time of order m/b . At times short compared with m/b , the first two terms of the Taylor series of the exponential give $v \approx gt$, which is free fall, because the body has not yet reached a speed at which the drag is comparable with its weight.

A Chain of Two Radioactive Decays

Suppose that a radioactive nucleus decays, with mean lifetime τ_1 , into a daughter nucleus that is itself radioactive, with mean lifetime τ_2 , and that a sample contains N_0 parent nuclei and no daughter nuclei at $t = 0$. The number of parent nuclei is $N_1 = N_0 e^{-t/\tau_1}$. The daughter nuclei, of number N_2 , are created at the rate N_1/τ_1 at which the parents decay and are lost at the rate N_2/τ_2 , so

$$\frac{dN_2}{dt} + \frac{N_2}{\tau_2} = \frac{N_0}{\tau_1} e^{-t/\tau_1}. \quad (5.16)$$

Multiplying by the integrating factor e^{t/τ_2} turns the left-hand side into $d(e^{t/\tau_2} N_2)/dt$. For $\tau_1 \neq \tau_2$, integrating both sides from 0 to t with $N_2(0) = 0$ and using $1/\tau_2 - 1/\tau_1 = (\tau_1 - \tau_2)/(\tau_1 \tau_2)$ gives

$$N_2 = \frac{N_0 \tau_2}{\tau_1 - \tau_2} (e^{-t/\tau_1} - e^{-t/\tau_2}). \quad (5.17)$$

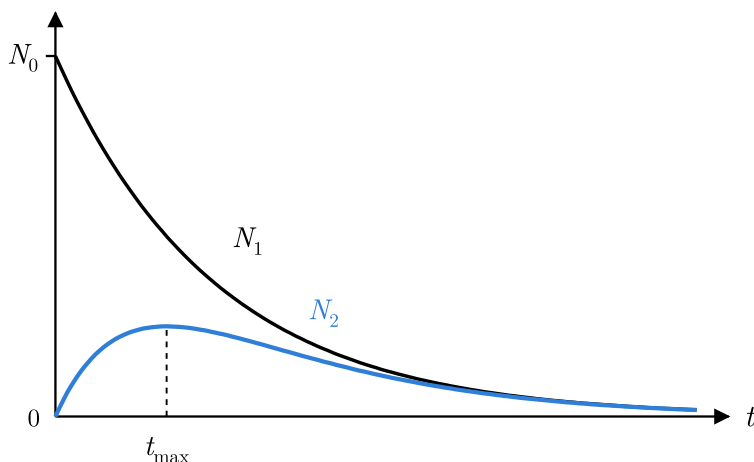


Figure 44. The numbers of parent nuclei, N_1 , and daughter nuclei, N_2 , in a chain of two radioactive decays, drawn for $\tau_1 = 2\tau_2$. The number of daughter nuclei rises from zero, reaches its maximum at $t = t_{\max}$, and then falls, approaching $\tau_2 N_1/(\tau_1 - \tau_2)$, which for this choice of lifetimes is equal to N_1 .

The number N_2 is positive whichever lifetime is the longer, since $\tau_1 - \tau_2$ and $e^{-t/\tau_1} - e^{-t/\tau_2}$ have the same sign. As Figure 44 shows,

it rises from zero, passes through a maximum, and then decays. If the parent lives much longer than the daughter, the term e^{-t/τ_2} dies away first, and after a few multiples of τ_2 we have $N_2 \approx (\tau_2/\tau_1)N_1$. The rates of decay N_1/τ_1 and N_2/τ_2 are then equal, so the daughter nuclei decay as fast as they are produced. You will find the time of the maximum, and the solution for $\tau_1 = \tau_2$, in Problem 3.

Exact Equations

A first-order equation can always be written in the form

$$P(x, y) dx + Q(x, y) dy = 0, \quad (5.18)$$

which is equivalent to $dy/dx = -P/Q$ wherever $Q \neq 0$. The equation is called exact if its left-hand side is the total differential $d\psi$ of a function $\psi(x, y)$, so that $P = \partial\psi/\partial x$ and $Q = \partial\psi/\partial y$. Along a solution curve $d\psi = 0$, so the solutions are the curves $\psi(x, y) = C$. By the test for exactness of Section 2.2, which rests on the equality of the mixed second derivatives of ψ , the equation is exact if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}. \quad (5.19)$$

To find ψ , we integrate P with respect to x , holding y constant, and add an arbitrary function of y , which is fixed by the requirement $\partial\psi/\partial y = Q$. As an example, consider $2xy dx + (x^2 + y^2) dy = 0$, which is exact, since $\partial P/\partial y = 2x = \partial Q/\partial x$. Integrating $2xy$ with respect to x gives $\psi = x^2 y + \chi(y)$, and the requirement $x^2 + \chi'(y) = x^2 + y^2$ gives $\chi = y^3/3$, so the solutions are the curves $x^2 y + \frac{1}{3}y^3 = C$.

An equation that is not exact can sometimes be made exact by multiplying it by a function μ , again called an integrating factor. For a factor $\mu(x)$ that depends on x alone, the condition $\partial(\mu P)/\partial y = \partial(\mu Q)/\partial x$ becomes, by the product rule, $\mu \partial P/\partial y = \mu' Q + \mu \partial Q/\partial x$, or

$$\frac{\mu'}{\mu} = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right), \quad (5.20)$$

which can be integrated if its right-hand side depends on x alone. As an example, consider $y dx - x dy = 0$, for which the right-hand side is $-2/x$, so that $\mu = 1/x^2$. The equation $(y/x^2) dx - (1/x) dy = 0$ is the differential of $\psi = -y/x$, and the solutions are the lines $y = Cx$.

The linear equation, written as $(py - f)dx + dy = 0$, has $Q = 1$, so that the right-hand side of the equation for μ'/μ is $p(x)$, which gives the integrating factor found earlier in this section.

Equations Reduced by a Substitution

Some equations can be brought into one of these forms by a change of variable. An equation of the form

$$\frac{dy}{dx} = G\left(\frac{y}{x}\right) \quad (5.21)$$

is called homogeneous, because its right-hand side is unchanged when x and y are multiplied by the same constant. With $w = y/x$, the product rule gives $dy/dx = w + x dw/dx$, and the equation becomes $x dw/dx = G(w) - w$, in which the variables can be separated. As an example, consider $dy/dx = (x^2 + y^2)/(2xy)$, for which $G(w) - w = (1 - w^2)/(2w)$, and

$$\int \frac{2w dw}{1 - w^2} = \int \frac{dx}{x}. \quad (5.22)$$

The numerator on the left is minus the derivative of the denominator, so the left-hand side is $-\ln|1 - w^2|$, and $(1 - w^2)x = C$. Substituting $w = y/x$ gives $x^2 - y^2 = Cx$, or $(x - C/2)^2 - y^2 = C^2/4$, a family of hyperbolas centred on the x -axis. The value $C = 0$ gives the lines $y = \pm x$, on which $1 - w^2 = 0$, so that the division by $1 - w^2$ in the separation was not allowed.

An equation of the form

$$\frac{dy}{dx} + p(x)y = f(x)y^n, \quad (5.23)$$

with $n \neq 0, 1$, is called a Bernoulli equation. With $u = y^{1-n}$, the chain rule gives $du/dx = (1 - n)y^{-n} dy/dx$, and multiplying the equation by $(1 - n)y^{-n}$ gives the linear equation

$$\frac{du}{dx} + (1 - n)p(x)u = (1 - n)f(x). \quad (5.24)$$

As an example, consider a body of mass m moving horizontally, with initial velocity v_0 , through a fluid that exerts a drag force $bv + cv^2$ opposing its motion, with drag coefficients b and c . Newton's second law, $m dv/dt = -bv - cv^2$, is a Bernoulli equation with $n = 2$.

The substitution $u = 1/v$ gives $du/dt - (b/m)u = c/m$, with the integrating factor $e^{-bt/m}$, and the solution with $u(0) = 1/v_0$ gives

$$v = \frac{1}{(1/v_0 + c/b)e^{bt/m} - c/b}. \quad (5.25)$$

When $c = 0$ this is $v = v_0 e^{-bt/m}$, the decay expected for linear drag, and at long times the velocity again decays as $e^{-bt/m}$, since once the body is slow the linear term in the drag is much larger than the quadratic term.

5.2 Linear Equations of Higher Order

Since Newton's second law involves the acceleration, the equations of mechanics are usually of the second order. In this section we first describe the structure shared by the solutions of every linear equation, then solve linear equations with constant coefficients, with the damped, driven oscillator as the running example, and finally turn to equations whose coefficients vary.

The Structure of the Solutions

The general linear equation of order n is

$$a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y = f(x), \quad (5.26)$$

where $y^{(n)}$ is the n th derivative of y . It is called homogeneous if $f = 0$ and inhomogeneous otherwise, and f is called the forcing term. This meaning of homogeneous differs from the one used in Section 5.1.

If y_1 and y_2 solve a homogeneous linear equation, so does $C_1 y_1 + C_2 y_2$ for any constants C_1 and C_2 , because the derivative of a sum is the sum of the derivatives. This result is called the principle of superposition, and it makes the solutions a vector space in the sense of Section 4.2. The space has dimension n , by the existence and uniqueness theorem, which states that, when the coefficients are continuous and $a_n \neq 0$, exactly one solution takes given values of $y, y', \dots, y^{(n-1)}$ at a point x_0 ; the proof of this theorem is beyond the scope of this course. Let y_k be the solution whose $(k-1)$ th derivative is 1 at x_0 and whose other derivatives up to order $n-1$ are 0 there. Any solution y equals $\sum_k y^{(k-1)}(x_0) y_k$, since both sides

are solutions with the same initial values at x_0 , and by the existence and uniqueness theorem only one solution has those values, and the y_k are independent, since $\sum_k C_k y_k$ has $(k-1)$ th derivative C_k at x_0 . The general solution therefore contains n arbitrary constants.

The difference of two solutions of the inhomogeneous equation solves the homogeneous one, so if y_p is any solution of the inhomogeneous equation, every solution is

$$y = y_c + y_p, \quad (5.27)$$

where y_c , the general solution of the homogeneous equation, is called the complementary function, and y_p is called a particular integral. The solution $x - 1 + Ce^{-x}$ found in Section 5.1 has this form.

The running example is a mass m on a spring of spring constant k , with a drag force $b\dot{x}$, as in Section 5.1, and an applied force $F(t)$. Newton's second law, $m\ddot{x} = -kx - b\dot{x} + F(t)$, can be rewritten as

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \frac{F(t)}{m}, \quad \gamma = \frac{b}{2m}, \quad \omega_0 = \sqrt{\frac{k}{m}}, \quad (5.28)$$

where γ is called the damping rate. With $\gamma = 0$ and $F = 0$ this is the equation of simple harmonic motion.

Equations with Constant Coefficients

The derivatives of an exponential are multiples of it, so the trial solution $y = e^{\alpha x}$ turns every term of $a_2 y'' + a_1 y' + a_0 y = 0$ into a multiple of $e^{\alpha x}$, and it is a solution if

$$a_2 \alpha^2 + a_1 \alpha + a_0 = 0, \quad (5.29)$$

which is called the auxiliary equation. There are three cases. If the auxiliary equation has different real roots α_1 and α_2 , the general solution is $y = Ae^{\alpha_1 x} + Be^{\alpha_2 x}$. If the roots are complex, they are a conjugate pair $\beta \pm i\zeta$, because the coefficients are real, and half the sum and $1/(2i)$ times the difference of the two complex solutions are, by Euler's formula from Section 1.2, the real solutions $e^{\beta x} \cos \zeta x$ and $e^{\beta x} \sin \zeta x$, so

$$y = e^{\beta x} (A \cos \zeta x + B \sin \zeta x). \quad (5.30)$$

If the root is repeated, $\alpha_1 = -a_1/(2a_2)$, a second solution is $xe^{\alpha_1 x}$. The first two derivatives of $xe^{\alpha_1 x}$, by the product rule, are $(1 +$

$\alpha_1 x)e^{\alpha_1 x}$ and $(2\alpha_1 + \alpha_1^2)x e^{\alpha_1 x}$, and substituting them turns the left-hand side of the equation into

$$[x(a_2\alpha_1^2 + a_1\alpha_1 + a_0) + (2a_2\alpha_1 + a_1)]e^{\alpha_1 x}, \quad (5.31)$$

in which the first bracket vanishes because α_1 is a root of the auxiliary equation, and the second because $\alpha_1 = -a_1/(2a_2)$, so the general solution is $y = (A + Bx)e^{\alpha_1 x}$. An equation of order n is treated in the same way, and a root that occurs r times contributes $e^{\alpha x}, xe^{\alpha x}, \dots, x^{r-1}e^{\alpha x}$.

The Three Kinds of Damping

For the unforced oscillator, the roots of $\alpha^2 + 2\gamma\alpha + \omega_0^2 = 0$ are $\alpha = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$. When $\gamma < \omega_0$ the roots are $-\gamma \pm i\omega_d$, with $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$, and

$$x = e^{-\gamma t} (A \cos \omega_d t + B \sin \omega_d t), \quad (5.32)$$

which describes an oscillation whose amplitude decays as $e^{-\gamma t}$; this case is called underdamped. When $\gamma > \omega_0$ the roots, $-\gamma \pm \Omega$ with $\Omega = \sqrt{\gamma^2 - \omega_0^2} < \gamma$, are both negative, and the mass returns to equilibrium without oscillating. This case is called overdamped. At long times the root $-\gamma + \Omega$ dominates, and for $\gamma \gg \omega_0$ the binomial expansion of Section 1.1 gives it as $-\omega_0^2/(2\gamma)$, so heavier damping makes the return slower, because the drag opposes the return as well. When $\gamma = \omega_0$ the root is repeated, $x = (A + Bt)e^{-\gamma t}$, and the case is called critically damped.

The three motions are compared in Figure 45 for a mass released from rest at $x = x_0$. For the underdamped case these conditions give $A = x_0$ and $B = \gamma x_0/\omega_d$, and for the critical case $x = x_0(1 + \gamma t)e^{-\gamma t}$, while you will find the overdamped motion in Problem 3. Of the motions that do not overshoot, the critical one returns fastest, since its decay rate is ω_0 , while the slower rate of an overdamped motion, $\gamma - \Omega = \omega_0^2/(\gamma + \Omega)$, is less than ω_0 .

The Wronskian

Let y_1 and y_2 solve $y'' + p(x)y' + q(x)y = 0$, the equation divided by its leading coefficient. If $C_1y_1 + C_2y_2 = 0$ throughout an interval, then also $C_1y_1' + C_2y_2' = 0$, and together these relations are two equations for C_1 and C_2 whose matrix of coefficients has the

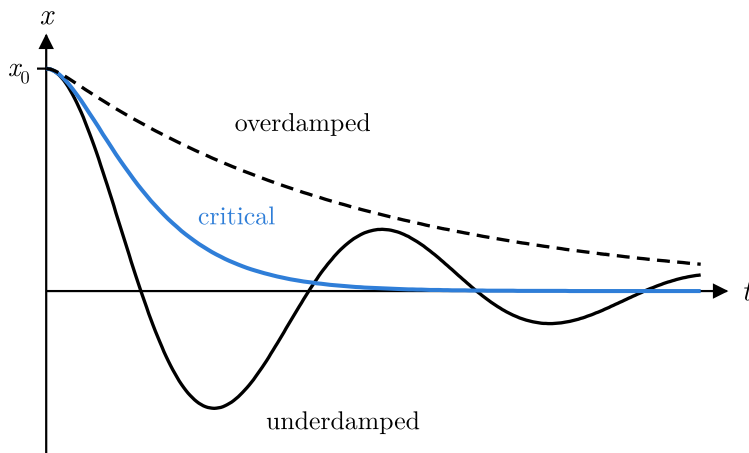


Figure 45. The three kinds of damping, for a mass released from rest at $x = x_0$. The underdamped curve is drawn for $\gamma = 0.2\omega_0$, the critical curve for $\gamma = \omega_0$, and the dashed overdamped curve for $\gamma = 3\omega_0$.

determinant

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1', \quad (5.33)$$

which is called the Wronskian of y_1 and y_2 . By the result of Section 4.1 on equations whose right-hand sides are all zero, if $W \neq 0$ at any point, then $C_1 = C_2 = 0$, and y_1 and y_2 are linearly independent.

For solutions the converse holds. If $W(x_0) = 0$, the result of Section 4.1 gives constants C_1 and C_2 , not both zero, for which $C_1 y_1 + C_2 y_2$ is a solution that vanishes with its derivative at x_0 . By the existence and uniqueness theorem this solution is zero everywhere, so y_1 and y_2 are dependent.

Differentiating W by the product rule, the terms $y_1' y_2'$ cancel, and replacing each second derivative by $-p y' - q y$ gives

$$W' = y_1 y_2'' - y_2 y_1'' = -p W. \quad (5.34)$$

This separable equation has the solution

$$W(x) = W(x_0) \exp \left(- \int_{x_0}^x p(x_1) dx_1 \right), \quad (5.35)$$

which is called Abel's formula. An exponential is never zero, so the Wronskian of two solutions is either zero everywhere or zero nowhere. As an example, $e^{\alpha_1 x}$ and $e^{\alpha_2 x}$ have $W = (\alpha_2 - \alpha_1) e^{(\alpha_1 + \alpha_2)x}$, which is not zero for different roots and agrees with Abel's formula, since $p = a_1/a_2$ and the roots of the auxiliary equation sum to $-a_1/a_2$.

Particular Integrals by Trial

When the coefficients are constant, a particular integral can often be found by trying a function of the same form as the forcing term, with coefficients to be determined. The trial functions most often used are the following:

1. for a polynomial of degree r , a polynomial of degree r , provided that $a_0 \neq 0$;
2. for $f_0 e^{\nu x}$, a multiple $D e^{\nu x}$, which gives $D = f_0 / (a_2 \nu^2 + a_1 \nu + a_0)$;
3. for $f_0 \cos \omega x$ or $f_0 \sin \omega x$, a combination of $\cos \omega x$ and $\sin \omega x$, or a complex exponential.

The list is not complete, and for a sum of such forcing terms, by linearity, the sum of their particular integrals is a particular integral. The oscillator with a constant force F_0 , for example, has $x_p = F_0/k$. The second trial function fails when ν is a root of the auxiliary equation, because $e^{\nu x}$ then gives zero on the left-hand side, and it is multiplied by x instead, or by x^2 for a repeated root, as with the solution $x e^{\alpha_1 x}$.

The Driven Oscillator and Resonance

Let the applied force be $F_0 \cos \omega t$, the real part of $F_0 e^{i\omega t}$. Since the equation has real coefficients, the real part of a solution of

$$\ddot{z} + 2\gamma \dot{z} + \omega_0^2 z = \frac{F_0}{m} e^{i\omega t} \quad (5.36)$$

solves the real equation. The trial solution $z = Ze^{i\omega t}$, with Z a complex constant, gives

$$Z = \frac{F_0/m}{\omega_0^2 - \omega^2 + 2i\gamma\omega}. \quad (5.37)$$

With $Z = |Z|e^{-i\delta}$, the particular integral is $x_p = |Z|\cos(\omega t - \delta)$, where

$$|Z| = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2}}, \quad \tan \delta = \frac{2\gamma\omega}{\omega_0^2 - \omega^2}, \quad (5.38)$$

with the phase lag δ between 0 and π , since the imaginary part $2\gamma\omega$ of the denominator of Z is positive when $\gamma > 0$ and $\omega > 0$. When $\gamma > 0$ the complementary function, called the transient, decays in each of the three cases, and the mass is left oscillating at the driving frequency with amplitude $|Z|$, independently of its initial conditions. This motion is called the steady state.

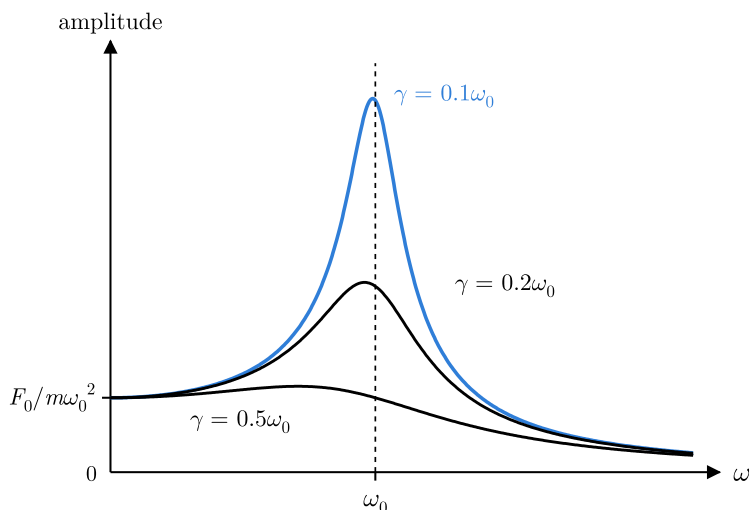


Figure 46. The steady-state amplitude $|Z|$ of a driven oscillator against the driving frequency ω , computed from the formula for $|Z|$ for three damping rates γ . The peak near ω_0 grows taller and narrower as γ decreases.

The amplitude is plotted in Figure 46. For $\omega \ll \omega_0$ it tends to F_0/k and δ to zero, so the mass follows the force as if it were steady. For $\omega \gg \omega_0$ the amplitude goes as $F_0/(m\omega^2)$ and δ tends to π , so the

mass moves opposite to the force, as a free mass does. At $\omega = \omega_0$ the lag is $\pi/2$ and the amplitude is $F_0/(2m\gamma\omega_0)$, large when the damping is light. Setting to zero the derivative, with respect to ω^2 , of the expression under the square root gives $-2(\omega_0^2 - \omega^2) + 4\gamma^2 = 0$, so the amplitude is greatest at

$$\omega_r = \sqrt{\omega_0^2 - 2\gamma^2}, \quad (5.39)$$

provided that $\gamma < \omega_0/\sqrt{2}$, where it is $F_0/(2m\gamma\omega_d)$. The large response of a lightly damped system driven close to its natural frequency is known as resonance.

With no damping and $\omega = \omega_0$, the denominator of Z vanishes, because $e^{i\omega_0 t}$ solves the homogeneous equation, and, following the subsection Particular Integrals by Trial, where a trial function that solves the homogeneous equation is multiplied by the independent variable, we try $z = Zte^{i\omega_0 t}$. By the product rule, $\ddot{z} = Z(2i\omega_0 - \omega_0^2 t)e^{i\omega_0 t}$, so the equation requires $2i\omega_0 Z = F_0/m$, and the real part of z is

$$x_p = \frac{F_0}{2m\omega_0} t \sin \omega_0 t. \quad (5.40)$$

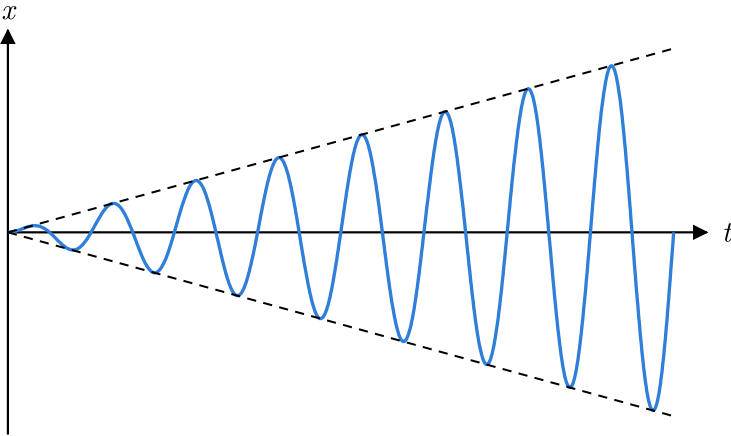


Figure 47. The particular integral of an undamped oscillator driven at its natural frequency, between the dashed lines $x = \pm[F_0/(2m\omega_0)]t$.

The amplitude grows in proportion to the time, as Figure 47 shows,

because the force is in step with the velocity and adds energy on every cycle, until, in a real system, damping stops the growth near the amplitude $F_0/(2m\gamma\omega_0)$.

Reduction of Order

For equations whose coefficients vary, no general method gives the solutions in closed form, but if one solution y_1 of $y'' + p y' + q y = 0$ is known, a second can be written in terms of integrals. Let $y = \xi(x) y_1(x)$. Substituting $y' = \xi' y_1 + \xi y_1'$ and $y'' = \xi'' y_1 + 2\xi' y_1' + \xi y_1''$, from the product rule, gives

$$\xi(y_1'' + p y_1' + q y_1) + \xi'' y_1 + \xi'(2y_1' + p y_1) = 0, \quad (5.41)$$

in which the first bracket vanishes because y_1 is a solution. What remains is a separable equation for ξ' , and dividing by $\xi' y_1$ and integrating gives

$$\xi' = \frac{C}{y_1^2} \exp\left(-\int p dx\right). \quad (5.42)$$

By Abel's formula the numerator is proportional to the Wronskian, in agreement with the identity $W = y_1^2 (y_2/y_1)'$, which follows from the quotient rule. The method is called reduction of order.

As an example, the repeated-root equation $y'' - 2\alpha_1 y' + \alpha_1^2 y = 0$ has $p = -2\alpha_1$ and $y_1 = e^{\alpha_1 x}$, so $\xi' = C$, which gives the second solution $x e^{\alpha_1 x}$ found above. As another example, consider $x y'' - (x+1) y' + y = 0$ for $x > 0$, which, as substitution shows, has the solution $y_1 = e^x$. Here $p = -1 - 1/x$, so $\exp(-\int p dx) = x e^x$ and $\xi' = C x e^{-x}$. Integrating by parts gives $\xi = -C(x+1)e^{-x}$, and the second solution is $x+1$.

Variation of Parameters

A particular integral of $y'' + p y' + q y = f$ can be found from the complementary function $A y_1 + B y_2$ for any forcing term, by replacing the constants with functions, $y_p = A(x) y_1 + B(x) y_2$. With two unknown functions and one equation to satisfy, we may impose the further condition

$$A' y_1 + B' y_2 = 0, \quad (5.43)$$

so that $y_p' = Ay_1' + By_2'$ contains no derivatives of A and B . Then $y_p'' = Ay_1'' + By_2'' + A'y_1' + B'y_2'$, and substituting into the equation, where the terms in A and B vanish because y_1 and y_2 are solutions, gives

$$A'y_1' + B'y_2' = f. \quad (5.44)$$

These two displayed conditions are linear equations for A' and B' , with the Wronskian as the determinant of their coefficients, and Cramer's rule from Section 4.1 gives

$$A' = -\frac{y_2 f}{W}, \quad B' = \frac{y_1 f}{W}, \quad (5.45)$$

which are integrated to give A and B . The method is called variation of parameters.

As an example, consider the undamped oscillator driven by any force $F(t)$. Here $y_1 = \cos \omega_0 t$, $y_2 = \sin \omega_0 t$, $W = \omega_0$, and $f = F/m$. Integrating A' and B' from 0 to t , with t_1 as the variable of integration, and combining the terms with the compound-angle formula, gives

$$x_p(t) = \frac{1}{m\omega_0} \int_0^t F(t_1) \sin \omega_0(t - t_1) dt_1. \quad (5.46)$$

This particular integral and, as you will show in Problem 6, its velocity vanish at $t = 0$. For a constant force F_0 switched on at $t = 0$ it gives $x_p = (F_0/k)(1 - \cos \omega_0 t)$, an oscillation about the new equilibrium position F_0/k .

The Euler Equation

An equation of the form

$$x^2 y'' + a_1 x y' + a_0 y = 0, \quad x > 0, \quad (5.47)$$

with constants a_1 and a_0 , is called an Euler equation, and the change of variable $x = e^\theta$ turns it into an equation with constant coefficients. Since $d\theta/dx = 1/x$, the chain rule gives $xy' = dy/d\theta$, and differentiating $y' = x^{-1} dy/d\theta$ again, by the product rule and the chain rule, gives $x^2 y'' = d^2 y/d\theta^2 - dy/d\theta$. The equation becomes

$$\frac{d^2 y}{d\theta^2} + (a_1 - 1) \frac{dy}{d\theta} + a_0 y = 0, \quad (5.48)$$

whose solutions are $e^{\alpha\theta} = x^\alpha$, with $\alpha(\alpha-1) + a_1\alpha + a_0 = 0$, and, for a repeated root, $\theta e^{\alpha\theta} = x^\alpha \ln x$. As an example, $x^2y'' + xy' - y = 0$ gives $\alpha^2 = 1$, and its general solution is $y = Ax + B/x$.

Problems

§5.1 First-Order Equations

1. Solve $dy/dx = -x/y$ and describe the solution curves. Solve $dy/dx = y^2$ with $y(0) = y_0 > 0$, and show that the solution grows without limit as x approaches a finite value, which you should find.
2. A body obeys Newton's law of cooling in surroundings at temperature $T_s = T_0 + \Theta \cos \omega t$. By writing $T_s = T_0 + \operatorname{Re}(\Theta e^{i\omega t})$ and looking for a solution $T = T_0 + \operatorname{Re}(Z e^{i\omega t})$ with a complex constant Z , show that after a time long compared with $1/\kappa$ the temperature of the body oscillates with amplitude $\kappa\Theta/\sqrt{\kappa^2 + \omega^2}$ and lags behind the surroundings by a phase δ with $\tan \delta = \omega/\kappa$.
3. For the chain of two decays, show that N_2 is greatest at $t_{\max} = \tau_1\tau_2 \ln(\tau_1/\tau_2)/(\tau_1 - \tau_2)$. When $\tau_1 = \tau_2 = \tau$, solve the equation for N_2 directly and show that $N_2 = N_0(t/\tau)e^{-t/\tau}$, and check that this is the limit of the general formula as $\tau_2 \rightarrow \tau_1$.
4. For the body falling with linear drag, find the distance fallen in a time t , and show that for $t \ll m/b$ it is approximately $\frac{1}{2}gt^2$.
5. Show that $(y^2 + 2x)dx + 2xydy = 0$ is exact, and solve it. Show that $Pdx + Qdy = 0$ has an integrating factor $\mu(y)$ if $(\partial Q/\partial x - \partial P/\partial y)/P$ depends on y alone, and use this result to solve $ydx + 2xdy = 0$.
6. Solve $dy/dx = (x + y)/x$ for $x > 0$, first as a homogeneous equation and then as a linear equation.
7. For the body moving with linear and quadratic drag, show that the total distance it travels is $(m/c) \ln(1 + cv_0/b)$, and that this tends to the distance for linear drag alone as $c \rightarrow 0$.

§5.2 Linear Equations of Higher Order

1. Find the general solution of $y'' - (\beta_1 + \beta_2)y' + \beta_1\beta_2 y = 0$, where $\beta_1 \neq \beta_2$ are constants, and of $y'''' - K^4 y = 0$, where $K > 0$ is a constant.
2. Find the Wronskian of $e^{-\gamma t} \cos \omega_d t$ and $e^{-\gamma t} \sin \omega_d t$, and check it against Abel's formula.
3. Show that the overdamped oscillator released from rest at $x = x_0$ moves as $x = x_0 [(\gamma + \Omega) e^{-(\gamma - \Omega)t} - (\gamma - \Omega) e^{-(\gamma + \Omega)t}] / (2\Omega)$, and that x never changes sign. Check that the critical solution $x_0(1 + \gamma t) e^{-\gamma t}$ satisfies the same initial conditions.
4. For the driven, damped oscillator in the steady state, show that the mean power delivered by the applied force over a period is $\gamma \omega^2 F_0^2 / (m [(\omega_0^2 - \omega^2)^2 + 4\gamma^2 \omega^2])$, that this equals the mean rate at which the drag removes energy, and that it is greatest at $\omega = \omega_0$ exactly.
5. Find a particular integral of $\ddot{x} + 2\gamma \dot{x} + \omega_0^2 x = F_1 t / m$, with F_1 a constant, and of $y'' - y = e^x$.
6. Using variation of parameters, find a particular integral of $y'' + y = \sec x$ for $|x| < \pi/2$. Show that the particular integral x_p of the driven undamped oscillator in the text has $\dot{x}_p(0) = 0$, and check the result for a constant force by the trial method.
7. The equation $x^2 y'' - x y' + y = 0$ has the solution $y_1 = x$. Find a second solution by reduction of order, and check it by solving the equation as an Euler equation.
8. Solve the Euler equation $x^2 y'' + x y' + K^2 y = 0$, where $K > 0$ is a constant.

Fourier Analysis and Integral Transforms

6.1 Fourier Series

In this chapter we learn how to write a function as a combination of sines and cosines, or of complex exponentials. This section treats periodic functions, and the sections that follow extend the idea to functions that are not periodic and introduce the delta function. In Section 5.2 we found the response of a driven oscillator to a force that varies sinusoidally in time. A periodic force need not be sinusoidal, and a mass on a spring could, for example, be pushed by a force that switches between two values at regular intervals. Since the equation of motion is linear, if such a force is a sum of sinusoidal forces, the response is the sum of the responses to each of them. The question that arises is whether a periodic function can be written as a sum of sines and cosines, and how their amplitudes are then found.

Periodic Functions and Orthogonality

A function $f(x)$ is said to be periodic, with period $2L$, if $f(x+2L) = f(x)$ for every x . The displacement of a mass oscillating on a spring, for example, is a periodic function of time. Such a function is

known everywhere once it is known on one interval of length $2L$, which we take to be $-L < x < L$. The functions $\cos(n\pi x/L)$ and $\sin(n\pi x/L)$, with $n = 1, 2, 3, \dots$, all have period $2L$, since increasing x by $2L$ increases their argument by $2n\pi$. They satisfy the following relations, for whole numbers $m, n \geq 1$:

$$\begin{aligned}\int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx &= L \delta_{mn}, \\ \int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx &= L \delta_{mn}, \\ \int_{-L}^L \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx &= 0,\end{aligned}\tag{6.1}$$

where δ_{mn} is the Kronecker delta of Section 4.2, which is 1 when $m = n$ and 0 otherwise. To prove the first relation, we use the identity $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$, which follows from adding the formulas for the cosine of a difference and of a sum. The integrand becomes $\frac{1}{2} \cos[(m - n)\pi x/L] + \frac{1}{2} \cos[(m + n)\pi x/L]$. For any non-zero whole number q ,

$$\int_{-L}^L \cos \frac{q\pi x}{L} dx = \frac{L}{q\pi} \left[\sin \frac{q\pi x}{L} \right]_{-L}^L = 0,\tag{6.2}$$

because $\sin(\pm q\pi) = 0$. When $m \neq n$, both $m - n$ and $m + n$ are non-zero, and the integral vanishes. When $m = n$, the first term is $\frac{1}{2}$, which integrates to L , and the second term integrates to zero, since $m + n$ is not zero. The second relation follows in the same way from $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$. For the third, we need the idea of an odd function, one for which $g(-x) = -g(x)$, such as $\sin x$. The integral of an odd function over an interval symmetric about the origin is zero, because the contributions from x and from $-x$ cancel, and the product of a sine and a cosine is odd, since the sine changes sign when x is replaced by $-x$ and the cosine does not. The same arguments show that each cosine and each sine integrates to zero over the interval.

A pair of functions whose product integrates to zero over an interval is said to be orthogonal on that interval. The name comes from the vector spaces of Section 4.2, which include the functions on an interval. The sines and cosines behave like an orthogonal basis, with the integral of the product of two functions in the place of

the scalar product of two vectors. Scalar products on spaces of functions are not treated in this course.

The Fourier Coefficients

Suppose that a function $f(x)$ of period $2L$ can be written as

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right], \quad (6.3)$$

which is called the Fourier series of f , with constants a_n and b_n that are called its Fourier coefficients. We multiply both sides by $\cos(m\pi x/L)$, with $m \geq 1$, and integrate from $-L$ to L , assuming that the series may be integrated term by term. By the orthogonality relations, every term on the right gives zero except the term with $n = m$ in the cosines, whose integral is La_m . Multiplying by $\sin(m\pi x/L)$ instead picks out Lb_m . Therefore

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx. \quad (6.4)$$

Integrating the series itself gives $\int_{-L}^L f dx = La_0$, which is the formula for a_n with $n = 0$. The factor $\frac{1}{2}$ in front of a_0 is there so that one formula serves for every a_n , and the constant term $\frac{1}{2}a_0$ is the mean value of f over a period.

The Square Wave

As an example, consider the function

$$f(x) = \begin{cases} -1, & -L < x < 0, \\ 1, & 0 < x < L, \end{cases} \quad (6.5)$$

repeated with period $2L$, which is called a square wave. The function is odd, so the product $f(x) \cos(n\pi x/L)$ is odd, since f changes sign when x is replaced by $-x$ and the cosine does not, and every a_n vanishes. The product $f(x) \sin(n\pi x/L)$ is unchanged when x is replaced by $-x$, since f and the sine each change sign, and its integral over $-L < x < L$ is twice its integral over $0 < x < L$, where $f = 1$. Hence

$$b_n = \frac{2}{L} \int_0^L \sin \frac{n\pi x}{L} dx = \frac{2}{n\pi} [1 - (-1)^n], \quad (6.6)$$

using $\cos n\pi = (-1)^n$. The coefficient is $4/n\pi$ when n is odd and zero when n is even, and the series is

$$f(x) = \frac{4}{\pi} \left[\sin \frac{\pi x}{L} + \frac{1}{3} \sin \frac{3\pi x}{L} + \frac{1}{5} \sin \frac{5\pi x}{L} + \cdots \right]. \quad (6.7)$$

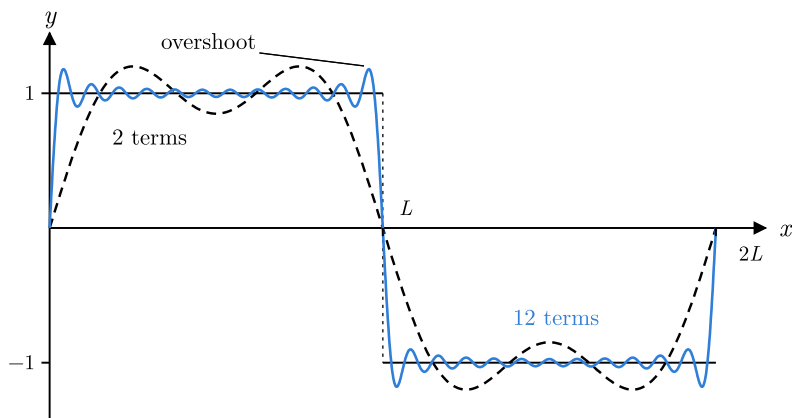


Figure 48. Partial sums of the Fourier series of the square wave, which equals 1 for $0 < x < L$ and -1 for $L < x < 2L$. The dashed curve is the sum of the first two non-zero terms, and the blue curve is the sum of the first twelve. Near each jump the partial sums overshoot the square wave, and as terms are added the overshoot becomes narrower while its height stays close to the same value.

Figure 48 shows the sums of the first few terms of this series, which are its partial sums in the sense of Section 1.1. At a jump, such as $x = 0$ or $x = L$, every term of the series is zero, so the series converges to zero there, and zero is the mean of the values -1 and 1 on either side of the jump.

This behaviour is general. Let $f(x+)$ and $f(x-)$ be the limits of f as x is approached from the right and from the left. A theorem due to Dirichlet states that the Fourier series of a periodic function f converges to $f(x)$ at every point where f is continuous, and to $\frac{1}{2}[f(x+) + f(x-)]$ at every point where f jumps, provided that, on one period,

1. f is bounded;

2. f has a finite number of maxima and minima;
3. f has a finite number of discontinuities.

These requirements are called the Dirichlet conditions. They are sufficient for convergence without being necessary, and they are met by the functions that describe most physical quantities. The proof of the theorem is beyond the scope of this course.

Figure 48 also shows that, just to the left of the jump at $x = L$, the sum of twelve terms rises above 1, and just to the right it falls below -1 . As more terms are added, this overshoot moves closer to the jump and becomes narrower, while its height tends to about 9 per cent of the size of the jump. This is called the Gibbs phenomenon, and you will find its limiting height in Problem 2. It might seem that an overshoot that never dies away contradicts Dirichlet's theorem. There is no contradiction, because at any fixed x away from the jump the overshoot eventually moves past x , and the partial sums at x then approach $f(x)$.

Even and Odd Functions

A function is said to be even if $g(-x) = g(x)$, such as $\cos x$. The product of two even functions or of two odd functions is even, the product of an even function and an odd function is odd, and the integral of an even function over $-L < x < L$ is twice its integral over $0 < x < L$. If f is even, every b_n vanishes, and the series contains only cosines and the constant term, with

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx. \quad (6.8)$$

If f is odd, every a_n vanishes, and b_n is given by the same formula with the cosine replaced by a sine. Since the square wave is odd, its series contains only sines.

Half-Range Series

A function given only on $0 < x < L$, such as the temperature along a bar of length L at one instant, can be written as a Fourier series once its values on $-L < x < 0$ have been chosen. Of the possible choices, two are in common use. The odd extension sets $f(-x) = -f(x)$ and

gives a series of sines only, which is called the half-range sine series of f . The even extension sets $f(-x) = f(x)$ and gives a series of cosines only, which is called the half-range cosine series. The coefficients of each series are given by the integrals over $0 < x < L$ of the previous subsection, and the series agree with f on $0 < x < L$ and differ outside it.

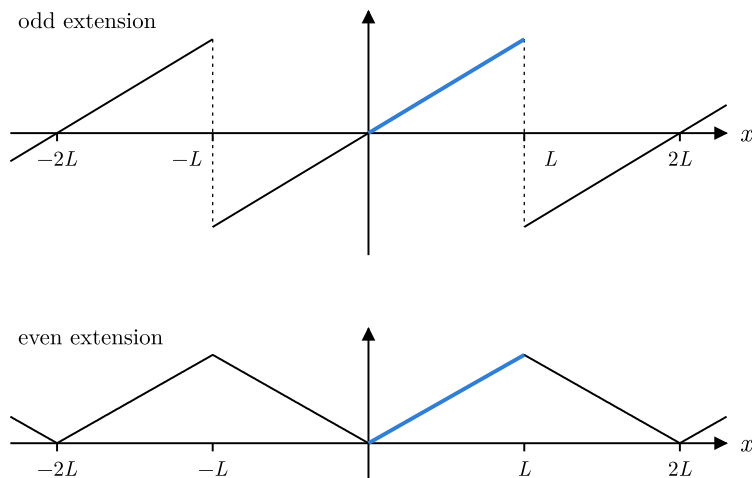


Figure 49. The function x , given on $0 < x < L$ (blue), extended to $-L < x < 0$ as an odd function (upper panel) and as an even function (lower panel), and then repeated with period $2L$. The odd extension is a sawtooth, which jumps at $x = \pm L$, and the even extension is a triangle wave, which is continuous.

As an example, consider $f(x) = x$ on $0 < x < L$, whose extensions are drawn in Figure 49. The odd extension is x itself on $-L < x < L$, repeated with period $2L$, which is called a sawtooth. Integrating by parts, with x as the factor to be differentiated, gives

$$\begin{aligned} \int_0^L x \sin \frac{n\pi x}{L} dx &= \left[-\frac{L}{n\pi} x \cos \frac{n\pi x}{L} \right]_0^L + \frac{L}{n\pi} \int_0^L \cos \frac{n\pi x}{L} dx \\ &= -\frac{L^2}{n\pi} (-1)^n, \end{aligned} \quad (6.9)$$

since the integral of the cosine is zero. The sine coefficients are $2/L$

times this integral, so $b_n = 2L(-1)^{n+1}/n\pi$, and

$$x = \frac{2L}{\pi} \left[\sin \frac{\pi x}{L} - \frac{1}{2} \sin \frac{2\pi x}{L} + \frac{1}{3} \sin \frac{3\pi x}{L} - \cdots \right], \quad -L < x < L. \quad (6.10)$$

At $x = L$ the series gives zero, the mean of the limits L and $-L$ on either side of the jump. The coefficients go as $1/n$, as those of the square wave do, and each of these functions has jumps.

Differentiation and Integration of Series

A Fourier series of a function that satisfies the Dirichlet conditions may be integrated term by term, and the result converges to the integral of the function. We assumed that a series may be integrated term by term when finding the coefficients, and the proof of this result is beyond the scope of this course. As an example, we integrate the square-wave series from 0 to x . The integral of f from 0 to x is $|x|$ for $-L < x < L$, which is the triangle wave, the even extension of Figure 49, and each term gives $\int_0^x \sin(n\pi x'/L) dx' = (L/n\pi)[1 - \cos(n\pi x/L)]$, where x' is the variable of integration. Therefore

$$|x| = \frac{4L}{\pi^2} \sum_{n \text{ odd}} \frac{1}{n^2} - \frac{4L}{\pi^2} \sum_{n \text{ odd}} \frac{1}{n^2} \cos \frac{n\pi x}{L}. \quad (6.11)$$

The constant term must be the mean value of $|x|$ over a period, which is $L/2$. This gives $\sum_{n \text{ odd}} 1/n^2 = \pi^2/8$, and the half-range cosine series of x is

$$|x| = \frac{L}{2} - \frac{4L}{\pi^2} \left[\cos \frac{\pi x}{L} + \frac{1}{9} \cos \frac{3\pi x}{L} + \frac{1}{25} \cos \frac{5\pi x}{L} + \cdots \right]. \quad (6.12)$$

These coefficients go as $1/n^2$, faster than the $1/n$ of the sine series of the same function. Integration introduced a factor $L/n\pi$ into each coefficient, and a function whose slope jumps, such as the triangle wave, is the integral of a function whose value jumps, so in general the smoother the periodic function, the faster its coefficients fall.

Term-by-term differentiation needs more care. Differentiating the sawtooth series gives $2[\cos(\pi x/L) - \cos(2\pi x/L) + \cdots]$, whose terms do not tend to zero, and a series whose terms do not tend to zero diverges, as we saw in Section 1.1. The cause is the jumps of the

sawtooth. Term-by-term differentiation is valid when the periodic function is continuous and its derivative satisfies the Dirichlet conditions. The triangle wave is an example, and its differentiated series is the series of the square wave.

The Complex Form of the Series

Euler's formula, from Section 1.2, gives $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ and $\sin \theta = (e^{i\theta} - e^{-i\theta})/2i$. Substituting these into the Fourier series gives

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}, \quad (6.13)$$

with $c_0 = \frac{1}{2}a_0$, $c_n = \frac{1}{2}(a_n - ib_n)$, and $c_{-n} = \frac{1}{2}(a_n + ib_n)$ for $n \geq 1$. For whole numbers m and n the exponentials satisfy the orthogonality relation

$$\int_{-L}^L e^{im\pi x/L} e^{-in\pi x/L} dx = 2L \delta_{mn}, \quad (6.14)$$

since for $m = n$ the integrand is 1, and for $m \neq n$ the integral is $2L \sin[(m-n)\pi]/[(m-n)\pi] = 0$, by Euler's formula. Multiplying the series by $e^{-im\pi x/L}$ and integrating therefore gives

$$c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx. \quad (6.15)$$

When f is real, $c_{-n} = c_n^*$. The complex form is often the more convenient, since a single formula gives every coefficient.

Parseval's Theorem

Let f have the complex series above, so that $f^*(x) = \sum_m c_m^* e^{-im\pi x/L}$. Multiplying the series for f^* and for f together and integrating over a period term by term gives

$$\begin{aligned} \frac{1}{2L} \int_{-L}^L |f(x)|^2 dx &= \sum_m \sum_n c_m^* c_n \frac{1}{2L} \int_{-L}^L e^{i(n-m)\pi x/L} dx \\ &= \sum_{n=-\infty}^{\infty} |c_n|^2, \end{aligned} \quad (6.16)$$

since the orthogonality relation makes every term with $m \neq n$ zero. This result is called Parseval's theorem. For a real function, $|c_0|^2 =$

$\frac{1}{4}a_0^2$ and $|c_n|^2 + |c_{-n}|^2 = \frac{1}{2}(a_n^2 + b_n^2)$, and the theorem becomes

$$\frac{1}{L} \int_{-L}^L f(x)^2 dx = \frac{1}{2}a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2). \quad (6.17)$$

The mean values of $a_n^2 \cos^2(n\pi x/L)$ and of $b_n^2 \sin^2(n\pi x/L)$ over a period are $\frac{1}{2}a_n^2$ and $\frac{1}{2}b_n^2$, and the cross term $2a_n b_n \cos(n\pi x/L) \sin(n\pi x/L)$ averages to zero, so the mean square of f is the mean square of its constant term $\frac{1}{2}a_0$ plus the sum of the mean squares of its harmonics, as the terms $a_n \cos(n\pi x/L) + b_n \sin(n\pi x/L)$ of the series are called.

As an application, consider the sawtooth, for which $a_n = 0$ and $b_n = 2L(-1)^{n+1}/n\pi$. The left-hand side of Parseval's theorem is $(1/L) \int_{-L}^L x^2 dx = 2L^2/3$, and the right-hand side is $\sum_n 4L^2/n^2\pi^2$. Equating the left-hand and right-hand sides gives

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \cdots = \frac{\pi^2}{6}. \quad (6.18)$$

In Section 1.1 we showed that this series converges, by comparing it with $\sum 2/[n(n+1)]$, and Parseval's theorem gives its sum.

6.2 Fourier Transforms and Convolution

Many functions met in physics are not periodic, such as a single pulse travelling along a rope, or the current through a resistor while a capacitor discharges through it. A function of this kind can also be written as a combination of oscillations, although the combination now contains every wave number, and the sum of a Fourier series becomes an integral. In this section we first obtain that integral, called the Fourier transform, from the complex series of Section 6.1, and work out the transforms of some simple functions. We then derive the properties of the transform, and finally prove the convolution theorem and Parseval's theorem for transforms.

From Series to Transform

Let $f(x)$ be a function defined for all x , which falls to zero as $|x| \rightarrow \infty$ fast enough for $\int_{-\infty}^{\infty} |f| dx$ to be finite. On the interval

$-L < x < L$, the complex series of Section 6.1 represents f as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{ik_n x}, \quad c_n = \frac{1}{2L} \int_{-L}^L f(x') e^{-ik_n x'} dx', \quad (6.19)$$

where $k_n = n\pi/L$, and x' is the variable of integration. The quantity k_n is a wave number, since $e^{ik_n x}$ is a wave of wavelength $2\pi/k_n$, and successive values of k_n are a distance $\Delta k = \pi/L$ apart. Let

$$F_L(k) = \frac{1}{\sqrt{2\pi}} \int_{-L}^L f(x') e^{-ikx'} dx', \quad (6.20)$$

so that $c_n = F_L(k_n) \Delta k / \sqrt{2\pi}$, since $1/2L = \Delta k / 2\pi$. The series then becomes

$$f(x) = \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} F_L(k_n) e^{ik_n x} \Delta k, \quad (6.21)$$

which is a sum of the areas of strips of width Δk under the curve $F_L(k) e^{ikx}$. As $L \rightarrow \infty$, the width of the strips tends to zero and the sum tends to an integral over k , which gives the pair of formulas

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx, \quad f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk. \quad (6.22)$$

The function $\tilde{f}(k)$, the limit of $F_L(k)$, is called the Fourier transform of $f(x)$, and the second formula is called the inversion formula. The passage to the limit makes the inversion formula plausible, although we have not justified replacing the sum by an integral. Where the coefficient c_n was the amplitude of the single wave $e^{ik_n x}$, the quantity $\tilde{f}(k) dk / \sqrt{2\pi}$ is the amplitude contributed by the waves with wave numbers between k and $k + dk$. The factors $1/\sqrt{2\pi}$ are shared equally between the transform and the inversion formula in this course, and other books often place them differently. When the variable is a time t , the transform is written as a function of an angular frequency ω , with x and k replaced by t and ω .

The Rectangular Pulse

As an example, consider the rectangular pulse of height 1 and width $2w$,

$$\Pi(x) = \begin{cases} 1, & |x| < w, \\ 0, & |x| > w, \end{cases} \quad (6.23)$$

which could, for example, describe the brightness across a slit of width $2w$ that is lit uniformly. Its transform is

$$\tilde{\Pi}(k) = \frac{1}{\sqrt{2\pi}} \int_{-w}^w e^{-ikx} dx = \frac{1}{\sqrt{2\pi}} \frac{e^{ikw} - e^{-ikw}}{ik} = \sqrt{\frac{2}{\pi}} \frac{\sin kw}{k}, \quad (6.24)$$

by Euler's formula in the form $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$. As $k \rightarrow 0$, $\sin kw \approx kw$, and $\tilde{\Pi}(0) = \sqrt{2/\pi} w$, which agrees with $(1/\sqrt{2\pi}) \int \Pi dx$. The transform is real and even, because Π is real and even.

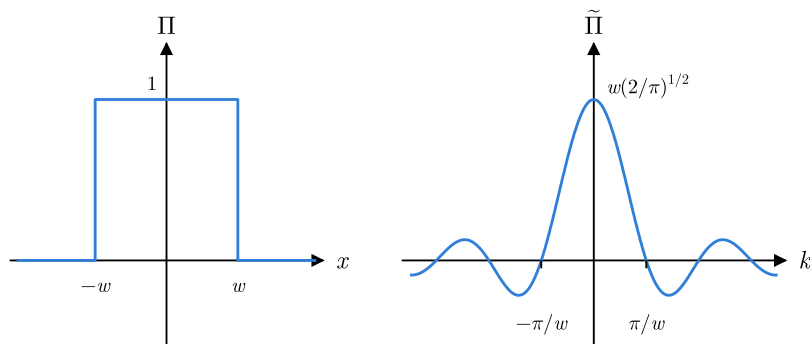


Figure 50. The rectangular pulse of height 1 and width $2w$ (left) and its Fourier transform $\tilde{\Pi}(k) = \sqrt{2/\pi} \sin(kw)/k$ (right). Most of the transform lies in the central peak between the zeros at $k = \pm\pi/w$.

The transform, drawn in Figure 50, is zero at $k = \pm\pi/w, \pm 2\pi/w$, and so on, and most of it lies in the central peak between $k = -\pi/w$ and $k = \pi/w$. A narrow pulse therefore has a wide transform, and a wide pulse a narrow one. The product of the width $2w$ of the pulse and the width $2\pi/w$ of the central peak is 4π , which does not depend on w .

The Decaying Exponential

As another example, consider a signal that is zero before $t = 0$ and decays exponentially afterwards,

$$f(t) = \begin{cases} 0, & t < 0, \\ e^{-\lambda t}, & t > 0, \end{cases} \quad (6.25)$$

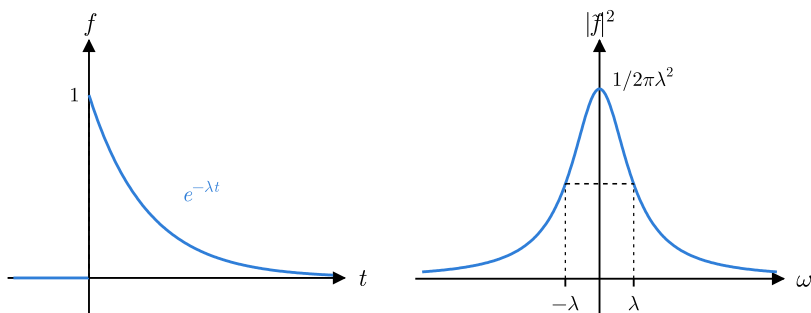


Figure 51. The signal that is zero for $t < 0$ and equal to $e^{-\lambda t}$ for $t > 0$ (left), and its power spectrum $|\tilde{f}(\omega)|^2 = 1/[2\pi(\lambda^2 + \omega^2)]$ (right), a Lorentzian, which falls to half its greatest value at $\omega = \pm\lambda$.

where λ is a positive constant, which could, for example, describe the current through a resistor when a capacitor begins to discharge through it at $t = 0$. The transform is

$$\tilde{f}(\omega) = \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-(\lambda+i\omega)t} dt = \frac{1}{\sqrt{2\pi}(\lambda + i\omega)}, \quad (6.26)$$

where the exponential vanishes at the upper limit because its modulus is $e^{-\lambda t}$. The transform is complex, because f is neither even nor odd. Its squared modulus, $|\tilde{f}|^2 = 1/[2\pi(\lambda^2 + \omega^2)]$, falls to half its largest value at $\omega = \pm\lambda$, and a curve of this shape, drawn in Figure 51, is called a Lorentzian. The signal lasts for a time of order $1/\lambda$, and its frequencies are spread over a range of order λ , so a signal that decays quickly contains a wide range of frequencies.

The Gaussian

Consider next the Gaussian $f(x) = e^{-\alpha x^2}$, where α is a positive constant. Its transform is

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-\alpha x^2} e^{-ikx} dx, \quad (6.27)$$

and the integrand has no antiderivative that can be written in terms of elementary functions. We can instead find a differential equation for $\tilde{f}$. Differentiating under the integral sign with respect to k , as in Section 2.2, which is allowed because the derivative of the integrand

has modulus $|x|e^{-\alpha x^2}$, whose integral over the whole line converges, gives

$$\frac{d\tilde{f}}{dk} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (-ix) e^{-\alpha x^2} e^{-ikx} dx. \quad (6.28)$$

By the chain rule, $-ix e^{-\alpha x^2} = (i/2\alpha) d(e^{-\alpha x^2})/dx$. Integrating by parts, with e^{-ikx} as the factor to be differentiated, gives

$$\begin{aligned} \frac{d\tilde{f}}{dk} &= \frac{i}{2\alpha} \frac{1}{\sqrt{2\pi}} \left\{ [e^{-\alpha x^2} e^{-ikx}]_{-\infty}^{\infty} + ik \int_{-\infty}^{\infty} e^{-\alpha x^2} e^{-ikx} dx \right\} \\ &= -\frac{k}{2\alpha} \tilde{f}(k), \end{aligned} \quad (6.29)$$

where the boundary term vanishes because $e^{-\alpha x^2}$ tends to zero at both limits while $|e^{-ikx}| = 1$. This is a first-order linear equation for $\tilde{f}$, and with the integrating factor $e^{k^2/4\alpha}$ of Section 5.1 it becomes $d(e^{k^2/4\alpha} \tilde{f})/dk = 0$, so that $\tilde{f}(k) = \tilde{f}(0) e^{-k^2/4\alpha}$. The Gaussian integral $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\pi/\alpha}$ of Section 2.3 gives $\tilde{f}(0) = 1/\sqrt{2\alpha}$, and

$$\tilde{f}(k) = \frac{1}{\sqrt{2\alpha}} e^{-k^2/4\alpha}. \quad (6.30)$$

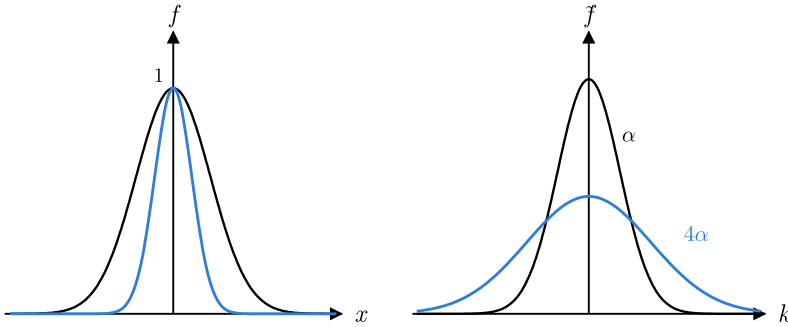


Figure 52. The Gaussians $e^{-\alpha x^2}$ and $e^{-4\alpha x^2}$ (left) and their Fourier transforms $e^{-k^2/4\alpha}/\sqrt{2\alpha}$ and $e^{-k^2/16\alpha}/\sqrt{8\alpha}$ (right). In each panel the blue curve belongs to the parameter 4α . The Gaussian that is narrower in x has the wider transform.

The transform of a Gaussian is therefore another Gaussian. The function f falls to $e^{-1/2}$ of its largest value at $x = \pm 1/\sqrt{2\alpha}$, and

$\tilde{f}$ does so at $k = \pm\sqrt{2\alpha}$, so the product of these half-widths is 1 for every α . Figure 52 shows that increasing α by a factor of 4 halves the width of f and doubles the width of $\tilde{f}$. With $\alpha = \frac{1}{2}$, the transform of $e^{-x^2/2}$ is $e^{-k^2/2}$, the same function of k as f is of x .

The pulse, the decaying exponential, and the Gaussian all show the same relation between the width of a function and the width of its transform, which is called the reciprocal relation. A function confined to a region of width Δx has a transform spread over a range Δk of wave numbers, where the product $\Delta x \Delta k$ cannot be made much smaller than one, so a short pulse of sound, for example, contains a wide range of frequencies. A precise form of this statement is not treated in this course.

The Spectrum of a Finite Burst

As a further example of the reciprocal relation, consider a signal that oscillates as $\cos \omega_0 t$ for $-T < t < T$ and is zero at other times, which could, for example, describe the voltage from an oscillator that is switched on for a time $2T$. Writing the cosine as $\frac{1}{2}(e^{i\omega_0 t} + e^{-i\omega_0 t})$, the transform is

$$\begin{aligned}\tilde{f}(\omega) &= \frac{1}{2\sqrt{2\pi}} \int_{-T}^T [e^{-i(\omega-\omega_0)t} + e^{-i(\omega+\omega_0)t}] dt \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{\sin(\omega - \omega_0)T}{\omega - \omega_0} + \frac{\sin(\omega + \omega_0)T}{\omega + \omega_0} \right],\end{aligned}\tag{6.31}$$

where each integral has been evaluated as for the rectangular pulse, with k replaced by $\omega \mp \omega_0$ and w by T .

The transform of a signal, regarded as a function of frequency, is called its spectrum. The spectrum of the burst, drawn in Figure 53, consists of two copies of the transform of the pulse, each of half the height, centred on $\omega = \omega_0$ and $\omega = -\omega_0$. When many oscillations fit into the burst, so that $\omega_0 T \gg 1$, the copies overlap very little, and each peak has height close to $T/\sqrt{2\pi}$ and a width of $2\pi/T$ between its first zeros. A long burst has a spectrum concentrated near ω_0 , while a burst that lasts only a few cycles contains a wide spread of frequencies about ω_0 , and the product of the duration $2T$ and the width $2\pi/T$ does not depend on T .

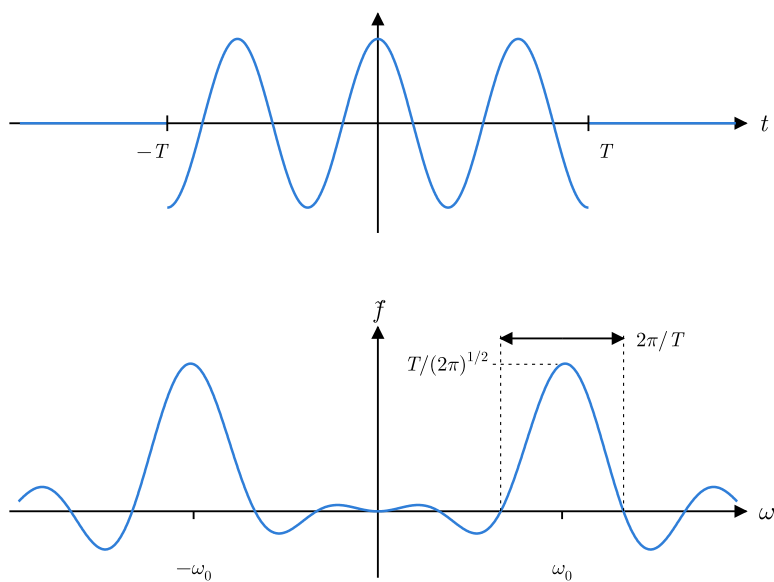


Figure 53. A burst of the oscillation $\cos \omega_0 t$ lasting from $t = -T$ to $t = T$ (upper panel), drawn for $\omega_0 T = 3\pi$, and its Fourier transform (lower panel). The transform has peaks of height close to $T/\sqrt{2\pi}$ at $\omega = \pm\omega_0$, and the zeros on either side of each peak are a distance $2\pi/T$ apart.

Properties of the Transform

Let $\tilde{f}(k)$ be the transform of $f(x)$, let a be a non-zero real constant, and let x_0 and k_0 be real constants. The transform has the following properties:

1. the shift property, which states that the transform of $f(x - x_0)$ is $e^{-ikx_0} \tilde{f}(k)$;
2. the scaling property, which states that the transform of $f(ax)$ is $\tilde{f}(k/a)/|a|$;
3. the derivative property, which states that the transform of df/dx is $ik\tilde{f}(k)$;
4. the modulation property, which states that the transform of $e^{ik_0x} f(x)$ is $\tilde{f}(k - k_0)$.

The shift property follows from the substitution $u = x - x_0$ in the integral for the transform, and it shows that delaying a signal changes the phase of each frequency and leaves $|\tilde{f}|$ unchanged. For the scaling property we substitute $u = ax$, which gives $\tilde{f}(k/a)/a$ when $a > 0$. When $a < 0$ the limits of the integral are exchanged, which changes the sign and turns $1/a$ into $1/|a|$. Compressing a function by a factor a therefore stretches its transform by the same factor, which is the reciprocal relation again. For the derivative property we integrate by parts:

$$\begin{aligned} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{df}{dx} e^{-ikx} dx &= \frac{1}{\sqrt{2\pi}} [f(x) e^{-ikx}]_{-\infty}^{\infty} \\ &\quad + \frac{ik}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx \\ &= ik \tilde{f}(k), \end{aligned} \tag{6.32}$$

where the boundary term vanishes because f tends to zero at both limits. Applied twice, the derivative property shows that the transform of d^2f/dx^2 is $-k^2\tilde{f}(k)$, so a linear differential equation with constant coefficients becomes an algebraic equation for the transform of its solution. You will use this result in Problem 2 for the driven oscillator. The modulation property follows from writing $e^{ik_0x} e^{-ikx} = e^{-i(k-k_0)x}$ in the integral, and the spectrum of the burst is an instance of the modulation property.

Convolution

The convolution of two functions f and g is defined as follows:

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x') g(x - x') dx'. \quad (6.33)$$

To form the integrand, the function g is reflected, so that $g(-x')$ is drawn against x' , and then shifted a distance x to the right. The substitution $x'' = x - x'$ shows that $f * g = g * f$. If, for instance, g is the pulse Π divided by its width $2w$, then $(f * g)(x)$ is the mean value of f over an interval of width $2w$ centred on x , which could, for example, describe the value recorded by an instrument that averages the quantity it measures over a finite range.

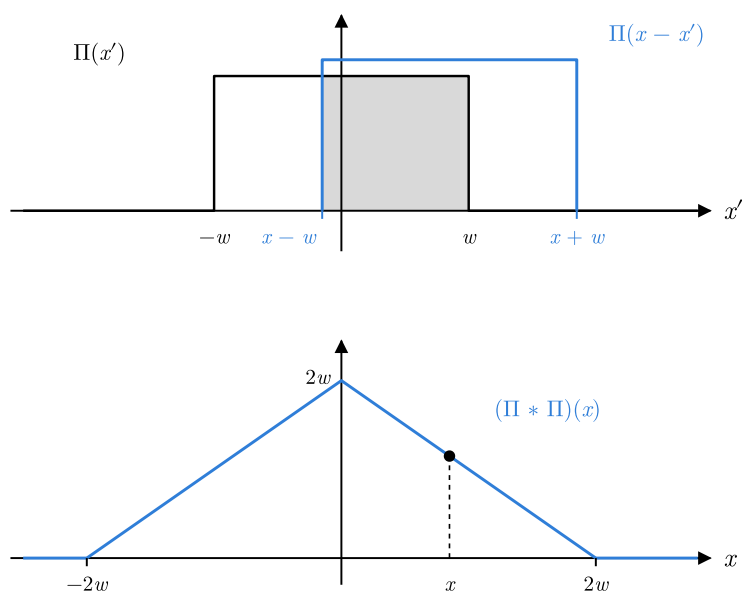


Figure 54. The convolution of the rectangular pulse Π with itself. The upper panel shows $\Pi(x')$ and the reflected and shifted pulse $\Pi(x - x')$ (blue, drawn a little higher than $\Pi(x')$ so that the pulses can be told apart), which is non-zero for $x - w < x' < x + w$. The value of $(\Pi * \Pi)(x)$ is the length of the shaded overlap. The lower panel shows the result, the triangle $2w - |x|$, with the point that corresponds to the shift in the upper panel.

As an example of a convolution, consider the pulse Π convolved with itself, as shown in Figure 54. The integrand $\Pi(x')\Pi(x-x')$ is 1 where $|x'| < w$ and $|x-x'| < w$, and zero elsewhere. For $0 < x < 2w$ both conditions are met on the interval $x-w < x' < w$, of length $2w-x$, and a similar argument holds for $-2w < x < 0$. Hence $(\Pi * \Pi)(x) = 2w - |x|$ for $|x| < 2w$, and zero for $|x| > 2w$, which is a triangle of height $2w$ and base $4w$.

The transform of a convolution is given by the convolution theorem. Let $h = f * g$, so that

$$\tilde{h}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x') g(x-x') dx' \right] e^{-ikx} dx. \quad (6.34)$$

We exchange the order of integration, as in Section 2.3, and in the integral over x at fixed x' we substitute $u = x - x'$ and write $e^{-ikx} = e^{-ikx'} e^{-iku}$. The double integral then becomes a product of two single integrals, each of which is $\sqrt{2\pi}$ times a transform, and

$$\begin{aligned} \tilde{h}(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x') e^{-ikx'} dx' \int_{-\infty}^{\infty} g(u) e^{-iku} du \\ &= \sqrt{2\pi} \tilde{f}(k) \tilde{g}(k). \end{aligned} \quad (6.35)$$

This result is called the convolution theorem, and you will check it for the triangle $\Pi * \Pi$ in Problem 3. A similar calculation, in which $g(x)$ is written with the inversion formula as $(1/\sqrt{2\pi}) \int \tilde{g}(k') e^{ik'x} dk'$ and the order of integration is exchanged, gives the transform of a product:

$$\begin{aligned} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) g(x) e^{-ikx} dx &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{g}(k') \tilde{f}(k-k') dk' \\ &= \frac{1}{\sqrt{2\pi}} (\tilde{f} * \tilde{g})(k). \end{aligned} \quad (6.36)$$

Parseval's Theorem for Transforms

The transform satisfies the relation

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\tilde{f}(k)|^2 dk, \quad (6.37)$$

which is called Parseval's theorem for transforms. To prove it, we write the factor $\tilde{f}(k)$ in $|\tilde{f}|^2 = \tilde{f}^* \tilde{f}$ as its defining integral, and

exchange the order of integration:

$$\begin{aligned}\int_{-\infty}^{\infty} \tilde{f}^*(k) \tilde{f}(k) dk &= \int_{-\infty}^{\infty} f(x) \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}^*(k) e^{-ikx} dk \right] dx \\ &= \int_{-\infty}^{\infty} f(x) f^*(x) dx,\end{aligned}\tag{6.38}$$

since the bracket is the complex conjugate of the inversion formula for $f(x)$.

If $f(t)$ is the voltage across a resistor, the energy dissipated is proportional to $\int f^2 dt$, and therefore also to $\int |\tilde{f}(\omega)|^2 d\omega$. The quantity $|\tilde{f}(\omega)|^2 d\omega$ is then proportional to the energy carried by the angular frequencies between ω and $\omega + d\omega$, and $|\tilde{f}(\omega)|^2$ is often called the power spectrum of the signal, although for a pulse of finite energy it is strictly an energy per unit angular frequency. For a real signal, $\tilde{f}(-\omega) = \tilde{f}^*(\omega)$, so the positive and negative frequencies contribute equally. We can check the theorem with the decaying exponential, for which the left-hand side is $\int_0^{\infty} e^{-2\lambda t} dt = 1/2\lambda$, and the right-hand side is

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{d\omega}{\lambda^2 + \omega^2} = \frac{1}{2\pi\lambda} \left[\arctan \frac{\omega}{\lambda} \right]_{-\infty}^{\infty} = \frac{1}{2\lambda}.\tag{6.39}$$

6.3 The Dirac Delta Function

Some physical quantities are concentrated at a single point or at a single instant. A point mass m has a density that is zero everywhere except at the point, and yet the integral of the density over any volume containing the point is m . A hammer blow exerts a very large force for a very short time, and its effect is fixed by the impulse $\int F dt$, which is the change of momentum it produces. The delta function describes quantities of this kind. We first introduce it as the limit of a sequence of functions and derive its properties, then relate it to the step function and to the Fourier transform, and finally use it to describe a point mass and a sudden impulse.

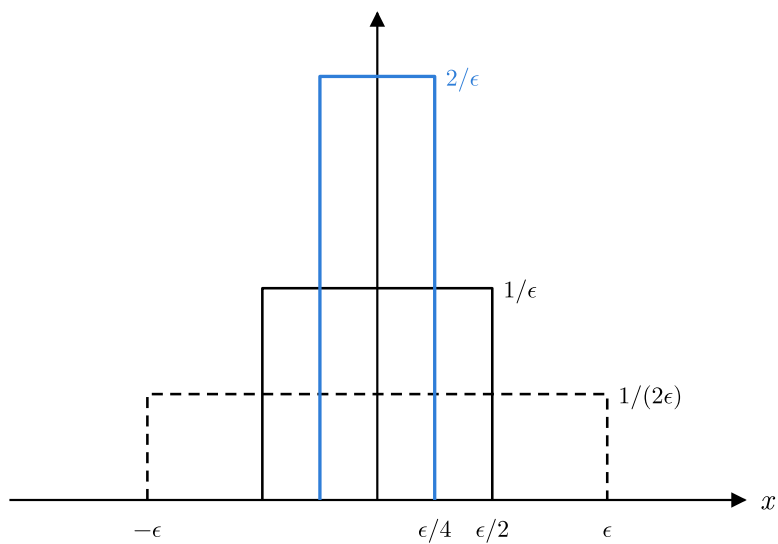


Figure 55. The functions $\delta_\epsilon(x)$ (dashed), $\delta_{\epsilon/2}(x)$, and $\delta_{\epsilon/4}(x)$ (blue), with heights $1/2\epsilon$, $1/\epsilon$, and $2/\epsilon$. Each encloses unit area, and as the half-width is reduced the functions become taller and narrower.

A Sequence of Tall, Narrow Functions

Let ε be a positive constant, and let

$$\delta_\varepsilon(x) = \begin{cases} 1/2\varepsilon, & |x| < \varepsilon, \\ 0, & |x| > \varepsilon. \end{cases} \quad (6.40)$$

Each of these functions encloses unit area, and as ε is reduced they become taller and narrower, as Figure 55 shows.

Let $f(x)$ be a function that is continuous at $x = 0$. Then

$$\int_{-\infty}^{\infty} f(x) \delta_\varepsilon(x) dx = \frac{1}{2\varepsilon} \int_{-\varepsilon}^{\varepsilon} f(x) dx, \quad (6.41)$$

which is the mean value of f over the interval $-\varepsilon < x < \varepsilon$. As $\varepsilon \rightarrow 0$ the interval shrinks onto the origin, and since f is continuous there, its mean value over the interval tends to $f(0)$. Therefore

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} f(x) \delta_\varepsilon(x) dx = f(0). \quad (6.42)$$

The limit of $\delta_\varepsilon(x)$ itself is not a function in the ordinary sense. It would be zero for every $x \neq 0$ and infinite at $x = 0$, and an ordinary function that is zero everywhere except at one point has zero integral, whereas the area under every δ_ε is 1. We therefore define the delta function $\delta(x)$ by its effect inside an integral:

$$\int_{-\infty}^{\infty} f(x) \delta(x) dx = f(0), \quad (6.43)$$

for every function f that is continuous at $x = 0$, where the integral is understood as the limit above. The function $\delta(x)$ is called the Dirac delta function. Taking $f = 1$ gives $\int \delta(x) dx = 1$, and taking f to be zero near the origin shows that the integral of $\delta(x)$ over any interval that does not contain the origin is zero.

Other sequences lead to the same delta function. The Gaussians $e^{-x^2/\varepsilon^2}/(\varepsilon\sqrt{\pi})$ have unit area, by the Gaussian integral of Section 2.3, and with the substitution $x = \varepsilon u$ the integral of $f(x)$ multiplied by one of them becomes $(1/\sqrt{\pi}) \int f(\varepsilon u) e^{-u^2} du$, which tends to $f(0)$ as $\varepsilon \rightarrow 0$. The general theory of objects of this kind, which are called generalised functions, is not treated in this course.

With the substitution $u = x - x_0$, where x_0 is a constant, the definition becomes

$$\int_{-\infty}^{\infty} f(x) \delta(x - x_0) dx = f(x_0), \quad (6.44)$$

and this is called the sifting property of the delta function, since the delta function picks out the value of f at the single point x_0 . The integral over a finite interval is $f(x_0)$ if the interval contains x_0 and zero if it does not. Since $f(x) \delta(x - x_0)$ and $f(x_0) \delta(x - x_0)$ have the same effect inside every integral, we may write $f(x) \delta(x - x_0) = f(x_0) \delta(x - x_0)$.

The Scaling Rule

Let a be a non-zero real constant, and consider $\delta(ax)$. For $a > 0$, the substitution $u = ax$ gives

$$\int_{-\infty}^{\infty} f(x) \delta(ax) dx = \frac{1}{a} \int_{-\infty}^{\infty} f\left(\frac{u}{a}\right) \delta(u) du = \frac{f(0)}{a}. \quad (6.45)$$

For $a < 0$, the same substitution sends $x \rightarrow -\infty$ to $u \rightarrow \infty$, and exchanging the limits introduces a minus sign, so the result is $f(0)/|a|$. For either sign of a , $\delta(ax)$ has the same effect as $\delta(x)/|a|$, and therefore

$$\delta(ax) = \frac{\delta(x)}{|a|}. \quad (6.46)$$

The rule is consistent with the dimensions of the delta function, which, since $\int \delta(x) dx = 1$, are those of the inverse of its argument. With $a = -1$ the scaling rule gives $\delta(-x) = \delta(x)$, so the delta function is even.

Let $g(x)$ be a function whose zeros are at the points x_i , and suppose that each zero is simple, that is, $g'(x_i) \neq 0$. The function $\delta(g(x))$ is zero except where $g(x) = 0$, so an integral containing it receives contributions only from small neighbourhoods of the zeros. Near the zero x_i , the Taylor series of Section 1.1 gives $g(x) \approx g'(x_i)(x - x_i)$, since $g(x_i) = 0$, and the curve is close to its tangent there, as Figure 56 shows. Replacing g by its tangent in each neighbourhood, and applying the scaling rule with $a = g'(x_i)$ and with $x - x_i$ in

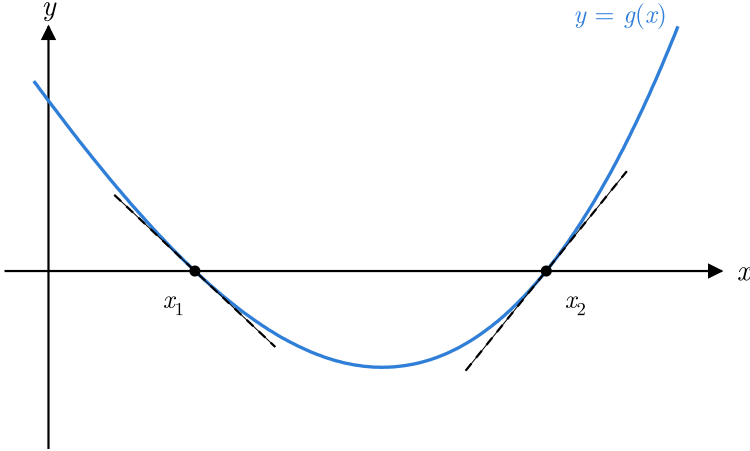


Figure 56. A function $g(x)$ with simple zeros at x_1 and x_2 . Near each zero the curve is close to its tangent (dashed), $g(x) \approx g'(x_i)(x - x_i)$, and only these neighbourhoods contribute to an integral that contains $\delta(g(x))$.

place of x , which the substitution $u = x - x_i$ allows, gives

$$\delta(g(x)) = \sum_i \frac{\delta(x - x_i)}{|g'(x_i)|}. \quad (6.47)$$

As an example, consider $g(x) = x^2 - x_0^2$, with $x_0 > 0$. The zeros are at $x = \pm x_0$, where $|g'| = 2x_0$, and so

$$\delta(x^2 - x_0^2) = \frac{\delta(x - x_0) + \delta(x + x_0)}{2x_0}. \quad (6.48)$$

The rule requires every zero to be simple. For $g(x) = x^2$, whose zero at the origin is not simple, $\delta_\varepsilon(x^2)$ equals $1/2\varepsilon$ for $|x| < \sqrt{\varepsilon}$, so its integral, $1/\sqrt{\varepsilon}$, grows without limit as $\varepsilon \rightarrow 0$, and $\delta(x^2)$ has no meaning.

The Step Function

The function

$$H(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0, \end{cases} \quad (6.49)$$

is called the Heaviside step function. It could, for example, describe the voltage across a circuit, with x as the time, when a cell of unit voltage is connected to it at $x = 0$. The integral of δ_ε from $-\infty$ to x is

$$\int_{-\infty}^x \delta_\varepsilon(x') dx' = \begin{cases} 0, & x < -\varepsilon, \\ (x + \varepsilon)/2\varepsilon, & |x| < \varepsilon, \\ 1, & x > \varepsilon, \end{cases} \quad (6.50)$$

which is a ramp that rises from 0 to 1 across the width of δ_ε .

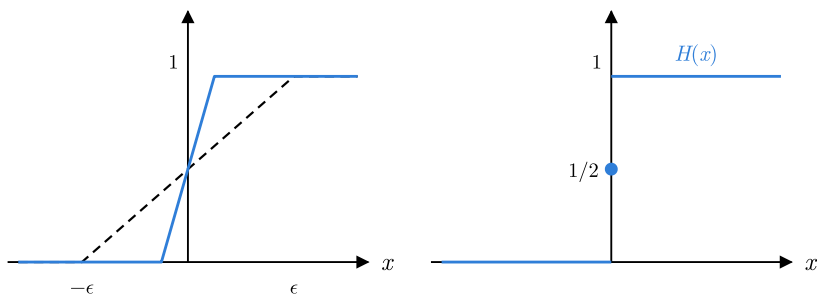


Figure 57. Left: the integrals $\int_{-\infty}^x \delta_\varepsilon(x') dx'$ (dashed) and $\int_{-\infty}^x \delta_{\varepsilon/4}(x') dx'$ (blue), which rise from 0 to 1 across the width of each function. Right: the Heaviside step function $H(x)$, which is the limit of the integrals as the width tends to zero, with the value $H(0) = \frac{1}{2}$ marked.

As $\varepsilon \rightarrow 0$ the ramp becomes steeper, as Figure 57 shows, and it tends to $H(x)$ at every $x \neq 0$. At $x = 0$ the ramp has the value $\frac{1}{2}$ for every ε . We therefore have

$$\int_{-\infty}^x \delta(x') dx' = H(x), \quad \frac{dH}{dx} = \delta(x). \quad (6.51)$$

The value of H at $x = 0$ is not fixed by its definition, and it is often taken to be $\frac{1}{2}$, which is the value of every ramp there, and also the value to which a Fourier series converges at a jump, as we found in Section 6.1.

We can check the relation $dH/dx = \delta(x)$ against the definition of the delta function. Let f be a differentiable function that tends to

zero as $|x| \rightarrow \infty$. Integrating by parts,

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) \frac{dH}{dx} dx &= [f(x) H(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{df}{dx} H(x) dx \\ &= 0 - \int_0^{\infty} \frac{df}{dx} dx \\ &= f(0), \end{aligned} \tag{6.52}$$

which is the effect of $\delta(x)$ inside the integral. The relation lets us differentiate a function that jumps. Suppose that f is smooth except at x_0 , where it jumps from $f(x_0-)$ to $f(x_0+)$. Then f is a smooth function plus $[f(x_0+) - f(x_0-)] H(x - x_0)$, and its derivative is the derivative of the smooth part plus $[f(x_0+) - f(x_0-)] \delta(x - x_0)$. In Problem 4 you will use this result to find the derivative of the square wave of Section 6.1, whose series diverges when it is differentiated term by term.

The Delta Function and the Fourier Transform

By the sifting property, the Fourier transform of $\delta(x - x_0)$ is

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x - x_0) e^{-ikx} dx = \frac{e^{-ikx_0}}{\sqrt{2\pi}}, \tag{6.53}$$

and the transform of $\delta(x)$ is the constant $1/\sqrt{2\pi}$. Every wave number is present with the same amplitude. This agrees with the reciprocal relation of Section 6.2, since a spike of zero width has a transform of unlimited width.

The transform of a constant cannot be found from the definition, since $\int e^{-ikx} dx$ does not converge, and we find it instead as a limit. The Gaussian $e^{-\eta x^2}$, where η is a positive constant, tends to 1 at every x as $\eta \rightarrow 0$, and by the transform of the Gaussian found in Section 6.2 its transform is

$$\frac{1}{\sqrt{2\eta}} e^{-k^2/4\eta}. \tag{6.54}$$

As $\eta \rightarrow 0$ this Gaussian in k becomes taller and narrower, and the area under it, found with the Gaussian integral of Section 2.3, is $\sqrt{4\pi\eta}/\sqrt{2\eta} = \sqrt{2\pi}$ for every η . This Gaussian in k is therefore $\sqrt{2\pi}$ times one of the Gaussian sequences for the delta function, with $\varepsilon = 2\sqrt{\eta}$, and the transform of the constant 1 is $\sqrt{2\pi} \delta(k)$. Writing

this out gives $\int e^{-ikx} dx = 2\pi \delta(k)$. Exchanging the names of the variables x and k , and then replacing k by $-k$ in the integral, which leaves its value unchanged, gives

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk, \quad (6.55)$$

which is called the integral representation of the delta function. The integral is to be understood as the limit found above, in which the integrand is multiplied by $e^{-\eta k^2}$ and $\eta \rightarrow 0$ after integrating. In the form $\int e^{ik(x-x')} dk = 2\pi \delta(x-x')$, the representation is the counterpart of the orthogonality relation $\int_{-L}^L e^{i(m-n)\pi x/L} dx = 2L \delta_{mn}$ of Section 6.1, with the delta function in the place of the Kronecker delta.

The integral representation lets us prove the inversion formula, which Section 6.2 obtained as the limit of a Fourier series without justifying the passage from a sum to an integral. Substituting the definition of $\tilde{f}$ into the right-hand side of the inversion formula and exchanging the order of integration, as in Section 2.3, gives

$$\begin{aligned} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk &= \int_{-\infty}^{\infty} f(x') \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x-x')} dk \right] dx' \\ &= \int_{-\infty}^{\infty} f(x') \delta(x-x') dx' \\ &= f(x), \end{aligned} \quad (6.56)$$

where the last step is the sifting property. The inversion formula therefore holds.

The same representation gives the transform of $\cos \omega_0 t$, for which the defining integral of Section 6.2 does not converge. Writing the cosine as $\frac{1}{2}(e^{i\omega_0 t} + e^{-i\omega_0 t})$, and using the integral representation in the form $\int e^{-i(\omega \mp \omega_0)t} dt = 2\pi \delta(\omega \mp \omega_0)$, we find

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \cos(\omega_0 t) e^{-i\omega t} dt = \sqrt{\frac{\pi}{2}} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)], \quad (6.57)$$

which consists of two spikes at the frequencies $\pm \omega_0$. The finite burst of Section 6.2, whose spectrum we found there by direct integration, is the product of the pulse $\Pi(t)$, with $w = T$, and $\cos \omega_0 t$. By the result of Section 6.2 for the transform of a product, its transform is

$1/\sqrt{2\pi}$ times the convolution of $\tilde{\Pi}$ with the transform of the cosine, $\sqrt{\pi/2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$, and by the sifting property this is $\frac{1}{2}[\tilde{\Pi}(\omega - \omega_0) + \tilde{\Pi}(\omega + \omega_0)]$, in agreement with the direct calculation. The convolution places a copy of the spectrum of the pulse at each frequency that the cosine contains.

A Point Mass and a Sudden Impulse

Consider first a point mass m at the point $\mathbf{r}_0 = (x_0, y_0, z_0)$. Its density is

$$\rho(\mathbf{r}) = m \delta(x - x_0) \delta(y - y_0) \delta(z - z_0), \quad (6.58)$$

which is zero everywhere except at $\mathbf{r}_0$. Integrating over all space as a repeated integral, as in Section 2.3, each delta function integrates to 1, and the total mass is m . The moment of inertia about the z -axis is

$$\int (x^2 + y^2) \rho dV = m (x_0^2 + y_0^2), \quad (6.59)$$

by the sifting property applied in x and in y , which is the mass multiplied by the square of its distance from the axis. A body such as a uniform rod carrying a small bead has a density that is the sum of a smooth part and a delta function, and its mass and moment of inertia are then given by a single integral.

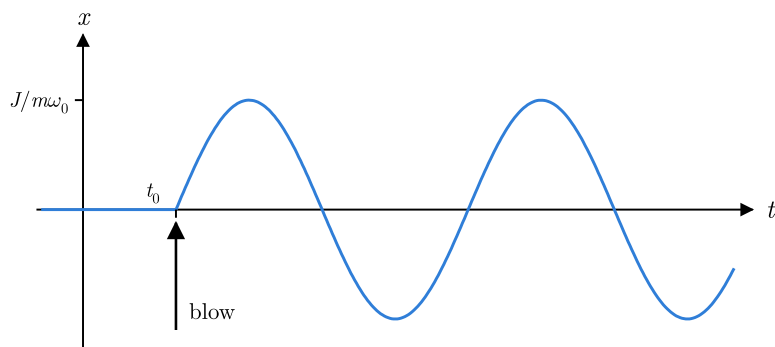


Figure 58. The displacement of a mass on a spring, at rest until it is struck at $t = t_0$ by a blow of impulse J . The displacement is continuous at t_0 , while the velocity jumps from 0 to J/m , after which the mass oscillates with amplitude $J/m\omega_0$.

Consider next a mass m on a spring, with natural angular frequency ω_0 , so that the spring constant is $m\omega_0^2$. The mass is at rest at its equilibrium position until the time t_0 , when it is struck by a blow of impulse J . The force of the blow is $J\delta(t - t_0)$, which acts only at t_0 and whose integral over time is J . Newton's second law and Hooke's law give

$$m\ddot{x} + m\omega_0^2 x = J\delta(t - t_0). \quad (6.60)$$

Let Δt be a short interval of time. Integrating both sides of the equation of motion from $t_0 - \Delta t$ to $t_0 + \Delta t$ gives

$$m[\dot{x}(t_0 + \Delta t) - \dot{x}(t_0 - \Delta t)] + m\omega_0^2 \int_{t_0 - \Delta t}^{t_0 + \Delta t} x \, dt = J. \quad (6.61)$$

The displacement is finite, so the integral tends to zero as $\Delta t \rightarrow 0$, and the velocity jumps by J/m at t_0 , which is the statement that the impulse equals the change of momentum. The displacement does not jump, since the velocity remains finite. After t_0 , the mass oscillates freely, starting from $x = 0$ with velocity J/m , and the solution for all t is

$$x(t) = \frac{J}{m\omega_0} \sin[\omega_0(t - t_0)] H(t - t_0), \quad (6.62)$$

which is drawn in Figure 58. We can check this solution by substitution. By the product rule, and with $dH(t - t_0)/dt = \delta(t - t_0)$,

$$\dot{x} = \frac{J}{m} \cos[\omega_0(t - t_0)] H(t - t_0) + \frac{J}{m\omega_0} \sin[\omega_0(t - t_0)] \delta(t - t_0), \quad (6.63)$$

and the second term is zero, because the sine vanishes at t_0 and $f(t)\delta(t - t_0) = f(t_0)\delta(t - t_0)$. Differentiating $\dot{x}$ in the same way, and using $\cos 0 = 1$, gives

$$\ddot{x} = -\omega_0^2 x + \frac{J}{m} \delta(t - t_0), \quad (6.64)$$

which is the equation of motion divided by m . The equation is linear, so the response to a force made of several blows is the sum of the responses to each blow, as you will see in Problem 5.

Problems

§6.1 Fourier Series

- Find the Fourier series of $f(x) = x^2$ on $-L < x < L$. By evaluating the series at $x = L$, show again that $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$, and by evaluating it at $x = 0$, show that $1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \cdots = \pi^2/12$.
- Let $S_N(x) = (4/\pi) \sum_{j=1}^N \sin[(2j-1)\pi x/L]/(2j-1)$ be the sum of the first N non-zero terms of the square-wave series. Using Euler's formula and the sum of a geometric series, show that $\sum_{j=1}^N \cos[(2j-1)\theta] = \sin(2N\theta)/(2\sin\theta)$, and hence that $dS_N/dx = (2/L) \sin(2N\pi x/L)/\sin(\pi x/L)$. Show that the first maximum of S_N to the right of $x = 0$ is at $x = L/2N$, and that the value of S_N there tends to $(2/\pi) \int_0^\pi (\sin u/u) du$ as $N \rightarrow \infty$. This integral cannot be written in terms of elementary functions, and its value is close to 1.85. What is the limiting overshoot, as a fraction of the size of the jump?
- Apply Parseval's theorem to the series of Problem 1 to show that $\sum_{n=1}^{\infty} 1/n^4 = \pi^4/90$.
- Show that the complex Fourier coefficients of $f(x) = e^{\beta x}$ on $-L < x < L$, where β is a real constant, are $c_n = (-1)^n \sinh(\beta L)/(\beta L - in\pi)$. To what value does the series converge at $x = L$? By pairing the terms with n and $-n$, deduce that $\sum_{n=1}^{\infty} 1/(y^2 + n^2\pi^2) = (y \coth y - 1)/2y^2$, where $y = \beta L$ and $\coth y = \cosh y / \sinh y$.
- Find the half-range cosine series of $\sin x$ on $0 < x < \pi$, taking $L = \pi$, and show that it is $|\sin x| = 2/\pi - (4/\pi) \sum_{m=1}^{\infty} \cos(2mx)/(4m^2 - 1)$. Check the result at $x = 0$ by summing $\sum_m 1/(4m^2 - 1)$ with partial fractions. Explain why the coefficients go as $1/m^2$.

§6.2 Fourier Transforms and Convolution

- Show that the Fourier transform of $e^{-\lambda|t|}$, where $\lambda > 0$, is $\sqrt{2/\pi} \lambda/(\lambda^2 + \omega^2)$. Check the inversion formula at $t = 0$.
- The displacement of a mass m on a spring, with natural angular

- frequency ω_0 and damping rate γ , driven by a force $F(t)$ that acts for a finite time, obeys $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = F(t)/m$, as in Section 5.2. Assuming that x and $\dot{x}$ tend to zero as $|t| \rightarrow \infty$, use the derivative property to show that $\tilde{x}(\omega) = \tilde{F}(\omega)/[m(\omega_0^2 - \omega^2 + 2i\gamma\omega)]$. When $\gamma \ll \omega_0$, near which frequencies is $|\tilde{x}|$ much larger than $|\tilde{F}|/m\omega_0^2$?
- By transforming the triangle $(\Pi * \Pi)(x)$ of Figure 54 directly, show that its transform is $(4/\sqrt{2\pi}) \sin^2(kw)/k^2$, and check that this agrees with the convolution theorem.
 - Show, by completing the square in the exponent, that $e^{-\alpha x^2} * e^{-\alpha x^2} = \sqrt{\pi/2\alpha} e^{-\alpha x^2/2}$, and obtain the same result from the convolution theorem. By what factor is the half-width of the convolution larger than that of each Gaussian?
 - Apply Parseval's theorem to the rectangular pulse to show that $\int_{-\infty}^{\infty} (\sin^2 u/u^2) du = \pi$.

§6.3 The Dirac Delta Function

- Use the rule for $\delta(g(x))$ to show that $\int_{-\infty}^{\infty} e^{-|x|} \delta(\sin x) dx = \coth(\pi/2)$.
- The derivative $\delta'(x)$ of the delta function is defined by the relation $\int f(x) \delta'(x - x_0) dx = -\int f'(x) \delta(x - x_0) dx$, for a differentiable function f , which is the result that integration by parts would give. Show that this integral equals $-f'(x_0)$, and that $x \delta'(x) = -\delta(x)$.
- Show that $(1/2\pi) \int_{-K}^K e^{ikx} dk = \sin(Kx)/\pi x$, where K is a positive constant. Let $I(\kappa) = \int_0^{\infty} e^{-\kappa u} (\sin u/u) du$ for $\kappa \geq 0$. By differentiating under the integral sign, show that $dI/d\kappa = -1/(1 + \kappa^2)$ for $\kappa > 0$, and deduce that $\int_{-\infty}^{\infty} (\sin u/u) du = \pi$. Hence show that each function $\sin(Kx)/\pi x$ encloses unit area.
- The square wave of Section 6.1 jumps from -1 to 1 at $x = 0$ and from 1 to -1 at $x = L$, and is repeated with period $2L$. Show that its derivative is $2 \sum_{m=-\infty}^{\infty} (-1)^m \delta(x - mL)$. By differentiating the square-wave series term by term, deduce that $\sum_{n \text{ odd}} \cos(n\pi x/L) = \frac{1}{2} L \sum_m (-1)^m \delta(x - mL)$, and check this by computing the cosine coefficients of the right-hand side over the period $-L/2 < x < 3L/2$.

5. The mass on a spring of Figure 58, initially at rest, receives a blow of impulse J at t_0 and a second blow of impulse J at $t_0 + \tau$. Find the smallest value of τ for which the mass is at rest after the second blow.

Partial Differential Equations and Boundary-Value Problems

7.1 The Classical Equations of Physics

A law of physics for a quantity spread through space, such as the displacement of a vibrating string or the temperature inside a body, relates the rates of change of the quantity with respect to position and time, and so takes the form of an equation that relates a function of several variables to its partial derivatives, which is called a partial differential equation. In this chapter we first derive the equations of this kind that occur most often in physics, with the conditions that fix their solutions, then solve them by separating the variables in Cartesian coordinates, and finally solve them in plane polar and spherical coordinates and on an infinite line.

Partial Differential Equations

The order of a partial differential equation is the order of the highest derivative in it. An equation is called linear if the unknown function and its derivatives appear in it only to the first power, each multiplied by a given function of the independent variables. A linear equation is called homogeneous if every term contains the unknown function or one of its derivatives. A term that is a given

function of the independent variables alone, such as the heat supplied by an electric heater buried inside a body, is called a source term, and an equation that contains one is called inhomogeneous.

All the equations of this chapter are linear and of second order. A linear homogeneous equation obeys the principle of superposition, which states that if u_1 and u_2 are solutions, then so is $\gamma_1 u_1 + \gamma_2 u_2$ for any constants γ_1 and γ_2 , because the derivative of a sum is the sum of the derivatives.

The Wave Equation for a Stretched String

Consider a string stretched along the x -axis with tension τ , and let ρ_ℓ be its mass per unit length. Let $u(x, t)$ be the displacement at time t of the point of the string whose undisturbed position is x . We assume that each point moves only perpendicular to the axis, that the weight of the string can be neglected, and that the slope $\partial u / \partial x$ is small compared with one everywhere.

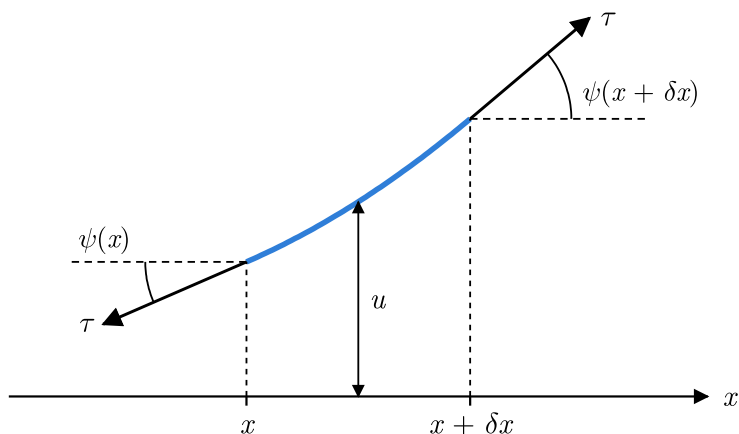


Figure 59. The element of a stretched string between x and $x + \delta x$, with its displacement u . The rest of the string pulls on each end of the element with a force of magnitude τ along the string, at the angles $\psi(x)$ and $\psi(x + \delta x)$ to the x -axis. The slopes are drawn much larger than the derivation assumes.

Figure 59 shows the element of string between x and $x + \delta x$, where $\psi(x)$ is the angle between the string and the x -axis. The components of the forces along the axis, $\tau \cos \psi$ at each end, balance, because the element does not move along the axis. As $\cos \psi$ differs from one only in the second order of the small angle, the tension has the same value τ all along the string. The resultant force in the direction of increasing u is $\tau \sin \psi(x + \delta x) - \tau \sin \psi(x)$, where for a small angle $\sin \psi$ is approximately $\tan \psi$, the slope $\partial u / \partial x$. The mass of the element is $\rho_\ell \delta x$, so Newton's second law gives

$$\rho_\ell \delta x \frac{\partial^2 u}{\partial t^2} = \tau \left[\frac{\partial u}{\partial x}(x + \delta x, t) - \frac{\partial u}{\partial x}(x, t) \right]. \quad (7.1)$$

When we divide both sides by $\rho_\ell \delta x$ and let δx tend to zero, the bracket divided by δx tends to $\partial^2 u / \partial x^2$, by the definition of the derivative, and we obtain

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 u}{\partial t^2}, \quad v = \sqrt{\frac{\tau}{\rho_\ell}}, \quad (7.2)$$

which is called the wave equation. We show below that v is the speed of waves on the string. With the Laplacian of Section 3.1 in place of $\partial^2 u / \partial x^2$, the same equation describes waves in two and three dimensions, such as sound waves in air.

The Diffusion Equation

In Section 3.3 we used Fourier's law of conduction, $\mathbf{j} = -\kappa \nabla T$, and the conservation of energy to show that the temperature $T(\mathbf{r}, t)$ in a body of density ρ , specific heat capacity c , and thermal conductivity κ satisfies the diffusion equation $\partial T / \partial t = D \nabla^2 T$, where $D = \kappa / \rho c$ is the thermal diffusivity. Suppose now that heat is also generated inside the body at a rate q per unit volume, for example in a wire that carries a current. The rate of increase of the heat in a fixed volume then gains the term $\int q dV$, and the argument of Section 3.3 gives $\rho c \partial T / \partial t = \kappa \nabla^2 T + q$, or

$$\frac{\partial T}{\partial t} = D \left(\nabla^2 T + \frac{q}{\kappa} \right), \quad (7.3)$$

in which Dq/κ is a source term.

Laplace's and Poisson's Equations

When the temperature has reached a steady state, $\partial T/\partial t = 0$, and the diffusion equation with a source becomes $\nabla^2 T = -q/\kappa$. An equation of the same form governs electrostatics. In Section 3.3 we obtained Gauss's law in the form $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$, where ρ is now the charge density, and a static field is related to the electric potential Φ by $\mathbf{E} = -\nabla\Phi$, the three-dimensional form of the school result that the field strength is minus the potential gradient. Substituting this relation into Gauss's law gives

$$\nabla^2 \Phi = -\frac{\rho}{\varepsilon_0}, \quad (7.4)$$

which is called Poisson's equation, a name also given to any equation of this form with a given function on the right-hand side. In a region that contains no charge, or for steady heat flow with no sources, the equation becomes $\nabla^2 \Phi = 0$, which is called Laplace's equation. The potential $Q/(4\pi\varepsilon_0 r)$ of a point charge Q at the origin, for example, satisfies Laplace's equation except at the origin, as we showed in Section 3.1 with the product rule and again in Section 3.4 with the spherical form of the Laplacian.

Boundary and Initial Conditions

A partial differential equation has a great many solutions; $\Phi = 1$, $\Phi = x$, and $\Phi = x^2 - y^2$ all satisfy Laplace's equation, for example. A physical problem selects one of them by conditions on the boundary of the region and, where time enters, at an initial instant. The conditions most often given on a boundary are of the following kinds:

1. a Dirichlet condition gives the value of the unknown function on the boundary, for example when a surface is held at a given temperature or a conductor at a given potential;
2. a Neumann condition gives the normal derivative $\partial T/\partial n = \hat{\mathbf{n}} \cdot \nabla T$ on the boundary, where $\hat{\mathbf{n}}$ is the outward unit normal, for example on an insulated surface, which carries no heat flux and so, by Fourier's law, has $\partial T/\partial n = 0$;
3. a mixed condition gives the value on part of the boundary and the normal derivative on the rest, for example for a bar whose

ends are held at fixed temperatures and whose sides are insulated.

For Poisson's or Laplace's equation, one of these conditions over the whole of a closed boundary fixes the solution, although a Neumann condition fixes it only up to an added constant. The diffusion equation needs, in addition, the temperature at the initial instant, and the wave equation, which contains the second derivative with respect to time, needs both the displacement and its rate of change $\partial u/\partial t$ at the initial instant. We prove the uniqueness of the solution for Poisson's equation at the end of this section, and you will prove it for the diffusion equation and for a string with fixed ends in Problems 5 and 6. On a region that extends to infinity, the boundary condition is replaced by a condition on the solution at large distances.

D'Alembert's Solution of the Wave Equation

On an infinite line the wave equation can be solved by a change of variables. Let $\xi = x - vt$ and $\eta = x + vt$, and regard u as a function of ξ and η . By the chain rule for several variables of Section 2.2,

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}, \quad \frac{\partial u}{\partial t} = -v \frac{\partial u}{\partial \xi} + v \frac{\partial u}{\partial \eta}. \quad (7.5)$$

Applying each of these relations a second time, and using the equality of the mixed second derivatives, we find

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 u}{\partial \xi^2} + 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}, \\ \frac{1}{v^2} \frac{\partial^2 u}{\partial t^2} &= \frac{\partial^2 u}{\partial \xi^2} - 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}. \end{aligned} \quad (7.6)$$

Subtracting the second of these equations from the first gives $4 \partial^2 u / \partial \xi \partial \eta$, so the wave equation is equivalent to $\partial^2 u / \partial \xi \partial \eta = 0$. Integrating this equation with respect to η shows that $\partial u / \partial \xi$ is a function of ξ alone, and integrating again with respect to ξ , with a constant of integration that may depend on η , gives

$$u(x, t) = f(x - vt) + g(x + vt), \quad (7.7)$$

where f and g are any functions that can be differentiated twice. The value of $f(x - vt)$ at the position x and time t is its value

at the position $x - vt$ at time zero, so the profile of f moves in the positive x -direction at speed v without changing its shape, and a disturbance of this kind is called a travelling wave. The term $g(x + vt)$ is a travelling wave moving in the negative x -direction.

Let the string have the displacement $u_0(x)$ and the velocity $\mathcal{U}(x)$ at $t = 0$, so that $f + g = u_0$ and $-vf' + vg' = \mathcal{U}$, where the primes denote derivatives. Integrating the second condition from 0 to x gives $g(x) - f(x) = v^{-1} \int_0^x \mathcal{U}(x') dx' + g(0) - f(0)$. Adding this equation to the first condition gives $2g$, subtracting it gives $2f$, and substituting the resulting f and g into the general solution gives

$$u(x, t) = \frac{1}{2} [u_0(x - vt) + u_0(x + vt)] + \frac{1}{2v} \int_{x-vt}^{x+vt} \mathcal{U}(x') dx', \quad (7.8)$$

in which the constants $g(0)$ and $f(0)$ have cancelled. This is called d'Alembert's solution.

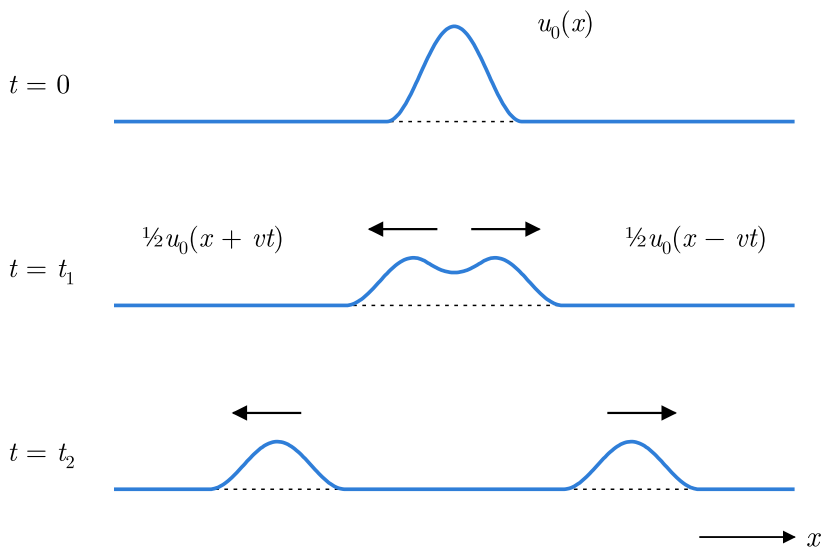


Figure 60. A string released from rest with the displacement $u_0(x)$. The initial pulse divides into two pulses of half its height, $\frac{1}{2}u_0(x + vt)$ and $\frac{1}{2}u_0(x - vt)$, which travel apart at speed v without changing their shapes. The times satisfy $0 < t_1 < t_2$.

As an example, consider a string released from rest, so that $\mathcal{U} = 0$,

with a single pulse of displacement $u_0(x)$. D'Alembert's solution is then half the pulse travelling to the right plus half travelling to the left, and, as Figure 60 shows, the two halves overlap at first and then separate.

Reflection at a Fixed End

Consider next a string that occupies $x > 0$ and is fixed at $x = 0$. The general solution holds for $x > 0$, and $u(0, t) = 0$ requires

$$f(-vt) = -g(vt) \quad \text{for all } t \geq 0, \quad (7.9)$$

so that the values of f for negative arguments are fixed by the values of g for positive arguments. Suppose that at $t = 0$ the only disturbance is a pulse $g(x + vt)$ travelling towards the fixed end, so that g is zero except near some point $x = x_c > 0$, and f is zero for positive arguments. At later times the argument $x - vt$ of f can be negative, and applying the condition at the fixed end with vt replaced by the positive quantity $vt - x$ gives $f(x - vt) = -g(vt - x)$, so that

$$u(x, t) = g(x + vt) - g(vt - x). \quad (7.10)$$

The second term, which vanishes at $t = 0$ for $x > 0$, is the incoming pulse reversed from left to right, turned upside down, and travelling in the positive x -direction. Figure 61 pictures it as an image pulse on a continuation of the string into $x < 0$, whose displacement cancels that of the real pulse at $x = 0$. After the real pulse reaches the end, the image pulse emerges onto the string as the reflected pulse, which is inverted, as you may have seen at school.

Uniqueness of the Solution of Poisson's Equation

We now prove that Poisson's equation with a Dirichlet condition has only one solution, using the following objects: a region $\mathcal{R}$ bounded by a closed surface S with outward unit normal $\hat{\mathbf{n}}$ and surface element $d\mathbf{S} = \hat{\mathbf{n}} dS$; a given charge density ρ in $\mathcal{R}$; and a given potential Φ_s on S . Suppose that Φ_1 and Φ_2 both satisfy $\nabla^2 \Phi = -\rho/\epsilon_0$ in $\mathcal{R}$ and $\Phi = \Phi_s$ on S , and let $w = \Phi_1 - \Phi_2$. Subtracting the equation for Φ_2 from the equation for Φ_1 gives $\nabla^2 w = 0$ in $\mathcal{R}$, and subtracting the boundary conditions gives $w = 0$ on S .

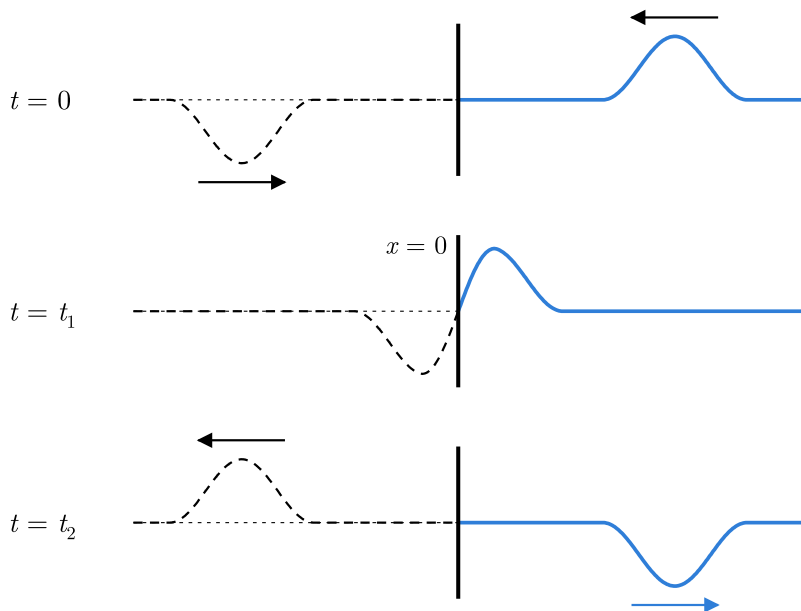


Figure 61. Reflection of a pulse at a fixed end. The string (solid) occupies $x > 0$ and is fixed at $x = 0$. The dashed curves in $x < 0$ are drawn on a continuation of the string beyond the end. At $t = 0$ the dashed curve is the inverted image pulse $-g(vt - x)$, which travels towards the end as the real pulse does, and at t_1 the real and image pulses overlap and the displacement at the end is zero. At t_2 the image pulse has emerged onto the string as the reflected pulse, which is inverted and travels away from the end, and the dashed curve is the incoming pulse continued beyond the end.

In Section 3.1 we proved the product rule $\nabla \cdot (w\mathbf{F}) = w \nabla \cdot \mathbf{F} + \mathbf{F} \cdot \nabla w$ for a scalar field w and a vector field $\mathbf{F}$. Taking $\mathbf{F} = \nabla w$ and using $\nabla^2 w = 0$ gives $\nabla \cdot (w \nabla w) = |\nabla w|^2$. We integrate both sides over $\mathcal{R}$ and apply the divergence theorem of Section 3.3 to the left-hand side, which gives

$$\int_{\mathcal{R}} |\nabla w|^2 dV = \oint_S w \nabla w \cdot d\mathbf{S} = 0, \quad (7.11)$$

since $w = 0$ on S . The integrand on the left is continuous and never negative, so $\nabla w = \mathbf{0}$ throughout $\mathcal{R}$. The difference w is therefore constant, and since it is zero on S , the constant is zero, and $\Phi_1 = \Phi_2$.

Since $\nabla w \cdot d\mathbf{S} = (\partial w / \partial n) dS$, the surface integral also vanishes for a Neumann or mixed condition, although with a Neumann condition w is only shown to be constant. By uniqueness, a solution that satisfies the boundary conditions is the solution, however it was found.

7.2 Separation of Variables in Cartesian Coordinates

Since the equations of Section 7.1 are linear, a complicated solution can be built by superposition from simple ones. The simple solutions are found by a method called separation of variables, which we apply first to a vibrating string, then to a cooling bar, and finally to the steady temperature in a rectangular plate. In each problem the simple solutions are combined into a Fourier series of the kind studied in Section 6.1, whose coefficients are chosen to satisfy the initial or boundary conditions.

Separating the Variables

Consider a string of length L whose ends, at $x = 0$ and $x = L$, are fixed, so that $u(0, t) = u(L, t) = 0$ for all t . Let us look for solutions of the wave equation that are a product of a function of x alone and a function of t alone,

$$u(x, t) = X(x) \Gamma(t). \quad (7.12)$$

Substituting the product into the wave equation and dividing both

sides by $X\Gamma$ gives

$$\frac{1}{X} \frac{d^2 X}{dx^2} = \frac{1}{v^2 \Gamma} \frac{d^2 \Gamma}{dt^2}. \quad (7.13)$$

The left-hand side depends on x alone and the right-hand side on t alone. If we change t while keeping x fixed, the left-hand side does not change, so the right-hand side cannot change either, and a similar argument holds when x is changed. The left-hand and right-hand sides must therefore be equal to the same constant, which is called the separation constant. We write it as $-k^2$, for a reason given below, and the partial differential equation is replaced by the two ordinary differential equations

$$\frac{d^2 X}{dx^2} = -k^2 X, \quad \frac{d^2 \Gamma}{dt^2} = -v^2 k^2 \Gamma. \quad (7.14)$$

The boundary conditions require $X(0)\Gamma(t) = 0$ for all t , and since Γ is not zero at every instant, $X(0) = 0$; in the same way $X(L) = 0$.

These conditions fix the sign of the separation constant. Had we written the constant as $+k^2$, with $k > 0$, the general solution for X would be $\gamma_1 e^{kx} + \gamma_2 e^{-kx}$, by the methods of Section 5.2; the condition at $x = 0$ gives $X = 2\gamma_1 \sinh kx$, and the condition at $x = L$ then requires $\gamma_1 = 0$, since $\sinh kL$ is not zero. A zero constant gives $X = \gamma_1 + \gamma_2 x$, and the two end conditions again require $\gamma_1 = \gamma_2 = 0$. With a negative constant, $-k^2$ with $k > 0$, the general solution is $X = \gamma_1 \cos kx + \gamma_2 \sin kx$. The condition $X(0) = 0$ gives $\gamma_1 = 0$, and the condition $X(L) = 0$ gives $\gamma_2 \sin kL = 0$. Taking $\gamma_2 = 0$ would leave no solution, so $\sin kL = 0$, and

$$k = k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots \quad (7.15)$$

Negative values of n add nothing new, since $\sin(-n\pi x/L) = -\sin(n\pi x/L)$ is a multiple of the solution for n . A problem of this kind, in which a differential equation with boundary conditions has solutions other than zero only for particular values of a constant, is called an eigenvalue problem, by analogy with the matrix equation $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ of Section 4.3. Taking the second derivative plays the part of multiplying by the matrix $\mathbf{A}$, since applied to the function $X_n(x) = \sin(n\pi x/L)$ it returns the multiple $-k_n^2 X_n$ of the same function, as $\mathbf{A}$ applied to $\mathbf{x}$ returns $\lambda\mathbf{x}$. The function X_n , which

is called an eigenfunction, plays the part of the eigenvector $\mathbf{x}$, and $-k_n^2$ the part of the eigenvalue λ .

For each k_n the equation for Γ is the equation of simple harmonic motion with angular frequency $\omega_n = vk_n$, so the separated solutions are

$$u_n(x, t) = \sin \frac{n\pi x}{L} (A_n \cos \omega_n t + B_n \sin \omega_n t), \quad \omega_n = \frac{n\pi v}{L}, \quad (7.16)$$

where A_n and B_n are constants.

The Vibrating String

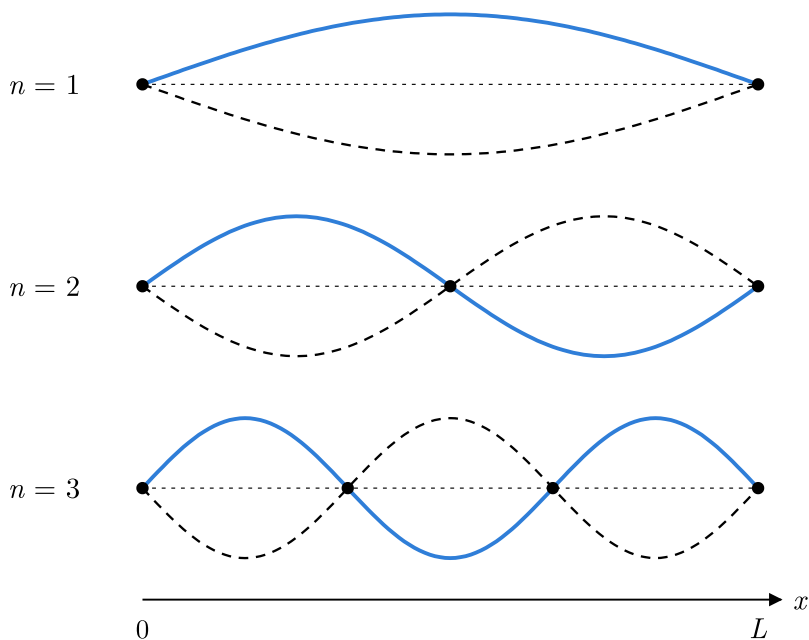


Figure 62. The first three normal modes of a string fixed at $x = 0$ and $x = L$. The solid and dashed curves show the string at the two extremes of its oscillation, half a period apart, and the dots mark the nodes.

Each separated solution u_n is a normal mode of the string in the sense of Section 4.3, since in the n th mode every point moves in simple harmonic motion with the same angular frequency ω_n . The amplitude of the motion varies along the string as $\sin(n\pi x/L)$, and

the points where this amplitude is zero stay at rest and are called nodes. The first three modes are drawn in Figure 62. The n th mode fits n half-wavelengths into the length of the string, which is the condition for a standing wave on a string that you know from school, and its frequency is n times the frequency $\omega_1/2\pi = v/2L$ of the lowest mode, which is called the fundamental. In Section 4.3 two masses coupled by springs had two normal modes; a string has infinitely many.

By the principle of superposition, the general motion of the string is a sum of normal modes,

$$u(x, t) = \sum_{n=1}^{\infty} \sin \frac{n\pi x}{L} (A_n \cos \omega_n t + B_n \sin \omega_n t). \quad (7.17)$$

The constants are fixed by the initial displacement $u_0(x)$ and the initial velocity $\mathcal{U}(x)$. Setting $t = 0$ in this sum and in its derivative with respect to t gives

$$u_0(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{L}, \quad \mathcal{U}(x) = \sum_{n=1}^{\infty} \omega_n B_n \sin \frac{n\pi x}{L}, \quad (7.18)$$

which are the half-range sine series of u_0 and $\mathcal{U}$ on $0 < x < L$ from Section 6.1. Such a series describes the function extended as an odd function and repeated with period $2L$, which is the extension that made the displacement vanish at a fixed end in Section 7.1, applied here at both ends. The orthogonality of the sines, from Section 6.1, gives the coefficients as

$$\begin{aligned} A_n &= \frac{2}{L} \int_0^L u_0(x) \sin \frac{n\pi x}{L} dx, & B_n \\ &= \frac{2}{\omega_n L} \int_0^L \mathcal{U}(x) \sin \frac{n\pi x}{L} dx. \end{aligned} \quad (7.19)$$

As an example, consider a string pulled aside at its midpoint to the displacement u_p and released from rest. The initial displacement is $u_0 = 2u_p x/L$ for $0 \leq x \leq L/2$ and $u_0 = 2u_p(L-x)/L$ for $L/2 \leq x \leq L$, and since $\mathcal{U} = 0$, every B_n is zero. The initial displacement is symmetric about the midpoint, and the formula for the sine of a difference gives $\sin[n\pi(L-x)/L] = -(-1)^n \sin(n\pi x/L)$, so the mode function is symmetric about the midpoint when n is odd and antisymmetric when n is even. For even n the contributions

of the two halves of the string to A_n therefore cancel, and for odd n they are equal, so that

$$A_n = \frac{4}{L} \int_0^{L/2} \frac{2u_p x}{L} \sin \frac{n\pi x}{L} dx, \quad n \text{ odd}. \quad (7.20)$$

Integrating by parts, with the sine integrated and x differentiated, gives

$$\begin{aligned} \int_0^{L/2} x \sin \frac{n\pi x}{L} dx &= \left[-\frac{Lx}{n\pi} \cos \frac{n\pi x}{L} \right]_0^{L/2} + \frac{L}{n\pi} \int_0^{L/2} \cos \frac{n\pi x}{L} dx \\ &= -\frac{L^2}{2n\pi} \cos \frac{n\pi}{2} + \frac{L^2}{n^2\pi^2} \sin \frac{n\pi}{2}, \end{aligned} \quad (7.21)$$

in which the first term is zero for odd n . Therefore $A_n = (8u_p/n^2\pi^2) \sin(n\pi/2)$ for odd n , and the motion of the string is

$$\begin{aligned} u(x, t) &= \frac{8u_p}{\pi^2} \left[\sin \frac{\pi x}{L} \cos \omega_1 t - \frac{1}{9} \sin \frac{3\pi x}{L} \cos \omega_3 t \right. \\ &\quad \left. + \frac{1}{25} \sin \frac{5\pi x}{L} \cos \omega_5 t - \dots \right]. \end{aligned} \quad (7.22)$$

The even modes are absent because each has a node at the midpoint, where the string was plucked, and the amplitudes of the odd modes go as $1/n^2$. As a check, at the midpoint at $t = 0$ every term in the bracket is positive, and the series gives $(8u_p/\pi^2) \sum_{n \text{ odd}} 1/n^2$, which is u_p , since in Section 6.1 we found from the series of a triangle wave that the sum is $\pi^2/8$.

The Cooling of a Bar

Consider a bar of length L with insulated sides, so that its temperature $T(x, t)$ satisfies $\partial T/\partial t = D \partial^2 T/\partial x^2$. The end $x = 0$ is held at the temperature T_1 , the end $x = L$ at T_2 , and the initial temperature is $T_i(x)$.

A separated solution $X(x)\Gamma(t)$ takes the value $X(0)\Gamma(t)$ at the end $x = 0$, which changes in time unless Γ is constant, so we first remove the non-zero boundary values. After a long time the bar reaches a steady state $T_\infty(x)$, which satisfies $d^2T_\infty/dx^2 = 0$ and

the boundary conditions, so that

$$T_{\infty}(x) = T_1 + (T_2 - T_1) \frac{x}{L}. \quad (7.23)$$

Let $U = T - T_{\infty}$. The steady state satisfies the diffusion equation, since both sides of it are zero, and the equation is linear, so U satisfies it as well, with the boundary conditions $U(0, t) = U(L, t) = 0$ and the initial condition $U(x, 0) = T_i(x) - T_{\infty}(x)$.

Substituting $U = X(x) \Gamma(t)$ into the diffusion equation and dividing by $DX\Gamma$ gives

$$\frac{1}{X} \frac{d^2 X}{dx^2} = \frac{1}{D\Gamma} \frac{d\Gamma}{dt} = -k^2, \quad (7.24)$$

where the equation for X and its boundary conditions are those of the string, so that $X = \sin(n\pi x/L)$ with $k = k_n = n\pi/L$. The equation for Γ describes exponential decay, with the solution $\Gamma = e^{-Dk_n^2 t}$, and adding the separated solutions gives

$$T(x, t) = T_{\infty}(x) + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} e^{-n^2 \pi^2 D t / L^2}, \quad (7.25)$$

and setting $t = 0$ shows that the b_n are the half-range sine coefficients of $T_i - T_{\infty}$,

$$b_n = \frac{2}{L} \int_0^L [T_i(x) - T_{\infty}(x)] \sin \frac{n\pi x}{L} dx. \quad (7.26)$$

Each mode decays with the time constant $L^2/(n^2\pi^2 D)$, so the higher modes, which carry the fine detail of the initial temperature, die away fastest, and after a time long compared with $L^2/(\pi^2 D)$ the temperature approaches the steady state with the shape of $\sin(\pi x/L)$. The time the bar takes to reach its steady state goes as L^2/D , so a bar twice as long takes four times as long.

Consider, for example, a bar at the uniform temperature T_0 whose ends are brought at $t = 0$ to the temperature T_1 and held there. The steady state is $T_{\infty} = T_1$, and the coefficients are

$$b_n = \frac{2(T_0 - T_1)}{L} \int_0^L \sin \frac{n\pi x}{L} dx = \frac{2(T_0 - T_1)}{n\pi} (1 - \cos n\pi), \quad (7.27)$$

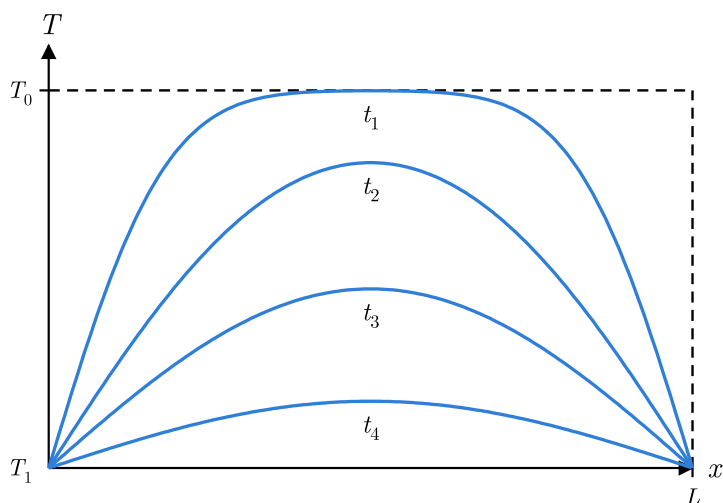


Figure 63. The temperature of a bar initially at T_0 whose ends are held at T_1 from $t = 0$, computed from the series with terms up to $n = 401$. The dashed lines show the initial temperature, which is T_0 inside the bar and jumps to T_1 at the ends, and the curves are drawn at the times t_1 , t_2 , t_3 , and t_4 , equal to 0.01, 0.045, 0.1, and 0.2 times L^2/D .

which is $4(T_0 - T_1)/(n\pi)$ for odd n and zero for even n . The temperature is therefore

$$T(x, t) = T_1 + \frac{4(T_0 - T_1)}{\pi} \sum_{n \text{ odd}} \frac{1}{n} \sin \frac{n\pi x}{L} e^{-n^2 \pi^2 D t / L^2}, \quad (7.28)$$

and Figure 63 shows it at several times. At $t = 0$ the sum is the square-wave series of Section 6.1, which converges to $T_0 - T_1$ inside the bar and, at the ends, where the odd extension jumps, to the mean of the one-sided limits, which is zero. At every positive time the exponential factors make the terms fall off rapidly with n , and the temperature is a smooth function of x . By the time t_4 the profile is close to a single sine curve.

Laplace's Equation in a Rectangle

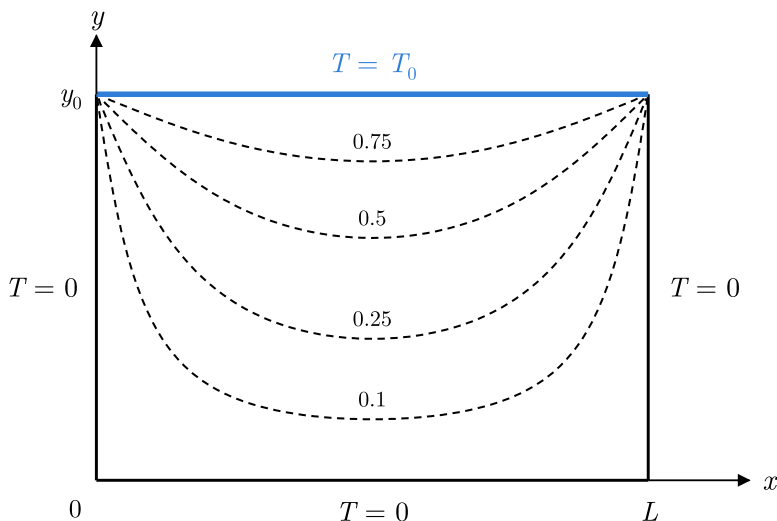


Figure 64. A rectangular plate $0 < x < L$, $0 < y < y_0$, with the edge $y = y_0$ held at the temperature T_0 and the other three edges at zero. The dashed curves are the isotherms $T/T_0 = 0.1, 0.25, 0.5$, and 0.75 , computed from the series for a plate with $y_0 = 0.7L$. All the isotherms meet at the two upper corners, where the temperature on the boundary jumps.

Consider a thin rectangular plate, $0 < x < L$, $0 < y < y_0$, with

insulated faces, whose steady temperature $T(x, y)$ satisfies

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0. \quad (7.29)$$

The edges $x = 0$, $x = L$, and $y = 0$ are held at a common temperature, from which we measure T , so that $T = 0$ on them, and the fourth edge, $y = y_0$, is held at the temperature $T_e(x)$. Figure 64 shows the plate for the case, treated below, in which the edge temperature is a constant T_0 .

Substituting $T = X(x)Y(y)$ into Laplace's equation and dividing by XY gives

$$\frac{1}{X} \frac{d^2 X}{dx^2} = -\frac{1}{Y} \frac{d^2 Y}{dy^2} = -k^2. \quad (7.30)$$

The function X must vanish at $x = 0$ and at $x = L$, as for the string, so the separation constant is negative and $X = \sin(n\pi x/L)$ with $k = k_n = n\pi/L$. The equation for Y is then $Y'' = k_n^2 Y$, whose general solution can be written with the hyperbolic functions of Section 1.2 as $Y = \gamma_1 \cosh k_n y + \gamma_2 \sinh k_n y$, and the condition $Y(0) = 0$ gives $\gamma_1 = 0$. Adding the separated solutions $\sin(n\pi x/L) \sinh(n\pi y/L)$ with coefficients chosen so that the sum is $T_e(x)$ on the edge $y = y_0$ gives

$$\begin{aligned} T(x, y) &= \sum_{n=1}^{\infty} b_n \frac{\sinh(n\pi y/L)}{\sinh(n\pi y_0/L)} \sin \frac{n\pi x}{L}, \quad b_n \\ &= \frac{2}{L} \int_0^L T_e(x) \sin \frac{n\pi x}{L} dx, \end{aligned} \quad (7.31)$$

since at $y = y_0$ the ratio of hyperbolic sines is one, and the sum becomes the half-range sine series of T_e .

Let us consider the case of an edge held at the uniform temperature T_0 . The coefficients are those of the bar in the previous subsection with $T_0 - T_1$ replaced by T_0 , namely $b_n = 4T_0/(n\pi)$ for odd n and zero for even n , and the isotherms computed from the resulting series are drawn in Figure 64.

When $n\pi y/L$ is large compared with one, each hyperbolic sine is approximately half of a growing exponential, and their ratio is approximately $e^{-n\pi(y_0-y)/L}$. The n th term therefore decays over a

distance $L/(n\pi)$ from the hot edge, so variations of the edge temperature on a short scale are felt only close to that edge.

When all four edges are held at given temperatures, the problem can be divided into four problems, in each of which one edge has its given temperature and the other three are at zero. By superposition the sum of their solutions satisfies Laplace's equation and takes the given value on every edge, and by the uniqueness theorem of Section 7.1, whose proof holds in two dimensions by the same argument, this sum is the steady temperature.

7.3 Separation of Variables in Polar and Spherical Coordinates

When the boundary of a region is a circle or a sphere, the variables are separated in polar or spherical coordinates. Using the Laplacians of Section 3.4, we solve Laplace's equation first in a disc, and then inside and outside a sphere for problems with axial symmetry. Finally we solve the diffusion equation on an infinite line, where the Fourier series is replaced by the Fourier transform of Section 6.2.

Laplace's Equation in Plane Polar Coordinates

Consider a thin disc of radius s_0 with insulated faces, whose rim is held at a temperature $T_e(\phi)$. The steady temperature $T(s, \phi)$ satisfies Laplace's equation, which in plane polar coordinates, from the cylindrical Laplacian of Section 3.4 with no dependence on z , is

$$\frac{1}{s} \frac{\partial}{\partial s} \left(s \frac{\partial T}{\partial s} \right) + \frac{1}{s^2} \frac{\partial^2 T}{\partial \phi^2} = 0. \quad (7.32)$$

Let us look for separated solutions $T = R(s) \Psi(\phi)$. Substituting this product and multiplying by $s^2/(R\Psi)$ gives

$$\frac{s}{R} \frac{d}{ds} \left(s \frac{dR}{ds} \right) = - \frac{1}{\Psi} \frac{d^2 \Psi}{d\phi^2}, \quad (7.33)$$

and, as in Section 7.2, each side must be equal to a constant, which we write as m^2 , so that $\Psi'' = -m^2 \Psi$. The angles ϕ and $\phi + 2\pi$ describe the same point of the disc, so $\Psi(\phi + 2\pi) = \Psi(\phi)$. The solutions $\cos m\phi$ and $\sin m\phi$ have this property only when m is an integer, which we may take to be $0, 1, 2, \dots$, and for $m = 0$ the solution $\Psi = \gamma_1 + \gamma_2 \phi$ is periodic only if $\gamma_2 = 0$. A negative

separation constant would give exponentials in ϕ , which are not periodic.

The radial equation is

$$s^2 \frac{d^2 R}{ds^2} + s \frac{dR}{ds} - m^2 R = 0, \quad (7.34)$$

which is an Euler equation of the kind solved in Section 5.2. Trying $R = s^p$ gives $p(p-1) + p - m^2 = 0$, so $p^2 = m^2$, and for $m \geq 1$ the solutions are s^m and s^{-m} . For $m = 0$ the equation reads $d(s dR/ds)/ds = 0$, so $s dR/ds$ is a constant, and integrating again gives $R = \gamma_1 + \gamma_2 \ln s$.

The temperature must be finite at the centre, where s^{-m} and $\ln s$ are infinite, so adding the remaining separated solutions gives

$$T(s, \phi) = \frac{1}{2}a_0 + \sum_{m=1}^{\infty} \left(\frac{s}{s_0} \right)^m (a_m \cos m\phi + b_m \sin m\phi), \quad (7.35)$$

where the factor s_0^{-m} has been included so that at the rim the sum becomes the Fourier series of $T_e(\phi)$, whose coefficients, from Section 6.1 with $L = \pi$, are

$$a_m = \frac{1}{\pi} \int_{-\pi}^{\pi} T_e(\phi) \cos m\phi d\phi, \quad b_m = \frac{1}{\pi} \int_{-\pi}^{\pi} T_e(\phi) \sin m\phi d\phi. \quad (7.36)$$

The temperature at the centre is $\frac{1}{2}a_0$, the mean temperature around the rim, and since the terms in $\cos m\phi$ and $\sin m\phi$ are weighted by $(s/s_0)^m$, rapid variations of the rim temperature are smoothed out towards the centre.

Consider, for instance, a disc whose rim is held at the temperature T_0 for $0 < \phi < \pi$ and at zero on the lower half, as in Figure 65. Then $a_0 = T_0$, and, for $m \geq 1$,

$$\begin{aligned} a_m &= \frac{T_0}{\pi} \int_0^{\pi} \cos m\phi d\phi \\ &= 0, \quad b_m \\ &= \frac{T_0}{\pi} \int_0^{\pi} \sin m\phi d\phi \\ &= \frac{T_0}{m\pi} (1 - \cos m\pi), \end{aligned} \quad (7.37)$$

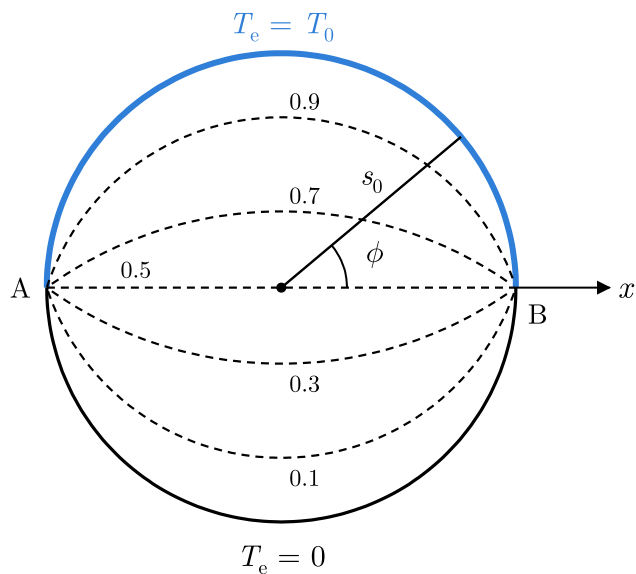


Figure 65. A disc of radius s_0 whose rim is held at T_0 for $0 < \phi < \pi$ and at zero for $-\pi < \phi < 0$. The dashed curves are the isotherms $T/T_0 = 0.1, 0.3, 0.5, 0.7$, and 0.9 , which are arcs of circles through the two points A and B where the rim temperature jumps. The isotherm $T = T_0/2$ is the diameter along the x -axis.

so that $b_m = 2T_0/(m\pi)$ for odd m and zero for even m . The steady temperature is therefore

$$T(s, \phi) = \frac{T_0}{2} + \frac{2T_0}{\pi} \sum_{m \text{ odd}} \frac{1}{m} \left(\frac{s}{s_0} \right)^m \sin m\phi. \quad (7.38)$$

On the diameter along the x -axis every sine is zero, and the temperature is $T_0/2$. The isotherms are drawn in Figure 65, and you will show in Problem 3 that they are arcs of circles through the two points where the rim temperature jumps.

Laplace's Equation in Spherical Coordinates

Consider next a potential $\Phi(r, \theta)$ with axial symmetry about the z -axis, such as the potential of a sphere of radius r_0 whose surface is held at a potential $\Phi_s(\theta)$, shown in Figure 66, which could, for example, be made of thin insulated rings held at different potentials. For a function independent of ϕ , the spherical Laplacian of Section 3.4 gives Laplace's equation as

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Phi}{\partial \theta} \right) = 0. \quad (7.39)$$

Substituting $\Phi = R(r) \Theta(\theta)$ and multiplying by $r^2/(R\Theta)$ gives

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = -\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = \lambda, \quad (7.40)$$

where λ is the separation constant. The radial equation, $r^2 R'' + 2rR' - \lambda R = 0$, is again an Euler equation, and trying $R = r^p$ gives

$$p(p+1) = \lambda. \quad (7.41)$$

The angular equation is simpler in terms of $\mu = \cos \theta$, which runs from 1 to -1 as θ runs from 0 to π . By the chain rule, $d\Theta/d\theta = -\sin \theta d\Theta/d\mu$, so that, using $\sin^2 \theta = 1 - \mu^2$, we have $\sin \theta d\Theta/d\theta = -(1 - \mu^2) d\Theta/d\mu$, and $(\sin \theta)^{-1} d/d\theta = -d/d\mu$. The angular equation becomes

$$\frac{d}{d\mu} \left[(1 - \mu^2) \frac{d\Theta}{d\mu} \right] + \lambda \Theta = 0, \quad (7.42)$$

which is called Legendre's equation. Carrying out the differentiation by the product rule, Legendre's equation can be rewritten as $(1 - \mu^2)\Theta'' - 2\mu\Theta' + \lambda\Theta = 0$.

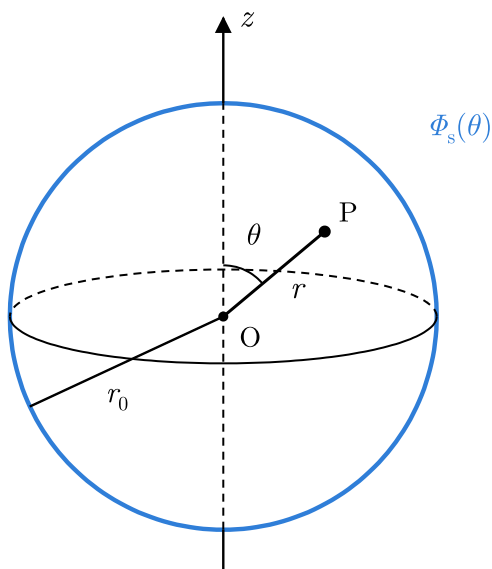


Figure 66. A sphere of radius r_0 centred on the origin O , whose surface is held at a potential $\Phi_s(\theta)$. A point P has the distance r from O and the polar angle θ , measured from the positive z -axis.

The Legendre Polynomials

A potential that is finite everywhere requires Θ to be finite at both poles, $\mu = \pm 1$, where the coefficient $1 - \mu^2$ of the highest derivative in Legendre's equation vanishes. We therefore try solutions that are polynomials in μ , which are finite everywhere. Suppose that Θ is a polynomial of degree l whose leading term is μ^l . Substituting it into the rewritten form of Legendre's equation, the terms of degree l on the left-hand side are $-l(l-1)\mu^l$ from $-\mu^2\Theta''$, $-2l\mu^l$ from $-2\mu\Theta'$, and $\lambda\mu^l$ from $\lambda\Theta$, while the term Θ'' contributes only powers below l . These terms of degree l must cancel, which requires

$$\lambda = l(l+1). \quad (7.43)$$

We can now find the polynomials of low degree by trial. For $l = 0$, $\lambda = 0$ and $\Theta = 1$ is a solution. For $l = 1$, $\lambda = 2$, and $\Theta = \mu$ gives $-2\mu + 2\mu = 0$. For $l = 2$, $\lambda = 6$, and we try $\Theta = \mu^2 + \gamma_1\mu + \gamma_0$ with constants to be determined. Substituting gives

$$(1-\mu^2) \cdot 2 - 2\mu(2\mu + \gamma_1) + 6(\mu^2 + \gamma_1\mu + \gamma_0) = (2+6\gamma_0) + 4\gamma_1\mu, \quad (7.44)$$

in which the terms in μ^2 have cancelled, and the remaining terms vanish for every μ only if $\gamma_1 = 0$ and $\gamma_0 = -\frac{1}{3}$. For $l = 3$, $\lambda = 12$, and a similar calculation with $\Theta = \mu^3 + \gamma_2\mu^2 + \gamma_1\mu + \gamma_0$, which you will carry out in Problem 4, gives $\gamma_2 = \gamma_0 = 0$ and $\gamma_1 = -\frac{3}{5}$. Each solution is usually multiplied by the constant that makes it equal to one at $\mu = 1$, and the resulting polynomials are called the Legendre polynomials $P_l(\mu)$:

$$P_0 = 1, \quad P_1 = \mu, \quad P_2 = \frac{1}{2}(3\mu^2 - 1), \quad P_3 = \frac{1}{2}(5\mu^3 - 3\mu), \quad (7.45)$$

and so on.

The polynomials are drawn in Figure 67, and a polynomial solution of degree l exists for every non-negative integer l . For any other value of λ , none of the solutions of Legendre's equation is finite at both poles, and for $\lambda = l(l+1)$ the second solution of the equation, which is independent of P_l , is infinite at the poles as well, so the Legendre polynomials are the only solutions that can describe a potential that is finite everywhere on the sphere. Any reasonable function of θ on $0 \leq \theta \leq \pi$ can also be expanded as a series of Legendre polynomials in $\cos\theta$, in the same way that a function on an interval can be expanded as a Fourier series. The proofs of

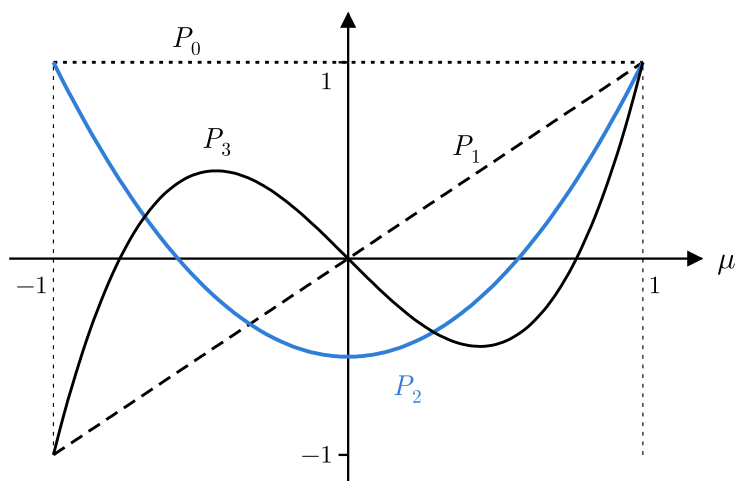


Figure 67. The Legendre polynomials P_0 to P_3 on $-1 \leq \mu \leq 1$. Each polynomial is equal to one at $\mu = 1$, the polynomial P_l has l zeros between -1 and 1 , and P_l is even or odd in μ according as l is even or odd.

these statements belong to the general theory of Legendre functions, which is not treated in this course.

The Legendre polynomials are orthogonal on $-1 \leq \mu \leq 1$, which means that

$$\int_{-1}^1 P_l(\mu) P_{l'}(\mu) d\mu = 0, \quad l' \neq l. \quad (7.46)$$

For the polynomials with $l \leq 2$ this can be checked directly. The products $P_0 P_1 = \mu$ and $P_1 P_2$ are odd functions, so their integrals over the symmetric interval vanish, and $\int_{-1}^1 P_0 P_2 d\mu = \frac{1}{2} \int_{-1}^1 (3\mu^2 - 1) d\mu = \frac{1}{2}(2 - 2) = 0$. The general result follows from Legendre's equation. Let P_l and $P_{l'}$ be two polynomial solutions, with $l' \neq l$, so that

$$\begin{aligned} \frac{d}{d\mu} [(1 - \mu^2)P_l'] &= -l(l+1)P_l, & \frac{d}{d\mu} [(1 - \mu^2)P_{l'}'] \\ &= -l'(l'+1)P_{l'}. \end{aligned} \quad (7.47)$$

We multiply the first equation by $P_{l'}$ and the second by P_l , subtract

the second from the first, and integrate from -1 to 1 . On the left-hand side we integrate each term by parts, differentiating $P_{l'}$ in the first term and P_l in the second. The boundary terms contain the factor $1 - \mu^2$, which vanishes at both limits, and the remaining integrals are both equal to $\int_{-1}^1 (1 - \mu^2) P_l' P_{l'}' d\mu$, so they cancel. This leaves

$$[l'(l' + 1) - l(l + 1)] \int_{-1}^1 P_l P_{l'} d\mu = 0. \quad (7.48)$$

Since $l(l+1)$ increases with l , the bracket is not zero when $l' \neq l$, and the integral must vanish. Direct integration gives $\int_{-1}^1 P_l^2 d\mu = 2, \frac{2}{3},$ and $\frac{2}{5}$ for $l = 0, 1,$ and 2 . These are the values of $2/(2l+1)$, which is the value of the integral for every l , although the proof of the general formula is beyond the scope of this course. In terms of θ , with $d\mu = -\sin \theta d\theta$, orthogonality states that $\int_0^\pi P_l(\cos \theta) P_{l'}(\cos \theta) \sin \theta d\theta = 0$ for $l' \neq l$.

The Potential of a Sphere

With $\lambda = l(l+1)$, the condition $p(p+1) = \lambda$ has the roots $p = l$ and $p = -(l+1)$. Inside the sphere the potential must be finite at the centre, so only r^l is kept, and

$$\Phi(r, \theta) = \sum_{l=0}^{\infty} \alpha_l \left(\frac{r}{r_0} \right)^l P_l(\cos \theta), \quad r \leq r_0. \quad (7.49)$$

Setting $r = r_0$, multiplying by $P_{l'}(\cos \theta) \sin \theta$, and integrating over θ leaves, by orthogonality, only the term with $l = l'$, so that

$$\alpha_l = \frac{\int_0^\pi \Phi_s(\theta) P_l(\cos \theta) \sin \theta d\theta}{\int_0^\pi [P_l(\cos \theta)]^2 \sin \theta d\theta}. \quad (7.50)$$

Outside the sphere the potential must tend to zero at large distances, so only $r^{-(l+1)}$ is kept, and the potential is $\sum_l \beta_l (r_0/r)^{l+1} P_l(\cos \theta)$, with $\beta_l = \alpha_l$ because both sums equal Φ_s at $r = r_0$.

As an illustration, consider a sphere whose surface is held at the potential $\Phi_s = \Phi_0 \cos^2 \theta$, where Φ_0 is a constant. Since $\mu^2 = \frac{1}{3}(2P_2 + 1)$, the surface potential is already a finite sum of

Legendre polynomials,

$$\Phi_s = \frac{\Phi_0}{3} P_0(\cos \theta) + \frac{2\Phi_0}{3} P_2(\cos \theta), \quad (7.51)$$

so $\alpha_0 = \Phi_0/3$, $\alpha_2 = 2\Phi_0/3$, and every other coefficient is zero, with no need for the integrals or for any result about infinite series. The potential inside the sphere is

$$\Phi = \frac{\Phi_0}{3} + \frac{\Phi_0 r^2}{3r_0^2} (3 \cos^2 \theta - 1), \quad r \leq r_0, \quad (7.52)$$

and outside it is

$$\Phi = \frac{\Phi_0 r_0}{3r} + \frac{\Phi_0 r_0^3}{3r^3} (3 \cos^2 \theta - 1), \quad r \geq r_0. \quad (7.53)$$

By the uniqueness theorem of Section 7.1, these functions are the potential. The potential at the centre, $\Phi_0/3$, is the mean value of $\Phi_0 \cos^2 \theta$ over the surface, as the temperature at the centre of the disc was the mean around its rim, and far from the sphere the outside potential is that of a point charge Q with $Q/(4\pi\epsilon_0) = \Phi_0 r_0/3$.

The Diffusion Equation on an Infinite Line

Consider an insulated bar, long enough to be treated as infinite, whose temperature at $t = 0$ is $T_i(x)$, measured from that of the surroundings, so that T tends to zero as $|x| \rightarrow \infty$. With no ends to pick out discrete values of k , we take the Fourier transform of Section 6.2 with respect to x at each time t ,

$$\tilde{T}(k, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} T(x, t) e^{-ikx} dx. \quad (7.54)$$

Differentiating under the integral sign, as in Section 2.2, shows that the transform of $\partial T/\partial t$ is $\partial \tilde{T}/\partial t$, and by the derivative property of Section 6.2 the transform of $\partial^2 T/\partial x^2$ is $-k^2 \tilde{T}$, since T and $\partial T/\partial x$ tend to zero at large $|x|$. The transform of the diffusion equation is therefore

$$\frac{\partial \tilde{T}}{\partial t} = -Dk^2 \tilde{T}, \quad (7.55)$$

which, for each k , is an ordinary differential equation in t , as it was for each mode of the bar in Section 7.2, with the solution $\tilde{T}(k, t) = \tilde{T}_i(k) e^{-Dk^2 t}$, where $\tilde{T}_i$ is the transform of the initial temperature.

To find T we take the inverse transform, substitute the definition of $\tilde{T}_i$, and exchange the order of the integrations over k and x' , which gives

$$\begin{aligned} T(x, t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-Dk^2 t} e^{ikx} \int_{-\infty}^{\infty} T_i(x') e^{-ikx'} dx' dk \\ &= \int_{-\infty}^{\infty} T_i(x') G(x - x', t) dx', \end{aligned} \quad (7.56)$$

where

$$G(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-Dtk^2} e^{ikx} dk. \quad (7.57)$$

In Section 6.2 we found the transform of a Gaussian, which for the Gaussian $e^{-Dt x^2}$ states that $\int_{-\infty}^{\infty} e^{-Dt x^2} e^{-ikx} dx = \sqrt{\pi/Dt} e^{-k^2/4Dt}$. The integral in G has this form with the roles of x and k exchanged, and the sign of ikx does not affect the result, since the Gaussian is even, so

$$G(x, t) = \frac{1}{\sqrt{4\pi Dt}} e^{-x^2/4Dt}. \quad (7.58)$$

The function G is often called the heat kernel, and the temperature is its convolution with the initial temperature.

As an example, suppose that at $t = 0$ an amount of heat $\mathcal{E}$ per unit area of cross-section is released in a thin slice of the bar at $x = 0$, which could, for example, be done with a brief flash of light. The heat per unit area in the bar is $\int \rho c T dx$, so the initial temperature is $T_i = (\mathcal{E}/\rho c) \delta(x)$, and the sifting property of Section 6.3 gives

$$T(x, t) = \frac{\mathcal{E}}{\rho c \sqrt{4\pi Dt}} e^{-x^2/4Dt}. \quad (7.59)$$

This spreading Gaussian is drawn in Figure 68. Its width goes as $\sqrt{Dt}$ and its height as $1/\sqrt{t}$, while its area stays the same, since the Gaussian integral of Section 2.3 gives $\int \rho c T dx = \mathcal{E}$ at every time, as the conservation of heat requires. As t tends to zero, G tends to the delta function, as the Gaussian sequences did in Section 6.3.

At every positive time G is different from zero at every x , so heat released at the origin raises the temperature, by however little, at any distance, because Fourier's law makes the heat flux respond at

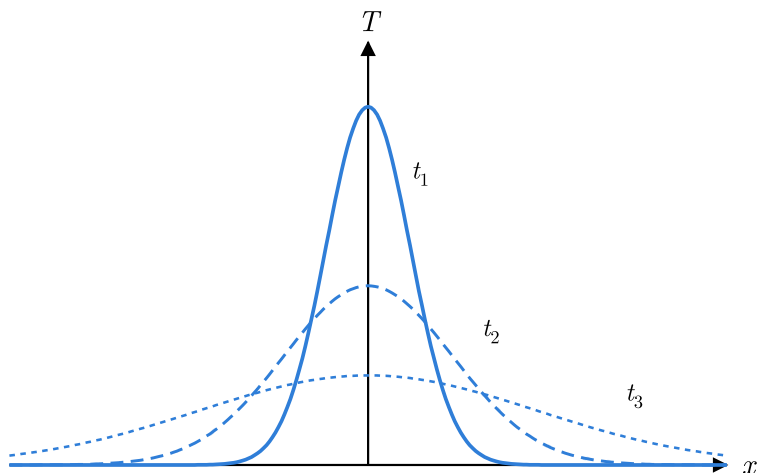


Figure 68. The temperature of an infinite bar after heat is released in a thin slice at $x = 0$, drawn at the times t_1 , $t_2 = 4t_1$, and $t_3 = 16t_1$. Each time the elapsed time is multiplied by four, the width of the Gaussian doubles and its height halves, and the area under each curve is the same.

the same instant to the temperature gradient. A disturbance on a string, by contrast, reaches the point x only after the time x/v , as d'Alembert's solution of Section 7.1 shows.

Problems

§7.1 The Classical Equations of Physics

1. Show by the chain rule that $u = f(x - vt)$ satisfies the wave equation for any function f that can be differentiated twice. Show also that $u = \sin kx \cos vkt$, where k is a constant, satisfies the wave equation, and write it as the sum of two travelling waves.
2. A long, thin wire of radius s_w loses heat from its surface to surroundings at the temperature T_a at the rate $\Lambda(T - T_a)$ per unit area, with Λ a constant, and its temperature $T(x, t)$ depends only on the distance x along it. By considering the heat in an element of the wire, show that $\partial T / \partial t = D \partial^2 T / \partial x^2 - [2D\Lambda / (\kappa s_w)](T - T_a)$. Show also that $T = T_a + e^{-2D\Lambda t / (\kappa s_w)} U(x, t)$ is a solution

of this equation whenever U satisfies the diffusion equation.

3. Concentric spheres of radii $r_1 < r_2$ are held at the potentials Φ_1 and Φ_2 . Using the Laplacian of Section 3.4, find the potential between them.
4. An infinite, straight string is struck so that at $t = 0$ the part $-x_s < x < x_s$ has the velocity $\mathcal{U}_s$ and the rest is at rest. Use d'Alembert's solution to find the displacement of the point $x = 0$ at later times.
5. Let T_I and T_{II} be solutions of the diffusion equation in a region $\mathcal{R}$ with the same initial temperature and the same Dirichlet condition on its surface S , and let $w = T_I - T_{II}$ and $\mathcal{J}(t) = \int_{\mathcal{R}} w^2 dV$. Show that $d\mathcal{J}/dt \leq 0$, and hence that $T_I = T_{II}$ at all times.
6. Let u_I and u_{II} be solutions of the wave equation for a string with fixed ends at $x = 0$ and $x = L$, with the same initial displacement and velocity, and let $w = u_I - u_{II}$. Show that $\mathcal{J}(t) = \int_0^L [(\partial w / \partial t)^2 + v^2 (\partial w / \partial x)^2] dx$, which is proportional to the energy of the motion described by w , does not change in time, and hence that $u_I = u_{II}$ at all times.
7. Show that Poisson's equation in a region $\mathcal{R}$ with a Neumann condition on its surface S can have a solution only if $\oint_S (\partial \Phi / \partial n) dS = -\varepsilon_0^{-1} \int_{\mathcal{R}} \rho dV$, and show that this condition is Gauss's law.

§7.2 Separation of Variables in Cartesian Coordinates

1. A string of length L with fixed ends is straight at $t = 0$, and every point of it is given the same velocity $\mathcal{U}_s$, which could, for example, model a string struck by a broad, flat hammer. Find the displacement $u(x, t)$.
2. A string of length L with fixed ends is pulled aside at the point $x = x_p$ to the displacement u_p , so that it forms two straight segments, and released from rest. Show that $A_n = 2u_p L^2 \sin(n\pi x_p / L) / [n^2 \pi^2 x_p (L - x_p)]$, check that this agrees with the text when $x_p = L/2$, and explain why the modes that have a node at x_p are absent.

3. Show that every motion of a string of length L with fixed ends repeats itself after the time $2L/v$.
4. A bar of length L with insulated sides also has insulated ends, so that $\partial T/\partial x = 0$ at $x = 0$ and $x = L$. Show that the separated solutions are $\cos(n\pi x/L) e^{-n^2\pi^2 Dt/L^2}$ with $n = 0, 1, 2, \dots$, and show that the temperature tends to the mean value of the initial temperature along the bar. Explain this result physically.
5. The bar of the text, with its ends held at T_1 and T_2 , is initially at the uniform temperature T_1 . Show that $b_n = 2(T_2 - T_1)(-1)^n/(n\pi)$. By evaluating the series at $x = L/2$ when $t = 0$, where it converges to T_1 , deduce that $1 - \frac{1}{3} + \frac{1}{5} - \dots = \pi/4$.
6. A square plate $0 < x < L$, $0 < y < L$ has one edge held at T_0 and the other three at zero. Show, by adding the solutions of the four problems obtained by rotating the square through right angles, that the temperature at the centre is $T_0/4$. Show also that the series of the text gives the temperature at the centre as $(2T_0/\pi) \sum_{n \text{ odd}} (-1)^{(n-1)/2} / [n \cosh(n\pi/2)]$, and deduce the value of this sum.

§7.3 Separation of Variables in Polar and Spherical Coordinates

1. A flat annulus $s_1 < s < s_2$ has its inner rim held at T_1 and its outer rim at T_2 . Find the steady temperature.
2. The rim of a disc of radius s_0 is held at $T_e = T_0 \cos^2 \phi$. Find the steady temperature, in polar and in Cartesian coordinates.
3. For the disc of Figure 65, let A be the point $\phi = \pi$ and B the point $\phi = 0$ of the rim, where the rim temperature jumps, and let χ be the angle APB at a point P in the upper half of the disc. Show that χ is the difference between the polar angles of P measured about B and about A, and hence that $T_0(\frac{3}{2} - \chi/\pi)$ satisfies Laplace's equation and takes the same values as the series of the text on the boundary of the upper half-disc. Deduce that the isotherms are arcs of circles through A and B.
4. Carry out the calculation of the text for $l = 3$ to show that $\Theta = \mu^3 - \frac{3}{5}\mu$. Show that P_3 is orthogonal to P_0 , P_1 , and P_2 , and

that $\int_{-1}^1 P_3^2 d\mu = \frac{2}{7}$.

5. The upper hemisphere of a sphere of radius r_0 is held at the potential Φ_0 and the lower hemisphere at $-\Phi_0$. Assuming that the surface potential can be expanded as a series of Legendre polynomials, show that $\alpha_l = 0$ for even l , find α_1 and α_3 , and write down the first two terms of the potential outside the sphere.
6. A sphere of radius r_0 , held at zero potential, is placed in a uniform electric field E_0 along the z -axis. Show that $\Phi = -E_0(r - r_0^3/r^2) \cos \theta$ satisfies Laplace's equation outside the sphere, vanishes on its surface, and tends to the potential $-E_0 z$ of the uniform field far from it.
7. Show by differentiation that $G(x, t) = (4\pi Dt)^{-1/2} e^{-x^2/4Dt}$ satisfies the diffusion equation for $t > 0$.
8. An infinite bar has the initial temperature T_0 for $x < 0$ and zero for $x > 0$. Show that $T(x, t) = \frac{1}{2}T_0[1 - \operatorname{erf}(x/\sqrt{4Dt})]$, where $\operatorname{erf}(\nu) = (2/\sqrt{\pi}) \int_0^\nu e^{-\nu'^2} d\nu'$ is called the error function, and hence that the temperature at $x = 0$ stays equal to $T_0/2$.

Probability, Statistics, and Stochastic Methods

8.1 Probability and Random Variables

Probability enters physics wherever the outcome of an experiment cannot be predicted in advance. For example, a Geiger counter registers the decays of a radioactive source at irregular times, and repeated measurements of the same length give slightly different values.

Probability at School

The ideas in this subsection should be familiar to you from school. The set of possible outcomes of an experiment is called the sample space S , and an event is a collection of outcomes, such as "the counter registers at least one decay in the next second". Each event A has a probability $P(A)$ between 0 and 1, with $P(S) = 1$. If A' denotes the event that A does not happen, and $A \cup B$ and $A \cap B$ denote the events " A or B " and " A and B ", then

$$P(A') = 1 - P(A), \quad P(A \cup B) = P(A) + P(B) - P(A \cap B). \quad (8.1)$$

Events A and B are called mutually exclusive when they cannot happen together, so that $P(A \cap B) = 0$, and independent when

$$P(A \cap B) = P(A)P(B).$$

A number attached to each outcome, such as a count of decays, is called a random variable. A discrete random variable x takes values x_j with probabilities P_j that add up to one, and its mean and variance are

$$\mu = \langle x \rangle = \sum_j x_j P_j, \quad \sigma^2 = \sum_j (x_j - \mu)^2 P_j = \langle x^2 \rangle - \langle x \rangle^2, \quad (8.2)$$

where σ is called the standard deviation. A continuous random variable, such as a measured length, is described by a probability density $p(x)$, defined so that the probability of finding x between a and b is $\int_a^b p(x) dx$. The density is normalised, $\int_{-\infty}^{\infty} p(x) dx = 1$, and the mean and variance are found by replacing the sums with integrals, so that $\langle x \rangle = \int x p(x) dx$.

If a trial succeeds with probability q and fails with probability $1 - q$, the number of successes k in n independent trials follows the binomial distribution,

$$P(k) = \binom{n}{k} q^k (1 - q)^{n-k}, \quad k = 0, 1, \dots, n, \quad (8.3)$$

whose mean is nq and whose variance is $nq(1 - q)$. The normal distribution with mean μ and variance σ^2 has the density

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right], \quad (8.4)$$

which is symmetric about μ . About two-thirds of its probability lies within one standard deviation of the mean, and about 95 per cent within two.

Conditional Probability and Bayes's Theorem

The probability of A given that B has occurred is called the conditional probability of A given B , and is defined as follows:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0. \quad (8.5)$$

In the Venn diagram of Figure 69, learning that B has occurred removes every outcome outside the circle B , so B becomes the new sample space, and dividing by $P(B)$ makes the probabilities within it add up to one. Multiplying the definition by $P(B)$ gives the

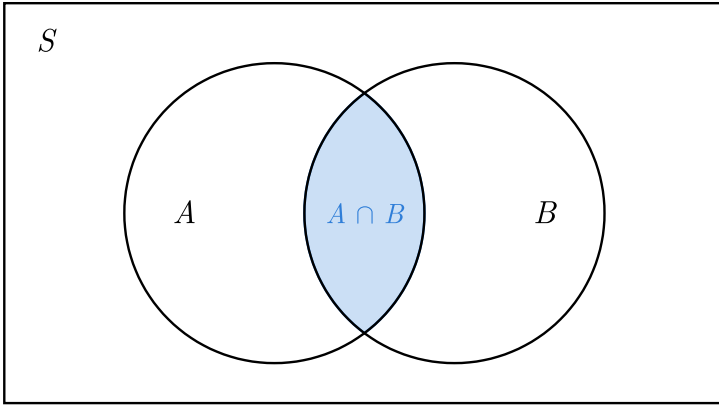


Figure 69. A Venn diagram of two events A and B in the sample space S . Once B is known to have occurred, the outcomes outside B are excluded, and the conditional probability $P(A|B)$ is the probability of the shaded region $A \cap B$ as a fraction of the probability of B .

multiplication rule $P(A \cap B) = P(A|B)P(B)$, and for independent events the definition gives $P(A|B) = P(A)$. Since $A \cap B$ is the same event as $B \cap A$, the multiplication rule also gives $P(A|B)P(B) = P(B|A)P(A)$, and dividing by $P(B)$ gives

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}, \quad (8.6)$$

which is called Bayes's theorem. The denominator is usually found by splitting B into the mutually exclusive events $B \cap A$ and $B \cap A'$, which gives $P(B) = P(B|A)P(A) + P(B|A')P(A')$.

As an example, consider a radiation monitor at the door of a laboratory. Let R be the event that a person passing through carries a radioactive source, with $P(R) = r$, and let G be the event that the monitor sounds its alarm. The alarm sounds with probability $P(G|R) = \eta_1$ when a source is present, and, because of the natural background radiation, with a smaller probability $P(G|R') = \eta_0$ when there is none. By Bayes's theorem, a person who sets off the alarm carries a source with probability

$$P(R|G) = \frac{\eta_1 r}{\eta_1 r + \eta_0(1 - r)}. \quad (8.7)$$

If sources are rare, so that r is small compared with η_0 , the second term in the denominator is the larger, and most alarms are false even when η_1 is close to one.

Moments of a Distribution

The mean of a function $\psi(x)$ of a random variable is $\langle\psi(x)\rangle = \sum_j \psi(x_j) P_j$, or $\int \psi(x) p(x) dx$ for a continuous variable. Since sums and integrals are linear, $\langle\psi_1 + C\psi_2\rangle = \langle\psi_1\rangle + C\langle\psi_2\rangle$ for functions ψ_1 and ψ_2 and a constant C . The mean $\langle x^n \rangle$ is called the n th moment of the distribution, and $\langle (x - \mu)^n \rangle$ the n th central moment. The variance is the second central moment, and expanding the square and using linearity gives $\langle (x - \mu)^2 \rangle = \langle x^2 \rangle - 2\mu\langle x \rangle + \mu^2 = \langle x^2 \rangle - \mu^2$, the second form of the variance quoted above. The third central moment divided by σ^3 is called the skewness. It measures how lopsided the distribution is, and it vanishes for a distribution that is symmetric about its mean, since the contributions from $x - \mu$ and $\mu - x$ cancel.

As an example, consider the time t at which a radioactive nucleus decays, measured from the moment we begin to watch it. By the law of radioactive decay, which we encountered when studying physics at school, the probability that the nucleus has not decayed by time t is $e^{-\gamma t}$, where γ is the decay constant. The probability that the nucleus decays between t and $t + dt$ is $e^{-\gamma t} - e^{-\gamma(t+dt)}$, which to first order in dt is $\gamma e^{-\gamma t} dt$, so the decay time has the density

$$p(t) = \gamma e^{-\gamma t}, \quad t \geq 0, \quad (8.8)$$

which is called the exponential distribution. Integrating by parts gives the mean,

$$\langle t \rangle = \int_0^\infty t \gamma e^{-\gamma t} dt = \left[-t e^{-\gamma t} \right]_0^\infty + \int_0^\infty e^{-\gamma t} dt = \frac{1}{\gamma}, \quad (8.9)$$

which is the mean lifetime of the nucleus. You will show in Problem 3 that the standard deviation is also $1/\gamma$ and that the skewness is 2.

The Binomial and Poisson Distributions

We now derive the binomial distribution by counting. By the multiplication rule for independent events, a particular sequence of n

trial	1	2	3	4
	S	S	F	F
	S	F	S	F
	S	F	F	S
	F	S	S	F
	F	S	F	S
	F	F	S	S

Figure 70. The orderings of $n = 4$ trials that contain $k = 2$ successes, S, and two failures, F. Each row is fixed by the choice of the two trials that are successes, and there are $\binom{4}{2} = 6$ such choices.

trials containing k successes has probability $q^k(1 - q)^{n-k}$, and every ordering of the same numbers of successes and failures has the same probability. An ordering is fixed by choosing which k of the n trials are the successes, as Figure 70 shows, and the number of such choices is the binomial coefficient $\binom{n}{k} = n!/[k!(n - k)!]$, which gives the distribution quoted above. By the binomial theorem the probabilities add up to $[q + (1 - q)]^n = 1$.

Consider now a radioactive source containing a very large number n of nuclei, whose decays we count for a time t_c that is short compared with the mean lifetime $1/\gamma$. Each nucleus decays in that time, independently of the others, with a small probability $q \approx \gamma t_c$, so the count k follows the binomial distribution. Let $\mu = nq$ be the mean count, and let $n \rightarrow \infty$ with μ held fixed. Writing $q = \mu/n$, the binomial distribution becomes

$$P(k) = \frac{\mu^k}{k!} \cdot \frac{n(n-1) \cdots (n-k+1)}{n^k} \cdot \left(1 - \frac{\mu}{n}\right)^{-k} \cdot \left(1 - \frac{\mu}{n}\right)^n. \quad (8.10)$$

At fixed k the second and third factors tend to one. The Taylor series of $\ln(1 + x)$ from Section 1.1 gives $n \ln(1 - \mu/n) = -\mu - \mu^2/2n - \dots$, which tends to $-\mu$. Since raising a number to a power

means that $(1 - \mu/n)^n = e^{n \ln(1 - \mu/n)}$, the last factor tends to $e^{-\mu}$, and the limit is

$$P(k) = \frac{\mu^k e^{-\mu}}{k!}, \quad k = 0, 1, 2, \dots, \quad (8.11)$$

which is called the Poisson distribution.

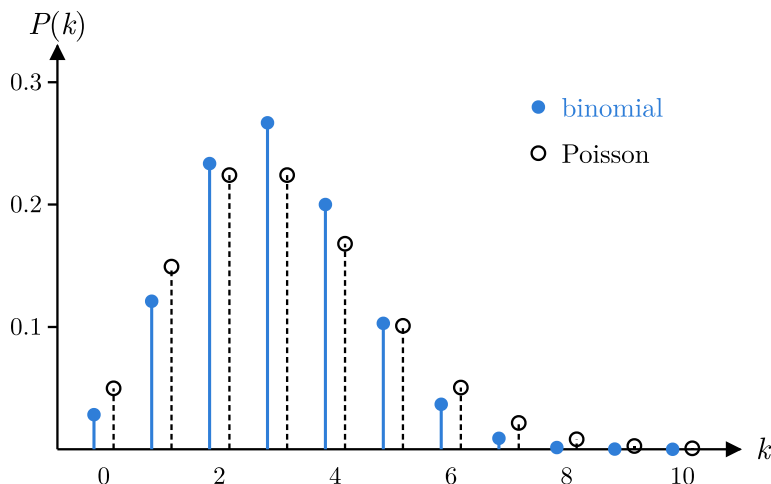


Figure 71. The binomial distribution with $n = 10$ and $q = 3/10$ (filled circles) and the Poisson distribution with the same mean, $\mu = 3$ (open circles), computed from their formulas. The Poisson distribution is slightly wider, since its variance μ exceeds the binomial variance $nq(1-q)$, and the binomial and Poisson distributions come closer together as q is reduced with nq held fixed.

The Poisson distribution describes the number of events in a fixed interval when the events occur independently and at a constant average rate, and it is compared with a binomial distribution in Figure 71. It is normalised because $\sum_k \mu^k/k! = e^\mu$, by the exponential series of Section 1.1. In the sum for the mean, the term with $k = 0$ vanishes, and cancelling k against the $k!$ gives

$$\langle k \rangle = \mu e^{-\mu} \sum_{k=1}^{\infty} \frac{\mu^{k-1}}{(k-1)!} = \mu. \quad (8.12)$$

Cancelling $k(k-1)$ against the $k!$ in the same way gives $\langle k(k-1) \rangle = \mu^2$, so the variance is $\langle k(k-1) \rangle + \langle k \rangle - \langle k \rangle^2 = \mu$. The standard

deviation of a Poisson count is therefore the square root of its mean, which is why a count of k decays is often quoted with an uncertainty of $\sqrt{k}$.

The Gaussian Distribution

The normal distribution is also called the Gaussian distribution, and its density can be shown to be normalised with the Gaussian integral $\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$, which we derived in Section 2.3 by squaring the integral. Changing the variable to $u = (x - \mu)/(\sqrt{2}\sigma)$, so that $dx = \sqrt{2}\sigma du$, gives

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right] dx = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-u^2} du = 1. \quad (8.13)$$

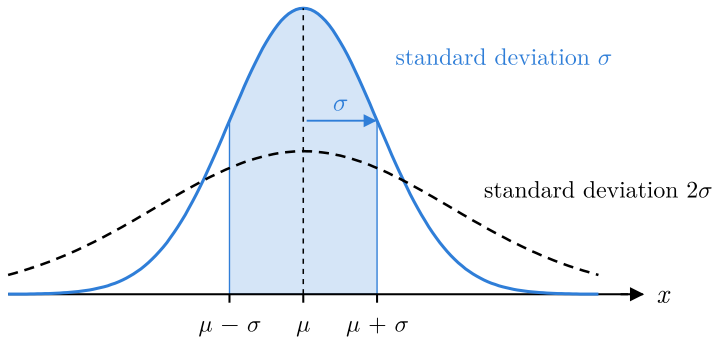


Figure 72. Gaussian densities with the same mean μ . The solid curve has standard deviation σ , and the shaded area within one standard deviation of the mean is about two-thirds of the whole. The arrow marks the half-width σ , which is the distance from the mean to each point of inflection. The dashed curve has standard deviation 2σ , and is twice as wide and half as tall.

The density is symmetric about $x = \mu$, so the mean is μ . With the same change of variable, and the integral $\int_{-\infty}^{\infty} u^2 e^{-u^2} du = \frac{1}{2}\sqrt{\pi}$, which we found in Section 2.3 by differentiating the Gaussian inte-

gral with respect to its parameter, the variance is

$$\langle (x - \mu)^2 \rangle = \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} u^2 e^{-u^2} du = \sigma^2, \quad (8.14)$$

so the parameter σ in the density is the standard deviation. Differentiating the density twice, by the chain rule and the product rule, gives $p''(x) = [(x - \mu)^2/\sigma^4 - 1/\sigma^2]p(x)$, which vanishes at $x = \mu \pm \sigma$, so the points of inflection lie one standard deviation from the mean, as marked in Figure 72.

Sums and Functions of Random Variables

Let x and y be independent random variables with densities $p_1(x)$ and $p_2(y)$, so that the probability of finding x in dx and y in dy is $p_1(x)p_2(y)dx dy$. The double integral for the mean of a function of x multiplied by a function of y then factorises into the product of their means. By linearity, the mean of $x + y$ is $\mu_x + \mu_y$. For the variance, expanding the square gives

$$\langle (x + y - \mu_x - \mu_y)^2 \rangle = \sigma_x^2 + \sigma_y^2 + 2 \langle (x - \mu_x)(y - \mu_y) \rangle. \quad (8.15)$$

The last mean is called the covariance of x and y , written $\text{cov}(x, y)$. For independent variables it factorises into $\langle x - \mu_x \rangle \langle y - \mu_y \rangle$, which is zero, and therefore

$$\sigma_{x+y}^2 = \sigma_x^2 + \sigma_y^2. \quad (8.16)$$

The sum $U_N = x_1 + x_2 + \cdots + x_N$ of N independent variables, each with mean μ and variance σ^2 , therefore has mean $N\mu$ and variance $N\sigma^2$, and its standard deviation, $\sigma\sqrt{N}$, grows more slowly than its mean. You will use this result in Problem 4 to derive the mean and variance of the binomial distribution.

Suppose next that $y = \psi(x)$, where ψ is an increasing function. The variable y lies between y and $y + dy$ exactly when x lies in the corresponding interval of width dx , so $p_y(y)dy = p_x(x)dx$. If ψ is decreasing, dx and dy have opposite signs, and in either case

$$p_y(y) = p_x(x) \left| \frac{dx}{dy} \right|, \quad (8.17)$$

with x written as a function of y . As an example, let x be distributed uniformly between 0 and 1, so that $p_x = 1$ there, and let $t = -(\ln x)/\gamma$. Then $x = e^{-\gamma t}$ and $|dx/dt| = \gamma e^{-\gamma t}$, so t has the

exponential density of the decay time. When ψ is not monotonic, several values of x give the same y , and their contributions are added, as you will do in Problem 6.

The Central Limit Theorem

Let $x_1, x_2, \dots, x_N$ be independent random variables with the same distribution, of mean μ and finite variance σ^2 , and let

$$z = \frac{U_N - N\mu}{\sigma\sqrt{N}}, \quad (8.18)$$

which is their sum U_N shifted and scaled to have mean zero and variance one. The central limit theorem states that, as $N \rightarrow \infty$, the distribution of z approaches the standard Gaussian distribution, with density $e^{-z^2/2}/\sqrt{2\pi}$, for any distribution of the individual variables with finite variance. Its proof is beyond the scope of this course.

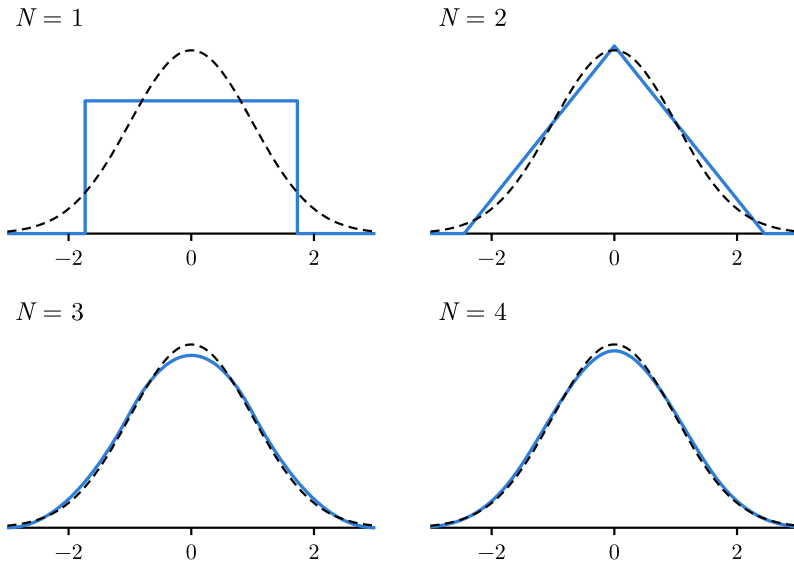


Figure 73. The density of $z = (U_N - N\mu)/(\sigma\sqrt{N})$, where U_N is the sum of N independent variables distributed uniformly between 0 and 1, for which $\mu = 1/2$ and $\sigma^2 = 1/12$ (solid curves). The densities were computed exactly by repeated convolution. The dashed curve in each panel is the standard Gaussian density.

Figure 73 illustrates the theorem for variables distributed uniformly between 0 and 1, whose mean is $\frac{1}{2}$ and whose variance is $\int_0^1 x^2 dx - \frac{1}{4} = \frac{1}{12}$. The density of the sum of two such variables is a triangle, as you will show in Problem 7, and the density of the sum of three is already close to a Gaussian.

Since the binomial count is a sum of n independent variables, each equal to 1 for a success and 0 for a failure, the theorem implies that for large n the binomial distribution is close to a Gaussian. The error in a measurement is often the sum of many small, independent disturbances, such as vibrations of the apparatus and variations in the experimenter's reaction time, and this is the usual reason for treating measurement errors as Gaussian.

8.2 Estimation and Hypothesis Testing

When the period of a pendulum is timed N times with a stopwatch, the values $x_1, x_2, \dots, x_N$ differ, because the experimenter's reaction time varies. We treat such a set of values as a sample, that is, as N independent values of a random variable whose distribution has an unknown mean μ , the quantity we want to measure, and an unknown variance σ^2 . This section asks first how μ and σ are estimated, then how uncertainties are carried through a calculation and how a straight line is fitted to data, and finally how to decide whether measurements agree with a model.

Estimators and the Standard Error of the Mean

A quantity calculated from the sample in order to estimate a property of the distribution is called an estimator. An estimator is said to be unbiased if its mean over many repetitions of the whole set of measurements equals the property it estimates. The sample mean

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad (8.19)$$

is an unbiased estimator of μ , since each x_i has mean μ . By Section 8.1, where we showed that the variances of independent variables add, the sum of the x_i has variance $N\sigma^2$, and dividing a variable

by N divides its variance by N^2 , so

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{N}. \quad (8.20)$$

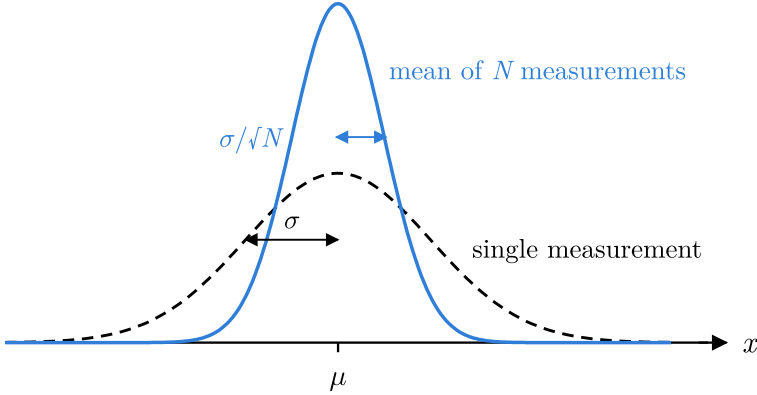


Figure 74. The distribution of a single measurement, with standard deviation σ (dashed), and the distribution of the mean of $N = 4$ measurements, with standard deviation $\sigma/\sqrt{N} = \sigma/2$ (solid). Both distributions are centred on the true mean μ .

The standard deviation of the sample mean, $\sigma/\sqrt{N}$, is called the standard error of the mean. As Figure 74 shows, the mean of four measurements has half the uncertainty of one.

The variance σ^2 has to be estimated from the scatter of the measurements. Writing $x_i - \bar{x} = (x_i - \mu) - (\bar{x} - \mu)$, expanding the square, and using $\sum_i (x_i - \mu) = N(\bar{x} - \mu)$, which follows from the definition of $\bar{x}$, gives

$$\sum_{i=1}^N (x_i - \bar{x})^2 = \sum_{i=1}^N (x_i - \mu)^2 - N(\bar{x} - \mu)^2. \quad (8.21)$$

Each term of the first sum has mean σ^2 , and $(\bar{x} - \mu)^2$ has mean σ^2/N , so the left-hand side has mean $(N-1)\sigma^2$, and

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \quad (8.22)$$

is an unbiased estimator of σ^2 , called the sample variance. The letter s is used for the sample standard deviation only in this chapter. The divisor is $N - 1$ because the deviations are measured from $\bar{x}$, which is calculated from the same data and lies closer to them than μ does; you will show in Problem 1 that $\bar{x}$ is the value of ξ that makes $\sum_i (x_i - \xi)^2$ smallest. The result of the measurement is often quoted as $\bar{x} \pm s/\sqrt{N}$.

Propagation of Uncertainty

A result is often calculated from several measured quantities; for example, the acceleration of free fall can be found from the length and the period of a pendulum. Let $f(x, y)$ be calculated from independent measured quantities x and y , with means μ_x and μ_y and small standard deviations σ_x and σ_y . Taylor's theorem in two variables, from Section 2.2, gives to first order

$$f(x, y) \approx f(\mu_x, \mu_y) + \frac{\partial f}{\partial x}(x - \mu_x) + \frac{\partial f}{\partial y}(y - \mu_y), \quad (8.23)$$

with the derivatives evaluated at the means. Since the deviations from the means have mean zero, the mean of f is $f(\mu_x, \mu_y)$ to this order, and squaring the deviation of f from this mean and taking the mean again gives

$$\sigma_f^2 = \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2 + 2 \frac{\partial f}{\partial x} \frac{\partial f}{\partial y} \text{cov}(x, y), \quad (8.24)$$

in which the covariance vanishes for independent x and y , as we showed in Section 8.1. For a function of any number of independent variables x_j , the same argument gives

$$\sigma_f^2 = \sum_j \left(\frac{\partial f}{\partial x_j}\right)^2 \sigma_j^2, \quad (8.25)$$

which is called the formula for the propagation of uncertainty. It requires f to be close to its tangent plane within a few standard deviations of the means; for a function of one variable, the second-order term $\frac{1}{2}f''\sigma_x^2$ must be small compared with the first-order term $f'\sigma_x$, that is, $\sigma_x \ll |2f'/f''|$.

Figure 75 shows how the tangent carries the distribution of x into a distribution of f that is $|f'|$ times as wide. For a sum or difference, $f = x \pm y$, the propagation formula gives $\sigma_f^2 = \sigma_x^2 + \sigma_y^2$, and the

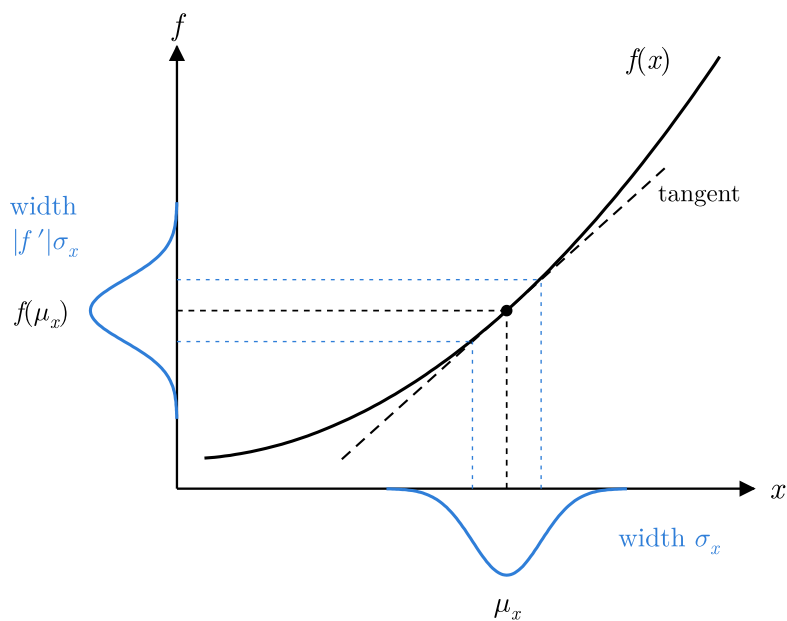


Figure 75. The propagation of uncertainty through a function of one variable. Near $x = \mu_x$ the curve $f(x)$ is close to its tangent, so a narrow distribution of x with width σ_x is carried into a distribution of f centred on $f(\mu_x)$ with width $|f'(\mu_x)| \sigma_x$.

uncertainties are said to add in quadrature. For a product of powers, $f = Kx^{\beta_1}y^{\beta_2}$ with constants K , β_1 , and β_2 , the derivatives are $\partial f/\partial x = \beta_1 f/x$ and $\partial f/\partial y = \beta_2 f/y$, and dividing the formula by f^2 gives

$$\left(\frac{\sigma_f}{f}\right)^2 = \beta_1^2 \left(\frac{\sigma_x}{x}\right)^2 + \beta_2^2 \left(\frac{\sigma_y}{y}\right)^2, \quad (8.26)$$

so the fractional uncertainties add in quadrature, each weighted by its power. Consider, for example, a simple pendulum of length L and period $T = 2\pi\sqrt{L/g}$. Then $g = 4\pi^2 L T^{-2}$, a product of powers with $\beta_1 = 1$ and $\beta_2 = -2$, and the fractional uncertainty in g is

$$\frac{\sigma_g}{g} = \left[\left(\frac{\sigma_L}{L}\right)^2 + 4 \left(\frac{\sigma_T}{T}\right)^2 \right]^{1/2}, \quad (8.27)$$

in which the fractional uncertainty of the period counts twice as heavily as that of the length, because the period enters squared.

The Method of Maximum Likelihood

Suppose that a length has been measured N times with different instruments, so that the measurement x_i is drawn from a Gaussian distribution with the common mean μ and a known standard deviation σ_i . The probability density of obtaining the whole set of values, regarded as a function of μ , is

$$\mathcal{L}(\mu) = \prod_{i=1}^N \frac{1}{\sigma_i \sqrt{2\pi}} \exp \left[-\frac{(x_i - \mu)^2}{2\sigma_i^2} \right], \quad (8.28)$$

and is called the likelihood. The method of maximum likelihood takes as the estimate of μ the value $\hat{\mu}$ that makes the likelihood largest, that is, the value for which the data actually obtained would have been most probable. Since the logarithm is an increasing function, we may instead maximise $\ln \mathcal{L}$, which apart from terms independent of μ is $-\frac{1}{2} \sum_i (x_i - \mu)^2 / \sigma_i^2$. Setting its derivative with respect to μ to zero and solving for $\hat{\mu}$ gives

$$\hat{\mu} = \frac{\sum_i w_i x_i}{\sum_i w_i}, \quad w_i = \frac{1}{\sigma_i^2}, \quad (8.29)$$

which is called the weighted mean. A measurement with a small uncertainty has a large weight and counts for more. Since $\partial \hat{\mu} / \partial x_i = w_i / \sum_j w_j$, the propagation formula gives the variance

of the weighted mean as $\sum_i w_i^2 \sigma_i^2 / (\sum_j w_j)^2 = 1 / \sum_i w_i$. When all the σ_i are equal, the weighted mean reduces to $\bar{x}$ and its standard deviation to $\sigma / \sqrt{N}$.

The method applies to any distribution, and you will use it for Poisson counts and for decay times in Problems 4 and 5. For Gaussian measurements, maximising the likelihood is the same as minimising the sum of squares $\sum_i (x_i - \mu)^2 / \sigma_i^2$, which is the principle of least squares.

The Least-Squares Straight Line

Let (x_i, y_i) , for $i = 1, \dots, N$, be points whose x_i are known exactly and whose y_i are drawn from Gaussian distributions centred on the line $y = mx + c$, all with the same standard deviation σ . The points could, for example, be the squares of the periods of a pendulum, $y_i = T_i^2$, plotted against its lengths, $x_i = L_i$, so that the slope is $m = 4\pi^2/g$. By the principle of least squares, the maximum-likelihood values of m and c minimise

$$Q(m, c) = \sum_{i=1}^N d_i^2, \quad d_i = y_i - mx_i - c, \quad (8.30)$$

where d_i , the vertical distance of the i th point from the line drawn in Figure 76, is called its residual.

At the minimum, as at any stationary point in Section 2.2, both partial derivatives of Q vanish. Setting $\partial Q / \partial c = -2 \sum_i (y_i - mx_i - c)$ to zero gives $c = \bar{y} - m\bar{x}$, so the line passes through $(\bar{x}, \bar{y})$. Setting $\partial Q / \partial m$ to zero gives $\sum_i x_i (y_i - mx_i - c) = 0$, which with this value of c becomes $\sum_i x_i (y_i - \bar{y}) = m \sum_i x_i (x_i - \bar{x})$. Since $\sum_i (y_i - \bar{y})$ and $\sum_i (x_i - \bar{x})$ both vanish, the factor x_i on each side may be replaced by $x_i - \bar{x}$, and the slope is

$$m = \frac{\sum_i (x_i - \bar{x}) y_i}{D}, \quad D = \sum_i (x_i - \bar{x})^2. \quad (8.31)$$

Since $Q_{cc} = 2N > 0$ and $Q_{mm}Q_{cc} - Q_{mc}^2 = 4ND > 0$, the test of Section 2.2 confirms a minimum.

The slope is a linear combination of the y_i , with coefficients $(x_i - \bar{x})/D$, and so is the intercept, $c = \sum_i [1/N - \bar{x}(x_i - \bar{x})/D] y_i$. The Taylor expansion of a linear function stops at the first-order terms,

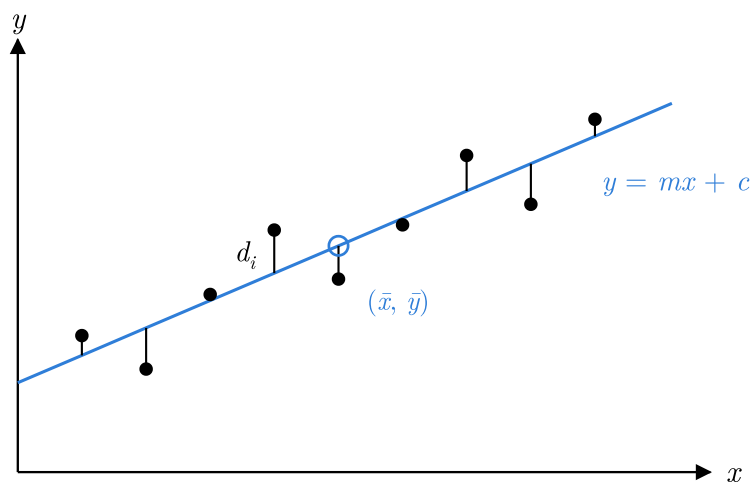


Figure 76. A straight line fitted by least squares. The vertical segments are the residuals d_i , and the line is chosen to make the sum of their squares as small as possible. The fitted line passes through the point $(\bar{x}, \bar{y})$, marked by the open circle.

so the propagation formula is exact here, and since each y_i has variance σ^2 ,

$$\sigma_m^2 = \frac{\sigma^2}{D}, \quad \sigma_c^2 = \sigma^2 \left(\frac{1}{N} + \frac{\bar{x}^2}{D} \right), \quad (8.32)$$

where the cross term in σ_c^2 has vanished because $\sum_i (x_i - \bar{x}) = 0$. If σ is not known, it is estimated from the scatter of the points about the line: you will show in Problem 7 that the mean of the minimum value of Q is $(N-2)\sigma^2$, so that $Q_{\min}/(N-2)$ is an unbiased estimator of σ^2 . The divisor is $N-2$ because two parameters have been fitted, as it is $N-1$ in the sample variance, where one has been fitted.

The Chi-Squared Statistic

Let $y_i \pm \sigma_i$, for $i = 1, \dots, N$, be measurements that a model predicts to have the values $F(x_i)$. The quantity

$$\chi^2 = \sum_{i=1}^N \frac{[y_i - F(x_i)]^2}{\sigma_i^2} \quad (8.33)$$

is called the chi-squared statistic, and it measures how far the data lie from the model in units of their uncertainties. If the model is correct and the errors are Gaussian, each term is the square of a standard Gaussian variable $z_i = [y_i - F(x_i)]/\sigma_i$, with $\langle z_i^2 \rangle = 1$, so the mean of χ^2 is N . As you showed in Problem 8 of Section 8.1, the variance of z_i^2 is 2, and since the terms are independent the variance of χ^2 is $2N$.

When parameters of the model are fitted to the same data by minimising χ^2 , the minimum lies below the value for the true parameters. For a constant fitted to data with equal uncertainties, $\chi_{\min}^2 = \sum_i (y_i - \bar{y})^2 / \sigma^2$ has mean $N-1$, by the result for the sample variance above, and for the straight line you will find the mean $N-2$ in Problem 7. In general, fitting N_p parameters on which the model depends linearly reduces the mean of $\chi_{\min}^2$ to $N - N_p$ and its variance to $2(N - N_p)$, a result whose general proof is not treated in this course. The number

$$\nu = N - N_p \quad (8.34)$$

is called the number of degrees of freedom, and the mean and variance of $\chi_{\min}^2$ are ν and 2ν , as they are for χ^2 with $\nu = N$ when

nothing is fitted.

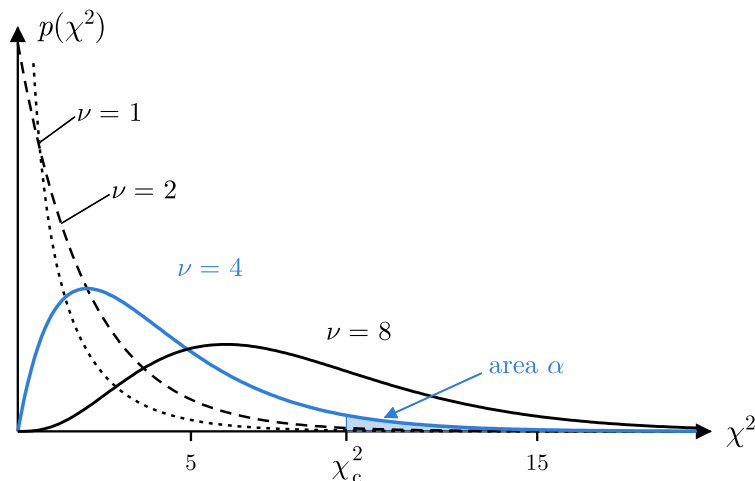


Figure 77. The probability density of χ^2 for $\nu = 1, 2, 4$, and 8 degrees of freedom. For $\nu = 4$ the shaded tail beyond the critical value χ_c^2 has area α , which is the probability that a correct model gives $\chi^2 > \chi_c^2$. Here α is one-twentieth.

The probability density of χ^2 depends only on ν , and is called the chi-squared distribution. It is plotted in Figure 77 and is widely tabulated; the derivation of the density is beyond the scope of this course. For large ν it approaches a Gaussian with mean ν and variance 2ν , as the central limit theorem suggests for a sum of many terms. A fit is therefore judged by comparing $\chi_{\min}^2$ with ν , allowing for a spread of about $\sqrt{2\nu}$. The ratio $\chi_{\min}^2/\nu$ is called the reduced chi-squared. A value much larger than one means that the model is wrong or the uncertainties have been underestimated, and a value much smaller than one usually means that they have been overestimated.

Testing a Hypothesis

A statement about the distribution of the data, such as "the points lie on a straight line with Gaussian errors of standard deviation σ ", is called a hypothesis, and the hypothesis under test is called the null hypothesis. A test of the null hypothesis is carried out as

follows:

1. choose a statistic, such as χ^2 , whose distribution is known when the null hypothesis is true;
2. choose a small probability α , called the significance level of the test, and find the critical value of the statistic that a correct hypothesis exceeds with probability α , such as the value χ_c^2 in Figure 77;
3. reject the null hypothesis if the value obtained from the data exceeds the critical value.

Commonly used significance levels are one-twentieth and one-hundredth, and a correct hypothesis is rejected with probability α . If the value does not exceed the critical value, the data are said to be consistent with the hypothesis, which does not prove it correct.

The simplest test compares a measured mean with a predicted value μ_0 , such as the accepted value of g . If the prediction is correct, the statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{N}} \quad (8.35)$$

has, by the central limit theorem, a distribution close to the standard Gaussian when N is large, and replacing an unknown σ by s is then a valid approximation. A Gaussian variable lies more than two standard deviations from its mean with a probability of about one-twentieth, so at that significance level the measurement is said to disagree with the prediction when $\bar{x}$ differs from μ_0 by more than about two standard errors.

Problems

§8.1 Probability and Random Variables

1. In the example of the radiation monitor, a person who sets off the alarm walks through the door a second time, and the alarm sounds again. Assuming that the alarms on the first and second passages are independent for a given person, find the probability that the person carries a source, and show that it is larger than $P(R|G)$ when $\eta_1 > \eta_0$.

2. Show that if the events A and B are independent, then so are A and B' .
3. For the exponential distribution $p(t) = \gamma e^{-\gamma t}$, show that $\langle t^n \rangle = n!/\gamma^n$. Hence show that the standard deviation of t is $1/\gamma$ and that the skewness is 2. Show also that the probability that a nucleus decays within a time τ after t_0 , given that it has not decayed by t_0 , does not depend on t_0 .
4. Derive the mean and variance of the binomial distribution by writing $k = x_1 + x_2 + \cdots + x_n$, where x_i is 1 if the i th trial is a success and 0 if it is a failure.
5. The counts from a pair of independent sources follow Poisson distributions with means μ_1 and μ_2 . Show that the total count follows the Poisson distribution with mean $\mu_1 + \mu_2$.
6. The variable x is distributed uniformly between $-a$ and a . Find the density of $y = x^2$, check that it is normalised, and find $\langle y \rangle$.
7. The variables x and y are independent, with densities $p_1(x)$ and $p_2(y)$. Show that the density of $\zeta = x+y$ is $\int_{-\infty}^{\infty} p_1(x) p_2(\zeta-x) dx$, an integral of the kind called a convolution in Section 6.2. If x and y are each distributed uniformly between 0 and 1, show that this density is ζ for $0 < \zeta < 1$ and $2 - \zeta$ for $1 < \zeta < 2$, and confirm that it becomes the curve in the panel of Figure 73 with $N = 2$.
8. Show that the fourth central moment of the Gaussian distribution is $3\sigma^4$, and hence that the variance of $(x - \mu)^2$ is $2\sigma^4$.

§8.2 Estimation and Hypothesis Testing

1. Show that $\sum_i (x_i - \xi)^2$ is smallest when $\xi = \bar{x}$, and hence that $\sum_i (x_i - \bar{x})^2 \leq \sum_i (x_i - \mu)^2$ for any sample.
2. A ball of mass M falls from rest through a height h in a time t , so that, the acceleration being uniform, its final speed is $2h/t$. Find the fractional uncertainty in its final kinetic energy in terms of the fractional uncertainties in M , h , and t , which are independent.
3. A quantity x has a Gaussian distribution with mean μ and stan-

dard deviation σ . Show that $\langle x^2 \rangle = \mu^2 + \sigma^2$ exactly, and that the variance of x^2 is $4\mu^2\sigma^2 + 2\sigma^4$. Compare these results with the first-order propagation formula, and state when that formula is a valid approximation.

4. A source is counted for N equal intervals, giving counts $k_1, \dots, k_N$ that follow a Poisson distribution with unknown mean μ . Show that the maximum-likelihood estimate of μ is the mean of the counts.
5. The decay times $t_1, \dots, t_N$ of N nuclei are recorded, each following the exponential distribution $\gamma e^{-\gamma t}$. Show that the maximum-likelihood estimate of the decay constant is $\hat{\gamma} = N / \sum_i t_i$.
6. Find the least-squares slope of a straight line $y = mx$ that is constrained to pass through the origin, for points with equal uncertainties σ in y , and show that its standard deviation is $\sigma / (\sum_i x_i^2)^{1/2}$.
7. Let $y_i = m_0 x_i + c_0 + e_i$, where m_0 and c_0 are the true slope and intercept and the errors e_i are independent, with mean zero and variance σ^2 . Show that $Q_{\min} = \sum_i (y_i - \bar{y})^2 - m^2 D$, where m is the fitted slope, and hence that $\langle Q_{\min} \rangle = (N - 2)\sigma^2$.
8. A quantity is measured with a pair of instruments, giving $x_1 \pm \sigma_1$ and $x_2 \pm \sigma_2$. Find their weighted mean and its standard deviation, and show that the value of χ^2 about the weighted mean is $(x_1 - x_2)^2 / (\sigma_1^2 + \sigma_2^2)$. How many degrees of freedom does this χ^2 have?

Numerical and Computational Mathematics

9.1 Floating-Point Arithmetic and Numerical Stability

Many equations of physics cannot be solved in closed form and are solved instead by a computer, which works with numbers stored to a fixed number of digits. In this chapter we first examine how errors arise and grow in such calculations, then derive methods for solving equations and fitting polynomials, and finally derive methods for approximating derivatives and integrals, analysing the error of each method symbolically.

Floating-Point Numbers

A computer stores a real number x in a fixed amount of memory, usually in the form

$$x = (-1)^s \times m \times 2^E, \quad (9.1)$$

which is called floating-point form. The sign bit s is 0 or 1, the exponent E is an integer lying between fixed limits $E_{\min}$ and $E_{\max}$, and the mantissa m is a binary fraction with a fixed number t of

binary digits,

$$m = 1.b_1b_2 \dots b_{t-1} = 1 + \frac{b_1}{2} + \frac{b_2}{2^2} + \dots + \frac{b_{t-1}}{2^{t-1}}, \quad (9.2)$$

where each digit b_k is 0 or 1. Every non-zero number is written with $1 \leq m < 2$, by choosing the exponent so that the leading digit is 1, and zero is stored as a special case. A format in wide use for scientific work is the IEEE 754 double-precision format, whose mantissa has 53 binary digits. The numbers that can be stored exactly in this way are called floating-point numbers, and every other real number is replaced by a floating-point number near it. A fraction whose denominator has a prime factor other than 2, such as $1/3$ or $1/10$, has a binary expansion that never ends, and so it is always stored with a small error.

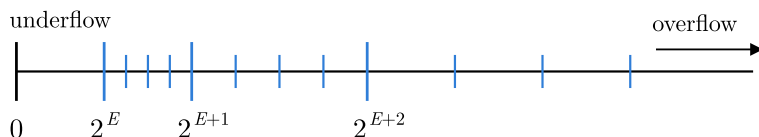


Figure 78. The non-negative floating-point numbers of a system with a mantissa of three binary digits and three values of the exponent, the smallest of which is E . Between successive powers of two there are four equally spaced numbers, and the spacing doubles at each power of two. Numbers larger than the largest floating-point number overflow, and non-zero numbers smaller than 2^E underflow.

The floating-point numbers are not spread evenly along the real line, as Figure 78 shows. For a given exponent E , the mantissa takes 2^{t-1} equally spaced values between 1 and 2, with spacing 2^{1-t} , so the floating-point numbers between 2^E and 2^{E+1} are equally spaced by $2^E 2^{1-t}$. The spacing is proportional to the size of the numbers, and it doubles each time the exponent increases by one. The quantity

$$\epsilon_m = 2^{1-t} \quad (9.3)$$

is called the machine epsilon. It is the gap between 1 and the next larger floating-point number, and in double precision it is 2^{-52} . The spacing of the floating-point numbers near x is between $\frac{1}{2}\epsilon_m|x|$ and

$\epsilon_m|x|$, as you will show in Problem 1.

Let $\text{fl}(x)$ be the floating-point number nearest to x ; when x lies exactly halfway between two floating-point numbers, the IEEE 754 standard chooses the one whose last binary digit is 0. Since x lies within half a spacing of $\text{fl}(x)$,

$$\text{fl}(x) = x(1 + \delta), \quad |\delta| \leq \frac{1}{2}\epsilon_m. \quad (9.4)$$

The standard also requires the result of each addition, subtraction, multiplication, and division of floating-point numbers to be the floating-point number nearest to the exact result, so each operation introduces a relative error of at most $\frac{1}{2}\epsilon_m$. The limits on the exponent set a largest and a smallest positive number. A result larger in magnitude than the largest number is said to overflow, and is replaced by a special value representing infinity. A non-zero result smaller in magnitude than $2^{E_{\min}}$ is said to underflow, and is replaced by zero or by a number with fewer significant digits.

Overflow and underflow can often be avoided by rewriting a formula. As an example, consider the length $\sqrt{x^2 + y^2}$ of a vector with components x and y . If x is larger than the square root of the largest floating-point number, x^2 overflows, even when the length itself could be stored. Supposing that $|x| \geq |y|$, the same length can be written as

$$\sqrt{x^2 + y^2} = |x|\sqrt{1 + (y/x)^2}, \quad (9.5)$$

in which the quantity under the square root lies between 1 and 2, so that nothing in the calculation overflows unless the result does. Similarly, a likelihood of the kind met in Section 8.2, which is a product of many probability densities, can underflow, and its logarithm, which is a sum, is usually computed in its place.

Round-Off Error and Subtractive Cancellation

The error introduced when a number is replaced by a nearby floating-point number is called round-off error. A single round-off error is small, of relative size ϵ_m , and the danger lies in operations that magnify it. One of the most common of these is the subtraction of two nearly equal numbers. Let x and y be stored with relative errors of order ϵ_m , so that their absolute errors are of order $\epsilon_m|x|$.

The difference $x - y$ carries an absolute error of the same order, and when x and y are close its relative error is of order

$$\epsilon_m \frac{|x|}{|x - y|}, \quad (9.6)$$

which is large. The leading digits of x and y cancel, and the digits that remain in the difference are mostly round-off. This loss of accuracy is called subtractive cancellation.

As an example, consider the roots of the quadratic equation $ax^2 + bx + c = 0$,

$$x_{\pm} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad (9.7)$$

in the case where $b > 0$ and $b^2 \gg |4ac|$. The square root is then close to b , and by the binomial expansion $\sqrt{b^2 - 4ac} \approx b - 2ac/b$, so the numerator of x_+ is approximately $-2ac/b$, a small difference of two large numbers. The computed square root has an absolute error of order $\epsilon_m b$, and dividing this error by the size $2|ac|/b$ of the numerator gives a relative error in x_+ of order

$$\frac{\epsilon_m b^2}{2|ac|}, \quad (9.8)$$

which grows as $b^2/|ac|$ grows, and can exceed one, in which case the computed root has no correct digits. The root x_- involves a sum of two numbers of the same sign and is computed accurately. The subtraction can be avoided by finding x_+ from the property that the product of the roots is c/a , which follows from $ax^2 + bx + c = a(x - x_+)(x - x_-)$. The following algorithm computes both roots, for real $a \neq 0$, b , and c with $b^2 \geq 4ac$, without subtracting nearly equal numbers:

```

input: a, b, c
d = sqrt(b*b - 4*a*c)
if b >= 0 then q = -(b + d)/2
else q = -(b - d)/2
if q = 0 then return 0, 0      (in this case b = c = 0)
x1 = q/a
x2 = c/q
return x1, x2

```

The quantity q is formed by adding b and d with the same sign, so no cancellation occurs, and $x_1 = q/a$ is the root of larger magnitude.

Other cancellations are removed by algebra of the same kind. For large x , for instance, $\sqrt{x+1}-\sqrt{x}$ is better computed as $1/(\sqrt{x+1}+\sqrt{x})$, and for small x , $1-\cos x$ is better computed as $2\sin^2(x/2)$, using the double-angle formula.

The Condition of a Problem

Some problems are sensitive to small changes in their data, however carefully they are solved. Let us consider the evaluation of a function f at a point x that is known only approximately. A small relative change $\Delta x/x$ in the data produces a relative change in the result of

$$\frac{\Delta f}{f} \approx \frac{f'(x) \Delta x}{f(x)} = \left(\frac{xf'(x)}{f(x)} \right) \frac{\Delta x}{x}, \quad (9.9)$$

to first order, by the definition of the derivative. The factor

$$\kappa = \left| \frac{xf'(x)}{f(x)} \right| \quad (9.10)$$

is called the condition number of the problem. It is the factor by which a relative error in the data is magnified in the result. A problem with a condition number of order one is said to be well conditioned, and a problem with a large condition number is said to be ill conditioned. For $f(x) = \sqrt{x}$ the condition number is $\frac{1}{2}$ for every x , and for $f(x) = x - 1$ it is $|x/(x-1)|$, which is large near $x = 1$, since there the result is a small difference of nearly equal numbers.

The roots of a polynomial can be ill conditioned. Consider, for example, the equation $(x - x_*)^2 = \eta$, which for $\eta = 0$ has a double root at x_* , and whose constant term is $x_*^2 - \eta$. Changing that constant term by a small amount $\eta > 0$ moves the roots to $x_* \pm \sqrt{\eta}$, as Figure 79 shows.

The displacement $\sqrt{\eta}$ of each root is much larger than the change η in the coefficient when η is small, and the ratio $1/\sqrt{\eta}$ grows without limit as $\eta \rightarrow 0$. If the constant term carries a round-off error of relative size ϵ_m , so that η is of order $\epsilon_m x_*^2$, the roots are uncertain by an amount of order $\sqrt{\epsilon_m} |x_*|$, and only about half of the digits of the computed roots are correct. The root x_+ of the previous subsection, by contrast, is well conditioned, since $x_+ \approx -c/b$ changes in proportion to c . The large error in its naive computation came

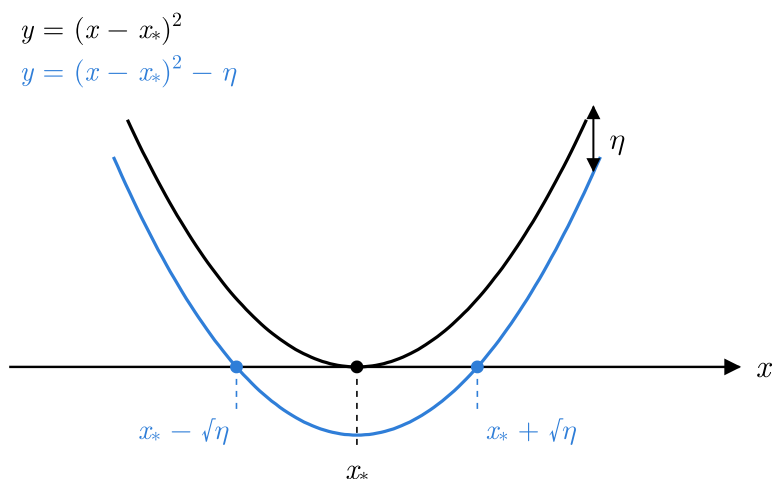


Figure 79. A double root is ill conditioned. Lowering the parabola $y = (x - x_*)^2$ by a small amount η splits the double root at x_* into roots at $x_* \pm \sqrt{\eta}$, which are much further apart than η when η is small.

from the formula used, and no change of formula can remove the error in a double root, which comes from the problem itself.

The Stability of an Algorithm

An algorithm is called numerically stable if the round-off errors introduced at each step are not magnified by much more than the condition number of the problem requires, and unstable if they are. The formula for x_+ with a subtraction is unstable for the problem of the previous subsection, while the algorithm with q is stable. A recurrence relation gives a clearer example, since the same relation can be stable when it is used in one direction and unstable in the other.

Let

$$\mathcal{J}_n = \int_0^1 x^n e^{x-1} dx, \quad n = 0, 1, 2, \dots \quad (9.11)$$

Integrating by parts, with x^n differentiated and e^{x-1} integrated,

gives

$$\mathcal{J}_n = \left[x^n e^{x-1} \right]_0^1 - n \int_0^1 x^{n-1} e^{x-1} dx = 1 - n\mathcal{J}_{n-1}, \quad (9.12)$$

for $n \geq 1$, with $\mathcal{J}_0 = 1 - e^{-1}$. Since e^{x-1} lies between e^{-1} and 1 on the interval, $\mathcal{J}_n$ lies between $1/[e(n+1)]$ and $1/(n+1)$, so the $\mathcal{J}_n$ are positive and decrease slowly towards zero. It seems as though the recurrence could be used to compute every $\mathcal{J}_n$ from $\mathcal{J}_0$. This turns out to fail in practice, and the reason is the way in which errors propagate.

Suppose that the stored value of $\mathcal{J}_0$ has a small error Δ_0 . The recurrence is linear, so the errors obey $\Delta_n = -n\Delta_{n-1}$ and therefore

$$\Delta_n = (-1)^n n! \Delta_0. \quad (9.13)$$

The factorial grows so quickly that, even with Δ_0 of order ϵ_m , the error soon exceeds the value $\mathcal{J}_n$ itself, which is smaller than $1/(n+1)$, and the computed values alternate in sign with growing magnitude. The forward recurrence is unstable. The values $\mathcal{J}_n$ themselves can be computed accurately by another algorithm, as we now show, so the instability belongs to the algorithm.

Now let us run the recurrence backwards, in the form $\mathcal{J}_{n-1} = (1 - \mathcal{J}_n)/n$, starting from a large index n_0 with the crude value $\mathcal{J}_{n_0} = 0$, whose error is at most $1/(n_0 + 1)$. Each backward step divides the error by the index, so the error in $\mathcal{J}_n$, for $n < n_0$, is

$$|\Delta_n| = \frac{|\Delta_{n_0}|}{n_0(n_0 - 1) \cdots (n + 1)}, \quad (9.14)$$

which becomes very small once n_0 is a modest amount larger than n . The backward recurrence is stable, and an error that the forward recurrence multiplies by $n!$ is divided by a similar factor when the recurrence is run the other way.

9.2 Root Finding, Interpolation, and Linear Systems

This section treats three problems that arise often in computation: finding a root x_* of an equation $f(x) = 0$ that cannot be solved in closed form, constructing a polynomial through a set of given points, and solving a system of linear equations, which we solved by hand in Section 4.1 and now treat as an algorithm.

Bisection

Let f be continuous on an interval $u_0 \leq x \leq v_0$, with $f(u_0)$ and $f(v_0)$ of opposite signs. The graph of a continuous function cannot pass from one side of the x -axis to the other without crossing it, a result called the intermediate value theorem, so f has a root in the interval. The method of bisection evaluates f at the midpoint and keeps the half of the interval at whose ends f still has opposite signs, as Figure 80 shows. The steps are as follows:

```

input: f, u, v with f(u) and f(v) of opposite signs, tolerance tau
while (v - u)/2 > tau
    mid = (u + v)/2
    if f(mid) = 0 then return mid
    if f(u) and f(mid) have opposite signs then v = mid
    else u = mid
return (u + v)/2

```

After n halvings the bracket has width $(v_0 - u_0)/2^n$, and its midpoint lies within half that width of the root. The method converges for every continuous function that changes sign on the interval, and it needs only the sign of f ; it gains, however, only one binary digit of accuracy with each evaluation of f .

The speed of an iteration is described by its order of convergence. If the errors $e_n = x_n - x_*$ satisfy

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^p} = K \quad (9.15)$$

for some positive constant K , the iteration is said to converge with order p . Convergence with $p = 1$ and $K < 1$ is called linear, and convergence with $p = 2$ is called quadratic. Bisection, whose bound on the error is halved at each step, is counted as linear with $K = \frac{1}{2}$.

The Newton–Raphson Method

Let x_n be an estimate of the root. The tangent to the curve $y = f(x)$ at x_n crosses the axis at

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad (9.16)$$

which is taken as the next estimate. This is called the Newton–Raphson method, and Figure 81 shows the construction.

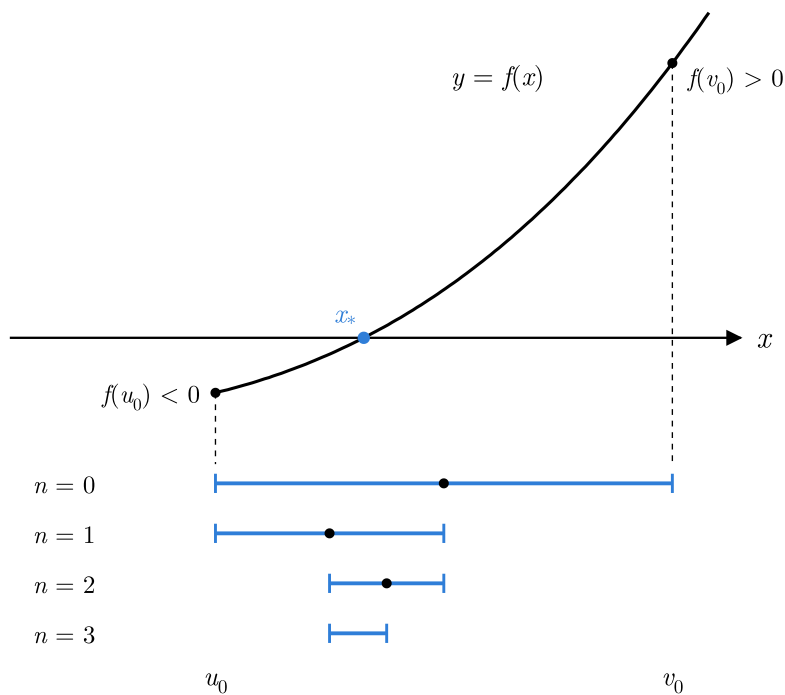


Figure 80. Bisection. The function has opposite signs at u_0 and v_0 . At each step the bracket is halved, keeping the half at whose ends f has opposite signs, and the dots mark the midpoints at which f is evaluated. The brackets close in on the root x_* .

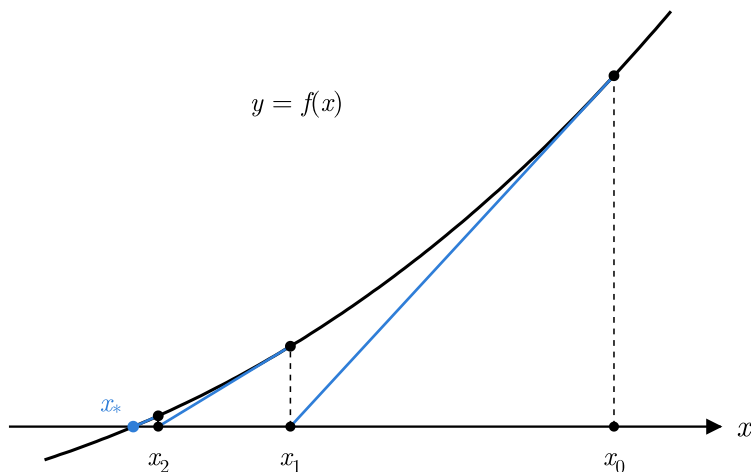


Figure 81. The Newton–Raphson method. From each estimate x_n the tangent to the curve is followed down to the axis, which gives x_{n+1} . The estimates x_0 , x_1 , and x_2 approach the root x_* rapidly.

Taylor’s theorem with the Lagrange form of the remainder, from Section 1.1, applied to f about x_n and evaluated at the root, gives

$$0 = f(x_n) + f'(x_n)(x_* - x_n) + \frac{1}{2}f''(\xi_n)(x_* - x_n)^2, \quad (9.17)$$

where ξ_n lies between x_n and x_* . Dividing by $f'(x_n)$ and using the Newton–Raphson formula in the form $f(x_n)/f'(x_n) = x_n - x_{n+1} = e_n - e_{n+1}$ gives

$$e_{n+1} = \frac{f''(\xi_n)}{2f'(x_n)} e_n^2. \quad (9.18)$$

If $f'(x_*) \neq 0$ and f'' is continuous, the fraction tends to $f''(x_*)/2f'(x_*)$ as the iterates approach the root, and the convergence is quadratic, so that once the error is small the number of correct digits roughly doubles at every step. Consider, for instance, the square root of a positive number C , the positive root of $f(x) = x^2 - C$, for which the method gives

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{C}{x_n} \right), \quad (9.19)$$

which is called Heron's method; you will show in Problem 4 that $e_{n+1} = e_n^2/2x_n$ exactly.

The method needs a starting value close enough to the root, and it can fail in several ways. If $f'(x_n) = 0$, the tangent never meets the axis. If $f'(x_*) = 0$, the root is multiple, and you will show in Problem 2 that the convergence is then only linear. The iterates can also cycle or diverge. For $f(x) = \operatorname{sgn}(x)|x|^{1/2}$, where $\operatorname{sgn}(x)$ is $+1$ for $x > 0$ and -1 for $x < 0$, we have $f/f' = 2x$, so $x_{n+1} = -x_n$ for any starting value, and for $f(x) = x^{1/3}$ we have $f/f' = 3x$, so $x_{n+1} = -2x_n$. The method is therefore used with a limit on the number of steps, as in the following listing, and it is sometimes combined with bisection, which keeps the root within a known bracket:

```
input: f, its derivative df, starting value x, tolerance tau, nmax
for n = 1 to nmax
  if df(x) = 0 then stop (the tangent is horizontal)
  step = f(x)/df(x)
  x = x - step
  if abs(step) < tau then return x
stop (no convergence after nmax steps)
```

The Secant Method

When the derivative is not available, the tangent can be replaced by the straight line through the points of the curve at the two most recent estimates, as in Figure 82, which crosses the axis at

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}. \quad (9.20)$$

This is called the secant method, and it needs one new evaluation of f per step.

To find its order, let f' and f'' denote the derivatives at x_* . To second order in the errors, $f(x_n) \approx f'e_n + \frac{1}{2}f''e_n^2$, and therefore $f(x_n) - f(x_{n-1}) \approx (e_n - e_{n-1})[f' + \frac{1}{2}f''(e_n + e_{n-1})]$. Subtracting x_* from both sides of the secant formula, substituting these expansions, and cancelling the factor $e_n - e_{n-1}$, which equals $x_n - x_{n-1}$,

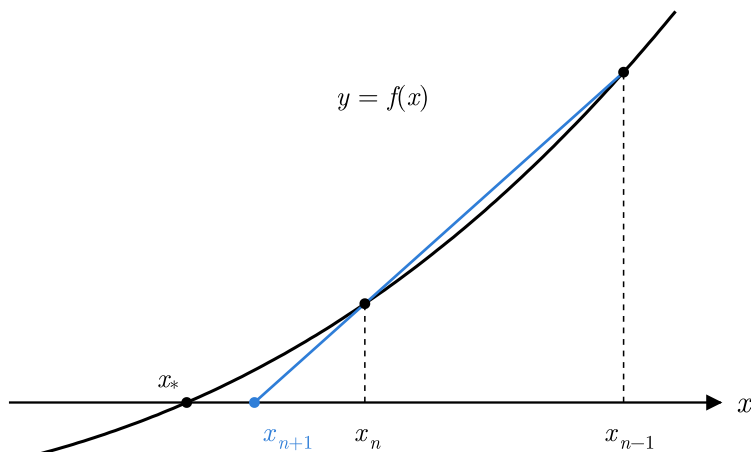


Figure 82. The secant method. The straight line through the points of the curve at x_{n-1} and x_n crosses the axis at x_{n+1} .

gives

$$e_{n+1} = e_n - \frac{e_n \left(1 + \frac{f''}{2f'} e_n \right)}{1 + \frac{f''}{2f'} (e_n + e_{n-1})} \approx \frac{f''}{2f'} e_n e_{n-1}, \quad (9.21)$$

where the denominator has been expanded by the binomial theorem and terms of third order dropped. Let us suppose that $|e_{n+1}| \approx K|e_n|^p$, so that $|e_{n-1}| \approx (|e_n|/K)^{1/p}$. Substituting these relations into the approximation $e_{n+1} \approx (f''/2f') e_n e_{n-1}$ gives $K|e_n|^p \propto |e_n|^{1+1/p}$, and the powers agree only if $p = 1 + 1/p$, whose positive root is

$$p = \frac{1 + \sqrt{5}}{2}, \quad (9.22)$$

a little above $\frac{3}{2}$. The secant method therefore converges more slowly per step than Newton–Raphson. Newton–Raphson, however, needs values of both f and f' at each step, so the secant method is often the faster when f' takes as long to evaluate as f .

Polynomial Interpolation

Let $(x_0, y_0), (x_1, y_1), \dots, (x_N, y_N)$ be $N + 1$ points with distinct x_j , which could, for example, be values $y_j = f(x_j)$ of a function that is slow to evaluate. For each j , let

$$L_j(x) = \prod_{k \neq j} \frac{x - x_k}{x_j - x_k}, \quad (9.23)$$

a polynomial of degree N that equals 1 at x_j and 0 at every other point x_k ; the points $x_0, x_1, \dots, x_N$ are called the nodes of the interpolation. The polynomial

$$q_N(x) = \sum_{j=0}^N y_j L_j(x) \quad (9.24)$$

then passes through all the points, since at $x = x_k$ only the term with $j = k$ is non-zero, and it is called the Lagrange form of the interpolating polynomial. It is the only polynomial of degree at most N through the points, since the difference of two such polynomials would vanish at $N + 1$ nodes, while a non-zero polynomial of degree at most N has at most N roots.

When f has $N + 1$ continuous derivatives, the error of the interpolation is

$$f(x) - q_N(x) = \frac{f^{(N+1)}(\xi)}{(N+1)!} w(x), \quad w(x) = \prod_{j=0}^N (x - x_j), \quad (9.25)$$

for some ξ in the smallest interval containing x and the nodes. To prove this error formula, we fix x at a point other than a node and let

$$g(x') = f(x') - q_N(x') - [f(x) - q_N(x)] \frac{w(x')}{w(x)}. \quad (9.26)$$

The function g vanishes at the $N + 1$ nodes and at $x' = x$. Between two zeros of a differentiable function its derivative has a zero, which is Rolle's theorem, so g' has at least $N + 1$ zeros, g'' at least N , and so on, and $g^{(N+1)}$ vanishes at some point ξ . Since q_N has degree at most N and $w(x') = x'^{N+1} + \dots$, we have $g^{(N+1)}(x') = f^{(N+1)}(x') - [f(x) - q_N(x)] (N+1)!/w(x)$, and setting $x' = \xi$ gives the error formula.

It might be expected that a polynomial through more points would always approximate f better. Figure 83 shows that this is not

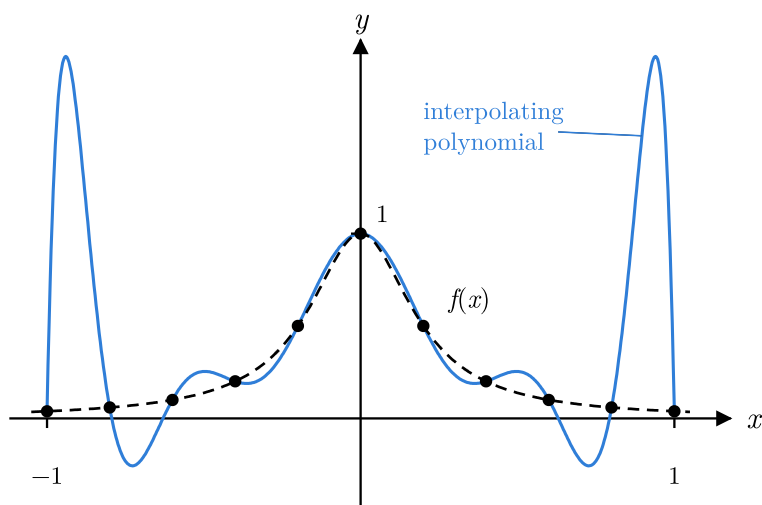


Figure 83. Runge's phenomenon. The dashed curve is $f(x) = 1/(1 + x^2/\beta^2)$ with $\beta = 1/5$, and the solid curve is the polynomial of degree 10 through 11 equally spaced points, marked by dots. The polynomial agrees with f at the nodes and oscillates wildly between the nodes near the ends of the interval.

so for equally spaced nodes. For the function $1/(1 + x^2/\beta^2)$ on $-1 \leq x \leq 1$, with β small, the interpolating polynomial agrees with f near the middle of the interval and develops large oscillations near its ends, which grow as the degree is increased; this is called Runge's phenomenon. The error formula contains the product $w(x)$, which for equally spaced nodes is largest near the ends of the interval, since a point there lies far from most of the nodes. The remedies for Runge's phenomenon are not treated in this course.

Gaussian Elimination as an Algorithm

In Section 4.1 we solved the system $\mathbf{Ax} = \mathbf{b}$ by Gaussian elimination, subtracting multiples of each equation from the equations below it until the matrix is upper triangular, and then solving the equations from the last upwards, which is called back substitution. For an $M \times M$ matrix with entries A_{ij} , the procedure is as follows, where step k removes x_k from the rows below row k :

```

input: M-by-M matrix a, right-hand side b
for k = 1 to M - 1
    find the row imax >= k for which abs(a[imax,k]) is largest
    swap rows k and imax of a, and entries k and imax of b
    for i = k + 1 to M
        l[i,k] = a[i,k]/a[k,k]
        for j = k + 1 to M
            a[i,j] = a[i,j] - l[i,k]*a[k,j]
        b[i] = b[i] - l[i,k]*b[k]
    for i = M down to 1
        x[i] = (b[i] - sum over j > i of a[i,j]*x[j])/a[i,i]
return x

```

The innermost line is carried out for $M - k$ rows and $M - k$ columns at step k , so, by the formula $\sum_{j=1}^{M-1} j^2 = \frac{1}{6}(M-1)M(2M-1)$ for the sum of the first squares, which you should be familiar with from school, for large M the number of multiplications is about

$$\sum_{k=1}^{M-1} (M-k)^2 \approx \frac{M^3}{3}, \quad (9.27)$$

while back substitution needs about $M^2/2$. Cramer's rule, with each determinant expanded along a row as in Section 4.1, needs a number of multiplications that grows as $M!$, far faster than any power of M .

The lines that find the row with the largest entry in column k and swap the rows are called partial pivoting, and the entry A_{kk} that divides the other entries of column k is called the pivot. As an example of their purpose, consider the system

$$\zeta x_1 + x_2 = b_1, \quad x_1 + x_2 = b_2, \quad (9.28)$$

with ζ smaller than ϵ_m and b_1 and b_2 of comparable size, whose solution is close to $x_1 = b_2 - b_1$, $x_2 = b_1$. Without pivoting, the multiplier is $1/\zeta$, and the second equation becomes $(1 - 1/\zeta)x_2 = b_2 - b_1/\zeta$. In floating point the 1 and the b_2 are lost beside the enormous terms, the computed x_2 is b_1 , and back substitution gives $x_1 = (b_1 - x_2)/\zeta = 0$. With the rows swapped, the multiplier is ζ , and back substitution gives $x_2 = b_1$ and $x_1 = b_2 - b_1$, which are correct to within round-off. Pivoting keeps every multiplier l_{ik} no larger than one in magnitude, so round-off errors are not multiplied by large factors.

Let $\mathbf{U}$ be the upper-triangular matrix left at the end, and $\mathbf{L}$ the lower-triangular matrix with ones on its diagonal and the multiplier l_{ik} in row i , column k . Suppose first that no rows are swapped. At step k , row i has l_{ik} times row k subtracted from it, and row k is not changed after step $k - 1$, so the row subtracted is row k of $\mathbf{U}$. Since row i is changed only at steps 1 to $i - 1$,

$$(\text{row } i \text{ of } \mathbf{A}) = \sum_{k < i} l_{ik} (\text{row } k \text{ of } \mathbf{U}) + (\text{row } i \text{ of } \mathbf{U}), \quad (9.29)$$

which is row i of the product $\mathbf{LU}$, and therefore $\mathbf{A} = \mathbf{LU}$. This is called the LU decomposition of $\mathbf{A}$. With pivoting the same argument applies to the matrix with its rows already in their final order, and gives $\mathbf{PA} = \mathbf{LU}$, where $\mathbf{P}$ is the matrix that performs the row swaps. Once the decomposition is known, $\mathbf{Ax} = \mathbf{b}$ is solved by solving $\mathbf{Ly} = \mathbf{b}$ from the top down and then $\mathbf{Ux} = \mathbf{y}$ from the bottom up, each in about $M^2/2$ operations. A system with many right-hand sides, which could, for example, describe the same circuit driven by many different sources, is therefore factorised once and solved for each right-hand side with little further work. Since the determinant of a product is the product of the determinants, and the determinant of a triangular matrix is the product of its diagonal entries, $\det \mathbf{A} = \pm \prod_k u_{kk}$, with the sign changed at each row swap.

Iterative Methods for Large Systems

Some systems have so many unknowns that $M^3/3$ operations are too many, while each equation involves only a few of the unknowns. The equations for the steady temperature at the points of a grid on a plate, for example, each link the temperature at one point to the temperatures at the neighbouring points. Such systems are often solved by iteration. Let $\mathbf{A} = \mathbf{D} + \mathbf{R}$, where $\mathbf{D}$ is the diagonal part of $\mathbf{A}$ and $\mathbf{R}$ the rest. The Jacobi method solves each equation for its diagonal unknown, using the previous estimates $\mathbf{x}^{(r)}$ for the others,

$$x_i^{(r+1)} = \frac{1}{A_{ii}} \left(b_i - \sum_{j \neq i} A_{ij} x_j^{(r)} \right), \quad (9.30)$$

which in matrix form is $\mathbf{D}\mathbf{x}^{(r+1)} = \mathbf{b} - \mathbf{R}\mathbf{x}^{(r)}$. The exact solution satisfies the same equation with $\mathbf{x}$ on both sides, and subtracting that equation from the iteration shows that the errors $\mathbf{e}^{(r)} = \mathbf{x}^{(r)} - \mathbf{x}$ obey

$$\mathbf{e}^{(r+1)} = \mathbf{G}\mathbf{e}^{(r)}, \quad \mathbf{G} = -\mathbf{D}^{-1}\mathbf{R}. \quad (9.31)$$

If $\mathbf{G}$ has a full set of eigenvectors, we may expand $\mathbf{e}^{(0)}$ in them, as in Section 4.3, and each component is multiplied by its eigenvalue at every step, so the errors tend to zero from any starting estimate if every eigenvalue of $\mathbf{G}$ has modulus less than one. A matrix is called strictly diagonally dominant if $|A_{ii}| > \sum_{j \neq i} |A_{ij}|$ in every row, and you will show in Problem 9 that the Jacobi method converges for such a matrix.

The Gauss–Seidel method differs only in using each new value as soon as it has been computed:

```

input: M-by-M matrix a, right-hand side b, starting estimate x, toleran
repeat
  change = 0
  for i = 1 to M
    new = (b[i] - sum over j not equal to i of a[i,j]*x[j])/a[i,i]
    change = max(change, abs(new - x[i]))
    x[i] = new
  until change < tau
return x

```

Because x_j for $j < i$ has already been updated when row i is reached, the method uses the most recent information, and it often converges

faster than the Jacobi method; you will compare the Jacobi and Gauss–Seidel methods for a system of two equations in Problem 8. When each row has only a few non-zero entries, a sweep of either method needs a number of operations proportional to M .

9.3 Numerical Differentiation and Quadrature

A derivative is the limit of a difference quotient, and an integral is the limit of a sum, so derivatives and integrals can both be approximated by a computer using values of the function at a finite set of points. The numerical evaluation of an integral is called quadrature. The questions are how large the error of each approximation is, and how the step size between the points should be chosen. In this section we look first at derivatives, then at integrals, and finally at a method that combines two approximations into a better one.

Finite Differences

Let f have as many continuous derivatives as we need, and let $h > 0$ be a small step. Taylor's theorem with the Lagrange form of the remainder, from Section 1.1, gives

$$f(x+h) = f(x) + hf'(x) + \frac{1}{2}h^2 f''(\xi), \quad (9.32)$$

for some ξ between x and $x+h$, and rearranging it gives

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{1}{2}hf''(\xi). \quad (9.33)$$

The quotient on the left is called the forward difference, and the error term $\frac{1}{2}hf''(\xi)$, which arises from truncating the Taylor series, is called the truncation error. It goes as h , and the forward difference is said to be accurate to first order. The backward difference, $[f(x) - f(x-h)]/h$, is also accurate to first order, as you will show in Problem 1.

Geometrically, the difference quotients are the slopes of chords, as Figure 84 shows, and the chord joining the points at $x-h$ and $x+h$ is much closer in slope to the tangent than either of the other chords. To see why, we expand $f(x \pm h)$ to third order,

$$f(x \pm h) = f(x) \pm hf'(x) + \frac{1}{2}h^2 f''(x) \pm \frac{1}{6}h^3 f'''(\xi_{\pm}), \quad (9.34)$$

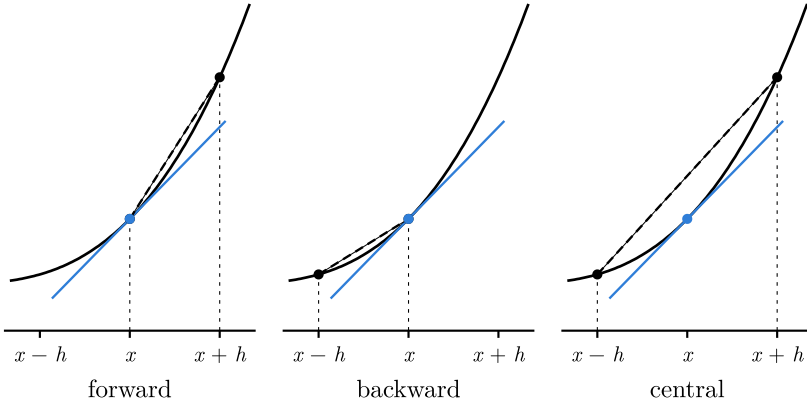


Figure 84. The forward, backward, and central differences at x are the slopes of the dashed chords, and the derivative $f'(x)$ is the slope of the tangent at x , drawn in blue. The central chord is nearly parallel to the tangent, although it does not pass through the point of contact.

with ξ_+ between x and $x+h$ and ξ_- between $x-h$ and x . Subtracting the expansion with the lower sign from the one with the upper sign, the terms in $f(x)$ and $f''(x)$ cancel, and dividing by $2h$ gives

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{1}{6}h^2 f'''(\xi), \quad (9.35)$$

which is called the central difference. Here we have used the intermediate value theorem to replace the mean of $f'''(\xi_+)$ and $f'''(\xi_-)$ by $f'''(\xi)$ at a single point between them. The truncation error goes as h^2 , and the central difference is accurate to second order. Adding the expansions, taken one order further, gives in a similar way the second difference

$$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + \frac{1}{12}h^2 f'''(\xi), \quad (9.36)$$

which you will derive in Problem 2.

Choosing the Step Size

The truncation error suggests making h as small as possible. This turns out to be a mistake, because the values of f are themselves computed with round-off errors. Let us suppose that each computed value of f near x carries a relative error of at most ϵ_m , so that its absolute error is at most $\epsilon_m|f|$; a function computed by a long

calculation may carry a larger error, and ϵ_m is then replaced by that relative error. In the central difference the errors in $f(x+h)$ and $f(x-h)$ may add, and dividing by $2h$ gives a round-off error of at most $\epsilon_m|f|/h$, which grows as h is reduced. The total error of the central difference is therefore bounded by

$$\mathcal{E}(h) = Ah^2 + \frac{B\epsilon_m}{h}, \quad A = \frac{1}{6}|f'''|, \quad B = |f|. \quad (9.37)$$

The first term falls and the second rises as h decreases, and Figure 85 shows the result on logarithmic axes. Setting $d\mathcal{E}/dh = 2Ah - B\epsilon_m/h^2 = 0$ gives the best step,

$$h_{\text{opt}} = \left(\frac{B\epsilon_m}{2A} \right)^{1/3} = \left(\frac{3\epsilon_m|f|}{|f'''|} \right)^{1/3}, \quad (9.38)$$

at which the round-off term $B\epsilon_m/h_{\text{opt}}$ equals $2Ah_{\text{opt}}^2$, so that the smallest total error is $\mathcal{E}(h_{\text{opt}}) = 3Ah_{\text{opt}}^2$.

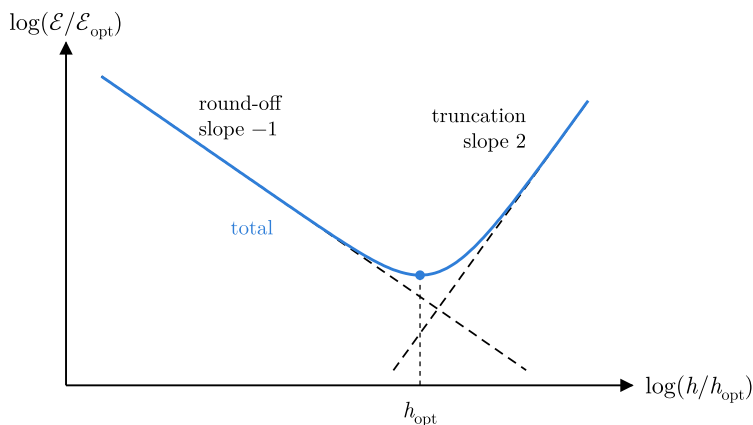


Figure 85. The bound $\mathcal{E}(h) = Ah^2 + B\epsilon_m/h$ on the total error of the central difference, on logarithmic axes, plotted as $\mathcal{E}/\mathcal{E}(h_{\text{opt}}) = (\hat{h}^2 + 2/\hat{h})/3$ against $\hat{h} = h/h_{\text{opt}}$. The dashed lines are the round-off term, of slope -1 , and the truncation term, of slope 2 . Reducing h below h_{opt} increases the error.

When f and its derivatives are of comparable size, h_{opt} is of order $\epsilon_m^{1/3}$ and the smallest error of order $\epsilon_m^{2/3}$, so the central difference gives about two-thirds of the digits that the computer holds. A

similar calculation for the forward difference, which you will carry out in Problem 3, gives a best step of order $\epsilon_m^{1/2}$ and an error of the same order, only half of the digits. On the logarithmic axes of Figure 85 the round-off and truncation terms appear as straight lines, and their slopes, -1 and 2 , can be used to recognise which error dominates when a computed derivative is examined as h is varied.

The Trapezium Rule

We now turn to integrals. Let $I = \int_{x_0}^{x_N} f(x) dx$. We divide the interval into N strips of width $h = (x_N - x_0)/N$, with nodes $x_j = x_0 + jh$ and values $f_j = f(x_j)$. On each strip we replace the curve by the straight line through its end points, as in the left panel of Figure 86, and the area of each trapezium is $\frac{1}{2}h(f_j + f_{j+1})$. Adding the areas gives

$$T(h) = h \left(\frac{1}{2}f_0 + f_1 + f_2 + \cdots + f_{N-1} + \frac{1}{2}f_N \right), \quad (9.39)$$

which is called the trapezium rule.

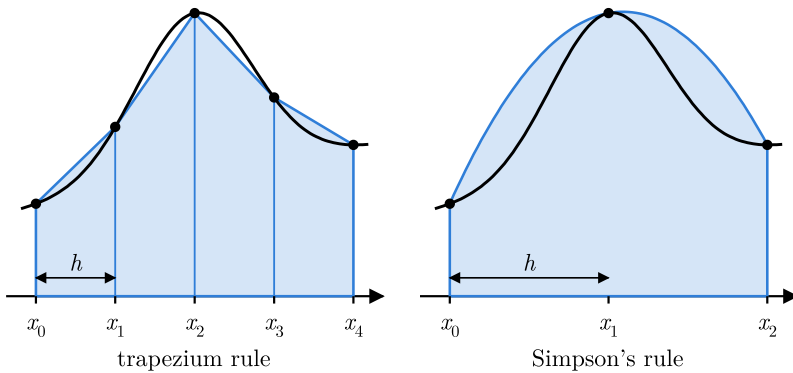


Figure 86. The trapezium rule replaces the curve on each strip of width h by a straight line (left). Simpson's rule replaces the curve on each pair of strips by the parabola through its three points (right). The shaded areas are the approximations to the integral.

Let us find the error on a single strip, which we centre on a point z , so that it runs from $z - \frac{1}{2}h$ to $z + \frac{1}{2}h$. Expanding f in a Taylor

series about z and integrating term by term, the odd powers of $x - z$ integrate to zero over the symmetric interval, and

$$\int_{z-h/2}^{z+h/2} f(x) dx = hf(z) + \frac{h^3}{24} f''(z) + \dots \quad (9.40)$$

The same expansion evaluated at the ends of the strip gives $\frac{1}{2}h[f(z - \frac{1}{2}h) + f(z + \frac{1}{2}h)] = hf(z) + \frac{1}{8}h^3 f''(z) + \dots$, and the difference between the exact integral and the trapezium approximation is the error of the single strip,

$$\int_{z-h/2}^{z+h/2} f(x) dx - \frac{1}{2}h [f(z - \frac{1}{2}h) + f(z + \frac{1}{2}h)] = -\frac{h^3}{12} f''(z) + \dots \quad (9.41)$$

Adding the errors of the N strips gives $-\frac{1}{12}h^2 \sum_j f''(z_j)$, where z_j is the centre of strip j , and the sum is an approximation to $\int f'' dx$, since it adds values of f'' multiplied by the strip width. The error of the trapezium rule is therefore

$$I - T(h) \approx -\frac{h^2}{12} [f'(x_N) - f'(x_0)] , \quad (9.42)$$

to leading order in h , so halving the step reduces the error by a factor of about four.

Simpson's Rule

A better approximation replaces the curve on each pair of adjacent strips by the parabola through its three points, as in the right panel of Figure 86. Let the pair of strips be centred on z , with points at $z - h$, z , and $z + h$, and let the parabola be $\alpha_0 + \alpha_1(x - z) + \alpha_2(x - z)^2$, with constants α_0 , α_1 , and α_2 to be determined. The integral of the parabola over the pair is $2h\alpha_0 + \frac{2}{3}h^3\alpha_2$, since the term in α_1 is odd about z . The parabola passes through the three points, so $\alpha_0 = f(z)$ and $f(z + h) + f(z - h) = 2\alpha_0 + 2h^2\alpha_2$, and substituting these conditions gives

$$\int_{z-h}^{z+h} f(x) dx \approx \frac{h}{3} [f(z - h) + 4f(z) + f(z + h)] . \quad (9.43)$$

Applying this formula to each pair of strips, with N even, gives

Simpson's rule,

$$S(h) = \frac{h}{3} (f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 2f_{N-2} + 4f_{N-1} + f_N) , \quad (9.44)$$

in which the nodes at the centres of the pairs have weight 4 and the nodes shared by two pairs have weight 2. The rule is written as an algorithm as follows:

```

input: f, x0, xN, N (even)
h = (xN - x0)/N
s = f(x0) + f(xN)
for j = 1 to N - 1
    if j is odd then s = s + 4*f(x0 + j*h)
    else s = s + 2*f(x0 + j*h)
return s*h/3
    
```

The error is found as for the trapezium rule. Integrating the Taylor series of f about z over the pair of strips gives $2hf + \frac{1}{3}h^3f'' + \frac{1}{60}h^5f''' + \dots$, with the derivatives evaluated at z , while Simpson's formula for the pair gives $2hf + \frac{1}{3}h^3f'' + \frac{1}{36}h^5f''' + \dots$. The terms in f''' vanish from the exact integral and from Simpson's formula by symmetry, so the formula is exact for cubic polynomials, and the error of the pair is

$$\int_{z-h}^{z+h} f(x) dx - \frac{h}{3} [f(z-h) + 4f(z) + f(z+h)] = -\frac{h^5}{90} f'''(z) + \dots . \quad (9.45)$$

Adding the errors of the $N/2$ pairs, each of width $2h$, gives

$$I - S(h) \approx -\frac{h^4}{180} [f'''(x_N) - f'''(x_0)] , \quad (9.46)$$

so halving the step reduces the error by a factor of about sixteen.

Richardson Extrapolation

The error formulas above tell us how an error changes with h , even when its size is unknown, and this knowledge can be used to remove the leading error. Suppose that an approximation $Q(h)$ to a quantity Q_* has the form

$$Q(h) = Q_* + \gamma h^p + \text{terms of higher order in } h , \quad (9.47)$$

with a known power p and an unknown constant γ . Then $Q(h/2) = Q_* + \gamma h^p/2^p + \dots$, and multiplying $Q(h/2)$ by 2^p and subtracting

$Q(h)$ removes the term in γ , which gives

$$Q_* = \frac{2^p Q(h/2) - Q(h)}{2^p - 1} + \text{terms of higher order in } h. \quad (9.48)$$

This combination is called Richardson extrapolation. The same two values also estimate the error, since subtracting the expansion of $Q(h/2)$ from that of $Q(h)$ gives $Q(h) - Q(h/2) \approx \gamma h^p(1 - 2^{-p})$, and therefore $Q(h/2) - Q_* \approx [Q(h) - Q(h/2)]/(2^p - 1)$.

As an illustration, consider the central difference $D(h) = [f(x + h) - f(x - h)]/2h$. It is unchanged when h is replaced by $-h$, so its Taylor series contains only even powers of h , and it has the form above with $p = 2$, the next term being of order h^4 . Richardson extrapolation gives

$$\begin{aligned} & \frac{4D(h/2) - D(h)}{3} \\ &= \frac{8[f(x + \frac{1}{2}h) - f(x - \frac{1}{2}h)] - [f(x + h) - f(x - h)]}{6h}, \end{aligned} \quad (9.49)$$

whose error goes as h^4 . The trapezium rule also has an error with $p = 2$, and you will show in Problem 7 that extrapolating it gives Simpson's rule. For a smooth function, the error of the trapezium rule contains only even powers of h , and extrapolation can then be repeated to remove the terms in h^4 , h^6 , and so on, which is called Romberg integration; the proof that only even powers occur is beyond the scope of this course.

Problems

§9.1 Floating-Point Arithmetic and Numerical Stability

1. Show that there are 2^{t-1} floating-point numbers x with $2^E \leq x < 2^{E+1}$, equally spaced by $2^E \epsilon_m$, and hence that the spacing near x is between $\frac{1}{2} \epsilon_m |x|$ and $\epsilon_m |x|$.
2. Using the rule that a result exactly halfway between two floating-point numbers is rounded to the one whose last binary digit is 0, show that $(1 + \frac{1}{2} \epsilon_m) + \frac{1}{2} \epsilon_m$ and $1 + (\frac{1}{2} \epsilon_m + \frac{1}{2} \epsilon_m)$ give different floating-point results, so that floating-point addition is not associative.

3. Find the condition numbers of $f(x) = e^x$, $f(x) = \ln x$, and $f(x) = \sin x$, and say where each problem is ill conditioned.
4. Show that $\sqrt{x+1} - \sqrt{x} = 1/(\sqrt{x+1} + \sqrt{x})$ and that $1 - \cos x = 2\sin^2(x/2)$, and explain why the right-hand sides are better for computation when x is large and when x is small respectively.
5. Let $Y_n = \int_0^1 x^n/(x+\Omega) dx$, where $\Omega > 1$ is a constant. Show that $Y_n = 1/n - \Omega Y_{n-1}$, and find how an error in Y_0 is propagated by this recurrence. In which direction should the recurrence be run?
6. In the backward recurrence for $\mathcal{J}_n$, show that starting from $\mathcal{J}_{n_0} = 0$ with $n_0 = n + k$ reduces the error in $\mathcal{J}_n$ to at most $1/[(n+1)^k(n_0+1)]$.

§9.2 Root Finding, Interpolation, and Linear Systems

1. Show that bisection guarantees an error no larger than τ after n steps provided that $n \geq \log_2[(v_0 - u_0)/\tau] - 1$.
2. Let $f(x) = (x - x_*)^m \psi(x)$, where $m \geq 2$ is an integer and $\psi(x_*) \neq 0$. Show that the Newton–Raphson method converges linearly, with $e_{n+1} \approx (1 - 1/m)e_n$, and that the modified iteration $x_{n+1} = x_n - m f(x_n)/f'(x_n)$ converges quadratically.
3. Show that the Newton–Raphson method applied to $f(x) = 1/x - C$ gives $x_{n+1} = x_n(2 - Cx_n)$, an iteration for $1/C$ that requires no division. Show that $1 - Cx_{n+1} = (1 - Cx_n)^2$, and find the starting values from which the iteration converges.
4. For Heron’s method, show that $e_{n+1} = e_n^2/2x_n$, where $e_n = x_n - \sqrt{C}$, and that $x_n \geq \sqrt{C}$ for every $n \geq 1$ for any positive starting value.
5. Show that $\sum_{j=0}^N L_j(x) = 1$ for every x .
6. Show that the error of linear interpolation between x_0 and $x_1 = x_0 + h$ is at most $\frac{1}{8}h^2 \max |f''|$ on that interval.
7. Find the LU decomposition of the matrix A with rows (A_{11}, A_{12}) and (A_{21}, A_{22}) , assuming $A_{11} \neq 0$, and check that $\det A = u_{11}u_{22}$.

8. For a system of two equations with matrix entries A_{ij} , show that the Jacobi iteration matrix has eigenvalues $\pm\sqrt{\Lambda}$, where $\Lambda = A_{12}A_{21}/(A_{11}A_{22})$, and that one sweep of the Gauss–Seidel method multiplies the error in x_2 by Λ . Deduce that when $|\Lambda| < 1$, one Gauss–Seidel sweep reduces the error as much as two Jacobi sweeps.
9. Let A be strictly diagonally dominant, let Γ be the largest of the ratios $\sum_{j \neq i} |A_{ij}|/|A_{ii}|$, and let $e_{\max}^{(r)}$ be the largest of the $|e_i^{(r)}|$. Show that $e_{\max}^{(r+1)} \leq \Gamma e_{\max}^{(r)}$ for the Jacobi method, and hence that the method converges.

§9.3 Numerical Differentiation and Quadrature

1. Show that $[f(x) - f(x - h)]/h = f'(x) - \frac{1}{2}hf''(\xi)$ for some ξ between $x - h$ and x .
2. Derive the second difference formula and its error term $\frac{1}{12}h^2f'''(\xi)$ by expanding $f(x \pm h)$ to fourth order.
3. For the forward difference, with round-off error at most $2\epsilon_m|f|/h$, show that the total error bound is smallest when $h = 2(\epsilon_m|f/f''|)^{1/2}$, and that the smallest bound is $2(\epsilon_m|ff''|)^{1/2}$.
4. Show that the trapezium rule is exact for linear functions, and that for $f(x) = x^2$ its error $I - T(h)$ is exactly $-\frac{1}{6}h^2(x_N - x_0)$. Compare this result with the leading-order error formula.
5. The midpoint rule approximates the integral over a strip by $hf(z)$, where z is the centre of the strip. Show that its error on the strip is $\frac{1}{24}h^3f''(z)$ to leading order. If T is the trapezium approximation on the same strip, show that $\frac{1}{3}[T + 2hf(z)]$ is Simpson's formula for the pair of half-strips.
6. Apply Simpson's formula to $\int_{-h}^h x^4 dx$, and check that its error agrees with $-h^5 f'''/90$.
7. Show that $\frac{1}{3}[4T(h/2) - T(h)] = S(h/2)$, where $T(h)$ is the trapezium rule with step h and $S(h/2)$ is Simpson's rule with step $h/2$.

Solutions

Chapter 1. Foundations of Mathematical Language

§1.1 Series, Limits, and Convergence

1. The ratio of successive moduli is $(n+1)|x|/n$, which tends to $|x|$, so the series converges for $|x| < 1$ and diverges for $|x| > 1$. At $x = \pm 1$ the terms have modulus n , which does not tend to zero.
2. For $\sum n! x^n$, $|c_n/c_{n+1}| = 1/(n+1) \rightarrow 0$, so $R = 0$. For $\sum x^n/n^2$, $|c_n/c_{n+1}| = (n+1)^2/n^2 \rightarrow 1$, so $R = 1$, and at $x = \pm 1$ the series converges absolutely by comparison with $\sum 1/n^2$.
3. For $p \geq 2$, $1/n^p \leq 1/n^2$, and $\sum 1/n^2$ converges. For $p \leq 1$, $1/n^p \geq 1/n$, and the harmonic series diverges.
4. Let $f(x) = \ln(1+x)$. Then $f^{(n)}(x) = (-1)^{n+1}(n-1)!/(1+x)^n$, which gives the coefficients $(-1)^{n+1}/n$. With $a = 0$ and $x = 1$ the remainder is $E_N(1) = (-1)^N/[(N+1)(1+\xi)^{N+1}]$ with $0 < \xi < 1$, so $|E_N(1)| < 1/(N+1)$, which tends to zero, and the partial sums at $x = 1$ tend to $\ln 2$.
5. The even partial sums increase towards S , and the odd partial sums $S_{2N+1} = S_{2N+2} + b_{2N+2}$ decrease towards S , since $S_{2N+3} - S_{2N+1} = -(b_{2N+2} - b_{2N+3}) \leq 0$. Every even partial sum is therefore at most S and every odd one at least S , so S lies between S_N and S_{N+1} , and $|S - S_N| \leq |S_{N+1} - S_N| = b_{N+1}$.
6. The Maclaurin series gives $(1 - e^{-\lambda t})/\lambda = t - \lambda t^2/2 + \dots$, which

tends to t . With l'Hôpital's rule, the derivatives of the numerator and the denominator with respect to λ are $te^{-\lambda t}$ and 1, whose ratio tends to t .

7. With $f(x) = (1+x)^{1/2}$, Taylor's theorem with $N = 1$ gives $\sqrt{1+x} = 1 + x/2 + f''(\xi)x^2/2$, where $f''(\xi) = -\frac{1}{4}(1+\xi)^{-3/2}$ and $\xi > 0$. Since $(1+\xi)^{-3/2} < 1$, the error is less than $x^2/8$.

§1.2 Complex Numbers and Phasors

1. The real part of $(\cos \theta + i \sin \theta)^4$ is $\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$, and substituting $\sin^2 \theta = 1 - \cos^2 \theta$ gives $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$. For the second part, $\sin^4 \theta = [(e^{i\theta} - e^{-i\theta})/2i]^4 = (e^{4i\theta} - 4e^{2i\theta} + 6 - 4e^{-2i\theta} + e^{-4i\theta})/16$, which is $(\cos 4\theta - 4 \cos 2\theta + 3)/8$.
2. Since $-1 = e^{i\pi}$, the solutions are $z = e^{i\pi/4} e^{ik\pi/2}$ for $k = 0, 1, 2, 3$, that is $e^{\pm i\pi/4}$ and $e^{\pm 3i\pi/4}$, at the corners of a square on the unit circle. Pairing each with its conjugate, $(z - e^{i\pi/4})(z - e^{-i\pi/4}) = z^2 - 2z \cos(\pi/4) + 1 = z^2 - \sqrt{2}z + 1$, and similarly the other pair gives $z^2 + \sqrt{2}z + 1$. The product of these quadratics is $(z^2 + 1)^2 - 2z^2 = z^4 + 1$.
3. By the sum of an arithmetic series, the product is $\zeta^{n(n-1)/2} = e^{i\pi(n-1)} = (-1)^{n+1}$.
4. Since $|-i| = 1$ and $\arg(-i) = -\pi/2$, $\ln(-i) = i(-\pi/2 + 2\pi k)$, with principal value $-i\pi/2$. For the power, $\ln(1+i) = \frac{1}{2} \ln 2 + i(\pi/4 + 2\pi k)$, and so $(1+i)^i = e^{i \ln(1+i)} = e^{-\pi/4 - 2\pi k} e^{\frac{i}{2} \ln 2}$, with principal value $e^{-\pi/4} e^{\frac{i}{2} \ln 2}$.
5. By the quotient rule and $\cosh^2 x - \sinh^2 x = 1$, $d(\tanh x)/dx = (\cosh^2 x - \sinh^2 x)/\cosh^2 x = 1/\cosh^2 x$. If $x = \sinh w$, then $e^{2w} - 2xe^w - 1 = 0$, a quadratic equation for e^w whose only positive root is $x + \sqrt{x^2 + 1}$, so $w = \ln(x + \sqrt{x^2 + 1})$.
6. The complex amplitude of the sum is $A \sum_{n=0}^{N-1} e^{in\delta} = A(1 - e^{iN\delta})/(1 - e^{i\delta})$, provided that δ is not a multiple of 2π ; when it is, every term is A and the amplitude is NA . Taking $e^{iN\delta/2}$ out of the numerator and $e^{i\delta/2}$ out of the denominator, and using the exponential form of the sine, gives $A e^{i(N-1)\delta/2} \sin(N\delta/2)/\sin(\delta/2)$, whose modulus is the stated amplitude.

7. Write $x_1 = \frac{1}{2}(C_1 e^{i\omega t} + C_1^* e^{-i\omega t})$, and similarly for x_2 . The product contains terms in $e^{\pm 2i\omega t}$, which average to zero over a period, and the constant terms $\frac{1}{4}(C_1 C_2^* + C_1^* C_2) = \frac{1}{2} \operatorname{Re}(C_1 C_2^*)$. Setting $C_2 = C_1$ gives $\frac{1}{2}|C_1|^2$.
8. At $t = 0$, $x_0 = \operatorname{Re} C$ and $v_0 = \operatorname{Re}(i\omega C) = -\omega \operatorname{Im} C$. Therefore $C = x_0 - iv_0/\omega$, and the amplitude is $|C| = \sqrt{x_0^2 + v_0^2/\omega^2}$.

Chapter 2. Calculus of Change and Accumulation

§2.1 Differentiation and Integration

1. Differentiating with respect to x gives $2x/a^2 + (2y/b^2) dy/dx = 0$, so the slope at (x_0, y_0) is $-b^2 x_0/(a^2 y_0)$. The tangent is $y - y_0 = -b^2 x_0(x - x_0)/(a^2 y_0)$. Multiplying by y_0/b^2 and rearranging gives $xx_0/a^2 + yy_0/b^2 = x_0^2/a^2 + y_0^2/b^2$, and the right-hand side is 1 because the point lies on the ellipse.
2. Let $y = \arctan x$, so $x = \tan y$ and $1 = \sec^2 y dy/dx$. Since $\sec^2 y = 1 + \tan^2 y = 1 + x^2$, the derivative is $1/(1+x^2)$. Similarly, if $y = \tanh^{-1} x$, then $1 = (1/\cosh^2 y) dy/dx$, using the derivative of $\tanh x$ that you found in Problem 5 of Section 1.2. Dividing $\cosh^2 y - \sinh^2 y = 1$ by $\cosh^2 y$ gives $1/\cosh^2 y = 1 - \tanh^2 y = 1 - x^2$, so the derivative is $1/(1-x^2)$.
3. Each derivative of $\sin x$ advances its argument by $\pi/2$, since $d \sin x/dx = \cos x = \sin(x + \pi/2)$, so the k th derivative is $\sin(x + k\pi/2)$. With $f = x$, only the terms $k = 0$ and $k = 1$ of Leibniz's rule are non-zero, giving $x \sin(x + n\pi/2) + n \cdot 1 \cdot \sin(x + (n-1)\pi/2)$.
4. Write $1/[x(x+a)^2] = A/x + B/(x+a) + C/(x+a)^2$, so that $1 = A(x+a)^2 + Bx(x+a) + Cx$. Setting $x = 0$ gives $A = 1/a^2$, setting $x = -a$ gives $C = -1/a$, and the coefficient of x^2 gives $A + B = 0$, so $B = -1/a^2$. Integrating term by term gives $(1/a^2) \ln|x/(x+a)| + 1/[a(x+a)] + \text{constant}$.
5. Multiplying the reduction formula by nK_{n-1} gives $nK_n K_{n-1} = (n-1)K_{n-1} K_{n-2}$, so the quantity $nK_n K_{n-1}$ takes the same value for every $n \geq 1$. For $n = 1$ it is $K_1 K_0 = \pi/2$, and therefore $K_n K_{n-1} = \pi/(2n)$.
6. On $0 \leq x \leq 1$ the integrand is finite, so only the range from

1 to infinity matters. There, $1/(1+x^p) < x^{-p}$, whose integral converges for $p > 1$. For $p \leq 1$ and $x \geq 1$ we have $x^p \leq x$, so $1/(1+x^p) \geq 1/(1+x) \geq 1/(2x)$, whose integral diverges. For the second integral, integrating by parts with $\ln x$ differentiated gives $\int_{\delta}^1 \ln x \, dx = [x \ln x - x]_{\delta}^1 = -1 - \delta \ln \delta + \delta$. With $\delta = e^{-u}$, so that $u \rightarrow \infty$ as $\delta \rightarrow 0$, the product $\delta \ln \delta$ is $-ue^{-u}$, and since $e^u > u^2/2$ for $u > 0$, as in the argument for J_n in the text, its modulus is less than $2/u$, which tends to zero. The integral is therefore -1 .

§2.2 Partial Differentiation

1. $\partial f/\partial x = yx^{y-1}$ and, writing $f = e^{y \ln x}$, $\partial f/\partial y = x^y \ln x$. Differentiating yx^{y-1} with respect to y and $x^y \ln x$ with respect to x both give $x^{y-1}(1 + y \ln x)$.
2. Here $\partial A/\partial y = \cos x = \partial B/\partial x$. Integrating A with respect to x gives $\mathcal{A} = y \sin x + x^2$, then $\chi'(y) = B - \partial \mathcal{A}/\partial y = 3y^2$, and $f = y \sin x + x^2 + y^3$, up to a constant.
3. With $\partial f/\partial s = 2s \sin 2\phi$ and $\partial f/\partial \phi = 2s^2 \cos 2\phi$, the formulas give $\partial f/\partial x = 2s(\cos \phi \sin 2\phi - \sin \phi \cos 2\phi) = 2s \sin \phi = 2y$ and $\partial f/\partial y = 2s(\sin \phi \sin 2\phi + \cos \phi \cos 2\phi) = 2s \cos \phi = 2x$, using the formulas for the sine and cosine of a difference.
4. From $x = (z/y)^{1/2}$, $y = z/x^2$, and $z = x^2 y$, the derivatives are $(\partial x/\partial y)_z = -x/(2y)$, $(\partial y/\partial z)_x = 1/x^2$, and $(\partial z/\partial x)_y = 2xy$, whose product is -1 .
5. The conditions are $f_x = y(a - 2x - y) = 0$ and $f_y = x(a - x - 2y) = 0$, which give the points $(0, 0)$, $(a, 0)$, $(0, a)$, and $(a/3, a/3)$. With $f_{xx} = -2y$, $f_{yy} = -2x$, and $f_{xy} = a - 2x - 2y$, the first three points have $f_{xx}f_{yy} - f_{xy}^2 = -a^2$ and are saddle points. At $(a/3, a/3)$, $f_{xx}f_{yy} - f_{xy}^2 = a^2/3 > 0$ and $f_{xx} < 0$, so the point is a maximum, where $f = a^3/27$.
6. We make xyz stationary subject to $g = 2(xy + yz + zx) - A_0 = 0$. The conditions are $yz = 2\lambda(y + z)$, $xz = 2\lambda(x + z)$, and $xy = 2\lambda(x + y)$. Multiplying the first condition by x , the second by y , and the third by z gives $xyz = 2\lambda(xy + xz) = 2\lambda(xy + yz) = 2\lambda(xz + yz)$, so $xz = yz$ and $xy = xz$, which give $x = y = z$.

7. The integrand vanishes at the upper limit $x = t$, and its derivative with respect to t is $\kappa(x)$, so Leibniz's rule gives $dF/dt = \int_0^t \kappa(x) dx$, and the fundamental theorem of calculus gives $d^2F/dt^2 = \kappa(t)$.
8. The derivative of the integrand with respect to α is $e^{-x} \cos \alpha x$, whose modulus is at most e^{-x} , so differentiating under the integral sign is allowed, and $dF/d\alpha = \operatorname{Re} \int_0^\infty e^{-(1-i\alpha)x} dx$. The derivative of $e^{-(1-i\alpha)x} = e^{-x} e^{i\alpha x}$ is $-(1-i\alpha)$ times itself, by the product rule and the derivative $de^{i\alpha x}/dx = i\alpha e^{i\alpha x}$ found in Section 1.2, and its modulus e^{-x} tends to zero, so the integral is $1/(1-i\alpha) = (1+i\alpha)/(1+\alpha^2)$, whose real part is $1/(1+\alpha^2)$. Integrating, with $F(0) = 0$, gives $F(\alpha) = \arctan \alpha$.

§2.3 Multiple Integrals and Change of Variables

1. The region is the triangle $0 \leq x \leq y \leq a$, above the line $y = x$. Taking horizontal strips, y runs from 0 to a and x from 0 to y , so the integral is $\int_0^a e^{-y^2} dy \int_0^y dx = \int_0^a y e^{-y^2} dy = \left[-\frac{1}{2} e^{-y^2}\right]_0^a = \frac{1}{2}(1 - e^{-a^2})$.
2. The Jacobian is $\partial(x, y)/\partial(\xi, \eta) = a \cdot b - 0 \cdot 0 = ab$, and the ellipse becomes the disc $\xi^2 + \eta^2 \leq 1$, whose area is π . The area of the ellipse is therefore πab .
3. The vectors are $\partial \mathbf{r}/\partial r = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}$, $\partial \mathbf{r}/\partial \theta = r(\cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}})$, and $\partial \mathbf{r}/\partial \phi = r \sin \theta(-\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}})$. The vector product $(\partial \mathbf{r}/\partial \theta) \times (\partial \mathbf{r}/\partial \phi)$, worked out component by component, is $r^2 \sin \theta (\sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}})$, using $\cos^2 \phi + \sin^2 \phi = 1$ in the z -component. Its scalar product with $\partial \mathbf{r}/\partial r$ is $r^2 \sin \theta (\sin^2 \theta + \cos^2 \theta) = r^2 \sin \theta$.
4. In cylindrical coordinates with the axis along z , $d_\perp = s$, so $I = \rho \int_0^a s^3 ds \int_0^{2\pi} d\phi \int_0^H dz = \rho \cdot (a^4/4) \cdot 2\pi H$. The mass is $M = \pi a^2 H \rho$, so $I = \frac{1}{2} M a^2$.
5. The mass is $M = 4\pi \rho_0 \int_0^a (1 - r/a) r^2 dr = 4\pi \rho_0 (a^3/3 - a^3/4) = \pi \rho_0 a^3/3$. With the diameter along the z -axis, $I = \rho_0 \int_0^a (1 - r/a) r^4 dr \int_0^\pi \sin^3 \theta d\theta \int_0^{2\pi} d\phi = \rho_0 (a^5/5 - a^5/6) \cdot (4/3) \cdot 2\pi =$

$4\pi\rho_0a^5/45$. Dividing by $Ma^2 = \pi\rho_0a^5/3$ gives $I/Ma^2 = 12/45 = 4/15$. The density is largest at the centre, so more of the mass lies close to the axis than in a uniform sphere, and the moment of inertia is smaller.

6. The volume is $\int_0^a dx \int_0^{b(1-x/a)} dy \int_0^{c(1-x/a-y/b)} dz$. The integral over z gives $c(1-x/a-y/b)$. Integrating this over y at fixed x gives $c[(1-x/a)y - y^2/2b]$ at $y = b(1-x/a)$, which is $\frac{1}{2}bc(1-x/a)^2$. Finally, $\frac{1}{2}bc \int_0^a (1-x/a)^2 dx = \frac{1}{2}bc \cdot a/3 = abc/6$.
7. Completing the square, $-\alpha x^2 + \beta x = -\alpha(x - \beta/2\alpha)^2 + \beta^2/4\alpha$. The substitution $x' = x - \beta/2\alpha$ leaves the limits unchanged, so the integral is $e^{\beta^2/4\alpha} \int_{-\infty}^{\infty} e^{-\alpha x'^2} dx' = \sqrt{\pi/\alpha} e^{\beta^2/4\alpha}$. Differentiating $\int x^2 e^{-\alpha x^2} dx = \frac{1}{2}\sqrt{\pi} \alpha^{-3/2}$ once more with respect to α gives $-\int x^4 e^{-\alpha x^2} dx = -\frac{3}{4}\sqrt{\pi} \alpha^{-5/2}$, which is the stated result.

Chapter 3. Vector Calculus and Field Geometry

§3.1 Gradient, Divergence, and Curl

1. The ellipsoid is a level surface of $f = x^2/a_1^2 + y^2/a_2^2 + z^2/a_3^2$, and the stated vector is $\frac{1}{2}\nabla f$. For a sphere of radius a_1 it is $\mathbf{r}/a_1^2$, which is radial.
2. With $\nabla f = yz\hat{\mathbf{x}} + xz\hat{\mathbf{y}} + xy\hat{\mathbf{z}}$, the directional derivative is $\hat{\mathbf{r}} \cdot \nabla f = 3xyz/r$. A step $d\ell$ along $\hat{\mathbf{r}}$ multiplies each coordinate by $1 + d\ell/r$, and so multiplies f by $(1 + d\ell/r)^3 \approx 1 + 3d\ell/r$.
3. From the text, $\nabla(r^n) = nr^{n-1}\hat{\mathbf{r}} = nr^{n-2}\mathbf{r}$. Taking $n = -1$ in $\nabla \cdot (r^n \mathbf{r}) = (n+3)r^n$ gives $\nabla \cdot \hat{\mathbf{r}} = 2/r$. By the third product rule, $\nabla \times (r^n \mathbf{r}) = r^n \nabla \times \mathbf{r} + nr^{n-2} \mathbf{r} \times \mathbf{r} = \mathbf{0}$.
4. By the product rule, the x -component of the left-hand side, $\partial(fA_z)/\partial y - \partial(fA_y)/\partial z$, is $f(\partial A_z/\partial y - \partial A_y/\partial z) + (A_z \partial f/\partial y - A_y \partial f/\partial z)$, which is the x -component of the right-hand side. The other components follow cyclically.
5. By the product rule, each term of $\partial(A_y B_z - A_z B_y)/\partial x + \partial(A_z B_x - A_x B_z)/\partial y + \partial(A_x B_y - A_y B_x)/\partial z$ splits into a part in which $\mathbf{A}$ is differentiated and a part in which $\mathbf{B}$ is. The first parts add up to $B_x(\partial A_z/\partial y - \partial A_y/\partial z) + B_y(\partial A_x/\partial z - \partial A_z/\partial x) + B_z(\partial A_y/\partial x - \partial A_x/\partial y) = \mathbf{B} \cdot (\nabla \times \mathbf{A})$, and the second parts are

the same expression with $\mathbf{A}$ and $\mathbf{B}$ exchanged and the sign reversed.

6. The components are $v_x = \omega_y z - \omega_z y$, $v_y = \omega_z x - \omega_x z$, and $v_z = \omega_x y - \omega_y x$. Since v_x does not depend on x , and similarly for the others, $\nabla \cdot \mathbf{v} = 0$. The x -component of the curl is $\partial v_z / \partial y - \partial v_y / \partial z = 2\omega_x$, and the others follow cyclically.
7. The field lines satisfy $dx/x = -dy/y$, so xy is constant and they are hyperbolas with the axes as asymptotes. The divergence is $1 - 1 = 0$, and $\partial F_y / \partial x - \partial F_x / \partial y = 0$. The box is stretched along x and squeezed along y at equal rates, so its volume does not change, and neither line of fluid turns.
8. Taking the divergence of $\nabla(f\psi) = f \nabla\psi + \psi \nabla f$ with the second product rule gives $f \nabla^2\psi + \nabla f \cdot \nabla\psi + \psi \nabla^2 f + \nabla\psi \cdot \nabla f$.
9. Using the definition of the curl twice and taking mixed derivatives in the other order, the x -component of the left-hand side is $\partial(\partial A_y / \partial y + \partial A_z / \partial z) / \partial x - \partial^2 A_x / \partial y^2 - \partial^2 A_x / \partial z^2$. Adding and subtracting $\partial^2 A_x / \partial x^2$ turns it into $\partial(\nabla \cdot \mathbf{A}) / \partial x - \nabla^2 A_x$, and the other components follow cyclically.
10. By the product rule, the derivative of $\mathbf{r} \times d\mathbf{r}/dt$ is $(d\mathbf{r}/dt) \times (d\mathbf{r}/dt) + \mathbf{r} \times (d^2\mathbf{r}/dt^2)$, and both vector products vanish, the first because its factors are equal and the second because they are parallel.

§3.2 Line, Surface, and Volume Integrals

1. The tangent vector has magnitude $R(2 - 2\cos t)^{1/2} = 2R \sin(t/2)$, using $1 - \cos t = 2 \sin^2(t/2)$, and the length is $\int_0^{2\pi} 2R \sin(t/2) dt = 8R$.
2. The only component of the curl is $2y - 2y = 0$, and the path of the text gives $f = xy^2$. On the line $\mathbf{r} = t(a\hat{\mathbf{x}} + b\hat{\mathbf{y}})$ the integrand is $3t^2 ab^2$, whose integral from 0 to 1 is ab^2 . On the second path the step along the x -axis gives nothing, since $F_x = 0$ there, and the second step gives $\int_0^b 2ay dy = ab^2$. Both line integrals equal $f(a, b, 0) - f(0, 0, 0)$.
3. The field has no z -component and does not depend on z , so only the z -component of the curl can be non-zero. By the quotient

rule, $\partial F_y/\partial x$ and $\partial F_x/\partial y$ both equal $(y^2 - x^2)/(x^2 + y^2)^2$, so the z -component of the curl vanishes. On the circle the field is $(-\sin t \hat{\mathbf{x}} + \cos t \hat{\mathbf{y}})/R$ and the tangent vector is R times the same unit vector, so the integrand is 1 and the circulation is 2π .

4. Since $\nabla(r^2) = 2\mathbf{r}$, as you showed in Problem 3 of Section 3.1, $\mathbf{F} = \nabla f$ with $f = -\frac{1}{2}k_s r^2$, and the work is $f(B) - f(A) = \frac{1}{2}k_s(r_A^2 - r_B^2)$.
5. On the sphere the integrand is $R \cos \theta \cdot \cos \theta \cdot R^2 \sin \theta$, and since $\int_0^\pi \cos^2 \theta \sin \theta d\theta = \frac{2}{3}$, the flux is $4\pi R^3/3$.
6. The area is $\int_0^{2\pi} \int_0^\alpha R^2 \sin \theta d\theta d\phi = 2\pi R^2(1 - \cos \alpha)$. The cap has height $R(1 - \cos \alpha)$, and the cylinder of radius R and that height has curved area $2\pi R \cdot R(1 - \cos \alpha)$.
7. With $\mathbf{r} = x \hat{\mathbf{x}} + y \hat{\mathbf{y}} + w \hat{\mathbf{z}}$, the tangent vectors are $\hat{\mathbf{x}} + (\partial w/\partial x) \hat{\mathbf{z}}$ and $\hat{\mathbf{y}} + (\partial w/\partial y) \hat{\mathbf{z}}$, whose vector product is $-(\partial w/\partial x) \hat{\mathbf{x}} - (\partial w/\partial y) \hat{\mathbf{y}} + \hat{\mathbf{z}}$, and its magnitude gives the result. For the paraboloid the partial derivatives are x/p and y/p , and with the polar area element $s ds d\phi$ of Section 2.3 the area is $2\pi \int_0^a (1 + s^2/p^2)^{1/2} s ds$. The substitution $\eta = 1 + s^2/p^2$ gives $(2\pi p^2/3) [(1 + a^2/p^2)^{3/2} - 1]$, which the binomial expansion reduces to πa^2 when $a \ll p$.
8. The volume per unit time is $2\pi v_0 \int_0^R (1 - s^2/R^2) s ds = \pi R^2 v_0/2$, which is πR^2 multiplied by $v_0/2$.

§3.3 The Integral Theorems of Gauss, Stokes, and Green

1. The volume integral of $\nabla \cdot \mathbf{r} = 3$ is $4\pi R^3$. On the sphere $\mathbf{r} \cdot \hat{\mathbf{r}} = R$, and the flux is $R \cdot 4\pi R^2 = 4\pi R^3$.
2. On C , $\mathbf{F} \cdot d\mathbf{r}/dt = R^2 \cos^2 t$, so the circulation is πR^2 . The curl is $\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}$, and its flux through the disc is πR^2 . Through the hemisphere the x - and y -components integrate to zero over ϕ , and the z -component gives $\int_0^{2\pi} \int_0^{\pi/2} \cos \theta R^2 \sin \theta d\theta d\phi = \pi R^2$.
3. Here $x dy - y dx = 3a^2 \sin^2 t \cos^2 t dt = \frac{3}{4}a^2 \sin^2 2t dt$, whose integral from $t = 0$ to $t = 2\pi$ is $\frac{3}{4}\pi a^2$, since $\sin^2 2t$ has mean value $\frac{1}{2}$ over that range, so the area is $3\pi a^2/8$.

4. By the second product rule of Section 3.1, $\nabla \cdot (f\nabla\psi) = f\nabla^2\psi + \nabla f \cdot \nabla\psi$, which gives the first result. Subtracting the same result with f and ψ exchanged gives the second.
5. The divergence of $\mathbf{c}$ is zero, so $\mathbf{c} \cdot \oint_S d\mathbf{S} = 0$. Taking $\mathbf{c}$ to be $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ in turn shows that each component of $\oint_S d\mathbf{S}$ is zero.
6. Since $\nabla \cdot \mathbf{r} = 3$, $\oint_S \mathbf{r} \cdot d\mathbf{S}$ is three times the volume enclosed by S . On the three faces in the coordinate planes $\mathbf{r}$ is perpendicular to the normal, and the sloping face gives $a^3/2$, so the volume is $a^3/6$, one third of the base area $a^2/2$ multiplied by the height a .
7. The flux through the sphere is $4\pi r^2 E(r)$. For $r < R$ the enclosed charge is $\frac{4}{3}\pi r^3 \rho$, so $\mathbf{E} = \rho \mathbf{r}/3\varepsilon_0$, whose divergence is ρ/ε_0 . For $r > R$ the enclosed charge is $q_{\text{tot}} = \frac{4}{3}\pi R^3 \rho$, and $E = q_{\text{tot}}/(4\pi\varepsilon_0 r^2)$, a multiple of the field $\mathbf{r}/r^3$, whose divergence is zero.
8. Since ρ does not depend on position, $\nabla \cdot (\rho \mathbf{v}) = \rho \nabla \cdot \mathbf{v} = 3k\rho$, so $d\rho/dt = -3k\rho$ and $\rho = \rho_0 e^{-3kt}$, with ρ_0 the density at $t = 0$. This agrees with Section 3.1, where a small volume grows at the rate $3k$ per unit volume, as e^{3kt} , while its mass is fixed.
9. By the conservation of energy, the rate of change of the heat energy is $-\oint_S \mathbf{j} \cdot d\mathbf{S}$, which vanishes because $\mathbf{j} \cdot \hat{\mathbf{n}} = 0$ on the insulated surface.

§3.4 Curvilinear Coordinates

1. The tangent vectors are $\partial \mathbf{r}/\partial u = u\hat{\mathbf{x}} + v\hat{\mathbf{y}}$, $\partial \mathbf{r}/\partial v = -v\hat{\mathbf{x}} + u\hat{\mathbf{y}}$, and $\partial \mathbf{r}/\partial z = \hat{\mathbf{z}}$. The scalar product of the first two is $-uv + vu = 0$, and both are perpendicular to $\hat{\mathbf{z}}$, so the coordinates are orthogonal. Their vector product is $(u^2 + v^2)\hat{\mathbf{z}}$, so the numbering is in the right order. The scale factors are $h_1 = h_2 = (u^2 + v^2)^{1/2}$ and $h_3 = 1$, and the volume element is $(u^2 + v^2) du dv dz$. In the Laplacian, $h_2 h_3/h_1 = h_3 h_1/h_2 = 1$ and $h_1 h_2/h_3 = u^2 + v^2$, which does not depend on z , so dividing by $h_1 h_2 h_3 = u^2 + v^2$ gives the stated result.
2. Differentiating $\hat{\mathbf{r}}$ in components with respect to θ gives $\cos\theta \cos\phi \hat{\mathbf{x}} + \cos\theta \sin\phi \hat{\mathbf{y}} - \sin\theta \hat{\mathbf{z}} = \hat{\boldsymbol{\theta}}$, and with respect to ϕ gives $\sin\theta(-\sin\phi \hat{\mathbf{x}} + \cos\phi \hat{\mathbf{y}}) = \sin\theta \hat{\boldsymbol{\phi}}$. The components of $\hat{\mathbf{r}}$

contain no r . With $\mathbf{r} = r\hat{\mathbf{r}}$, the product rule and the chain rule give $d\mathbf{r}/dt = (dr/dt)\hat{\mathbf{r}} + r[(\partial\hat{\mathbf{r}}/\partial\theta)(d\theta/dt) + (\partial\hat{\mathbf{r}}/\partial\phi)(d\phi/dt)]$, which is the stated velocity.

3. With $f = r^n$, $r^2 \partial f / \partial r = nr^{n+1}$, whose derivative with respect to r is $n(n+1)r^n$, and dividing by r^2 gives $n(n+1)r^{n-2}$. This vanishes for $n = 0$ and $n = -1$, so the only such powers are the constant and $1/r$.
4. For $\hat{\mathbf{s}}$, $F_s = 1$ and the divergence is $(1/s) \partial s / \partial s = 1/s$. For $\hat{\phi}$, $F_\phi = 1$ does not depend on ϕ , so the divergence is zero, and the only non-zero term of the curl is $(1/s) \partial(s) / \partial s \hat{\mathbf{z}} = \hat{\mathbf{z}}/s$. In Cartesian components $\hat{\mathbf{s}} = (x\hat{\mathbf{x}} + y\hat{\mathbf{y}})/(x^2 + y^2)^{1/2}$. By the product rule $\partial[x(x^2 + y^2)^{-1/2}]/\partial x = y^2/(x^2 + y^2)^{3/2}$, the corresponding derivative with respect to y is $x^2/(x^2 + y^2)^{3/2}$, and their sum is $1/(x^2 + y^2)^{1/2} = 1/s$.
5. The only component is $F_r = G(r)$, which does not depend on θ or ϕ . In the formula for the curl every term contains F_θ , F_ϕ , or a derivative of F_r with respect to θ or ϕ , so every term vanishes. The divergence is $r^{-2} d(r^2 G)/dr$. Setting it equal to K gives $d(r^2 G)/dr = Kr^2$, so $r^2 G = Kr^3/3 + \beta$, where β is a constant, and $G = Kr/3 + \beta/r^2$. The field inside the charged ball has $K = \rho/\epsilon_0$ and $\beta = 0$, and β must be zero there because the term β/r^2 would become infinite at the centre, where there is no point charge.
6. With $f = r \cos \theta$, $\partial f / \partial r = \cos \theta$, $\partial f / \partial \theta = -r \sin \theta$, and $\partial f / \partial \phi = 0$, so $\nabla f = \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\theta}$. From the components of the unit vectors, $\hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = \cos \theta$, $\hat{\mathbf{z}} \cdot \hat{\theta} = -\sin \theta$, and $\hat{\mathbf{z}} \cdot \hat{\phi} = 0$, so this vector is $\hat{\mathbf{z}}$, as it must be, since $r \cos \theta = z$.
7. With T independent of ϕ and z , the Laplacian in cylindrical coordinates has only its first term, and the equation is $\partial T / \partial t = (D/s) \partial(s \partial T / \partial s) / \partial s$, where D is the thermal diffusivity.

Chapter 4. Linear Algebra and Vector Spaces

§4.1 Matrices, Determinants, and Linear Equations

1. The product has $\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 = \cos(\theta_1 + \theta_2)$ on the diagonal, $-\sin(\theta_1 + \theta_2)$ at the top right, and $\sin(\theta_1 + \theta_2)$ at the

bottom left, which are the entries of $R(\theta_1 + \theta_2)$. With $\theta_1 = -\theta$ and $\theta_2 = \theta$ the product is $R(0) = I$. Since the cosine is even and the sine odd, $R(-\theta)$ is the transpose of $R(\theta)$.

2. Both $R(-\theta)P$ and $PR(\theta)$ have rows $(\cos \theta, -\sin \theta)$ and $(-\sin \theta, -\cos \theta)$. Then $(PR(\theta))^2 = R(-\theta)P^2R(\theta) = I$, since $P^2 = I$. The map $PR(\theta)$ carries $(\cos \psi, \sin \psi)$ into $(\cos(\psi + \theta), -\sin(\psi + \theta))$, which is the original vector when $\psi = -\theta/2$, so it is the reflection in the line at an angle $-\theta/2$ to the x_1 -axis.
3. Both expansions multiply out to $A_{11}A_{22}A_{33} - A_{11}A_{23}A_{32} - A_{12}A_{21}A_{33} + A_{12}A_{23}A_{31} + A_{13}A_{21}A_{32} - A_{13}A_{22}A_{31}$. Each term contains one entry from each row and each column, and the set of terms is unchanged when the two indices of every entry are exchanged, which is the transpose.
4. $((AB)^T)_{ik} = \sum_j A_{kj}B_{ji} = \sum_j (B^T)_{ij}(A^T)_{jk}$. Taking the transpose of $AA^{-1} = I$ gives $(A^{-1})^T A^T = I$.
5. Expanding down the first column leaves A_{11} times the determinant of a smaller upper triangular matrix, and repeating the expansion for each smaller matrix in turn gives $A_{11}A_{22} \cdots A_{nn}$. Elimination reaches upper triangular form using row exchanges, which change the sign, and additions of multiples of rows, which change nothing, so the determinant is $\pm$ the product of the final diagonal entries.
6. Cramer's rule gives $x_1 = (ed - bf)/(ad - bc)$ and $x_2 = (af - ce)/(ad - bc)$, which is also what the two-by-two inverse gives when it multiplies (e, f) .
7. The determinant $(1 - \kappa)^2 - 1$ vanishes for $\kappa = 0$ and $\kappa = 2$. For $\kappa = 0$ both equations become $x_1 + x_2 = 0$, with solutions $\tau(1, -1)$ for any number τ , and for $\kappa = 2$ both equations become $x_2 = x_1$, with solutions $\tau(1, 1)$.

§4.2 Vector Spaces and Linear Maps

1. (a) Sums and multiples of such vectors keep $x_1 + x_2 + x_3 = 0$, so the set is a subspace. Every member is $x_1(1, -1, 0) + (x_1 + x_2)(0, 1, -1)$, and these two vectors are independent, so

- the dimension is two. (b) The zero vector is not in the set, which is not a vector space. (c) If $p(1) = q(1) = 0$ then $(\alpha p + \beta q)(1) = 0$, so the set is a subspace. Any member is $p(x) - p(1) = c_1(x - 1) + c_2(x^2 - 1) + c_3(x^3 - 1)$, with c_k the coefficient of x^k in p , and these three polynomials have different degrees and are independent, so the dimension is three. (d) The set is a subspace, as in (c). It contains the functions $x^k(L - x)$ for every $k \geq 1$, and a combination of them, $x(L - x)$ times a polynomial, vanishes on the interval only if the polynomial does, so they are independent and the space is infinite-dimensional.
- The components in the basis $1, x, x^2$ are $(1, 0, 0)$, $(1, 1, 0)$, and $(1, 1, 1)$, which form an upper triangular matrix with determinant one, so the polynomials are independent. They also span the space, since by the argument of the text any fourth polynomial makes the set dependent, and its coefficient in the relation cannot be zero. Since $x^2 = (1 + x + x^2) - (1 + x)$, its components are $(0, -1, 1)$.
 - The derivative sends $\cos x$ to $-\sin x$ and $\sin x$ to $\cos x$, so the columns of D are $(0, -1)$ and $(1, 0)$, and $D^2 = -I$. The two-by-two inverse has columns $(0, 1)$ and $(-1, 0)$, sending $\cos x$ to $\sin x$ and $\sin x$ to $-\cos x$, which are integrals; adding a non-zero constant would take them out of the space.
 - Both traces equal $\sum_i \sum_j A_{ij} B_{ji}$. Then $\text{tr}((S^{-1}A)S) = \text{tr}(S(S^{-1}A)) = \text{tr} A$.
 - The rows are $(2, 0, \frac{2}{3})$ and $(1, 1, 1)$. The conditions $2c_0 + \frac{2}{3}c_2 = 0$ and $c_0 + c_1 + c_2 = 0$ give $c_2 = -3c_0$ and $c_1 = 2c_0$, so p is a multiple of $1 + 2x - 3x^2$.
 - If $\alpha e^x + \beta e^{-x} = 0$ on an interval, its derivative $\alpha e^x - \beta e^{-x}$ vanishes too, and adding and subtracting these two equations gives $\alpha = \beta = 0$. The derivative exchanges $\cosh x$ and $\sinh x$, so its matrix A in that basis is the reflection matrix of the text, and in the basis e^x, e^{-x} it is diagonal with entries 1 and -1 . Since $e^{\pm x} = \cosh x \pm \sinh x$, the matrix S has columns $(1, 1)$ and $(1, -1)$, with $S^{-1} = \frac{1}{2}S$, and multiplying out $S^{-1}AS$ gives the diagonal matrix.

§4.3 Eigenvalues, Diagonalisation, and Normal Modes

1. The characteristic equation $(A_{11} - \lambda)(A_{22} - \lambda) - A_{12}A_{21} = 0$ expands to the stated form. Comparing it with $(\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2$ gives the sum and product. For an $n \times n$ matrix, $\det(\mathbf{A} - \lambda\mathbf{I}) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$, since the coefficient of λ^n on the left is $(-1)^n$, and setting $\lambda = 0$ gives $\det \mathbf{A} = \lambda_1 \cdots \lambda_n$.
2. For $\lambda = e^{i\theta}$ and $\sin \theta \neq 0$, the equations reduce to $x_2 = -ix_1$, so the eigenvectors are multiples of $(1, -i)$; as a check, $\mathbf{R}(\theta)$ carries $(1, -i)$ into $e^{i\theta}(1, -i)$. For $e^{-i\theta}$ they are multiples of $(1, i)$.
3. The characteristic equation is $-\lambda^3 + 2\lambda = 0$, with roots 0 and $\pm\sqrt{2}$. The normalised eigenvectors are $(1, 0, -1)/\sqrt{2}$ for 0 and $(1, \pm\sqrt{2}, 1)/2$ for $\pm\sqrt{2}$, and their scalar products are $1 - 1 = 0$ and $1 - 2 + 1 = 0$. The orthogonal matrix has these vectors as its columns.
4. With $\mathbf{S}$ having columns $(1, 1)$ and $(1, -1)$, $\mathbf{S}^{-1} = \frac{1}{2}\mathbf{S}$, and multiplying out gives diagonal entries $\frac{1}{2}[(\mu + \nu)^k + (\mu - \nu)^k]$ and off-diagonal entries $\frac{1}{2}[(\mu + \nu)^k - (\mu - \nu)^k]$.
5. The eigenvalues are $1 + \gamma$ and $1 - \gamma$, along $(1, 1)$ and $(-1, 1)$. For $\gamma > 1$ the curve $(1 + \gamma)y_1^2 - (\gamma - 1)y_2^2 = 1$ is a hyperbola crossing the first axis at $y_1 = \pm 1/\sqrt{1 + \gamma}$ and never meeting the second. For $\gamma = 1$ the equation is $(x_1 + x_2)^2 = 1$, the parallel lines $x_1 + x_2 = \pm 1$, which cross the first axis at $y_1 = \pm 1/\sqrt{2}$ and run parallel to the second.
6. The matrix, with rows $(k + k', -k')$ and $(-k', k + k')$, is the first example with $\mu = k + k'$ and $\nu = -k'$, with eigenvalues k and $k + 2k'$ along $(1, 1)$ and $(1, -1)$. The identities $\cos \omega_1 t + \cos \omega_2 t = 2 \cos \bar{\omega} t \cos \frac{1}{2} \Delta t$ and $\cos \omega_1 t - \cos \omega_2 t = 2 \sin \bar{\omega} t \sin \frac{1}{2} \Delta t$, which follow from the compound-angle formulas, give the stated forms. When $k' \ll k$, $\Delta \ll \bar{\omega}$, and each mass oscillates at $\bar{\omega}$ with a slowly changing amplitude. The amplitude of the first mass falls from d to zero in a time π/Δ while that of the second rises to d , and the motion then passes back again. This phenomenon is known as beating.

Chapter 5. Ordinary Differential Equations

§5.1 First-Order Equations

1. Separating gives $x^2 + y^2 = C$, circles centred on the origin. For the second equation, $-1/y = x - 1/y_0$, so $y = y_0/(1 - y_0x)$, which grows without limit as $x \rightarrow 1/y_0$, although the right-hand side of the equation is well behaved everywhere.
2. The real part of a complex solution solves the equation with the real part of the forcing. Substituting $T - T_0 = Ze^{i\omega t}$ gives $(\kappa + i\omega)Z = \kappa\Theta$, so $|Z| = \kappa\Theta/\sqrt{\kappa^2 + \omega^2}$ and $\arg Z = -\delta$ with $\tan \delta = \omega/\kappa$, and $T = T_0 + |Z| \cos(\omega t - \delta)$ plus a term $Ce^{-\kappa t}$, which decays.
3. Setting $dN_2/dt = 0$ gives $e^{-t/\tau_1}/\tau_1 = e^{-t/\tau_2}/\tau_2$, and taking logarithms gives $t_{\max}$. For equal lifetimes, $d(e^{t/\tau}N_2)/dt = N_0/\tau$, so $N_2 = N_0(t/\tau)e^{-t/\tau}$. In the general formula, let $\tau_2 = \tau_1 - \epsilon$. Since the derivative of $e^{-t/\tau}$ with respect to τ is $(t/\tau^2)e^{-t/\tau}$, the bracket is approximately $\epsilon(t/\tau_1^2)e^{-t/\tau_1}$, and multiplying by $N_0\tau_2/\epsilon$ and letting $\epsilon \rightarrow 0$ gives the same result.
4. Integrating the velocity gives the distance fallen, $L = (mg/b)t - (m^2g/b^2)(1 - e^{-bt/m})$. With $1 - e^{-bt/m} \approx bt/m - \frac{1}{2}(bt/m)^2$, the terms in t cancel and $L \approx \frac{1}{2}gt^2$.
5. Here $\partial P/\partial y = 2y = \partial Q/\partial x$, and $\psi = xy^2 + x^2$, so the solutions are $xy^2 + x^2 = C$. For $\mu(y)$, the condition for exactness becomes $\mu'P + \mu \partial P/\partial y = \mu \partial Q/\partial x$, so $\mu'/\mu = (\partial Q/\partial x - \partial P/\partial y)/P$. For $y dx + 2x dy = 0$ this is $1/y$, so $\mu = y$, and $y^2 dx + 2xy dy$ is the differential of xy^2 , giving $xy^2 = C$.
6. With $w = y/x$, the equation becomes $xw' = 1$, so $y = x \ln x + Cx$. As a linear equation, $y' - y/x = 1$ has the integrating factor $1/x$, so $(y/x)' = 1/x$, with the same result.
7. With $A_0 = 1/v_0 + c/b$ and $B_0 = c/b$, substituting $r = e^{-bt/m}$, for which $dt = -(m/b) dr/r$, gives the distance $(m/b) \int_0^1 dr/(A_0 - B_0r) = (m/c) \ln(A_0/(A_0 - B_0))$, and $A_0 - B_0 = 1/v_0$. For small c the logarithm is about cv_0/b , and the distance tends to mv_0/b , the integral of $v_0e^{-bt/m}$.

§5.2 Linear Equations of Higher Order

1. The auxiliary equation $(\alpha - \beta_1)(\alpha - \beta_2) = 0$ gives $y = Ae^{\beta_1 x} + Be^{\beta_2 x}$. The roots of $\alpha^4 = K^4$ are $\pm K$ and $\pm iK$, so $y = Ae^{Kx} + Be^{-Kx} + C \cos Kx + D \sin Kx$.
2. In $y_1 \dot{y}_2 - y_2 \dot{y}_1$ the terms in γ cancel, and $\cos^2 \omega_d t + \sin^2 \omega_d t = 1$ leaves $W = \omega_d e^{-2\gamma t}$. Abel's formula with $p = 2\gamma$ gives $W(0) e^{-2\gamma t}$, with $W(0) = \omega_d$.
3. With $x = Ae^{(-\gamma+\Omega)t} + Be^{(-\gamma-\Omega)t}$, the conditions $A + B = x_0$ and $(-\gamma + \Omega)A - (\gamma + \Omega)B = 0$ give the stated coefficients. The first term has the larger coefficient and decays more slowly, so it always dominates and x keeps the sign of x_0 . For the critical solution, $\dot{x} = -x_0 \gamma^2 t e^{-\gamma t}$, which vanishes at $t = 0$.
4. With $x = |Z| \cos(\omega t - \delta)$, the mean of

$$F\dot{x} = -F_0 \omega |Z| \cos \omega t \sin(\omega t - \delta) \quad (\text{S.1})$$

over a period is $\frac{1}{2} F_0 \omega |Z| \sin \delta$, since $\langle \cos \omega t \sin \omega t \rangle = 0$ and $\langle \cos^2 \omega t \rangle = \frac{1}{2}$. With $D_\omega = (\omega_0^2 - \omega^2)^2 + 4\gamma^2 \omega^2$, we have $\sin \delta = 2\gamma\omega / \sqrt{D_\omega}$ and $|Z| = F_0 / (m\sqrt{D_\omega})$, which give the stated result. The mean of $b\dot{x}^2$, with $b = 2m\gamma$, is $\frac{1}{2} b \omega^2 |Z|^2$, the same. Dividing numerator and denominator by ω^2 leaves the denominator $(\omega_0^2/\omega - \omega)^2 + 4\gamma^2$, which is least at $\omega = \omega_0$.

5. Trying $D_1 t + D_0$ gives $D_1 = F_1 / (m\omega_0^2)$ and $D_0 = -2\gamma F_1 / (m\omega_0^4)$. For the second equation $\nu = 1$ is a root of $\alpha^2 - 1 = 0$, so we try Dxe^x , whose second derivative is $D(x+2)e^x$, which gives $D = \frac{1}{2}$.
6. With $y_1 = \cos x$, $y_2 = \sin x$, and $W = 1$, we have $A' = -\tan x$ and $B' = 1$, so $y_p = \cos x \ln \cos x + x \sin x$. For the oscillator, A and B vanish at $t = 0$, and the imposed condition gives $\dot{x}_p = A\dot{y}_1 + B\dot{y}_2$, which also vanishes. For a constant force, the trial method gives $F_0/k + C_1 \cos \omega_0 t + C_2 \sin \omega_0 t$, and $x(0) = \dot{x}(0) = 0$ give $C_1 = -F_0/k$ and $C_2 = 0$.
7. Here $p = -1/x$, so $\xi' = C/x$ and $y_2 = x \ln x$. As an Euler equation, $\alpha(\alpha - 1) - \alpha + 1 = (\alpha - 1)^2 = 0$ has the repeated root 1, which gives x and $x \ln x$.
8. The roots of $\alpha^2 + K^2 = 0$ are $\pm iK$, and $x^{\pm iK} = e^{\pm iK \ln x}$, so $y = A \cos(K \ln x) + B \sin(K \ln x)$.

Chapter 6. Fourier Analysis and Integral Transforms

§6.1 Fourier Series

1. The function is even, and $a_0 = 2L^2/3$. For $n \geq 1$, integrating $\int_0^L x^2 \cos(n\pi x/L) dx$ by parts, with a boundary term that vanishes, reduces it to $-(2L/n\pi) \int_0^L x \sin(n\pi x/L) dx = 2L^3(-1)^n/n^2\pi^2$, by the integral worked out in the text. Hence $a_n = 4L^2(-1)^n/n^2\pi^2$, and $x^2 = L^2/3 + (4L^2/\pi^2) \sum_n (-1)^n \cos(n\pi x/L)/n^2$. The periodic extension is continuous at $x = L$, and $\cos n\pi = (-1)^n$, so $L^2 = L^2/3 + (4L^2/\pi^2) \sum_n 1/n^2$, which gives $\pi^2/6$. At $x = 0$, $0 = L^2/3 + (4L^2/\pi^2) \sum_n (-1)^n/n^2$, which gives $\pi^2/12$.
2. The sum is the real part of the geometric series $\sum_j e^{i(2j-1)\theta} = e^{i\theta}(1 - e^{2iN\theta})/(1 - e^{2i\theta}) = i(1 - e^{2iN\theta})/(2\sin\theta)$, whose real part is $\sin(2N\theta)/(2\sin\theta)$. Differentiating S_N term by term and setting $\theta = \pi x/L$ gives the stated derivative. For $0 < x < L$ the denominator is positive, and the numerator first vanishes at $x = L/2N$, where S_N has its maximum. There $S_N = (2/\pi) \sum_j (\sin u_j/u_j) \Delta u$, with $u_j = (2j-1)\pi/2N$ and $\Delta u = \pi/N$, which is a sum of the areas of strips of width Δu between 0 and π and tends to the stated integral. The limit is about 1.18, an overshoot of about 0.18, which is about 9 per cent of the jump of 2.
3. The left-hand side is $(1/L) \int_{-L}^L x^4 dx = 2L^4/5$, and the right-hand side is $2L^4/9 + (16L^4/\pi^4) \sum_n 1/n^4$. Hence $\sum_n 1/n^4 = (\pi^4/16)(2/5 - 2/9) = \pi^4/90$.
4. We have $c_n = (1/2L) \int_{-L}^L e^{(\beta - in\pi/L)x} dx = [e^{\beta L} e^{-in\pi} - e^{-\beta L} e^{in\pi}]/[2L(\beta - in\pi/L)]$, and $e^{\pm in\pi} = (-1)^n$ gives the stated result. At $x = L$ the periodic extension jumps from $e^{\beta L}$ to $e^{-\beta L}$, so the series converges to $\cosh \beta L$. The series at $x = L$ is $\sum_n (-1)^n c_n = \sinh(y) \sum_n 1/(y - in\pi)$, in which the term with $n = 0$ is $1/y$ and the terms with n and $-n$ add to $2y/(y^2 + n^2\pi^2)$. Setting the series equal to $\cosh y$ and dividing by $\sinh y$ gives $1/y + 2y \sum_{n \geq 1} 1/(y^2 + n^2\pi^2) = \coth y$, which rearranges to the stated result.

5. Using $\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$, $a_n = (1/\pi) \int_0^\pi \{\sin[(1 + n)x] + \sin[(1 - n)x]\} dx$. This is zero for $n = 1$, and for $n \neq 1$ it is $(1/\pi)[1 + (-1)^n]2/(1 - n^2)$, since $\cos[(1 \pm n)\pi] = -(-1)^n$. So $a_0 = 4/\pi$, $a_n = 4/[\pi(1 - n^2)]$ for even n , and $a_n = 0$ for odd n , and writing $n = 2m$ gives the stated series. At $x = 0$ the series requires $\sum_m 1/(4m^2 - 1) = \frac{1}{2}$, and with $1/(4m^2 - 1) = \frac{1}{2}[1/(2m - 1) - 1/(2m + 1)]$ the terms of the sum cancel in pairs, leaving $\frac{1}{2}$. The coefficients go as $1/m^2$ because $|\sin x|$ is continuous, and only its slope jumps.

§6.2 Fourier Transforms and Convolution

1. The transform is $(1/\sqrt{2\pi})[1/(\lambda - i\omega) + 1/(\lambda + i\omega)]$, from the integrals over $t < 0$ and $t > 0$, which is the stated result. At $t = 0$ the inversion formula gives $(\lambda/\pi) \int d\omega/(\lambda^2 + \omega^2) = (1/\pi)[\arctan(\omega/\lambda)]_{-\infty}^{\infty} = 1$, which is e^0 .
2. By the derivative property, the transforms of $\dot{x}$ and $\ddot{x}$ are $i\omega\tilde{x}$ and $-\omega^2\tilde{x}$, so the equation becomes $(\omega_0^2 - \omega^2 + 2i\gamma\omega)\tilde{x} = \tilde{F}/m$. The modulus of the bracket is $[(\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2]^{1/2}$, which is much smaller than ω_0^2 only when ω is close to $\pm\omega_0$, where it is about $2\gamma\omega_0$. The frequencies in the force near the natural frequency are therefore strongly amplified, which is the resonance found for the driven oscillator in Section 5.2.
3. The triangle is even, so its transform is $(2/\sqrt{2\pi}) \int_0^{2w} (2w - x) \cos kx dx$. Integrating by parts, the boundary term vanishes and the integral is $(1/k) \int_0^{2w} \sin kx dx = (1 - \cos 2kw)/k^2 = 2\sin^2(kw)/k^2$, which gives the stated transform. The convolution theorem gives $\sqrt{2\pi} \tilde{\Pi}^2 = \sqrt{2\pi} (2/\pi) \sin^2(kw)/k^2$, and $\sqrt{2\pi} (2/\pi) = 4/\sqrt{2\pi}$.
4. The exponent in $\int e^{-\alpha u^2} e^{-\alpha(x-u)^2} du$ is $-2\alpha(u - x/2)^2 - \alpha x^2/2$, and the Gaussian integral gives $\sqrt{\pi/2\alpha} e^{-\alpha x^2/2}$. By the convolution theorem, the transform of the convolution is $\sqrt{2\pi} e^{-k^2/2\alpha}/2\alpha$, which is $\sqrt{\pi/2\alpha}$ times the transform $e^{-k^2/2\alpha}/\sqrt{\alpha}$ of $e^{-\alpha x^2/2}$. Replacing α by $\alpha/2$ increases the half-width $1/\sqrt{2\alpha}$ by a factor of $\sqrt{2}$.
5. The left-hand side is $\int \Pi^2 dx = 2w$. The right-hand

side is $(2/\pi) \int \sin^2(kw)/k^2 dk$, which with $u = kw$ is $(2w/\pi) \int \sin^2 u/u^2 du$, and equating the left-hand and right-hand sides gives the stated result.

§6.3 The Dirac Delta Function

1. The zeros of $\sin x$ are at $x = n\pi$ for every whole number n , and at each of them $|\cos x| = 1$, so $\delta(\sin x) = \sum_n \delta(x - n\pi)$. The integral is therefore $\sum_n e^{-|n|\pi} = 1 + 2 \sum_{n \geq 1} e^{-n\pi}$. Summing the geometric series gives $1 + 2e^{-\pi}/(1 - e^{-\pi}) = (1 + e^{-\pi})/(1 - e^{-\pi})$, and multiplying the numerator and the denominator by $e^{\pi/2}$ gives $\cosh(\pi/2)/\sinh(\pi/2) = \coth(\pi/2)$.
2. The first result is the sifting property. With $x_0 = 0$ and $f(x)$ replaced by $xf(x)$, the integral $\int f(x) x \delta'(x) dx$ is minus the derivative of $xf(x)$ at $x = 0$, which by the product rule is $-f(0)$, the effect of $-\delta(x)$.
3. The first integral is $(e^{iKx} - e^{-iKx})/(2\pi i x) = \sin(Kx)/\pi x$, by Euler's formula. For $\kappa > 0$, differentiating I under the integral sign is allowed by the condition of Section 2.2, since the derivative of the integrand has modulus at most $e^{-\kappa u}$, and it gives $-\int_0^\infty e^{-\kappa u} \sin u du$. Writing $\sin u$ as the imaginary part of e^{iu} , as in Problem 8 of Section 2.2, this integral is the imaginary part of $1/(\kappa - i) = (\kappa + i)/(1 + \kappa^2)$, so $dI/d\kappa = -1/(1 + \kappa^2)$. Hence $I(\kappa) = C - \arctan \kappa$ for $\kappa > 0$, and since $|I| \leq \int_0^\infty e^{-\kappa u} du = 1/\kappa$ tends to zero as $\kappa \rightarrow \infty$, $C = \pi/2$. The value at $\kappa = 0$ follows if $I(\kappa)$ tends to $I(0)$ as $\kappa \rightarrow 0$, which is true, although the proof is beyond the scope of this course. So $\int_0^\infty (\sin u/u) du = \pi/2$, and the integral over the whole line, of an even function, is π . With $u = Kx$, $\int \sin(Kx)/(\pi x) dx = (1/\pi) \int (\sin u/u) du = 1$.
4. The square wave is constant between its jumps, and at $x = mL$ it jumps by $2(-1)^m$, which gives the stated derivative. Differentiating the series term by term gives $(4/L) \sum_{n \text{ odd}} \cos(n\pi x/L)$, and setting this equal to the derivative gives the stated result. Over the period $-L/2 < x < 3L/2$ the right-hand side contains $\frac{1}{2}L \delta(x)$ and $-\frac{1}{2}L \delta(x - L)$. Since the right-hand side is periodic, its coefficients may be found by integrating over any interval of length $2L$, and by the sifting property its coefficient a_n is $(1/L)[\frac{1}{2}L - \frac{1}{2}L \cos n\pi] = \frac{1}{2}[1 - (-1)^n]$, which is 1 for odd n and

0 for even n , including $n = 0$. Its sine coefficients vanish, since $\sin 0 = \sin n\pi = 0$. These are the coefficients of the left-hand side.

- By linearity, the displacement after the second blow is $(J/m\omega_0)\{\sin[\omega_0(t - t_0)] + \sin[\omega_0(t - t_0 - \tau)]\}$. For this to vanish at every such t , the sine produced by the second blow must cancel the sine produced by the first, which requires $\omega_0\tau$ to be an odd multiple of π . The smallest such value is $\tau = \pi/\omega_0$, half a period of the oscillation. The second blow then arrives when the mass is passing back through its equilibrium position with velocity $-J/m$, and brings it to rest.

Chapter 7. Partial Differential Equations and Boundary-Value Problems

§7.1 The Classical Equations of Physics

- With $\xi = x - vt$, the chain rule gives $\partial^2 u / \partial x^2 = f''(\xi)$ and $\partial^2 u / \partial t^2 = v^2 f''(\xi)$; for the second function $\partial^2 u / \partial x^2 = -k^2 u$ and $\partial^2 u / \partial t^2 = -v^2 k^2 u$. The identity $\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$ gives $u = \frac{1}{2} \sin k(x + vt) + \frac{1}{2} \sin k(x - vt)$.
- The heat in the element, $\rho c \pi s_w^2 \delta x T$, increases at the rate $\pi s_w^2 \kappa (\partial^2 T / \partial x^2) \delta x - 2\pi s_w \delta x \Lambda(T - T_a)$; dividing by $\rho c \pi s_w^2 \delta x$ and using $1/\rho c = D/\kappa$ gives the equation. With $\gamma_1 = 2D\Lambda/(\kappa s_w)$, the product rule gives $\partial T / \partial t = e^{-\gamma_1 t} (\partial U / \partial t - \gamma_1 U)$, while the right-hand side is $e^{-\gamma_1 t} (D \partial^2 U / \partial x^2 - \gamma_1 U)$, and the left-hand and right-hand sides are equal because U satisfies the diffusion equation.
- For a function of r alone the Laplacian is $r^{-2} d(r^2 d\Phi/dr)/dr$, which vanishes when $r^2 d\Phi/dr$ is constant, as it is for $1/r$, and so for $\Phi = \gamma_1 + \gamma_2/r$. The conditions at the spheres give $\Phi = [r_2 \Phi_2(r - r_1) - r_1 \Phi_1(r - r_2)]/[r(r_2 - r_1)]$.
- With $u_0 = 0$, $u(0, t) = (2v)^{-1} \int_{-vt}^{vt} \mathcal{U}(x') dx'$. For $vt < x_s$ this is $\mathcal{U}_s t$, and for $vt \geq x_s$ it is the constant $\mathcal{U}_s x_s / v$: the midpoint rises steadily until the disturbances from the ends of the struck part reach it.
- The difference w satisfies the diffusion equation, vanishes on S ,

and is zero at $t = 0$, and differentiating under the integral sign gives $d\mathcal{J}/dt = 2D \int_{\mathcal{R}} w \nabla^2 w \, dV$. By the product rule of the text, $w \nabla^2 w = \nabla \cdot (w \nabla w) - |\nabla w|^2$, and by the divergence theorem the first term integrates to zero, so $d\mathcal{J}/dt = -2D \int_{\mathcal{R}} |\nabla w|^2 \, dV \leq 0$. Since $\mathcal{J}(0) = 0$ and $\mathcal{J}$ is never negative, $\mathcal{J} = 0$ and $w = 0$ for all t .

6. With subscripts for partial derivatives, $d\mathcal{J}/dt = 2 \int_0^L [w_t w_{tt} + v^2 w_x w_{xt}] \, dx$. Integrating the second term by parts gives $-v^2 \int_0^L w_{xx} w_t \, dx$, the boundary term vanishing because $w_t = 0$ at the fixed ends. Therefore $d\mathcal{J}/dt = 2 \int_0^L w_t (w_{tt} - v^2 w_{xx}) \, dx = 0$ by the wave equation. At $t = 0$, w and w_t vanish everywhere, so w_x does too, and $\mathcal{J}(0) = 0$. Hence $\mathcal{J} = 0$ at all times, so w_t and w_x vanish, w is constant, and since it is zero at the ends, $w = 0$.
7. Integrating Poisson's equation over $\mathcal{R}$ and applying the divergence theorem gives the condition, whose left-hand side is minus the flux of $\mathbf{E}$ out of S .

§7.2 Separation of Variables in Cartesian Coordinates

1. Every A_n is zero, and $\omega_n B_n = (2\mathcal{U}_s/L) \int_0^L \sin(n\pi x/L) \, dx$, which is $4\mathcal{U}_s/(n\pi)$ for odd n and zero for even n . With $\omega_n = n\pi v/L$, $u = (4\mathcal{U}_s L/\pi^2 v) \sum_{n \text{ odd}} n^{-2} \sin(n\pi x/L) \sin \omega_n t$.
2. With $k_n = n\pi/L$, integrating by parts once, with a boundary term that vanishes because u_0 is zero at both ends, gives $\int_0^L u_0 \sin k_n x \, dx = k_n^{-1} \int_0^L u'_0 \cos k_n x \, dx$. The slope u'_0 is u_p/x_p on the first segment and $-u_p/(L - x_p)$ on the second, and since $\sin k_n L = 0$, the last integral is $k_n^{-1} u_p L \sin k_n x_p / [x_p (L - x_p)]$. Multiplying by $2/L$ gives the stated A_n , which for $x_p = L/2$ is $(8u_p/n^2\pi^2) \sin(n\pi/2)$. A mode with a node at x_p has $\sin(n\pi x_p/L) = 0$; the string is plucked at a point where that mode does not move.
3. Increasing t by $2L/v$ increases $\omega_n t = n\pi v t/L$ by $2n\pi$, which leaves every term of the general solution unchanged.
4. With $X = \gamma_1 \cos kx + \gamma_2 \sin kx$, the condition $X'(0) = 0$ gives $\gamma_2 = 0$, and $X'(L) = 0$ requires $\sin kL = 0$, so $k = n\pi/L$; a

zero separation constant allows $X = 1$, the case $n = 0$. The general solution is $T = \frac{1}{2}a_0 + \sum_{n \geq 1} a_n \cos(n\pi x/L) e^{-n^2\pi^2 Dt/L^2}$, with the half-range cosine coefficients of T_i , and it tends to $\frac{1}{2}a_0 = L^{-1} \int_0^L T_i dx$. No heat leaves the bar, so the total heat is constant, and it spreads until the temperature is uniform.

5. Here $T_i - T_\infty = -(T_2 - T_1)x/L$. Integrating by parts, $\int_0^L x \sin(n\pi x/L) dx = -(L^2/n\pi) \cos n\pi = (-1)^{n+1} L^2/(n\pi)$, which gives the stated b_n . At $t = 0$ the series is the half-range sine series of $-(T_2 - T_1)x/L$, which is continuous at $x = L/2$, so by Dirichlet's theorem of Section 6.1 the series converges there to $T_i(L/2) - T_\infty(L/2) = -\frac{1}{2}(T_2 - T_1)$. At $x = L/2$ only odd $n = 2j + 1$ contribute to the sum $\sum b_n \sin(n\pi x/L)$, with $(-1)^n = -1$ and $\sin(n\pi/2) = (-1)^j$, so the sum is $-[2(T_2 - T_1)/\pi] \sum_{j \geq 0} (-1)^j/(2j + 1)$. Equating the two expressions for the value of the series at $x = L/2$ gives $\sum_{j \geq 0} (-1)^j/(2j + 1) = 1 - \frac{1}{3} + \frac{1}{5} - \dots = \pi/4$.
6. The four rotated problems have the same temperature at the centre, and the sum of their solutions satisfies Laplace's equation and is T_0 on every edge, so by uniqueness it is the constant T_0 , and each has $T_0/4$ at the centre. With $y_0 = L$ and $x = y = L/2$, the series gives $T = \sum_{n \text{ odd}} (4T_0/n\pi) \sin(n\pi/2) \sinh(n\pi/2)/\sinh(n\pi)$. Using $\sinh n\pi = 2 \sinh(n\pi/2) \cosh(n\pi/2)$ and $\sin(n\pi/2) = (-1)^{(n-1)/2}$ for odd n gives the stated series, and equating it to $T_0/4$ shows that $\sum_{n \text{ odd}} (-1)^{(n-1)/2}/[n \cosh(n\pi/2)] = \pi/8$.

§7.3 Separation of Variables in Polar and Spherical Coordinates

1. Only the $m = 0$ terms are present, and fitting $T = \gamma_1 + \gamma_2 \ln s$ to the rims gives $T = T_1 + (T_2 - T_1) \ln(s/s_1)/\ln(s_2/s_1)$.
2. Since $\cos^2 \phi = \frac{1}{2}(1 + \cos 2\phi)$, $T = \frac{1}{2}T_0 + \frac{1}{2}T_0(s/s_0)^2 \cos 2\phi$, and since $s^2 \cos 2\phi = x^2 - y^2$, this is $\frac{1}{2}T_0 + T_0(x^2 - y^2)/(2s_0^2)$.
3. The polar angle ϕ is independent of s and has $\partial^2 \phi / \partial \phi^2 = 0$, so it satisfies Laplace's equation, and so does the polar angle about any other point. With ϕ_A and ϕ_B the polar angles of P about A and B, the angles PAB and PBA of the triangle APB are ϕ_A

and $\pi - \phi_B$, so $\chi = \phi_B - \phi_A$, a solution by superposition. On the upper rim $\chi = \pi/2$, the angle in a semicircle, so $T = T_0$, and on AB $\chi = \pi$, so $T = T_0/2$. By uniqueness, $T_0(\frac{3}{2} - \chi/\pi)$ is therefore the series, and since angles subtended by a chord in the same segment of a circle are equal, the points where χ is constant lie on an arc of a circle through A and B.

4. The left-hand side of Legendre's equation becomes $6\gamma_2\mu^2 + (6 + 10\gamma_1)\mu + (2\gamma_2 + 12\gamma_0)$, which vanishes only if $\gamma_2 = \gamma_0 = 0$ and $\gamma_1 = -\frac{3}{5}$. The products P_3P_0 and P_3P_2 are odd, and $\int_{-1}^1 P_3P_1 d\mu = \frac{1}{2} \int_{-1}^1 (5\mu^4 - 3\mu^2) d\mu = 0$. Finally, $\int_{-1}^1 P_3^2 d\mu = \frac{1}{4} \int_{-1}^1 (25\mu^6 - 30\mu^4 + 9\mu^2) d\mu = \frac{1}{4}(\frac{50}{7} - 12 + 6) = \frac{2}{7}$.
5. The surface potential is an odd function of μ , so for even l the numerator of α_l vanishes. For odd l the numerator is $2\Phi_0 \int_0^1 P_l d\mu$, which is Φ_0 for $l = 1$ and $-\frac{1}{4}\Phi_0$ for $l = 3$, and the denominators are $\frac{2}{3}$ and $\frac{2}{7}$, so $\alpha_1 = \frac{3}{2}\Phi_0$ and $\alpha_3 = -\frac{7}{8}\Phi_0$. Outside the sphere, $\Phi = \frac{3}{2}\Phi_0(r_0/r)^2 \cos\theta - \frac{7}{8}\Phi_0(r_0/r)^4 P_3(\cos\theta) + \dots$
6. The function combines the separated solutions $rP_1(\cos\theta)$ and $r^{-2}P_1(\cos\theta)$, the bracket vanishes at $r = r_0$, and at large r its second term is negligible, so $\Phi \rightarrow -E_0 r \cos\theta = -E_0 z$.
7. By the product and chain rules, $\partial G/\partial t = G[-1/(2t) + x^2/(4Dt^2)]$ and $\partial^2 G/\partial x^2 = G[x^2/(4D^2t^2) - 1/(2Dt)]$, or $D^{-1} \partial G/\partial t$.
8. The convolution, with the substitution $\nu' = (x' - x)/\sqrt{4Dt}$, gives $(T_0/\sqrt{\pi}) \int_{-\infty}^{-x/\sqrt{4Dt}} e^{-\nu'^2} d\nu'$. The part of the range below $\nu' = 0$ gives $\frac{1}{2}\sqrt{\pi}$, by the Gaussian integral of Section 2.3, and the rest gives $-\frac{1}{2}\sqrt{\pi} \operatorname{erf}(x/\sqrt{4Dt})$, since the integrand is even; $\operatorname{erf}(0) = 0$.

Chapter 8. Probability, Statistics, and Stochastic Methods

§8.1 Probability and Random Variables

1. Bayes's theorem gives $\eta_1^2 r / [\eta_1^2 r + \eta_0^2 (1-r)] = 1/[1 + (\eta_0/\eta_1)^2 (1-r)/r]$, while $P(R|G) = 1/[1 + (\eta_0/\eta_1)(1-r)/r]$. When $\eta_0/\eta_1 < 1$ its square is smaller, so the probability is larger.

2. The event A is made up of the mutually exclusive events $A \cap B$ and $A \cap B'$, so $P(A \cap B') = P(A) - P(A)P(B) = P(A)P(B')$.
3. With the substitution $u = \gamma t$, $\langle t^n \rangle = \gamma^{-n} \int_0^\infty u^n e^{-u} du = n!/\gamma^n$, by the integral $J_n = n!$ found in Section 2.1, and the variance is $2/\gamma^2 - 1/\gamma^2 = 1/\gamma^2$. With $\mu = 1/\gamma$, the third central moment $\langle t^3 \rangle - 3\mu\langle t^2 \rangle + 3\mu^2\langle t \rangle - \mu^3$ is $(6 - 6 + 3 - 1)/\gamma^3 = 2\sigma^3$. The conditional probability of no decay by $t_0 + \tau$, given none by t_0 , is $e^{-\gamma(t_0+\tau)}/e^{-\gamma t_0} = e^{-\gamma\tau}$, which does not depend on t_0 .
4. Each x_i has mean q , and since $x_i^2 = x_i$, its variance is $q - q^2$. The means add to nq , and since the trials are independent the variances add to $nq(1 - q)$.
5. The total is k when the sources give j and $k - j$, so $P(k) = \sum_{j=0}^k [\mu_1^j e^{-\mu_1}/j!] [\mu_2^{k-j} e^{-\mu_2}/(k-j)!] = (e^{-(\mu_1+\mu_2)}/k!) \sum_j \binom{k}{j} \mu_1^j \mu_2^{k-j}$, and by the binomial theorem the sum is $(\mu_1 + \mu_2)^k$.
6. The density of x is $1/2a$. For $0 < y < a^2$ the values $x = \pm\sqrt{y}$ each contribute $(1/2a)(1/2\sqrt{y})$, so $p_y(y) = 1/(2a\sqrt{y})$, whose integral from 0 to a^2 is 1. The mean is $\int_0^{a^2} \sqrt{y}/(2a) dy = a^2/3$.
7. For a given x , the sum lies between ζ and $\zeta + d\zeta$ when y lies between $\zeta - x$ and $\zeta - x + d\zeta$, which has probability $p_2(\zeta - x) d\zeta$; multiplying by $p_1(x) dx$ and integrating over x gives the result. For the uniform densities the integrand is 1 where $0 < x < 1$ and $\zeta - 1 < x < \zeta$, and the integral is the length of that overlap, which is ζ or $2 - \zeta$. The variance is $\frac{1}{6}$, twice $\frac{1}{12}$, so $z = (\zeta - 1)\sqrt{6}$, and the density of z is $(1 - |z|/\sqrt{6})/\sqrt{6}$ for $|z| < \sqrt{6}$, the triangle in the figure.
8. With $u = (x - \mu)/(\sqrt{2}\sigma)$, the moment is $(4\sigma^4/\sqrt{\pi}) \int u^4 e^{-u^2} du$, and $\int u^4 e^{-u^2} du = \frac{3}{4}\sqrt{\pi}$, as you showed in Problem 7 of Section 2.3, so the moment is $3\sigma^4$. The variance of $(x - \mu)^2$ is then $3\sigma^4 - \sigma^4 = 2\sigma^4$.

§8.2 Estimation and Hypothesis Testing

1. The derivative of $\sum_i (x_i - \xi)^2$ with respect to ξ is $-2\sum_i (x_i - \xi)$, which vanishes at $\xi = \bar{x}$, and the second derivative, $2N$, is

- positive. Setting $\xi = \mu$ gives a value at least as large as this minimum.
- The kinetic energy is $E = \frac{1}{2}M(2h/t)^2 = 2Mh^2t^{-2}$, so $(\sigma_E/E)^2 = (\sigma_M/M)^2 + 4(\sigma_h/h)^2 + 4(\sigma_t/t)^2$.
 - Let $x = \mu + v$, where v has mean zero and variance σ^2 . Then $\langle x^2 \rangle = \mu^2 + \sigma^2$. The odd moments of v vanish, and $\langle v^4 \rangle = 3\sigma^4$ by Problem 8 of Section 8.1, so $\langle x^4 \rangle = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$, and subtracting $\langle x^2 \rangle^2$ gives $4\mu^2\sigma^2 + 2\sigma^4$. The first-order formula gives μ^2 for the mean of x^2 and $4\mu^2\sigma^2$ for its variance, and the corrections are small when $\sigma \ll |\mu|$, which is the condition for the first-order formula to be a valid approximation.
 - Here $\ln \mathcal{L} = \sum_i k_i \ln \mu - N\mu - \sum_i \ln k_i!$, whose derivative $\sum_i k_i/\mu - N$ vanishes at $\hat{\mu} = \sum_i k_i/N$.
 - Here $\ln \mathcal{L} = N \ln \gamma - \gamma \sum_i t_i$, whose derivative $N/\gamma - \sum_i t_i$ vanishes at $\hat{\gamma} = N/\sum_i t_i$, the reciprocal of the mean decay time.
 - The derivative of $\sum_i (y_i - mx_i)^2$ with respect to m vanishes when $m = \sum_i x_i y_i / \sum_i x_i^2$. This is a linear combination of the y_i with coefficients $x_i / \sum_j x_j^2$, so $\sigma_m^2 = \sigma^2 / \sum_i x_i^2$. A line constrained to pass through the origin is the appropriate fit for the pendulum, whose line $T^2 = (4\pi^2/g)L$ passes through the origin.
 - With $c = \bar{y} - m\bar{x}$, the residuals are $d_i = (y_i - \bar{y}) - m(x_i - \bar{x})$, and since $\sum_i (x_i - \bar{x})(y_i - \bar{y}) = mD$ by the formula for the slope, $Q_{\min} = \sum_i (y_i - \bar{y})^2 - m^2 D$. Now $y_i - \bar{y} = m_0(x_i - \bar{x}) + (e_i - \bar{e})$, where $\bar{e}$ is the mean error, so by the result for the sample variance $\langle \sum_i (y_i - \bar{y})^2 \rangle = m_0^2 D + (N-1)\sigma^2$. The slope is $m = m_0 + \sum_i (x_i - \bar{x})e_i/D$, so $\langle m^2 \rangle = m_0^2 + \sigma^2/D$, and therefore $\langle Q_{\min} \rangle = m_0^2 D + (N-1)\sigma^2 - (m_0^2 + \sigma^2/D)D = (N-2)\sigma^2$.
 - With $w_1 = 1/\sigma_1^2$ and $w_2 = 1/\sigma_2^2$, the weighted mean is $\hat{\mu} = (w_1 x_1 + w_2 x_2)/(w_1 + w_2)$, with standard deviation $(w_1 + w_2)^{-1/2} = \sigma_1 \sigma_2 / (\sigma_1^2 + \sigma_2^2)^{1/2}$. Since $x_1 - \hat{\mu} = w_2(x_1 - x_2)/(w_1 + w_2)$ and $x_2 - \hat{\mu} = -w_1(x_1 - x_2)/(w_1 + w_2)$, we find $\chi^2 = (x_1 - x_2)^2 w_1 w_2 / (w_1 + w_2) = (x_1 - x_2)^2 / (\sigma_1^2 + \sigma_2^2)$. There are two measurements and one fitted parameter, so $\nu = 1$.

Chapter 9. Numerical and Computational Mathematics

§9.1 Floating-Point Arithmetic and Numerical Stability

1. For a fixed exponent E , the mantissa takes the 2^{t-1} values $1 + j2^{1-t}$, for $j = 0, 1, \dots, 2^{t-1} - 1$, which are equally spaced by $2^{1-t} = \epsilon_m$ and lie in $1 \leq m < 2$. Multiplying by 2^E gives numbers in $2^E \leq x < 2^{E+1}$ spaced by $2^E \epsilon_m$. Since 2^E lies between $\frac{1}{2}|x|$ and $|x|$, the spacing lies between $\frac{1}{2}\epsilon_m|x|$ and $\epsilon_m|x|$.
2. The number $1 + \frac{1}{2}\epsilon_m$ lies exactly halfway between 1, whose last digit is 0, and $1 + \epsilon_m$, whose last digit is 1, so it is rounded to 1. Adding $\frac{1}{2}\epsilon_m$ to this result rounds to 1 again. In the other order, $\frac{1}{2}\epsilon_m + \frac{1}{2}\epsilon_m = \epsilon_m$ exactly, and $1 + \epsilon_m$ is a floating-point number, so the result is $1 + \epsilon_m$.
3. For e^x , $\kappa = |xe^x/e^x| = |x|$, which is large only for large $|x|$. For $\ln x$, $\kappa = |x(1/x)/\ln x| = 1/|\ln x|$, which is large near $x = 1$, where the logarithm is close to zero. For $\sin x$, $\kappa = |x \cos x / \sin x|$, which is large near the zeros $x = n\pi$ of the sine with $n \neq 0$; near $x = 0$ it tends to one.
4. Multiplying $\sqrt{x+1} - \sqrt{x}$ by $(\sqrt{x+1} + \sqrt{x})/(\sqrt{x+1} + \sqrt{x})$ and using the difference of two squares gives the first identity, and the second is the double-angle formula $\cos x = 1 - 2\sin^2(x/2)$. For large x the square roots on the left are nearly equal, and for small x the cosine is close to 1, so each left-hand side suffers subtractive cancellation, while each right-hand side involves only additions, multiplications, and a division of numbers that are not nearly equal.
5. Since $x^n + \Omega x^{n-1} = x^{n-1}(x + \Omega)$, adding Y_n and ΩY_{n-1} gives $\int_0^1 x^{n-1} dx = 1/n$. An error Δ_0 in Y_0 becomes $(-\Omega)^n \Delta_0$ in Y_n , which grows when $\Omega > 1$, while the Y_n themselves lie between $1/[(n+1)(1+\Omega)]$ and $1/[(n+1)\Omega]$, because $1/(x+\Omega)$ lies between $1/(1+\Omega)$ and $1/\Omega$ on the interval. The recurrence should be run backwards, as $Y_{n-1} = (1/n - Y_n)/\Omega$, which divides the error by Ω at each step.
6. The starting error is at most $1/(n_0 + 1)$, and it is divided by $n_0(n_0 - 1) \cdots (n + 1)$, a product of k factors each at least $n + 1$,

which gives the stated bound.

§9.2 Root Finding, Interpolation, and Linear Systems

1. After n steps the midpoint is within $(v_0 - u_0)/2^{n+1}$ of the root, which is at most τ when $2^{n+1} \geq (v_0 - u_0)/\tau$.
2. By the product rule, $f' = (x - x_*)^{m-1}[m\psi + (x - x_*)\psi']$, so $f(x_n)/f'(x_n) = e_n\psi/(m\psi + e_n\psi')$, which is e_n/m to first order, and $e_{n+1} = e_n - e_n/m$. For the modified iteration, $e_{n+1} = e_n - me_n\psi/(m\psi + e_n\psi') = e_n^2\psi'/(m\psi + e_n\psi')$, which is of order e_n^2 .
3. Since $f' = -1/x^2$, $x_{n+1} = x_n + x_n^2(1/x_n - C) = x_n(2 - Cx_n)$, and $1 - Cx_{n+1} = 1 - 2Cx_n + C^2x_n^2 = (1 - Cx_n)^2$. Hence $1 - Cx_n = (1 - Cx_0)^{2^n}$, which tends to zero if and only if $|1 - Cx_0| < 1$, that is, for $C > 0$, if and only if $0 < x_0 < 2/C$.
4. Subtracting $\sqrt{C}$ from Heron's formula gives $e_{n+1} = (x_n - \sqrt{C})^2/2x_n$, which is never negative when $x_n > 0$, so every iterate after the first is at least $\sqrt{C}$.
5. The interpolating polynomial of the constant function 1 is the function itself, by uniqueness, and with every $y_j = 1$ the Lagrange form gives $\sum_j L_j(x) = 1$.
6. With $N = 1$, the error is $\frac{1}{2}f''(\xi)(x - x_0)(x - x_1)$, and $|(x - x_0)(x - x_1)|$ is largest at the midpoint, where it equals $h^2/4$.
7. The multiplier is $l_{21} = A_{21}/A_{11}$, so L has rows $(1, 0)$ and $(l_{21}, 1)$, and U has rows (A_{11}, A_{12}) and $(0, A_{22} - l_{21}A_{12})$. Multiplying them returns A , and $u_{11}u_{22} = A_{11}A_{22} - A_{12}A_{21}$.
8. The matrix G has rows $(0, -A_{12}/A_{11})$ and $(-A_{21}/A_{22}, 0)$, so its characteristic equation is $\lambda^2 = \Lambda$, and multiplying G by itself gives $G^2 = \Lambda I$. In a Gauss–Seidel sweep the new errors, marked with primes, are $e'_1 = -(A_{12}/A_{11})e_2$ and $e'_2 = -(A_{21}/A_{22})e'_1 = \Lambda e_2$. One Gauss–Seidel sweep therefore multiplies the error by Λ , as two Jacobi sweeps do.
9. The error equation gives $e_i^{(r+1)} = -\sum_{j \neq i} (A_{ij}/A_{ii})e_j^{(r)}$, so $|e_i^{(r+1)}| \leq \sum_{j \neq i} |A_{ij}/A_{ii}| e_{\max}^{(r)} \leq \Gamma e_{\max}^{(r)}$ for every i . Hence $e_{\max}^{(r)} \leq \Gamma^r e_{\max}^{(0)}$, which tends to zero since $\Gamma < 1$.

§9.3 Numerical Differentiation and Quadrature

1. Taylor's theorem with the Lagrange remainder gives $f(x-h) = f(x) - hf'(x) + \frac{1}{2}h^2 f''(\xi)$. Rearranging and dividing by h gives the result.
2. Expanding to fourth order, $f(x \pm h) = f \pm hf' + \frac{1}{2}h^2 f'' \pm \frac{1}{6}h^3 f''' + \frac{1}{24}h^4 f''''(\xi_{\pm})$. Adding the expansions of $f(x+h)$ and $f(x-h)$ removes the odd terms, so $f(x+h) + f(x-h) - 2f(x) = h^2 f'' + \frac{1}{24}h^4[f''''(\xi_+) + f''''(\xi_-)]$. Dividing by h^2 and using the intermediate value theorem on the mean of the fourth derivatives gives the result.
3. The bound is $\mathcal{E}(h) = \frac{1}{2}h|f''| + 2\epsilon_m|f|/h$. Setting $d\mathcal{E}/dh = \frac{1}{2}|f''| - 2\epsilon_m|f|/h^2 = 0$ gives $h^2 = 4\epsilon_m|f|/|f''|$. At this step each term of $\mathcal{E}$ equals $(\epsilon_m|ff''|)^{1/2}$, so the smallest bound is twice that value.
4. For a linear function the straight line on each strip is the function itself. For $f = x^2$, the Taylor series about z stops at the second derivative, $f'' = 2$, so the single-strip error $-\frac{1}{12}h^3 f''$ is exact and equals $-\frac{1}{6}h^3$. Adding the N strips gives $-\frac{1}{6}Nh^3 = -\frac{1}{6}h^2(x_N - x_0)$. The leading-order formula gives $-\frac{1}{12}h^2[2x_N - 2x_0]$, which is the same.
5. The integral over the strip is $hf(z) + \frac{1}{24}h^3 f''(z) + \dots$, and subtracting $hf(z)$ gives the error. The trapezium error is $-\frac{1}{12}h^3 f''$, which is minus twice the midpoint error, so in $\frac{1}{3}[T + 2hf(z)]$ the terms in h^3 cancel. Writing it out, $\frac{1}{3}\{\frac{1}{2}h[f(z-\frac{1}{2}h) + f(z+\frac{1}{2}h)] + 2hf(z)\} = \frac{1}{6}h[f(z-\frac{1}{2}h) + 4f(z) + f(z+\frac{1}{2}h)]$, which is Simpson's formula with step $\frac{1}{2}h$.
6. The exact integral is $\frac{2}{5}h^5$, and Simpson's formula gives $\frac{1}{3}h(h^4 + 0 + h^4) = \frac{2}{3}h^5$. The difference is $\frac{2}{5}h^5 - \frac{2}{3}h^5 = -\frac{4}{15}h^5$, and with $f'''' = 24$ the formula gives $-\frac{24}{96}h^5 = -\frac{4}{15}h^5$, which agrees exactly, because the Taylor series of x^4 stops at the fourth derivative.
7. It suffices to check one strip of width h from x_j to x_{j+1} , with midpoint $x_{j+1/2}$. There $T(h) = \frac{1}{2}h(f_j + f_{j+1})$ and $T(h/2) = \frac{1}{4}h(f_j + 2f_{j+1/2} + f_{j+1})$, so $4T(h/2) - T(h) = \frac{1}{2}h(f_j + 4f_{j+1/2} + f_{j+1})$. Dividing by 3 gives $\frac{1}{3}(\frac{1}{2}h)(f_j + 4f_{j+1/2} + f_{j+1})$, which is Simpson's formula with step $\frac{1}{2}h$ on this pair of half-strips.